

Topological Energy Condensate Theory (TECT): 19

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(Dated: March, 2026)

I. NUMERICAL CONTINUATION OF THE REDUCED FLAVOR TOY MODEL

We continued the reduced pre-EWSB flavor analysis at the purely numerical level, keeping the common family-space logic

$$\mathcal{H}_{\text{fam}} \cong \mathbb{C}^3, \quad V_{\text{CKM}} = U_u^\dagger U_d,$$

and asking only the following concrete question:

Can the current reduced two-texture toy ansatz produce CKM-like mixing, in particular the Cabibbo angle?

Since the original exploratory scripts were not available in the present session, we reconstructed a minimal midpoint-overlap toy model consistent with the master contents. Let $X_a \in \mathbb{R}^3$ ($a = 1, 2, 3$) be the family centers, let σ be the Gaussian width, and define the overlap factor

$$S_{ab} = \exp\left(-\frac{|X_a - X_b|^2}{4\sigma^2}\right), \quad m_{ab} := \frac{X_a + X_b}{2}.$$

We used the radial defect profile factor

$$f_R(x) = \frac{4R^2}{(|x|^2 + R^2)^2},$$

and the reconstructed kernels

$$(B_0)_{ab} = S_{ab} f_R(m_{ab}), \tag{1}$$

$$(B_1)_{ab} = \kappa_Y S_{ab} f_R(m_{ab}) \exp\left(-\frac{|m_{ab} - C_Y|^2}{2\ell_Y^2}\right), \tag{2}$$

$$(B_H)_{ab} = \kappa_H S_{ab} f_R(m_{ab}) \exp\left(-\frac{|m_{ab} - C_H|^2}{2\ell_H^2}\right). \tag{3}$$

Thus

$$H_u = B_0 - \frac{2}{3}B_1, \quad H_d = B_0 + \frac{1}{3}B_1, \quad M_u = H_u + B_H, \quad M_d = H_d - B_H.$$

We then diagonalized

$$U_u^\dagger M_u U_u = \text{diag}(m_u, m_c, m_t), \quad U_d^\dagger M_d U_d = \text{diag}(m_d, m_s, m_b),$$

and read off

$$V_{\text{CKM}} = U_u^\dagger U_d.$$

The numerical target values were taken from the 2025 PDG global-fit magnitudes:

$$|V_{us}| = 0.22501, \quad |V_{cb}| = 0.04183, \quad |V_{ub}| = 0.003732.$$

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A constrained random-plus-refinement search produced the best reconstructed fit

$$|V_{\text{CKM}}|_{\text{toy}} \approx \begin{pmatrix} 0.97373 & 0.22770 & 0.00312 \\ 0.22762 & 0.97281 & 0.04269 \\ 0.00669 & 0.04228 & 0.99908 \end{pmatrix}.$$

Hence

$$|V_{us}|_{\text{toy}} = 0.22770, \quad |V_{cb}|_{\text{toy}} = 0.04269, \quad |V_{ub}|_{\text{toy}} = 0.00312.$$

This has the following immediate consequences.

First,

$$|V_{us}|_{\text{toy}}$$

reaches the observed Cabibbo scale. Therefore the current two-texture reduced ansatz is *not* structurally incapable of producing the Cabibbo angle.

Second, the full CKM hierarchy

$$|V_{us}| \gg |V_{cb}| \gg |V_{ub}|$$

is reproduced numerically in the reduced toy model.

Third, the best-fit matrices are genuinely misaligned. For the best point,

$$B_0 \approx \begin{pmatrix} 0.02479 & 0.06702 & 0.12594 \\ 0.06702 & 0.09241 & 0.29356 \\ 0.12594 & 0.29356 & 0.49214 \end{pmatrix},$$

$$B_1 \approx \begin{pmatrix} 1.19 \times 10^{-10} & 5.41 \times 10^{-6} & 5.89 \times 10^{-6} \\ 5.41 \times 10^{-6} & 1.03 \times 10^{-3} & 6.54 \times 10^{-3} \\ 5.89 \times 10^{-6} & 6.54 \times 10^{-3} & 5.20 \times 10^{-3} \end{pmatrix},$$

$$B_H \approx \begin{pmatrix} 0.00513 & 0.01234 & 0.00942 \\ 0.01234 & 0.00026 & 0.00095 \\ 0.00942 & 0.00095 & 0.00055 \end{pmatrix},$$

with

$$\|[B_0, B_1]\|_F \approx 2.79 \times 10^{-3}, \quad \|[M_u, M_d]\|_F \approx 2.83 \times 10^{-2}.$$

So the CKM-like mixing is produced by genuine noncommutativity, not by a numerical artifact of nearly degenerate matrices.

However, the current ansatz still has two clear limitations.

First, although $|V_{us}|$ and $|V_{cb}|$ can be matched well, $|V_{ub}|$ tends to come out slightly too small. This indicates that the present two-texture ansatz has enough freedom to generate the dominant 1–2 and 2–3 mixings, but likely not enough independent control over the 1–3 sector.

Second, in the present reconstruction all matrices are real symmetric, so the model can fit only CKM magnitudes, not the physical Dirac CP phase. A genuinely complex texture is required for that.

Therefore the honest status is:

the current reduced two-texture toy model is sufficient to produce Cabibbo-scale mixing and the CKM hierarchy,

but

it is not yet sufficient for a full precision CKM fit with CP violation.

The most economical next enlargement is to keep the same family-space reduction but replace the single B_H texture by either

$$B_H \rightarrow B_H^{(1)} + B_H^{(2)},$$

with two displaced Higgs-like kernels, or

$$B_1 \rightarrow B_1^{(1)} + e^{i\phi} B_1^{(2)},$$

so that the 1–3 sector and the physical CKM phase can be adjusted independently.

II. FINAL SUMMARY

The reduced TECT flavor problem has now passed a nontrivial numerical test. Using a reconstructed midpoint-overlap toy realization of

$$M_u = \left(B_0 - \frac{2}{3}B_1\right) + B_H, \quad M_d = \left(B_0 + \frac{1}{3}B_1\right) - B_H, \quad V_{\text{CKM}} = U_u^\dagger U_d,$$

we find explicit parameter points with

$$|V_{us}| \approx 0.228, \quad |V_{cb}| \approx 0.0427, \quad |V_{ub}| \approx 0.0031,$$

which is already close to the observed CKM hierarchy.

Therefore the present two-texture reduced ansatz is sufficient to generate:

1. nontrivial quark-sector misalignment,
2. Cabibbo-scale mixing,
3. the qualitative CKM hierarchy

$$|V_{us}| \gg |V_{cb}| \gg |V_{ub}|.$$

So the statement

“the current toy ansatz cannot generate Cabibbo”

is false.

The honest remaining limitations are:

1. $|V_{ub}|$ tends to be slightly underestimated;
2. the present real-symmetric reconstruction does not generate a physical CKM CP phase.

Hence the best current conclusion is

two-texture TECT flavor reduction is sufficient at the hierarchy/Cabibbo level,

but

an enlarged complex texture is likely needed for full CKM precision and CP violation.

III. INTERPRETATION OF THE CURRENT CKM REFINEMENT RUN

We analyze the current best point obtained from the exploratory refinement fitter:

$$|V_{us}| = 0.208440, \quad |V_{cb}| = 0.045425, \quad |V_{ub}| = 0.003730,$$

with target values

$$|V_{us}|^{\text{tar}} = 0.225000, \quad |V_{cb}|^{\text{tar}} = 0.041000, \quad |V_{ub}|^{\text{tar}} = 0.003500.$$

The deviations are

$$\Delta|V_{us}| = -0.016560, \quad \Delta|V_{cb}| = +0.004425, \quad \Delta|V_{ub}| = +0.000230,$$

or in relative form

$$\frac{\Delta|V_{us}|}{|V_{us}|^{\text{tar}}} \approx -7.36\%, \quad \frac{\Delta|V_{cb}|}{|V_{cb}|^{\text{tar}}} \approx +10.79\%, \quad \frac{\Delta|V_{ub}|}{|V_{ub}|^{\text{tar}}} \approx +6.58\%.$$

Thus the present best point already reproduces the qualitative CKM hierarchy, but the dominant remaining mismatch is clearly the Cabibbo entry:

$|V_{us}|$ is still too small.

The full absolute CKM matrix is

$$|V_{\text{CKM}}| = \begin{pmatrix} 0.978028 & 0.20844021 & 0.00373027 \\ 0.20806076 & 0.97706053 & 0.04542511 \\ 0.01311312 & 0.04365091 & 0.99896078 \end{pmatrix}.$$

This shows that the model is far from the trivial identity limit and already produces a realistic hierarchical mixing pattern.

The misalignment diagnostics are

$$\|[B_0, B_1]\|_F = 2.76446644 \times 10^{-2}, \quad \|[M_u, M_d]\|_F = 3.62498081 \times 10^{-2},$$

while the numerical unitarity error is

$$\|V^\dagger V - \mathbf{1}\| \sim 5.62 \times 10^{-16}.$$

Therefore the mixing is genuine and numerically stable: it is produced by actual noncommutativity of the sectoral family operators, not by diagonalization noise.

The family centers are

$$X_1 = -1.42549, \quad X_2 = -0.28092, \quad X_3 = 2.85749, \quad \sigma = 1.488486.$$

Hence the pairwise separations are

$$d_{12} \approx 1.14457, \quad d_{23} \approx 3.13841, \quad d_{13} \approx 4.28298.$$

Using the Gaussian overlap estimate

$$S_{ab} \sim \exp\left(-\frac{d_{ab}^2}{4\sigma^2}\right),$$

one finds approximately

$$S_{12} \approx 0.863, \quad S_{23} \approx 0.329, \quad S_{13} \approx 0.126.$$

This is exactly the geometric pattern expected for hierarchical flavor mixing:

$$1-2 \text{ overlap is strong}, \quad 2-3 \text{ overlap is moderate}, \quad 1-3 \text{ overlap is suppressed}.$$

Hence the current best point strongly supports the interpretation that the reduced TECT flavor ansatz naturally produces CKM-like hierarchy from family-space localization plus sectoral misalignment.

The sectoral eigenvalues are

$$m_u^{\text{toy}} \sim (0.01922, 0.10373, 0.80661), \quad m_d^{\text{toy}} \sim (0.01327, 0.06098, 0.85649).$$

Although these are not yet directly meaningful as physical quark masses, they do show that the toy model already generates a nondegenerate hierarchical spectrum.

The honest interpretation is therefore:

the current reduced ansatz successfully produces nontrivial hierarchical CKM-like mixing,

but

it still does not achieve a precise simultaneous fit, mainly because $|V_{us}|$ remains too small.

A plausible structural interpretation is that the present texture family does not yet have enough independent freedom to increase the 1–2 mixing angle without simultaneously distorting the 2–3 sector. However, this should be treated as a hypothesis suggested by the present point, not yet as a theorem.

Therefore the immediate next numerical task should not be another blind global run. Instead, one should perform a local sensitivity audit around the current best point:

1. freeze the family geometry (X_a, σ, R) near the present optimum;
2. vary only the texture parameters $(\epsilon, \delta, \text{amps}_H, \text{phases}_H)$;
3. measure

$$\frac{\partial |V_{us}|}{\partial p_i}, \quad \frac{\partial |V_{cb}|}{\partial p_i}, \quad \frac{\partial |V_{ub}|}{\partial p_i};$$

4. identify whether there exists a local direction in parameter space that increases $|V_{us}|$ while leaving $|V_{cb}|$ nearly fixed.

If no such direction exists in a robust neighborhood of the present optimum, then the correct conclusion will be that the current ansatz is too rigid and must be enlarged, for example by adding:

$$B_H \rightarrow B_H^{(1)} + B_H^{(2)} + B_H^{(3)},$$

or by allowing a genuinely complex hypercharge-sensitive texture

$$B_1 \rightarrow B_1^{(1)} + e^{i\phi} B_1^{(2)}.$$

Such an enlargement would provide additional control over the 1–2 and 2–3 sectors and, eventually, over the physical CKM phase as well.

IV. FINAL SUMMARY

The current refinement run gives

$$|V_{us}| = 0.208440, \quad |V_{cb}| = 0.045425, \quad |V_{ub}| = 0.003730,$$

with excellent numerical unitarity and clearly nonzero commutator diagnostics:

$$\|[B_0, B_1]\|_F > 0, \quad \|[M_u, M_d]\|_F > 0.$$

Therefore the reduced TECT flavor model has passed the following nontrivial test:

it genuinely produces hierarchical CKM-like mixing from family-space misalignment.

What is not yet achieved is a precise simultaneous fit. The dominant remaining problem is

$|V_{us}|$ is still too small.

The current best-fit geometry has strong 1–2 overlap, moderate 2–3 overlap, and suppressed 1–3 overlap, which explains why the CKM hierarchy already emerges naturally.

Hence the honest current status is:

hierarchy-level success is real,

precision-level Cabibbo fitting remains open.

The immediate next task is a local sensitivity audit around the present best point. If $|V_{us}|$ cannot be increased locally without spoiling $|V_{cb}|$, then the present ansatz should be enlarged by an additional independent texture mode.

V. COMPLEX-TEXTURE EXTENSION FOR CKM PHASE GENERATION

We now enlarge the reduced flavor toy ansatz by adding a single complex Hermitian texture mode. The goal is to test whether the reduced TECT flavor model can generate not only CKM-like magnitudes but also a nontrivial CP-violating phase.

We keep the previous real textures and replace

$$B_1 \longrightarrow B_1^{(\text{real})} + B_1^{(\text{cx})},$$

where the added complex mode is taken to be

$$\boxed{(B_1^{(\text{cx})})_{ab} = A_{ab} \left[\cos(q_{\text{cp}}(x_a - x_b)) + i \chi_{\text{cp}} \sin(q_{\text{cp}}(x_a - x_b)) \right]}.$$

Here A_{ab} is a real symmetric envelope,

$$A_{ab} = \epsilon_{\text{cp}} S_{ab} f_R(m_{ab}) \exp\left(-\frac{(m_{ab} - c_{\text{cp}})^2}{2\ell_{\text{cp}}^2}\right),$$

with S_{ab} the Gaussian family overlap factor and m_{ab} the midpoint coordinate. Because the cosine term is even and the sine term is odd under $a \leftrightarrow b$, one has

$$(B_1^{(\text{cx})})_{ba} = \overline{(B_1^{(\text{cx})})_{ab}},$$

so $B_1^{(\text{cx})}$ is exactly Hermitian.

The quark-sector matrices are still defined by

$$M_u = \left(B_0 - \frac{2}{3}B_1\right) + B_H, \quad M_d = \left(B_0 + \frac{1}{3}B_1\right) - B_H,$$

and are therefore Hermitian as well. We diagonalize them as

$$U_u^\dagger M_u U_u = \text{diag}(m_u, m_c, m_t), \quad U_d^\dagger M_d U_d = \text{diag}(m_d, m_s, m_b),$$

and define

$$V_{\text{CKM}} = U_u^\dagger U_d.$$

To quantify CP violation we use the Jarlskog invariant

$$J = \text{Im}(V_{ud}V_{cs}V_{us}^*V_{cd}^*),$$

and reconstruct the standard-phase quantity δ from

$$J = c_{12}c_{23}c_{13}^2 s_{12}s_{23}s_{13} \sin \delta,$$

with

$$s_{13} = |V_{ub}|, \quad s_{12} = \frac{|V_{us}|}{\sqrt{1 - |V_{ub}|^2}}, \quad s_{23} = \frac{|V_{cb}|}{\sqrt{1 - |V_{ub}|^2}}.$$

The default numerical target is taken from the PDG 2025 global-fit CKM review:

$$|V_{us}| = 0.22501, \quad |V_{cb}| = 0.04183, \quad |V_{ub}| = 0.003732,$$

$$|V_{td}| = 0.00858, \quad |V_{ts}| = 0.04111, \quad J = 3.12 \times 10^{-5}, \quad \delta = 1.147 \text{ rad}.$$

A representative best balanced run gives

$$|V_{\text{CKM}}| \approx \begin{pmatrix} 0.973044 & 0.230574 & 0.004563 \\ 0.230501 & 0.972253 & 0.039918 \\ 0.007379 & 0.039494 & 0.999193 \end{pmatrix},$$

together with

$$\delta \approx 0.912742 \text{ rad}, \quad J \approx 3.230584 \times 10^{-5}.$$

This result is significant for two reasons.

First, the model now enters a genuinely CP-violating regime:

$$\boxed{J \neq 0, \quad \delta \neq 0.}$$

Hence the reduced TECT flavor model with one additional complex Hermitian texture is capable of producing a physical CKM-phase surrogate, not merely real mixing angles.

Second, the fitted Jarlskog invariant is already close to the phenomenological target:

$$J_{\text{toy}} \approx 3.23 \times 10^{-5} \quad \text{vs.} \quad J_{\text{target}} \approx 3.12 \times 10^{-5}.$$

Thus the overall size of CP violation can already be reproduced with only one extra complex mode.

However, the same run also shows the remaining limitation of the ansatz. Although the fit is qualitatively good, the simultaneous precision fit of

$$(|V_{us}|, |V_{cb}|, |V_{ub}|, \delta)$$

is not yet achieved. In the current best balanced point,

$$|V_{us}| \text{ is slightly high,} \quad |V_{cb}| \text{ is slightly low,} \quad |V_{ub}| \text{ is somewhat high,} \quad \delta \text{ is somewhat low.}$$

Moreover, another search basin was found in which

$$\delta \approx 1.144 \text{ rad}$$

is almost perfect, but

$$|V_{ub}| \approx 0.0064$$

becomes too large. This indicates that the present ansatz has enough freedom to generate either nearly correct phase or nearly correct magnitudes, but not yet enough freedom to fit all observables simultaneously with high precision.

The commutator diagnostics remain clearly nonzero,

$$\|[B_0, B_1]\|_F \approx 1.07 \times 10^{-1}, \quad \|[M_u, M_d]\|_F \approx 1.12 \times 10^{-1},$$

while the numerical unitarity error is tiny,

$$\|V^\dagger V - \mathbf{1}\| \approx 7.1 \times 10^{-16}.$$

Therefore the complex phase is produced by genuine sectoral noncommutativity, not by numerical instability.

The honest current conclusion is

$$\boxed{\text{one extra complex Hermitian texture is sufficient to generate a nontrivial CKM phase and a realistic } J,}$$

but

$$\boxed{\text{it is still not sufficient for a fully precise simultaneous fit of all CKM magnitudes and the phase.}}$$

The most economical next step is not a radical reparametrization, but a minimally enlarged ansatz such as

$$B_H \rightarrow B_H^{(1)} + B_H^{(2)} + B_H^{(3)},$$

or

$$B_1 \rightarrow B_1^{(\text{real})} + B_1^{(\text{cx},1)} + B_1^{(\text{cx},2)},$$

so that the magnitude sector and the CP phase can be tuned more independently.

VI. FINAL SUMMARY

We extended the reduced TECT flavor toy model by adding one complex Hermitian texture mode to B_1 . This preserves Hermiticity of M_u and M_d , while allowing genuinely complex flavor misalignment.

The main result is:

$$\boxed{\text{with one extra complex mode, the model can generate } J \neq 0 \text{ and } \delta \neq 0.}$$

A representative balanced fit gives

$$|V_{us}| \approx 0.2306, \quad |V_{cb}| \approx 0.0399, \quad |V_{ub}| \approx 0.00456,$$

$$\delta \approx 0.913 \text{ rad}, \quad J \approx 3.23 \times 10^{-5}.$$

Thus the model already reproduces the correct order of magnitude of CP violation:

$$J_{\text{toy}} \sim J_{\text{exp}}.$$

However, a precise simultaneous fit of all CKM magnitudes and the phase is still not achieved. The current ansatz is therefore strong enough to demonstrate genuine emergent CP structure, but not yet strong enough for precision phenomenology.

Hence the honest status is

$$\boxed{\text{CP violation is now numerically present in the reduced TECT flavor model,}}$$

but

$$\boxed{\text{one additional complex texture mode beyond this is likely needed for a precision CKM+phase fit.}}$$

VII. LOCAL SENSITIVITY AUDIT OF THE ONE-COMPLEX-MODE CKM ANSATZ

We now analyze the local structure of the reduced TECT flavor ansatz with one additional complex Hermitian texture mode. The purpose is not another blind global search, but a direct local audit of whether the current ansatz can simultaneously improve the CKM magnitudes and the CP-violating phase near the current best point.

The current best balanced fit is

$$|V_{\text{CKM}}| \approx \begin{pmatrix} 0.973044 & 0.230574 & 0.004563 \\ 0.230501 & 0.972253 & 0.039918 \\ 0.007379 & 0.039494 & 0.999193 \end{pmatrix},$$

with

$$\delta \approx 0.912742, \quad J \approx 3.230584 \times 10^{-5}, \quad \text{score} = 4.548907.$$

To test the local flexibility of the ansatz, we performed two complementary scans:

1. a one-parameter finite scan, in which each parameter was perturbed by a small $\pm$ step and the responses of

$$|V_{us}|, |V_{cb}|, |V_{ub}|, \delta, J, \text{ score}$$

were recorded;

2. a local random-neighborhood scan consisting of 6000 small perturbations around the current best point.

The most important result is the following.

Among the 6000 local random perturbations, the best improved point has

$$\text{score} = 2.788096,$$

with

$$|V_{us}| = 0.219799, \quad |V_{cb}| = 0.044385, \quad |V_{ub}| = 0.003374,$$

$$\delta = 1.172249, \quad J = 2.956551 \times 10^{-5}.$$

Thus the score can indeed be improved locally, and the phase can even become more realistic. However, this improvement occurs only at the price of reducing the Cabibbo entry:

$$|V_{us}| : 0.230574 \longrightarrow 0.219799.$$

This already suggests a local tension: the dominant nearby descent directions improve the phase sector and $|V_{ub}|$, but move $|V_{us}|$ in the wrong direction.

This conclusion becomes much stronger when one inspects the whole neighborhood sample. In the local scan we find

$$\boxed{\# \{ \Delta \text{score} < 0 \text{ and } \Delta |V_{us}| > 0 \} = 0.}$$

That is, among all 6000 tested local perturbations, there is not a single point that both improves the total score and increases $|V_{us}|$.

This is not a proof of impossibility, but it is strong numerical evidence that the present ansatz has reached a local Pareto wall:

near the current optimum, improving the global fit tends to require lowering $|V_{us}|$.

Conversely, local points with increased $|V_{us}|$ do exist, but they worsen the overall fit. The best local point with

$$\Delta |V_{us}| > 0$$

has

$$\text{score} = 8.214184,$$

with

$$|V_{us}| = 0.231139, \quad |V_{cb}| = 0.046548, \quad |V_{ub}| = 0.004584,$$

$$\delta = 1.421370, \quad J = 4.7 \times 10^{-5},$$

which shows that the increase of the Cabibbo entry is accompanied by noticeable distortions in other observables.

The one-parameter finite scan supports the same interpretation. The strongest score-improving one-step moves are approximately:

$$q_{H,2} \downarrow, \quad q_{H,1} \uparrow, \quad \kappa_H \uparrow, \quad \text{amp}_{H,1} \uparrow, \quad \phi_{H,2} \uparrow.$$

But these directions almost uniformly decrease $|V_{us}|$. Hence the local best-response directions of the current ansatz are not aligned with the remaining Cabibbo improvement that is still needed.

A linearized local influence analysis also shows that the same small set of parameters controls both the magnitude and phase sectors. The dominant local drivers of $|V_{us}|$ are

$$\kappa_Y, \quad x_3, \quad \delta_1, \quad \ell_{\text{cp}}, \quad \epsilon_2, \quad \text{amp}_{\text{cp}}, \quad q_{\text{cp}},$$

while the dominant local drivers of δ are

$$x_3, \quad \epsilon_2, \quad \kappa_Y, \quad x_2, \quad \ell_{\text{cp}}, \quad q_{\text{cp}}.$$

In particular, the overlap

$$\{ \kappa_Y, \epsilon_2, \ell_{\text{cp}}, q_{\text{cp}}, x_3 \}$$

shows that the phase-sensitive and Cabibbo-sensitive directions are still strongly entangled.

Therefore the honest conclusion is:

one additional complex Hermitian texture mode is sufficient to generate $J \neq 0$ and $\delta \neq 0$,

but

it is not locally flexible enough to improve the full CKM+phase fit without Cabibbo tension.

The most natural next step is therefore not a larger blind search, but a minimal ansatz enlargement that decouples the Cabibbo-sensitive sector from the CP-phase-sensitive sector. The most economical possibilities are:

$$B_1 \rightarrow B_1^{(\text{real})} + B_1^{(\text{cx},1)} + B_1^{(\text{cx},2)},$$

or

$$B_H \rightarrow B_H^{(1)} + B_H^{(2)} + B_H^{(3)},$$

so that the 1–2 mixing, the 1–3 leakage, and the CP phase can be tuned more independently.

VIII. FINAL SUMMARY

We performed a local sensitivity audit around the current best point of the reduced TECT flavor ansatz with one additional complex Hermitian texture mode.

The audit shows three clear facts.

First,

$$J \neq 0 \quad \text{and} \quad \delta \neq 0$$

are robustly present, so the ansatz genuinely enters a CP-violating CKM regime.

Second, local score improvements do exist, but they systematically tend to reduce the Cabibbo entry:

$$|V_{us}| \downarrow \quad \text{when} \quad \text{score} \downarrow.$$

Third, in a neighborhood scan of 6000 perturbations, no sample was found with

$$\Delta \text{score} < 0 \quad \text{and} \quad \Delta |V_{us}| > 0.$$

This is strong numerical evidence that the present ansatz has reached a local Pareto wall.

Hence the honest current status is:

one complex texture mode is enough for emergent CP violation,

but

it is still too rigid for a locally unconstrained precision CKM+phase fit.

Therefore the next correct move is a minimal ansatz enlargement that separates Cabibbo control from CP-phase control.

IX. TWO-COMPLEX-MODE CONTINUATION OF THE REDUCED CKM+PHASE SEARCH

We extended the reduced TECT flavor toy ansatz by allowing two independent complex Hermitian texture modes in B_1 :

$$B_1 = B_1^{(\text{real})} + B_1^{(\text{cx},1)} + B_1^{(\text{cx},2)}.$$

Each complex mode is taken in the Hermitian form

$$(B_1^{(\text{cx},i)})_{ab} = A_{ab}^{(i)} [\cos(q_i(x_a - x_b)) + i \chi_i \sin(q_i(x_a - x_b))], \quad i = 1, 2,$$

with real symmetric envelope

$$A_{ab}^{(i)} = \epsilon_i S_{ab} f_R(m_{ab}) \exp\left(-\frac{(m_{ab} - c_i)^2}{2\ell_i^2}\right).$$

Hence B_1 remains Hermitian, and so do

$$M_u = \left(B_0 - \frac{2}{3}B_1\right) + B_H, \quad M_d = \left(B_0 + \frac{1}{3}B_1\right) - B_H.$$

The purpose of this continuation was to test whether the previously observed tension between the Cabibbo entry and the phase sector is genuinely removed by the second complex mode.

The first important result is that the search does find points with substantially improved score relative to the previous one-complex-mode best fit

$$\text{score}_{\text{old}} = 4.548907.$$

The best overall point found in the present continuation has

$$\text{score}_{\text{new}} = 2.385929,$$

which is an improvement of approximately

$$\frac{\text{score}_{\text{old}} - \text{score}_{\text{new}}}{\text{score}_{\text{old}}} \approx 47.5\%.$$

Its observables are

$$|V_{us}| = 0.225964, \quad |V_{cb}| = 0.046134, \quad |V_{ub}| = 0.003968,$$

$$|V_{td}| = 0.010131, \quad |V_{ts}| = 0.045182,$$

$$\delta = 1.307556, \quad J = 3.88615 \times 10^{-5}.$$

However, the crucial structural fact is that at this best overall point

$$\boxed{\text{amp}_{\text{cp},2} = 0.}$$

Therefore the best fit found in the enlarged search space is effectively a retuned one-complex-mode point.

This has an important methodological consequence. It shows that the previously suspected local wall was not absolute: better basins do exist. But it does *not* yet prove that the second complex mode is structurally required at the optimum.

The second important result is that there also exist genuinely two-mode points, with nonzero second complex amplitude, that improve over the previous one-mode best. The best genuine two-mode point found in this continuation has

$$\text{score} = 3.015898,$$

with

$$|V_{us}| = 0.220669, \quad |V_{cb}| = 0.038485, \quad |V_{ub}| = 0.003479,$$

$$|V_{td}| = 0.008308, \quad |V_{ts}| = 0.037738,$$

$$\delta = 1.314586, \quad J = 2.78523 \times 10^{-5},$$

and

$$\text{amp}_{\text{cp},2} = 0.148203.$$

Thus the second complex mode is not inert. It does open nontrivial improved directions in parameter space. Nevertheless, in the current search the globally best point still collapses back to an effective one-mode optimum. The honest interpretation is therefore the following.

First,

the reduced TECT flavor model robustly generates nontrivial CKM magnitudes and nonzero CP structure.

Second,

the previously suspected local obstruction was not global: better basins exist.

Third,

the current evidence does not yet show that a second complex mode is strictly necessary at the optimum.

This means that the true current bottleneck is likely mixed: it is not only the number of complex modes, but also the detailed parameterization and optimization strategy.

In particular, the present numerical evidence suggests the following refined conclusion:

mode enlargement helps exploration, but structural necessity of the second mode remains unproven.

Hence the next mathematically nonredundant task is not simply to add even more modes, but to separate two questions:

1. Does the one-complex-mode ansatz already admit a substantially better optimum under a stronger search?
2. If one enforces $\text{amp}_{\text{cp},2} \geq \epsilon_0 > 0$, does the best achievable score beat the best one-mode optimum?

Only after this constrained comparison can one honestly claim that the second complex mode is or is not structurally necessary.

X. FINAL SUMMARY

We performed a two-complex-mode continuation of the reduced TECT CKM+phase toy search. The main numerical findings are:

1. The previous best score

$$4.548907$$

was improved substantially to

$$2.385929.$$

2. The best overall point has

$$|V_{us}| = 0.225964, \quad |V_{cb}| = 0.046134, \quad |V_{ub}| = 0.003968, \quad \delta = 1.307556, \quad J = 3.88615 \times 10^{-5},$$

but also

$$\text{amp}_{\text{cp},2} = 0,$$

so it is effectively a retuned one-complex-mode optimum.

3. The best genuinely two-mode point has

$$\text{score} = 3.015898,$$

with nonzero second complex amplitude, and it also improves over the previous one-mode best.

Therefore the correct current conclusion is

better CKM+phase basins do exist, so the previously suspected local wall was not absolute.

But also

the present evidence still does not prove that the second complex mode is structurally required at the optimum.

Thus the next decisive step is a constrained comparison:

$$\text{best one-mode optimum} \quad \text{vs.} \quad \text{best two-mode optimum under } \text{amp}_{\text{cp},2} \geq \epsilon_0.$$

Only that comparison can settle whether the second mode is genuinely necessary.

XI. CONSTRAINED TWO-MODE SEARCH AND REVISED STRUCTURAL INTERPRETATION

We performed a constrained continuation of the reduced TECT CKM+phase toy search by imposing

$$\text{amp}_{\text{cp},2} \geq \epsilon_0 > 0,$$

so that the second complex Hermitian mode cannot collapse to zero.

The primary purpose was to test the previously open question:

is a genuinely nonzero second complex mode structurally necessary for the best CKM+phase fit?

We scanned the constrained branches

$$\epsilon_0 \in \{0.01, 0.02, 0.05, 0.10, 0.15\}.$$

The best constrained points found are:

$$\epsilon_0 = 0.01 : \quad \text{score} \approx 3.4434, \quad |V_{us}| \approx 0.22588, \quad |V_{cb}| \approx 0.04165, \quad |V_{ub}| \approx 0.00232, \quad \delta \approx 0.9770.$$

$$\epsilon_0 = 0.02 : \quad \text{score} \approx 2.8529, \quad |V_{us}| \approx 0.22342, \quad |V_{cb}| \approx 0.03652, \quad |V_{ub}| \approx 0.00366, \quad \delta \approx 1.2691.$$

$$\epsilon_0 = 0.05 : \quad \text{score} \approx 2.7937, \quad |V_{us}| \approx 0.21950, \quad |V_{cb}| \approx 0.04281, \quad |V_{ub}| \approx 0.00417, \quad \delta \approx 1.1967.$$

$$\epsilon_0 = 0.10 : \quad \text{score} \approx 1.9933, \quad |V_{us}| \approx 0.22172, \quad |V_{cb}| \approx 0.04031, \quad |V_{ub}| \approx 0.00289, \quad \delta \approx 1.0986.$$

$$\epsilon_0 = 0.15 : \quad \text{score} \approx 2.2591, \quad |V_{us}| \approx 0.22294, \quad |V_{cb}| \approx 0.03986, \quad |V_{ub}| \approx 0.00487, \quad \delta \approx 1.2587.$$

Thus the most promising constrained branch is

$$\epsilon_0 \approx 0.10.$$

We then took the $\epsilon_0 = 0.10$ best point and performed additional local refinements. The best refined point found is

$$|V_{us}| \approx 0.223279, \quad |V_{cb}| \approx 0.039836, \quad |V_{ub}| \approx 0.004172,$$

$$|V_{td}| \approx 0.008056, \quad |V_{ts}| \approx 0.039235,$$

$$\delta \approx 1.131806, \quad J \approx 3.27119 \times 10^{-5},$$

with

$$\boxed{\text{score} \approx 0.717543.}$$

This point is substantially better than the previously recorded baselines:

$$\text{score}_{\text{old one-mode}} \approx 4.5489, \quad \text{score}_{\text{old unconstrained overall}} \approx 2.3859, \quad \text{score}_{\text{old best genuine two-mode}} \approx 3.0159.$$

Hence the constrained continuation proves that much better CKM+phase basins exist.

However, the structural interpretation is subtle and important. At the new best refined point we find

$$\boxed{\text{amp}_{\text{cp},1} = 0, \quad \text{amp}_{\text{cp},2} \approx 0.533.}$$

Therefore the best current point is not a truly two-active-mode optimum. It is effectively a *single active complex mode* solution.

This implies that the current evidence does *not* prove that two simultaneously active complex modes are structurally necessary. Instead, it supports the more cautious conclusion:

$$\boxed{\text{the two-mode search was valuable because it opened access to a much better basin,}}$$

but

$$\boxed{\text{the resulting optimum itself is still effectively one-complex-mode.}}$$

Therefore the present bottleneck is likely not the minimal number of active complex modes, but rather the optimizer basin structure and the detailed parameterization of the single complex mode sector.

The revised honest conclusion is:

1. Nontrivial CKM hierarchy remains robust.
2. Nonzero CP structure ($J \neq 0$, $\delta \neq 0$) remains robust.
3. The previously suspected local obstruction was too strong a statement.
4. The current best point does not demonstrate necessity of two active complex modes.
5. The evidence now suggests that a stronger one-mode search or a reparameterized one-mode ansatz may already achieve a near-precision fit.

What remains visibly imperfect in the current best refined point is mainly the pair

$$|V_{cb}| \text{ slightly low,} \quad |V_{ub}| \text{ slightly high.}$$

So the next numerical objective is no longer to “turn on more CP modes” blindly, but to separate the 2–3 control from the 1–3 leakage while preserving the already good $|V_{us}|$, δ , and J .

XII. FINAL SUMMARY

We forced the second complex texture mode to remain nonzero by imposing

$$\text{amp}_{\text{cp},2} \geq \epsilon_0 > 0,$$

and scanned several constrained branches.

The best constrained branch occurs near

$$\epsilon_0 \approx 0.10,$$

and after additional local refinement yields

$$|V_{us}| \approx 0.223279, \quad |V_{cb}| \approx 0.039836, \quad |V_{ub}| \approx 0.004172,$$

$$\delta \approx 1.131806, \quad J \approx 3.27119 \times 10^{-5}, \quad \text{score} \approx 0.717543.$$

This is substantially better than all previously recorded fits.
However, the same best point satisfies

$$\text{amp}_{\text{cp},1} = 0, \quad \text{amp}_{\text{cp},2} \neq 0,$$

so it is still effectively a one-active-mode solution.

Therefore the correct current conclusion is:

two-mode search is a powerful basin-finding tool,

but

the current evidence still does not prove that two simultaneously active complex modes are necessary.

The strongest current interpretation is that the previous one-mode search underexplored the parameter space, and that a better one-mode-like optimum exists.

The immediate remaining numerical tension is

$$|V_{cb}| \text{ slightly low}, \quad |V_{ub}| \text{ slightly high},$$

while $|V_{us}|$, δ , and J are already close to target.

XIII. DUAL-ACTIVE AMPLITUDE CONSTRAINT AND THE DISTINCTION BETWEEN AMPLITUDE AND ACTUAL OPERATOR ACTIVITY

We now imposed the stronger numerical constraint

$$\text{amp}_{\text{cp},1} \geq \epsilon, \quad \text{amp}_{\text{cp},2} \geq \epsilon,$$

with

$$\epsilon \in \{0.01, 0.03, 0.05, 0.08, 0.10, 0.15, 0.25, 0.45\},$$

in order to test whether two simultaneously active complex texture modes are genuinely required.

The first result is that high-quality CKM+phase fits remain available under this dual floor. Across all tested ϵ , the best point repeatedly converges to essentially the same observable values:

$$|V_{us}| \approx 0.223279, \quad |V_{cb}| \approx 0.039836, \quad |V_{ub}| \approx 0.004172,$$

$$|V_{td}| \approx 0.008056, \quad |V_{ts}| \approx 0.039235,$$

$$\delta \approx 1.131806, \quad J \approx 3.27119 \times 10^{-5},$$

with

$$\text{score} \approx 0.717543.$$

This is substantially better than the previous baselines:

$$\text{score}_{\text{old one-mode}} \approx 4.5489, \quad \text{score}_{\text{old unconstrained two-mode}} \approx 2.3859, \quad \text{score}_{\text{old constrained amp}_{\text{cp},2}} \approx 1.9933.$$

Hence the reduced TECT flavor toy model remains highly viable even when both complex-mode amplitudes are forced to be nonzero.

However, the decisive structural diagnostic is not the amplitudes themselves, but the actual operator sizes of the two complex modes. For the best point at $\epsilon = 0.45$, one finds

$$\text{amp}_{\text{cp},1} = 0.45, \quad \text{amp}_{\text{cp},2} \approx 0.532758,$$

but

$$\|B_1^{(\text{cp},1)}\|_F \approx 9.20 \times 10^{-12}, \quad \|B_1^{(\text{cp},2)}\|_F \approx 5.07 \times 10^{-2}.$$

Therefore

$$\|B_1^{(\text{cp},1)}\|_F \ll \|B_1^{(\text{cp},2)}\|_F,$$

and in practice the first complex mode is numerically negligible even though its amplitude is strictly nonzero.

This happens because the first complex mode is localized far away from the actual family-overlap region. Its envelope parameters are approximately

$$c_{\text{cp},1} \approx 1.94, \quad \ell_{\text{cp},1} \approx 0.333,$$

whereas the relevant midpoint coordinates satisfy

$$m_{ab} \in [-1.89, -0.46].$$

Hence the Gaussian envelope factor

$$\exp\left(-\frac{(m_{ab} - c_{\text{cp},1})^2}{2\ell_{\text{cp},1}^2}\right)$$

is exponentially suppressed on the entire family-support region.

Thus the dual amplitude floor does *not* prove that two complex modes are simultaneously operative in the effective flavor matrix. It only proves that both amplitudes are nonzero in parameter space. This is a crucial distinction.

The correct conclusion is therefore:

a dual amplitude floor is insufficient as a test of genuine two-mode necessity.

What the calculation actually shows is the following.

First,

the reduced TECT flavor toy model robustly supports CKM hierarchy and nonzero CP structure.

Second,

good fits persist even when both complex amplitudes are forced to be positive.

Third,

the current optimum is still effectively a one-contributing-complex-mode solution.

Therefore the next mathematically honest test of genuine two-mode necessity is not

$$\text{amp}_{\text{cp},1}, \text{amp}_{\text{cp},2} > 0,$$

but rather a lower bound on the actual operator strengths, for example

$$\|B_1^{(\text{cp},1)}\|_F \geq \eta_1, \quad \|B_1^{(\text{cp},2)}\|_F \geq \eta_2,$$

or on their relative contribution,

$$\frac{\|B_1^{(\text{cp},1)}\|_F}{\|B_1^{(\text{cp},2)}\|_F} \geq r_0 > 0.$$

Only such a constraint would truly test whether two independently contributing complex textures are structurally required for the CKM+phase fit.

XIV. FINAL SUMMARY

We forced both complex texture amplitudes to remain nonzero:

$$\text{amp}_{\text{cp},1} \geq \epsilon, \quad \text{amp}_{\text{cp},2} \geq \epsilon.$$

The best fit found under this dual floor is

$$|V_{us}| \approx 0.223279, \quad |V_{cb}| \approx 0.039836, \quad |V_{ub}| \approx 0.004172,$$

$$\delta \approx 1.131806, \quad J \approx 3.27119 \times 10^{-5}, \quad \text{score} \approx 0.717543.$$

This is substantially better than all previously recorded baselines.

However, the dual-floor test also reveals the key subtlety: although both amplitudes are nonzero, the first complex mode can be made effectively invisible in the actual flavor operator:

$$\|B_1^{(\text{cp},1)}\|_F \approx 9.20 \times 10^{-12}, \quad \|B_1^{(\text{cp},2)}\|_F \approx 5.07 \times 10^{-2}.$$

Therefore the current best solution is still effectively a one-contributing-complex-mode solution. Hence the honest current conclusion is:

forcing both amplitudes to be nonzero does not prove genuine two-mode necessity.

What is needed next is a stronger structural constraint on the actual operator contribution of each mode, such as

$$\|B_1^{(\text{cp},1)}\|_F \geq \eta_1, \quad \|B_1^{(\text{cp},2)}\|_F \geq \eta_2,$$

or a nontrivial lower bound on their ratio.

XV. OPERATOR-NORM-FLOOR TEST FOR GENUINELY CONTRIBUTING TWO-COMPLEX-MODE SOLUTIONS

The previous amplitude-floor test was not sufficient, because a nonzero amplitude does not guarantee a non-negligible contribution to the actual flavor operator. Therefore we replaced the amplitude-floor condition by the stronger operator-level constraint

$$\|B_1^{(\text{cp},1)}\|_F \geq \eta, \quad \|B_1^{(\text{cp},2)}\|_F \geq \eta.$$

This is the correct numerical test of whether two complex texture modes are both genuinely contributing. We performed a first completed quick sweep for

$$\eta \in \{0.003, 0.005, 0.010\}.$$

In all three cases, the best feasible point converged to essentially the same basin, with observables

$$|V_{us}| \approx 0.227728, \quad |V_{cb}| \approx 0.049833, \quad |V_{ub}| \approx 0.003578,$$

$$|V_{td}| \approx 0.013887, \quad |V_{ts}| \approx 0.047993,$$

$$\delta \approx 0.852922, \quad J \approx 2.97388 \times 10^{-5},$$

and total score

$$\text{score} \approx 8.940321.$$

The corresponding mode strengths are

$$\|B_1^{(\text{cp},1)}\|_F \approx 0.038206, \quad \|B_1^{(\text{cp},2)}\|_F \approx 4.528798.$$

Hence this solution is genuinely two-contributing at the operator level: both complex modes are truly present in the effective family operator.

However, the fit quality is dramatically worse than the previously obtained effective one-contributing-mode solution, for which

$$\text{score} \approx 0.717543.$$

Thus

$$0.717543 \longrightarrow 8.940321$$

under the operator-norm-floor constraint.

This is the key result of the present stage.

The honest interpretation is:

within the current toy parameterization, excellent CKM+phase fits appear to rely strongly on an effectively one-dominant-com

Equivalently, once both complex modes are forced to contribute non-negligibly at the operator level, the fit deteriorates strongly.

This does *not* prove that no good genuinely two-contributing solution exists. It only shows that, in the present completed search, no such high-quality solution has been found. So the result is numerical evidence of a strong tension, not a theorem of impossibility.

The pattern of the best feasible norm-floor point is also informative:

$$|V_{us}| \text{ remains acceptable,} \quad |V_{ub}| \text{ remains acceptable,}$$

but

$$|V_{cb}|, |V_{td}|, |V_{ts}| \text{ become too large,} \quad \delta \text{ becomes too small.}$$

Thus the genuinely two-contributing regime seems to overdrive the 2–3 sector and the third-row leakage.

The current strongest honest conclusion is therefore

the present reduced TECT flavor ansatz robustly supports CKM hierarchy and nonzero CP structure,

but also

its best fits are presently concentrated near an effectively one-contributing-complex-mode geometry.

To go beyond this point, one should not merely increase the search time. The ansatz itself must be refined so that the two complex modes control more orthogonal directions in flavor space. A natural next option is to impose a mode-separation condition, for example by requiring distinct midpoint support regions:

$$|c_{\text{cp},1} - c_{\text{cp},2}| \geq \Delta_c > 0,$$

together with comparable norm constraints

$$r_{\min} \leq \frac{\|B_1^{(\text{cp},1)}\|_F}{\|B_1^{(\text{cp},2)}\|_F} \leq r_{\max}.$$

Only then can we honestly test whether two structurally distinct contributing complex modes are beneficial, rather than allowing one mode to dominate and the other merely to survive.

XVI. FINAL SUMMARY

We replaced the previous amplitude-floor constraint by the stronger operator-level condition

$$\|B_1^{(\text{cp},1)}\|_F \geq \eta, \quad \|B_1^{(\text{cp},2)}\|_F \geq \eta,$$

which is the correct test of genuinely contributing two-complex-mode solutions.

In the completed quick sweep for

$$\eta \in \{0.003, 0.005, 0.010\},$$

the best feasible solution is

$$|V_{us}| \approx 0.227728, \quad |V_{cb}| \approx 0.049833, \quad |V_{ub}| \approx 0.003578,$$

$$\delta \approx 0.852922, \quad J \approx 2.97388 \times 10^{-5}, \quad \text{score} \approx 8.940321,$$

with

$$\|B_1^{(\text{cp},1)}\|_F \approx 0.038206, \quad \|B_1^{(\text{cp},2)}\|_F \approx 4.528798.$$

So this point is genuinely two-contributing at the operator level.

However, it is much worse than the previous effective one-contributing-mode best fit

$$\text{score} \approx 0.717543.$$

Hence the current strongest honest conclusion is:

excellent fits in the present toy ansatz are still concentrated near an effectively one-dominant-complex-mode structure.

This is not a proof that no good genuinely two-contributing solution exists. It is strong numerical evidence that, within the current parameterization and completed search, such solutions are under strong tension.

XVII. SUPPORT-SEPARATION AND NORM-RATIO CONSTRAINED SEARCH

To test whether two genuinely distinct complex texture modes are beneficial, we imposed two additional structural constraints on the reduced TECT flavor ansatz.

First, a support-separation condition:

$$|c_{\text{cp},1} - c_{\text{cp},2}| \geq \Delta_c.$$

Second, a norm-ratio condition:

$$r_{\min} \leq \frac{\|B_1^{(\text{cp},1)}\|_F}{\|B_1^{(\text{cp},2)}\|_F} \leq r_{\max}.$$

Together with the previous operator-level floor

$$\|B_1^{(\text{cp},1)}\|_F \geq \eta, \quad \|B_1^{(\text{cp},2)}\|_F \geq \eta,$$

this forces the two complex modes to be simultaneously present, to contribute at comparable size, and to have distinct support in family-midpoint space.

A first completed scan was carried out for several representative choices of

$$(\eta, \Delta_c, r_{\min}, r_{\max}).$$

The best completed branch was found at

$$\eta = 0.01, \quad \Delta_c = 0.55, \quad r_{\min} = 0.25, \quad r_{\max} = 4.0,$$

with

$$\|B_1^{(\text{cp},1)}\|_F \approx 0.012055, \quad \|B_1^{(\text{cp},2)}\|_F \approx 0.011200,$$

$$|c_{\text{cp},1} - c_{\text{cp},2}| \approx 0.983531, \quad \frac{\|B_1^{(\text{cp},1)}\|_F}{\|B_1^{(\text{cp},2)}\|_F} \approx 1.0764.$$

Hence this branch is genuinely two-mode, support-separated, and norm-comparable.

Its CKM+phase observables are

$$|V_{us}| \approx 0.220165, \quad |V_{cb}| \approx 0.038517, \quad |V_{ub}| \approx 0.004944,$$

$$\delta \approx 0.990784, \quad J \approx 3.418413 \times 10^{-5},$$

with total score

$$\boxed{\text{score} \approx 4.567234.}$$

For comparison, the previously best effective one-dominant-mode solution had

$$\text{score} \approx 0.717543.$$

Therefore the support-separated, norm-comparable two-mode branch is substantially worse.

The other completed branches show the same pattern:

$$(\eta, \Delta_c, r_{\min}, r_{\max}) = (0.01, 0.45, 0.20, 5.0) \Rightarrow \text{score} \approx 5.788167,$$

$$(\eta, \Delta_c, r_{\min}, r_{\max}) = (0.01, 0.65, 0.33, 3.0) \Rightarrow \text{score} \approx 12.104031.$$

Thus stronger separation and stronger comparability constraints tend to worsen the fit further.

Additional local refinements around the support-separated branch did not reverse this conclusion. The best follow-up local point found reached only

$$\text{score} \approx 6.964825,$$

with

$$|V_{us}| \approx 0.224259, \quad |V_{cb}| \approx 0.035561, \quad |V_{ub}| \approx 0.005537, \quad \delta \approx 0.921224.$$

The honest interpretation is therefore:

within the current toy parameterization, high-quality CKM+phase fits remain concentrated near an effectively one-dominant-mode solution.

This is not a theorem of impossibility for genuinely two-mode solutions. It is strong numerical evidence that, in the present ansatz, forcing two complex modes to contribute at comparable strength and distinct support creates substantial tension.

Equivalently,

the current reduced TECT flavor ansatz robustly produces CKM hierarchy and nonzero CP structure,

but

its best solutions are not presently realized as two separated, norm-comparable complex modes.

Hence the bottleneck is no longer merely search depth. Rather, the present parameterization appears too rigid to let two structurally distinct complex modes cooperate efficiently. The next natural improvement is therefore to enlarge the ansatz so that the two complex modes act on more orthogonal sectors of flavor space, for example by assigning one mode primarily to the 1–2 sector and the other to the 1–3/2–3 sector through a more structured envelope.

XVIII. FINAL SUMMARY

We tested whether two genuinely distinct complex texture modes can improve the reduced TECT CKM+phase fit by imposing three simultaneous constraints:

$$\|B_1^{(\text{cp},1)}\|_F \geq \eta, \quad \|B_1^{(\text{cp},2)}\|_F \geq \eta,$$

$$|c_{\text{cp},1} - c_{\text{cp},2}| \geq \Delta_c,$$

$$r_{\min} \leq \frac{\|B_1^{(\text{cp},1)}\|_F}{\|B_1^{(\text{cp},2)}\|_F} \leq r_{\max}.$$

The best completed support-separated, norm-comparable branch gives

$$|V_{us}| \approx 0.220165, \quad |V_{cb}| \approx 0.038517, \quad |V_{ub}| \approx 0.004944,$$

$$\delta \approx 0.990784, \quad J \approx 3.418413 \times 10^{-5}, \quad \text{score} \approx 4.567234.$$

This is much worse than the previously best effective one-dominant-mode fit

$$\text{score} \approx 0.717543.$$

Therefore the strongest honest current conclusion is

within the present toy ansatz, excellent CKM+phase fits remain concentrated near an effectively one-dominant-complex-mode :

So the issue is no longer simply under-searching. The present parameterization itself appears too rigid to make two separated, comparable complex modes cooperate efficiently.

XIX. ASSESSMENT OF A COMPLEX MODE ACTING ONLY THROUGH THE SECOND FAMILY

The proposal “a single complex mode acting only on family 2” has two inequivalent interpretations, and they lead to very different conclusions.

XX. INTERPRETATION A: PURELY DIAGONAL ACTION ON FAMILY 2

If the proposal is interpreted literally as

$$B_{\text{cp}}^{(2-\text{diag})} = z P_2, \quad P_2 = |2\rangle\langle 2|,$$

then Hermiticity requires

$$B_{\text{cp}}^{(2-\text{diag})\dagger} = B_{\text{cp}}^{(2-\text{diag})} \implies z = \bar{z},$$

so

$$z \in \mathbb{R}.$$

Hence

$$B_{\text{cp}}^{(2-\text{diag})} = \alpha P_2, \quad \alpha \in \mathbb{R}.$$

Therefore this term is not genuinely complex in the Hermitian flavor ansatz. It can shift the second-family level and modify mixing angles indirectly, but it cannot by itself generate a physical CKM phase. In particular, if all other matrices remain real symmetric, then

$$J = 0, \quad \delta = 0$$

still holds.

So a strictly diagonal family-2-only complex mode is not a good CP-generating candidate.

XXI. INTERPRETATION B: FAMILY-2-ANCHORED OFF-DIAGONAL COMPLEX MODE

A much more promising interpretation is that the complex mode acts only through the channels involving family 2:

$$1 \leftrightarrow 2, \quad 2 \leftrightarrow 3, \quad 1 \not\leftrightarrow 3 \quad (\text{directly in the complex sector}).$$

The minimal Hermitian ansatz is

$$B_{\text{cp}}^{(2\text{-anchor})} = \begin{pmatrix} 0 & \rho_{12}e^{i\phi_{12}} & 0 \\ \rho_{12}e^{-i\phi_{12}} & 0 & \rho_{23}e^{i\phi_{23}} \\ 0 & \rho_{23}e^{-i\phi_{23}} & 0 \end{pmatrix},$$

or equivalently, in a minimal imaginary form,

$$B_{\text{cp}}^{(2\text{-anchor})} = i\eta_{12}(E_{12} - E_{21}) + i\eta_{23}(E_{23} - E_{32}).$$

This ansatz has three attractive properties.

First, it directly targets the phenomenologically important channels

$$1\text{--}2 \quad \text{and} \quad 2\text{--}3,$$

which are the main carriers of V_{us} and V_{cb} .

Second, it suppresses direct complex 1–3 leakage, so it may help reduce the tendency of the toy model to overproduce $|V_{ub}|$.

Third, it matches the numerical evidence obtained so far: the best fits appear near an effectively one-dominant complex mode, while forcing several comparable complex modes tends to overdrive the 1–3 and 2–3 sectors. A family-2-anchored complex mode is therefore a natural way to make the complex structure more selective.

XXII. NECESSARY CAVEAT: CP IS NOT AUTOMATIC

However, a 2-anchored complex mode does not automatically guarantee physical CP violation. For a nontrivial phase invariant to appear, one still needs an effective triangular carrier in flavor space.

In practical terms, if the complex mode contributes only to

$$(12) \quad \text{and} \quad (23),$$

then the remaining real sector must provide a nonzero

$$(13)$$

entry. Thus one should require at least one of

$$(B_0)_{13} \neq 0 \quad \text{or} \quad (B_H)_{13} \neq 0.$$

This is the correct minimal design principle:

$$\boxed{\text{complex part on } 12, 23 \text{ only; real } 13 \text{ carrier from } B_0 \text{ or } B_H.}$$

XXIII. RECOMMENDED NEXT ANSATZ

The most economical next model is therefore

$$B_1 = B_1^{(\text{real})} + B_{\text{cp}}^{(2\text{-anchor})},$$

with

$$B_{\text{cp}}^{(2\text{-anchor})} = \begin{pmatrix} 0 & a e^{i\phi_{12}} & 0 \\ a e^{-i\phi_{12}} & 0 & b e^{i\phi_{23}} \\ 0 & b e^{-i\phi_{23}} & 0 \end{pmatrix}, \quad a, b \geq 0,$$

while keeping B_0 and B_H real symmetric, but requiring

$$(B_0)_{13} \neq 0 \quad \text{or} \quad (B_H)_{13} \neq 0.$$

This ansatz is structurally attractive because it gives a selective complex source for the 1–2 and 2–3 sectors, while leaving the 1–3 channel mostly controlled by the real background. It is therefore the most natural next test if the goal is to improve

$$|V_{us}|, \quad |V_{cb}|, \quad |V_{ub}|, \quad \delta$$

simultaneously with minimal extra freedom.

XXIV. FINAL SUMMARY

The proposal “a complex mode acting only on family 2” must be split into two cases.

If it means a purely diagonal family-2 projector,

$$B_{\text{cp}} = z P_2,$$

then Hermiticity forces

$$z \in \mathbb{R},$$

so this does not provide a genuine CP-carrying complex mode.

If instead it means a family-2-anchored off-diagonal mode acting only on

$$1 \leftrightarrow 2 \quad \text{and} \quad 2 \leftrightarrow 3,$$

then the idea is highly promising. The minimal Hermitian form is

$$B_{\text{cp}}^{(2\text{-anchor})} = \begin{pmatrix} 0 & a e^{i\phi_{12}} & 0 \\ a e^{-i\phi_{12}} & 0 & b e^{i\phi_{23}} \\ 0 & b e^{-i\phi_{23}} & 0 \end{pmatrix}.$$

This ansatz is attractive because it can control V_{us} and V_{cb} directly while suppressing direct complex 1–3 leakage, which may help with the current $|V_{ub}|$ tension.

However, physical CP violation then still requires a real 1–3 carrier from B_0 or B_H .

Therefore the strongest current recommendation is:

do not use a purely diagonal family-2 complex mode; use a family-2-anchored 12/23-only complex mode instead.

XXV. FAMILY-2-ANCHORED COMPLEX MODE: NUMERICAL TEST

We implemented and tested the proposal that the complex flavor mode should act only through the second family, namely through the channels

$$1 \leftrightarrow 2, \quad 2 \leftrightarrow 3, \quad 1 \not\leftrightarrow 3 \quad (\text{directly in the complex sector}).$$

Two versions were tested.

XXVI. GENERAL ANCHORED COMPLEX ANSATZ

The first implementation used the minimal Hermitian form

$$B_{\text{cp}}^{(2\text{-anchor})} = \begin{pmatrix} 0 & a e^{i\phi_{12}} & 0 \\ a e^{-i\phi_{12}} & 0 & b e^{i\phi_{23}} \\ 0 & b e^{-i\phi_{23}} & 0 \end{pmatrix},$$

with Gaussian midpoint envelopes on the 12 and 23 channels.

A moderate multi-seed search gave the best point

$$|V_{us}| \approx 0.22221, \quad |V_{cb}| \approx 0.04311, \quad |V_{ub}| \approx 0.00459,$$

$$|V_{td}| \approx 0.01281, \quad |V_{ts}| \approx 0.04142,$$

$$\delta \approx 0.91153, \quad J \approx 3.38 \times 10^{-5},$$

with

$$\boxed{\text{score} \approx 3.8889.}$$

This shows that the anchored ansatz can generate nontrivial CKM hierarchy and nonzero CP structure, but it still tends to leak too much into the 1–3 sector.

XXVII. PURE-IMAGINARY ANCHORED ANSATZ

The second implementation used the stricter form

$$\boxed{B_{\text{cp}}^{(2\text{-anchor, imag})} = i\eta_{12}(E_{12} - E_{21}) + i\eta_{23}(E_{23} - E_{32}),}$$

again with Gaussian midpoint envelopes, but with no direct complex 13 entry and no extra real part in the complex sector.

This produced the best result obtained in the present anchored study:

$$|V_{us}| \approx 0.22127, \quad |V_{cb}| \approx 0.04147, \quad |V_{ub}| \approx 0.00310,$$

$$|V_{td}| \approx 0.00777, \quad |V_{ts}| \approx 0.04085,$$

$$\delta \approx 0.93409, \quad J \approx 2.23 \times 10^{-5},$$

with

$$\boxed{\text{score} \approx 2.76111.}$$

This is better than the support-separated, norm-comparable two-mode branches studied earlier, but still substantially worse than the previously best effective one-dominant-mode solution

$$\text{score} \approx 0.71754.$$

XXVIII. INTERPRETATION

The anchored idea is therefore structurally promising, but the present minimal anchored ansatz is still too weak to beat the best previous fit.

The numerical pattern is clear:

1. The 12/23-only complex structure helps suppress direct complex 1–3 leakage, so $|V_{ub}|$ does not overshoot badly.
2. However, the same restriction tends to weaken the CP-carrying triangular structure, so δ and especially J become too small unless the anchored mode becomes stronger.

For the best pure-imaginary anchored point, one finds

$$\|B_{\text{cp}}^{(2\text{-anchor})}\|_F \approx 1.39 \times 10^{-3},$$

while the real 13 carrier satisfies

$$|(B_0 + B_H)_{13}| \approx 7.14 \times 10^{-2}.$$

Thus the present failure mode is not the absence of a real 13 carrier. Rather, it is the insufficient strength of the anchored complex 12/23 source.

Hence the honest current conclusion is

the family-2-anchored idea is directionally correct,

but

the current minimal anchored parameterization does not yet provide enough independent control over the complex source versus

The most natural next step is therefore not to abandon the anchored idea, but to promote the real 13 carrier to an explicit controllable texture while keeping the complex sector restricted to 12/23. This would separate the roles

complex (12, 23) source and real (13) carrier,

which is precisely what the present numerical bottleneck suggests is needed.

XXIX. FINAL SUMMARY

We tested the proposal that the complex flavor mode should act only through family 2.

A strictly 12/23-only anchored complex mode is numerically viable and indeed helps suppress direct complex 1–3 leakage. The best pure-imaginary anchored fit obtained is

$$|V_{us}| \approx 0.22127, \quad |V_{cb}| \approx 0.04147, \quad |V_{ub}| \approx 0.00310,$$

$$\delta \approx 0.93409, \quad J \approx 2.23 \times 10^{-5}, \quad \text{score} \approx 2.76111.$$

Thus the idea is not wrong. In fact, it performs better than the previously forced genuinely two-mode branches. However, it still does not beat the current best effective one-dominant-mode solution

$$\text{score} \approx 0.71754.$$

The bottleneck is now clear: the real 13 carrier is present, but the anchored complex 12/23 source is too weak. So the next correct ansatz is:

keep the complex sector 12/23-only, but introduce an explicit independent real 13 carrier control.

XXX. ASSESSMENT OF THE EXPLORATORY FLAVOR SEARCH AND THE CORRECT TRANSITION TO A PRINCIPLED DERIVATION

The current question is whether the ongoing exploratory CKM/phase search still has genuine scientific value, or whether its main purpose has already been achieved and the program should now return to the principled derivation from the microscopic TECT construction.

The correct answer is the following.

A. What has already been established

The exploratory reduced flavor search has already achieved its primary goal. At the reduced matrix level, it has provided strong numerical evidence for the following three statements:

(i) nontrivial hierarchical CKM-like mixing exists in the reduced TECT flavor problem,

(ii) CP-violating branches with $J \neq 0$ and $\delta \neq 0$ also exist,

(iii) no single ad hoc complex texture appears to be uniquely necessary.

This means that the exploratory phase has already passed the key existential test. The reduced flavor problem is not empty, not overconstrained, and not obviously dependent on a unique arbitrary ansatz.

In that precise sense, the exploratory search has already been meaningful and successful.

B. What has *not* been established

However, the exploratory search has *not* established any of the following:

(a) a unique microscopic derivation of the flavor operator from the full defect PDE,

(b) a theorem that a particular texture class is necessary,

(c) a first-principles CKM prediction independent of residual parameter choices.

Thus the present numerical exploration should not be mistaken for a derivation. It is an existence-and-nonuniqueness study, not yet a microscopic theorem.

C. Why further blind fitting now has diminishing value

Once the existential question has been answered positively, continued blind ansatz-searching has sharply diminishing scientific return.

The reason is simple: the dominant open problem is no longer

“does a viable reduced fit exist?”

but rather

“which flavor operator structure is actually generated by the microscopic TECT construction?”

Therefore reducing the score from, say, 0.7 to 0.5 inside a still-ad hoc toy space would add very little structural knowledge. It would improve numerical cosmetics, but not the derivational status of the theory.

Hence the correct transition point has effectively already been reached.

D. What “proceeding in the proper way” now means

At the present stage, a proper derivational program should consist of four steps.

a. Step 1: replace texture ansätze by an operator-basis decomposition. Instead of guessing Gaussian textures directly, one should first expand the effective family operators in a Hermitian basis of 3×3 matrices:

$$B_0 = \sum_{A=0}^8 b_A \Lambda_A, \quad B_1 = \sum_{A=0}^8 y_A \Lambda_A, \quad B_H = \sum_{A=0}^8 h_A \Lambda_A,$$

where $\{\Lambda_A\}$ is a basis of $\mathfrak{u}(3)$.

This removes inessential texture language and asks the correct structural question: which operator channels are actually generated, and which are suppressed?

b. Step 2: classify the reduced CP-carrying mechanisms at the theorem level. The numerical exploration has already strongly suggested several structural facts that should now be converted into exact reduced-level propositions. Examples include:

1. a purely diagonal complex mode cannot generate a physical CKM phase in a Hermitian ansatz;
2. a complex mode restricted to the 12 and 23 channels can generate CP violation provided a real 13 carrier is present;
3. a direct complex 13 entry is not obviously necessary;
4. the best reduced fits tend to lie near an effectively one-dominant-complex-mode geometry.

These statements should now be formalized in terms of basis-independent invariants such as

$$\det([M_u, M_d]), \quad J, \quad \text{Tr}([M_u, M_d]^3).$$

c. Step 3: derive the allowed operator channels from the microscopic TECT PDE / Schur complement. This is the real next frontier.

The heavy-sector elimination should produce an effective family operator

$$\mathcal{Y}_{\text{eff}}$$

or its sectoral descendants, and the task is to determine which reduced matrix components are actually generated:

$$(12), \quad (23), \quad (13), \quad \text{diagonal splittings}, \quad \text{complex antisymmetric parts}.$$

Only this step can turn the reduced flavor stage from an existence study into a derivation.

d. Step 4: perform fitting only after the allowed operator structure has been derived. Once the microscopic derivation has reduced the admissible operator space, one may then perform a final phenomenological fit inside that derived space.

This is the correct order. Fitting before derivation is exploratory. Fitting after derivation is explanatory.

E. The strongest honest present conclusion

Therefore the correct present conclusion is:

the exploratory reduced flavor search has already done enough work to justify returning to the principled derivation.

Equivalently,

continued blind ansatz-searching should no longer be the main research direction.

But the numerical search results should not be discarded. They now serve as empirical boundary conditions for the microscopic derivation.

In particular, the exploratory phase suggests the following operator-level hypotheses to test first in the full derivation:

1. purely diagonal complex sectors are weak CP carriers;
2. direct complex 13 terms are not obviously necessary;

3. 12/23-anchored complex structure plus a real 13 carrier is a viable reduced CP mechanism;
4. good fits tend to concentrate near an effectively one-dominant-complex-mode geometry;
5. two separated norm-comparable complex modes are disfavored in the present reduced ansatz.

These are no longer arbitrary guesses. They are the correct reduced-level hypotheses that the microscopic theory must now confront.

XXXI. FINAL SUMMARY

The exploratory CKM/phase search in the reduced TECT flavor problem has already achieved its primary purpose. It has provided strong numerical evidence that:

hierarchical CKM-like mixing exists in the reduced model,

CP-violating branches with $J \neq 0$ and $\delta \neq 0$ also exist,

no single ad hoc complex texture appears uniquely necessary.

Therefore the existential stage is essentially complete.

What remains open is not the existence of viable reduced fits, but the derivation of the effective flavor operator from the microscopic TECT construction. Hence further blind ansatz-searching now has sharply diminishing value.

The correct next move is:

replace texture fitting by operator-basis classification and microscopic derivation.

So the strongest honest current conclusion is:

the exploratory stage has done enough; it is now correct to proceed in the principled derivational direction.

XXXII. GENERAL HERMITIAN BASIS FOR THE REDUCED FLAVOR OPERATORS

We now abandon ad hoc texture language and pass to the canonical Hermitian basis of 3×3 family operators. Let E_{ab} denote the elementary matrix with a 1 in the (a, b) entry and zeros elsewhere. Define the Hermitian basis

$$D_0 = \mathbf{1}, \quad D_1 = \text{diag}(1, -1, 0), \quad D_2 = \text{diag}(1, 1, -2),$$

$$S_{12} = E_{12} + E_{21}, \quad S_{13} = E_{13} + E_{31}, \quad S_{23} = E_{23} + E_{32},$$

$$A_{12} = i(E_{12} - E_{21}), \quad A_{13} = i(E_{13} - E_{31}), \quad A_{23} = i(E_{23} - E_{32}).$$

Then every Hermitian 3×3 matrix M admits the unique expansion

$$M = \alpha_0 D_0 + \alpha_1 D_1 + \alpha_2 D_2 + s_{12} S_{12} + s_{13} S_{13} + s_{23} S_{23} + a_{12} A_{12} + a_{13} A_{13} + a_{23} A_{23},$$

with all coefficients

$$\alpha_0, \alpha_1, \alpha_2, s_{12}, s_{13}, s_{23}, a_{12}, a_{13}, a_{23} \in \mathbb{R}.$$

Thus the reduced TECT family operators can be written as

$$B_0 = \sum_{\Xi} b_{\Xi} \Xi, \quad B_1 = \sum_{\Xi} y_{\Xi} \Xi, \quad B_H = \sum_{\Xi} h_{\Xi} \Xi,$$

where

$$\Xi \in \{D_0, D_1, D_2, S_{12}, S_{13}, S_{23}, A_{12}, A_{13}, A_{23}\}.$$

At the toy post-EWSB level,

$$M_u = \left(B_0 - \frac{2}{3}B_1\right) + B_H, \quad M_d = \left(B_0 + \frac{1}{3}B_1\right) - B_H,$$

so

$$M_u = \sum_{\Xi} u_{\Xi} \Xi, \quad M_d = \sum_{\Xi} d_{\Xi} \Xi,$$

with

$$u_{\Xi} = b_{\Xi} - \frac{2}{3}y_{\Xi} + h_{\Xi}, \quad d_{\Xi} = b_{\Xi} + \frac{1}{3}y_{\Xi} - h_{\Xi}.$$

Therefore the reduced flavor problem is, in full generality, a problem of determining the coefficient vectors

$$(u_{\Xi}), \quad (d_{\Xi})$$

in this Hermitian basis.

XXXIII. CP-EVEN AND CP-ODD SUBSPACES

In the above basis, complex conjugation acts as

$$D_i^* = D_i, \quad S_{ab}^* = S_{ab}, \quad A_{ab}^* = -A_{ab}.$$

Hence the Hermitian operator space splits as

$$\mathfrak{u}(3)_{\text{Herm}} = \mathcal{D} \oplus \mathcal{S} \oplus \mathcal{A},$$

where

$$\mathcal{D} = \text{span}\{D_0, D_1, D_2\}, \quad \mathcal{S} = \text{span}\{S_{12}, S_{13}, S_{23}\}, \quad \mathcal{A} = \text{span}\{A_{12}, A_{13}, A_{23}\}.$$

The sectors $\mathcal{D}$ and $\mathcal{S}$ are real under complex conjugation, while $\mathcal{A}$ is odd. Therefore:

In a Hermitian flavor ansatz, all genuinely CP-odd carrier content lives in $\mathcal{A} = \text{span}\{A_{12}, A_{13}, A_{23}\}$.

This yields the first exact reduced-level proposition.

a. Proposition 1 (diagonal “complex mode” is impossible). A purely diagonal Hermitian flavor term cannot carry a physical complex phase.

b. Proof. A diagonal Hermitian matrix belongs to $\mathcal{D}$, hence has the form

$$M_{\text{diag}} = \alpha_0 D_0 + \alpha_1 D_1 + \alpha_2 D_2$$

with real coefficients $\alpha_i \in \mathbb{R}$. So a would-be “complex diagonal mode” is forced by Hermiticity to be real. Therefore it cannot by itself generate a CKM CP phase. $\square$

XXXIV. BASIS-INDEPENDENT CP INVARIANT

Because M_u and M_d are Hermitian in the reduced model, define

$$H_u = M_u^2, \quad H_d = M_d^2.$$

These are positive Hermitian matrices and transform covariantly under family-basis changes.

The standard basis-independent CP diagnostic is

$$\mathcal{I}_{\text{CP}} := \frac{1}{3i} \text{Tr}([H_u, H_d]^3).$$

This quantity is invariant under

$$M_u \mapsto U^\dagger M_u U, \quad M_d \mapsto U^\dagger M_d U,$$

and vanishes whenever H_u and H_d can be made simultaneously real. For three nondegenerate families, it is the standard necessary-and-sufficient Jarlskog criterion:

$$\mathcal{I}_{\text{CP}} \neq 0 \iff \text{physical CP violation.}$$

Thus, at the reduced flavor level, the real mathematical question is not “which texture looks complex,” but rather:

which coefficient patterns in $\mathcal{D} \oplus \mathcal{S} \oplus \mathcal{A}$ make $\mathcal{I}_{\text{CP}} \neq 0$?

XXXV. USEFUL COMMUTATOR ALGEBRA

The mechanism is governed by a small set of exact commutators. For $D(\mu) = \text{diag}(\mu_1, \mu_2, \mu_3)$, one has

$$[D(\mu), A_{12}] = i(\mu_1 - \mu_2)S_{12}, \quad [D(\mu), A_{23}] = i(\mu_2 - \mu_3)S_{23},$$

$$[D(\mu), S_{12}] = -i(\mu_1 - \mu_2)A_{12}, \quad [D(\mu), S_{23}] = -i(\mu_2 - \mu_3)A_{23}.$$

The inter-channel commutators needed for the anchored mechanism are

$$[S_{13}, A_{12}] = iS_{23}, \quad [S_{13}, A_{23}] = -iS_{12}, \quad [S_{12}, A_{23}] = iS_{13},$$

$$[A_{12}, A_{23}] = -A_{13}, \quad [S_{12}, S_{23}] = -iA_{13}.$$

These relations already show the basic structural fact:

anchored (12, 23) complex content can propagate into the rest of flavor space through real off-diagonal carriers.

But they do *not* imply that every such anchored pattern is automatically CP-violating. This distinction is crucial.

XXXVI. EXACT REDUCED-LEVEL PROPOSITIONS

a. Proposition 2 (two-family reduction cannot carry CKM CP violation). Suppose both M_u and M_d are supported in a single 2×2 family block, e.g.

$$M_u, M_d \in \text{span}\{D_0, D_1, D_2, S_{12}, A_{12}\}.$$

Then

$$\mathcal{I}_{\text{CP}} = 0.$$

b. Proof. In this case both H_u and H_d are block-diagonal of type

$$2 \oplus 1.$$

Hence their commutator $[H_u, H_d]$ also has vanishing third row and column. Therefore $[H_u, H_d]^3$ has zero trace, so

$$\text{Tr}([H_u, H_d]^3) = 0.$$

Equivalently, a genuine three-family CP invariant cannot arise in a reduced two-family block. $\square$

This is the exact statement behind the informal slogan:

a single active 2×2 channel cannot produce CKM CP violation.

c. *Proposition 3 (direct complex 13 support is not necessary).* There exist Hermitian reduced flavor pairs with

$$a_{13}^{(u)} = a_{13}^{(d)} = 0$$

but nevertheless

$$\mathcal{I}_{\text{CP}} \neq 0.$$

d. *Proof by explicit witness.* Consider

$$M_u = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix} = D_u + S_{12} + S_{13},$$

$$M_d = \begin{pmatrix} 0 & i & 0 \\ -i & 1 & i \\ 0 & -i & 3 \end{pmatrix} = D_d + A_{12} + A_{23}.$$

Here the complex sector is anchored entirely in the 12 and 23 channels; there is no A_{13} term.

Now define

$$H_u = M_u^2, \quad H_d = M_d^2.$$

A direct computation gives

$$\text{Tr}([H_u, H_d]^3) = 1260 i \neq 0.$$

Hence

$$\mathcal{I}_{\text{CP}} \neq 0.$$

So direct complex 13 support is not necessary for physical CP violation. $\square$

This proposition is important because it upgrades the earlier numerical hint to an exact reduced-level fact:

$A_{13} \text{ is not a necessary CP carrier.}$

e. *Proposition 4 (anchored complex 12/23 plus real 13 alone is not automatically sufficient).* Consider the family of Hermitian matrices

$$M_u = D_u + x S_{13}, \quad M_d = D_d + a A_{12} + b A_{23},$$

with arbitrary real D_u, D_d, x, a, b . Then

$$\mathcal{I}_{\text{CP}} = 0$$

identically.

f. *Proof.* Define

$$H_u = M_u^2, \quad H_d = M_d^2.$$

A direct symbolic computation yields

$$\text{Tr}([H_u, H_d]^3) = 0$$

for all values of the parameters. Hence

$$\mathcal{I}_{\text{CP}} = 0.$$

So a real 13 carrier, together with anchored A_{12}, A_{23} , is *not by itself* a sufficient condition for CP violation. $\square$

This corrects an overly naive version of the anchored intuition. The correct statement is weaker and more precise:

$\text{anchored } A_{12}, A_{23} \text{ are algebraically viable CP carriers,}$

but

$\text{a real } S_{13} \text{ carrier alone does not guarantee } \mathcal{I}_{\text{CP}} \neq 0.$

g. Proposition 5 (anchored complex structure becomes viable once the real skeleton is sufficiently rich). There exist Hermitian reduced flavor pairs with complex support only in the anchored channels

$$A_{12}, A_{23},$$

but with no A_{13} , for which $\mathcal{I}_{\text{CP}} \neq 0$.

h. Proof. This is already established by Proposition 3. In that witness pair,

$$M_u = D_u + S_{12} + S_{13}, \quad M_d = D_d + A_{12} + A_{23},$$

and

$$\mathcal{I}_{\text{CP}} \neq 0.$$

Thus the anchored mechanism is genuinely viable, provided the real skeleton is richer than a single S_{13} carrier. $\square$

XXXVII. WHAT IS NOW RIGOROUSLY CLOSED AND WHAT REMAINS OPEN

The following reduced-level facts are now mathematically closed:

1. A diagonal Hermitian sector cannot carry a physical complex phase.
2. A genuine CKM CP invariant cannot arise in a single reduced 2×2 family block.
3. Direct complex 13 support is not necessary.
4. Anchored 12/23 complex structure is algebraically viable.
5. Anchored 12/23 complex structure plus real 13 alone is not automatically sufficient.

Therefore the reduced TECT flavor problem has now been upgraded from texture intuition to an operator-level classification:

the CP problem is a classification problem in $\mathcal{D} \oplus \mathcal{S} \oplus \mathcal{A}$, controlled by $\mathcal{I}_{\text{CP}}$.

What remains open is the genuinely microscopic question:

which coefficients u_{Ξ} , d_{Ξ} are actually generated by the TECT defect PDE / Schur complement?

That is the correct next frontier.

XXXVIII. FINAL SUMMARY

We replaced ad hoc flavor textures by the full Hermitian basis

$$\{D_0, D_1, D_2, S_{12}, S_{13}, S_{23}, A_{12}, A_{13}, A_{23}\}$$

of 3×3 family operators and classified the CP-carrying sectors.

The rigorous reduced-level conclusions are:

1. Physical complex-phase carrier content lives only in

$$\mathcal{A} = \text{span}\{A_{12}, A_{13}, A_{23}\}.$$

2. A purely diagonal Hermitian flavor term cannot carry CKM CP violation.
3. Any effectively two-family block has

$$\mathcal{I}_{\text{CP}} = 0.$$

4. Direct complex 13 support is not necessary: there exist explicit Hermitian witness pairs with only anchored complex support in 12 and 23, but with

$$\mathcal{I}_{\text{CP}} \neq 0.$$

5. However, anchored complex support in 12 and 23, together with a single real 13 carrier, is not automatically sufficient: for

$$M_u = D_u + xS_{13}, \quad M_d = D_d + aA_{12} + bA_{23},$$

one finds

$$\mathcal{I}_{\text{CP}} = 0$$

identically.

6. Therefore the anchored mechanism is viable, but only when the real skeleton is sufficiently rich.

So the reduced flavor problem is now properly reformulated as an operator-basis classification problem controlled by the basis-independent invariant

$$\mathcal{I}_{\text{CP}} = \frac{1}{3i} \text{Tr}([H_u, H_d]^3).$$

The next real question is no longer texture fitting, but:

which operator coefficients are actually generated by the microscopic TECT defect PDE / Schur complement?

XXXIX. MICROSCOPIC BLOCK ELIMINATION AND THE GENERAL REDUCED FLAVOR OPERATOR

We now derive the correct general formula for the reduced family operator from the microscopic defect/heavy-sector problem.

XL. LIGHT/HEAVY DECOMPOSITION OF THE MICROSCOPIC OPERATOR

Let the full microscopic Hilbert space decompose as

$$\mathcal{H}_{\text{micro}} = \mathcal{H}_L \oplus \mathcal{H}_H,$$

where

$$\mathcal{H}_L = \text{span}\{\phi_1, \phi_2, \phi_3\}$$

is the light family subspace, and $\mathcal{H}_H$ is the heavy sector.

Let $\mathbb{O}(\lambda)$ denote the microscopic Hermitian operator relevant for the spectral problem. With respect to the decomposition $\mathcal{H}_L \oplus \mathcal{H}_H$, write

$$\mathbb{O}(\lambda) = \begin{pmatrix} L(\lambda) & K(\lambda) \\ K^\dagger(\lambda) & H(\lambda) \end{pmatrix},$$

where

$$L : \mathcal{H}_L \rightarrow \mathcal{H}_L, \quad K : \mathcal{H}_H \rightarrow \mathcal{H}_L, \quad H : \mathcal{H}_H \rightarrow \mathcal{H}_H.$$

Assume that $H(\lambda)$ is invertible in the spectral region of interest.

XLI. EXACT SCHUR-COMPLEMENT FORMULA

Let

$$\Psi = \begin{pmatrix} \psi_L \\ \psi_H \end{pmatrix} \in \mathcal{H}_L \oplus \mathcal{H}_H$$

satisfy

$$\mathbb{O}(\lambda)\Psi = 0.$$

Then the heavy component obeys

$$K^\dagger \psi_L + H \psi_H = 0 \implies \psi_H = -H^{-1} K^\dagger \psi_L.$$

Substituting back into the light equation gives

$$L \psi_L + K \psi_H = 0 \implies (L - K H^{-1} K^\dagger) \psi_L = 0.$$

Therefore the exact reduced light-sector operator is

$$\boxed{\mathcal{Y}_{\text{eff}}(\lambda) = L(\lambda) - K(\lambda) H(\lambda)^{-1} K(\lambda)^\dagger.}$$

If one prefers the spectral-parameter form with a fixed heavy operator H_0 , then one writes

$$\boxed{\mathcal{Y}_{\text{eff}}(\lambda) = L - K(H_0 - \lambda)^{-1} K^\dagger.}$$

This is the precise Schur-complement origin of the reduced flavor matrix.

XLII. LOW-ENERGY EXPANSION

If $\|\lambda H_0^{-1}\| < 1$, then

$$(H_0 - \lambda)^{-1} = H_0^{-1} + \lambda H_0^{-2} + \lambda^2 H_0^{-3} + \dots$$

Hence

$$\boxed{\mathcal{Y}_{\text{eff}}(\lambda) = L - K H_0^{-1} K^\dagger - \lambda K H_0^{-2} K^\dagger - \lambda^2 K H_0^{-3} K^\dagger - \dots}$$

So the leading reduced operator is

$$\mathcal{Y}_{\text{eff}}^{(0)} = L - K H_0^{-1} K^\dagger,$$

and the higher powers of H_0^{-1} generate systematic subleading corrections.

XLIII. KERNEL REPRESENTATION FOR THE PDE PROBLEM

Suppose the heavy sector is represented by a differential operator $\mathbb{H}$ with Green kernel

$$G_H(z, w; \lambda) = \langle z | (\mathbb{H} - \lambda)^{-1} | w \rangle.$$

Let the light-heavy mixing kernel be $\mathbb{K}(x, z)$. Then the effective light kernel is

$$\boxed{\mathcal{Y}_{\text{eff}}(x, y; \lambda) = L(x, y; \lambda) - \iint \mathbb{K}(x, z) G_H(z, w; \lambda) \mathbb{K}^\dagger(w, y) dz dw.}$$

Projecting onto the family basis $\{\phi_a\}_{a=1}^3$, one gets

$$(\mathcal{Y}_{\text{eff}})_{ab} = \iint \phi_a^\dagger(x) \mathcal{Y}_{\text{eff}}(x, y; \lambda) \phi_b(y) dx dy.$$

Define the source profile induced by the a -th family mode:

$$J_a(z; \lambda) := \int \mathbb{K}^\dagger(z, x; \lambda) \phi_a(x) dx.$$

Then

$$(\mathcal{Y}_{\text{eff}})_{ab} = L_{ab} - \iint J_a^\dagger(z; \lambda) G_H(z, w; \lambda) J_b(w; \lambda) dz dw.$$

This is the general microscopic matrix-element formula. It is the exact place where the reduced family coefficients come from.

XLIV. FINITE HEAVY-MODE EXPANSION

Let $\{\chi_\alpha\}$ be an orthonormal heavy basis, so that

$$H_{\alpha\beta} = \langle \chi_\alpha, \mathbb{O} \chi_\beta \rangle, \quad K_{a\alpha} = \langle \phi_a, \mathbb{O} \chi_\alpha \rangle, \quad L_{ab} = \langle \phi_a, \mathbb{O} \phi_b \rangle.$$

Then

$$(\mathcal{Y}_{\text{eff}})_{ab} = L_{ab} - \sum_{\alpha, \beta} K_{a\alpha} (H^{-1})_{\alpha\beta} \overline{K_{b\beta}}.$$

Thus the entire reduced flavor problem has been reduced to the microscopic quantities

$$L_{ab}, \quad K_{a\alpha}, \quad H_{\alpha\beta}.$$

XLV. PROJECTION ONTO THE CANONICAL HERMITIAN BASIS

We now project the effective 3×3 Hermitian operator onto the canonical basis

$$D_0 = \mathbf{1}, \quad D_1 = \text{diag}(1, -1, 0), \quad D_2 = \text{diag}(1, 1, -2),$$

$$S_{12} = E_{12} + E_{21}, \quad S_{13} = E_{13} + E_{31}, \quad S_{23} = E_{23} + E_{32},$$

$$A_{12} = i(E_{12} - E_{21}), \quad A_{13} = i(E_{13} - E_{31}), \quad A_{23} = i(E_{23} - E_{32}).$$

Every Hermitian 3×3 matrix M has the unique expansion

$$M = \alpha_0 D_0 + \alpha_1 D_1 + \alpha_2 D_2 + s_{12} S_{12} + s_{13} S_{13} + s_{23} S_{23} + a_{12} A_{12} + a_{13} A_{13} + a_{23} A_{23}.$$

The coefficients are

$$\alpha_0 = \frac{1}{3} \text{Tr} M, \quad \alpha_1 = \frac{1}{2} \text{Tr}(MD_1) = \frac{M_{11} - M_{22}}{2}, \quad \alpha_2 = \frac{1}{6} \text{Tr}(MD_2) = \frac{M_{11} + M_{22} - 2M_{33}}{6},$$

$$s_{12} = \frac{1}{2} \text{Im}(MS_{12}) = \Re M_{12}, \quad s_{13} = \frac{1}{2} \text{Im}(MS_{13}) = \Re M_{13}, \quad s_{23} = \frac{1}{2} \text{Im}(MS_{23}) = \Re M_{23},$$

$$a_{12} = \frac{1}{2} \text{Re}(MA_{12}) = \Im M_{12}, \quad a_{13} = \frac{1}{2} \text{Re}(MA_{13}) = \Im M_{13}, \quad a_{23} = \frac{1}{2} \text{Re}(MA_{23}) = \Im M_{23}.$$

Applying this to

$$M = \mathcal{Y}_{\text{eff}},$$

we obtain the exact reduced coefficients

$$\alpha_i^{\text{eff}}, \quad s_{ab}^{\text{eff}}, \quad a_{ab}^{\text{eff}}$$

directly from the microscopic block data.

XLVI. ORIGIN OF THE CP-ODD COEFFICIENTS

From

$$(\mathcal{Y}_{\text{eff}})_{ab} = L_{ab} - \sum_{\alpha,\beta} K_{a\alpha}(H^{-1})_{\alpha\beta}\overline{K_{b\beta}},$$

it follows that

$$a_{ab}^{\text{eff}} = \Im L_{ab} - \Im \left[\sum_{\alpha,\beta} K_{a\alpha}(H^{-1})_{\alpha\beta}\overline{K_{b\beta}} \right].$$

Thus, if the bare light block L is real in the chosen family basis,

$$a_{ab}^{\text{eff}} = -\Im \left[\sum_{\alpha,\beta} K_{a\alpha}(H^{-1})_{\alpha\beta}\overline{K_{b\beta}} \right].$$

Now assume H is positive definite. Define the heavy-coupling vectors

$$u_a := H^{-1/2}K^\dagger e_a \in \mathcal{H}_H,$$

where e_a is the a -th standard basis vector of $\mathbb{C}^3$. Then

$$\sum_{\alpha,\beta} K_{a\alpha}(H^{-1})_{\alpha\beta}\overline{K_{b\beta}} = u_a^\dagger u_b.$$

Hence

$$(\mathcal{Y}_{\text{eff}})_{ab} = L_{ab} - u_a^\dagger u_b,$$

and therefore

$$a_{ab}^{\text{eff}} = \Im L_{ab} - \Im(u_a^\dagger u_b).$$

This formula is conceptually decisive: the CP-odd reduced coefficients are generated by the imaginary overlaps of the heavy-dressed coupling vectors u_a .

XLVII. REALITY CRITERION

We now obtain a strict no-go criterion.

a. Proposition. If there exists a light/heavy basis in which

$$L = L^*, \quad K = K^*, \quad H = H^*,$$

then

$$\mathcal{Y}_{\text{eff}} = \mathcal{Y}_{\text{eff}}^*,$$

hence

$$a_{12}^{\text{eff}} = a_{13}^{\text{eff}} = a_{23}^{\text{eff}} = 0.$$

b. Proof. If L, K, H are all real in the same basis, then H^{-1} is real as well, and

$$KH^{-1}K^\dagger$$

is real symmetric. Therefore

$$\mathcal{Y}_{\text{eff}} = L - KH^{-1}K^\dagger$$

is real symmetric. Hence all imaginary off-diagonal coefficients vanish. $\square$

So CP-odd reduced coefficients require a genuine microscopic complex obstruction. This is an exact statement.

XLVIII. HOW TO IDENTIFY

B_0 , B_1 , B_H from microscopic spurions

At the microscopic level, suppose the full block operator depends on small spurions ϵ_I (for example, sector-blind, hypercharge-sensitive, Higgs-like, anisotropy-like deformations):

$$L = L^{(0)} + \sum_I \epsilon_I L^{(I)} + \mathcal{O}(\epsilon^2),$$

$$K = K^{(0)} + \sum_I \epsilon_I K^{(I)} + \mathcal{O}(\epsilon^2),$$

$$H = H^{(0)} + \sum_I \epsilon_I H^{(I)} + \mathcal{O}(\epsilon^2).$$

Expanding the Schur complement to first order gives

$$\mathcal{Y}_{\text{eff}} = \mathcal{Y}_{\text{eff}}^{(0)} + \sum_I \epsilon_I \mathcal{Y}_{\text{eff}}^{(I)} + \mathcal{O}(\epsilon^2),$$

with

$$\mathcal{Y}_{\text{eff}}^{(0)} = L^{(0)} - K^{(0)}(H^{(0)})^{-1}(K^{(0)})^\dagger,$$

and

$$\mathcal{Y}_{\text{eff}}^{(I)} = L^{(I)} - K^{(I)}(H^{(0)})^{-1}(K^{(0)})^\dagger - K^{(0)}(H^{(0)})^{-1}(K^{(I)})^\dagger + K^{(0)}(H^{(0)})^{-1}H^{(I)}(H^{(0)})^{-1}(K^{(0)})^\dagger.$$

This is the exact first-order formula one needs for the microscopic identification of

$$B_0, \quad B_1, \quad B_H.$$

For example, if the spurions are chosen as

$$\epsilon_0 = 1, \quad \epsilon_Y, \quad \epsilon_H,$$

corresponding respectively to the sector-blind, hypercharge-sensitive, and Higgs-like microscopic deformations, then one sets

$$B_0 = \mathcal{Y}_{\text{eff}}^{(0)}, \quad B_1 = \mathcal{Y}_{\text{eff}}^{(Y)}, \quad B_H = \mathcal{Y}_{\text{eff}}^{(H)},$$

up to the normalization convention of the microscopic spurion parameters.

Thus B_0, B_1, B_H are not guessed textures. They are the coefficients of the microscopic Schur-complement expansion.

XLIX. WHAT IS NOW RIGOROUSLY CLOSED

The following facts are now exact at the reduced-operator level:

1. The reduced family operator is the Schur complement

$$\mathcal{Y}_{\text{eff}} = L - KH^{-1}K^\dagger.$$

2. In the PDE picture, its matrix elements are

$$(\mathcal{Y}_{\text{eff}})_{ab} = L_{ab} - \iint J_a^\dagger G_H J_b.$$

3. The Hermitian basis coefficients are read exactly by

$$\alpha_0, \alpha_1, \alpha_2, s_{ab}, a_{ab}.$$

4. The CP-odd coefficients satisfy

$$a_{ab}^{\text{eff}} = \Im L_{ab} - \Im(u_a^\dagger u_b).$$

5. If the microscopic data can all be made real in a common basis, then

$$a_{12}^{\text{eff}} = a_{13}^{\text{eff}} = a_{23}^{\text{eff}} = 0.$$

6. The first-order microscopic origin of

$$B_0, B_1, B_H$$

is given by the spurion-expanded Schur-complement formula above.

L. WHAT REMAINS OPEN

What is *not* yet closed is the actual evaluation of the block data

$$L^{(I)}, \quad K^{(I)}, \quad H^{(I)}$$

from the specific TECT defect PDE and its heavy-sector spectrum.

That is the real next frontier.

The correct remaining task is therefore:

compute the microscopic matrix elements L_{ab} , $K_{a\alpha}$, $H_{\alpha\beta}$ from the TECT defect background,

and then project them onto the basis

$$\{D_i, S_{ab}, A_{ab}\}.$$

Only that step will decide which reduced operator channels are actually generated by the microscopic theory.

LI. FINAL SUMMARY

We derived the correct microscopic origin of the reduced family flavor operator.

Given the light/heavy block decomposition

$$\mathbb{O} = \begin{pmatrix} L & K \\ K^\dagger & H \end{pmatrix},$$

the exact reduced light-sector operator is the Schur complement

$\mathcal{Y}_{\text{eff}} = L - KH^{-1}K^\dagger.$

In the PDE/kernel representation, its family matrix elements are

$(\mathcal{Y}_{\text{eff}})_{ab} = L_{ab} - \iint J_a^\dagger(z) G_H(z, w) J_b(w) dz dw.$

Projecting onto the canonical Hermitian 3×3 basis

$$\{D_0, D_1, D_2, S_{12}, S_{13}, S_{23}, A_{12}, A_{13}, A_{23}\},$$

the coefficients are

$$\alpha_0 = \frac{1}{3}M, \quad \alpha_1 = \frac{M_{11} - M_{22}}{2}, \quad \alpha_2 = \frac{M_{11} + M_{22} - 2M_{33}}{6},$$

$$s_{ab} = \Re M_{ab}, \quad a_{ab} = \Im M_{ab}.$$

Therefore the CP-odd carrier coefficients satisfy

$$a_{ab}^{\text{eff}} = \Im L_{ab} - \Im \left[\sum_{\alpha, \beta} K_{a\alpha} (H^{-1})_{\alpha\beta} \overline{K_{b\beta}} \right].$$

Equivalently, if $H > 0$ and $u_a = H^{-1/2} K^\dagger e_a$, then

$$a_{ab}^{\text{eff}} = \Im L_{ab} - \Im(u_a^\dagger u_b).$$

Hence the reduced CP structure is generated by the imaginary overlaps of the heavy-dressed coupling vectors. We also derived the exact no-go statement: if L, K, H can all be made real in a common basis, then

$$a_{12}^{\text{eff}} = a_{13}^{\text{eff}} = a_{23}^{\text{eff}} = 0.$$

Finally, if the microscopic operator depends on spurions ϵ_I , then the first-order reduced operators are

$$\mathcal{Y}_{\text{eff}}^{(0)} = L^{(0)} - K^{(0)}(H^{(0)})^{-1}(K^{(0)})^\dagger,$$

$$\mathcal{Y}_{\text{eff}}^{(I)} = L^{(I)} - K^{(I)}(H^{(0)})^{-1}(K^{(0)})^\dagger - K^{(0)}(H^{(0)})^{-1}(K^{(I)})^\dagger + K^{(0)}(H^{(0)})^{-1}H^{(I)}(H^{(0)})^{-1}(K^{(0)})^\dagger.$$

These are the correct microscopic formulas from which B_0, B_1, B_H must be extracted. So the exploratory stage is over: the next real task is to compute the microscopic block data

$$L_{ab}, \quad K_{a\alpha}, \quad H_{\alpha\beta}$$

from the actual TECT defect PDE and then project them onto

$$\{D_i, S_{ab}, A_{ab}\}.$$

LII. DEFINITION OF THE LIGHT FAMILY SUBSPACE AND THE CORRECT MICROSCOPIC COMPUTATION PIPELINE

We now formulate the reduced TECT flavor problem in a fully principled way. The goal is to define the family light sector directly from the microscopic defect operator, and then derive the reduced flavor matrices by block elimination rather than by texture ansätze.

LIII. MICROSCOPIC REFERENCE OPERATOR AND DEFECT BACKGROUND

Let

$$\Phi_{\text{def}}$$

denote a static TECT defect background solving the appropriate microscopic field equations. Let the microscopic one-particle operator on the corresponding background be

$$\mathbb{O}(\epsilon) = \mathbb{O}^{(0)} + \sum_I \epsilon_I \mathbb{V}^{(I)} + \mathcal{O}(\epsilon^2),$$

acting on a Hilbert space

$$\mathcal{H}_{\text{micro}} = L^2(\mathbb{R}^3) \otimes \mathcal{H}_{\text{int}},$$

where $\mathcal{H}_{\text{int}}$ denotes the internal/spinor/gauge fiber relevant to the reduced problem.

Here:

- $\mathbb{O}^{(0)}$ is the reference sector-blind operator on the chosen defect background;
- $\mathbb{V}^{(I)}$ are microscopic deformations (hypercharge-sensitive, Higgs-like, anisotropy-like, etc.);
- ϵ_I are small bookkeeping spurions.

We assume throughout that $\mathbb{O}^{(0)}$ is self-adjoint and that its low spectrum contains an isolated three-dimensional spectral sector.

LIV. SPECTRAL DEFINITION OF THE FAMILY LIGHT SPACE

Let

$$\sigma_L = \{\lambda_1^{(0)}, \lambda_2^{(0)}, \lambda_3^{(0)}\}$$

be three isolated low eigenvalues of $\mathbb{O}^{(0)}$, separated from the rest of the spectrum by a gap

$$\Delta_H > 0.$$

Let

$$P_L = \mathbf{1}_{\sigma_L}(\mathbb{O}^{(0)})$$

be the spectral projector onto this low-energy sector, and define

$$\boxed{\mathcal{H}_L := \text{Ran}(P_L).}$$

Then

$$\dim \mathcal{H}_L = 3.$$

This is the correct mathematical definition of the family space.

The heavy sector is the orthogonal complement

$$\boxed{\mathcal{H}_H := \text{Ran}(P_H), \quad P_H := 1 - P_L.}$$

This definition is basis-independent and does not rely on any Gaussian approximation.

a. Important point. The fundamental object is the projector P_L , not a specific basis. Any orthonormal basis

$$\{\phi_1, \phi_2, \phi_3\} \subset \mathcal{H}_L$$

is admissible. If one wants a localized family interpretation, one may choose a Wannier-type or defect-centered basis inside $\mathcal{H}_L$, but the reduced operator itself is defined independently of that choice.

LV. LOCALIZED FAMILY BASIS AS A UNITARY GAUGE CHOICE INSIDE $\mathcal{H}_L$

Let

$$\{u_i\}_{i=1}^3$$

be an orthonormal eigenbasis of $\mathbb{O}^{(0)}|_{\mathcal{H}_L}$:

$$\mathbb{O}^{(0)}u_i = \lambda_i^{(0)}u_i.$$

A localized family basis is then any unitary rotation

$$\phi_a = \sum_{i=1}^3 U_{ia} u_i, \quad U \in U(3).$$

Thus

$$\{\phi_a\}_{a=1}^3$$

can be chosen to optimize defect localization near three defect cores X_a , but this is a basis choice inside the already well-defined light space.

So the logical order is:

$$\boxed{\text{first define } P_L \text{ spectrally, then choose a convenient basis } \phi_a \text{ inside } \mathcal{H}_L.}$$

LVI. EXACT LIGHT/HEAVY BLOCK DECOMPOSITION

Because P_L is a spectral projector of $\mathbb{O}^{(0)}$, one has

$$[\mathbb{O}^{(0)}, P_L] = 0, \quad [\mathbb{O}^{(0)}, P_H] = 0.$$

Hence the reference operator is exactly block-diagonal:

$$\mathbb{O}^{(0)} = \begin{pmatrix} L^{(0)} & 0 \\ 0 & H^{(0)} \end{pmatrix}$$

with respect to

$$\mathcal{H}_{\text{micro}} = \mathcal{H}_L \oplus \mathcal{H}_H.$$

For the full operator

$$\mathbb{O}(\epsilon) = \mathbb{O}^{(0)} + \mathbb{V}(\epsilon), \quad \mathbb{V}(\epsilon) = \sum_I \epsilon_I \mathbb{V}^{(I)} + \mathcal{O}(\epsilon^2),$$

the exact block form is

$$\mathbb{O}(\epsilon) = \begin{pmatrix} L^{(0)} + V_{LL} & V_{LH} \\ V_{HL} & H^{(0)} + V_{HH} \end{pmatrix},$$

where

$$V_{LL} = P_L \mathbb{V} P_L, \quad V_{LH} = P_L \mathbb{V} P_H, \quad V_{HL} = P_H \mathbb{V} P_L, \quad V_{HH} = P_H \mathbb{V} P_H.$$

LVII. EXACT REDUCED OPERATOR AND PERTURBATIVE EXPANSION

The exact reduced light-sector operator is the Schur complement

$$\boxed{\mathcal{Y}_{\text{eff}} = L^{(0)} + V_{LL} - V_{LH} (H^{(0)} + V_{HH})^{-1} V_{HL}.}$$

Expanding in small ϵ_I , one gets to second order

$$\boxed{\mathcal{Y}_{\text{eff}} = L^{(0)} + V_{LL} - V_{LH} (H^{(0)})^{-1} V_{HL} + \mathcal{O}(\epsilon^3),}$$

because V_{LL} is first order and $V_{LH} (H^{(0)})^{-1} V_{HL}$ is second order.

Thus the first two nontrivial layers are:

- first-order direct light projection:

$$\mathcal{Y}_{\text{eff}}^{(1)} = V_{LL};$$

- second-order heavy-mediated correction:

$$\mathcal{Y}_{\text{eff}}^{(2)} = -V_{LH} (H^{(0)})^{-1} V_{HL}.$$

This is the correct hierarchy. The reduced flavor structure must be computed in this order.

LVIII. DEFINITION OF THE MATRIX ELEMENTS L, K, H

Choose an orthonormal family basis

$$\{\phi_a\}_{a=1}^3 \subset \mathcal{H}_L$$

and an orthonormal heavy basis

$$\{\chi_\alpha\}_{\alpha \in \mathcal{A}} \subset \mathcal{H}_H.$$

Then the microscopic block matrix elements are

$$L_{ab}^{(0)} = \langle \phi_a, \mathbb{O}^{(0)} \phi_b \rangle,$$

$$V_{ab}^{(I)} = \langle \phi_a, \mathbb{V}^{(I)} \phi_b \rangle,$$

$$K_{a\alpha}^{(I)} = \langle \phi_a, \mathbb{V}^{(I)} \chi_\alpha \rangle,$$

$$H_{\alpha\beta}^{(0)} = \langle \chi_\alpha, \mathbb{O}^{(0)} \chi_\beta \rangle.$$

Because $\phi_a \in \mathcal{H}_L$ and $\chi_\alpha \in \mathcal{H}_H$, the reference light-heavy mixing vanishes identically:

$$\langle \phi_a, \mathbb{O}^{(0)} \chi_\alpha \rangle = 0.$$

This is why the correct computation order is:

(1) solve the reference spectral problem, (2) define P_L, P_H , (3) compute only perturbative V_{LL}, V_{LH}, V_{HH} .

LIX. HEAVY EIGENBASIS AND THE SECOND-ORDER FORMULA

Now choose the heavy basis χ_α to diagonalize the reference heavy block:

$$\mathbb{O}^{(0)} \chi_\alpha = E_\alpha \chi_\alpha, \quad E_\alpha \geq \Delta_H > 0.$$

Then

$$H_{\alpha\beta}^{(0)} = E_\alpha \delta_{\alpha\beta}, \quad (H^{(0)})_{\alpha\beta}^{-1} = E_\alpha^{-1} \delta_{\alpha\beta}.$$

Therefore the second-order correction becomes

$$(\mathcal{Y}_{\text{eff}}^{(2)})_{ab} = - \sum_{I,J} \epsilon_I \epsilon_J \sum_{\alpha} \frac{K_{a\alpha}^{(I)} \overline{K_{b\alpha}^{(J)}}}{E_\alpha}.$$

In particular, if one temporarily focuses on a single perturbation channel $\mathbb{V}$, then

$$(\mathcal{Y}_{\text{eff}}^{(2)})_{ab} = - \sum_{\alpha} \frac{K_{a\alpha} \overline{K_{b\alpha}}}{E_\alpha}.$$

This is the microscopic formula that must be evaluated in the TECT defect background.

LX. COORDINATE-SPACE FORMULAS

If the operator acts on wavefunctions in space, then

$$V_{ab}^{(I)} = \int d^3x \phi_a^\dagger(x) (\mathbb{V}^{(I)} \phi_b)(x),$$

$$K_{a\alpha}^{(I)} = \int d^3x \phi_a^\dagger(x) (\mathbb{V}^{(I)} \chi_\alpha)(x).$$

For a local potential-type deformation

$$\mathbb{V}^{(I)} = U_I(x),$$

these reduce to

$$V_{ab}^{(I)} = \int d^3x \phi_a^\dagger(x) U_I(x) \phi_b(x), \quad K_{a\alpha}^{(I)} = \int d^3x \phi_a^\dagger(x) U_I(x) \chi_\alpha(x).$$

For a first-order Dirac-type perturbation

$$\mathbb{V}^{(I)} = \Gamma^i A_i^{(I)}(x) + M_I(x),$$

one has

$$V_{ab}^{(I)} = \int d^3x \phi_a^\dagger(x) (\Gamma^i A_i^{(I)}(x) + M_I(x)) \phi_b(x),$$

and similarly for $K_{a\alpha}^{(I)}$. All derivative terms must be treated with the Hermitian domain conditions of the microscopic operator.

LXI. PROJECTION ONTO THE HERMITIAN FAMILY BASIS

After computing

$$M := \mathcal{Y}_{\text{eff}}|_{\mathcal{H}_L},$$

one expands it in the canonical Hermitian basis

$$\{D_0, D_1, D_2, S_{12}, S_{13}, S_{23}, A_{12}, A_{13}, A_{23}\}.$$

The coefficients are

$$\alpha_0 = \frac{1}{3}M, \quad \alpha_1 = \frac{M_{11} - M_{22}}{2}, \quad \alpha_2 = \frac{M_{11} + M_{22} - 2M_{33}}{6},$$

$$s_{12} = \Re M_{12}, \quad s_{13} = \Re M_{13}, \quad s_{23} = \Re M_{23},$$

$$a_{12} = \Im M_{12}, \quad a_{13} = \Im M_{13}, \quad a_{23} = \Im M_{23}.$$

Thus the actual computational pipeline is

$$\Phi_{\text{def}} \rightarrow \mathbb{O}^{(0)} \rightarrow P_L, P_H \rightarrow \phi_a, \chi_\alpha \rightarrow V_{ab}^{(I)}, K_{a\alpha}^{(I)}, E_\alpha \rightarrow \mathcal{Y}_{\text{eff}} \rightarrow (\alpha_i, s_{ab}, a_{ab}).$$

LXII. MICROSCOPIC ORIGIN OF THE CP-ODD COEFFICIENTS

From the second-order formula,

$$(\mathcal{Y}_{\text{eff}}^{(2)})_{ab} = - \sum_{I,J} \epsilon_I \epsilon_J \sum_{\alpha} \frac{K_{a\alpha}^{(I)} \overline{K_{b\alpha}^{(J)}}}{E_{\alpha}},$$

it follows immediately that

$$a_{ab}^{(2)} = - \sum_{I,J} \epsilon_I \epsilon_J \sum_{\alpha} \frac{\Im \left(K_{a\alpha}^{(I)} \overline{K_{b\alpha}^{(J)}} \right)}{E_{\alpha}}.$$

This formula is decisive. It shows that the reduced CP-odd coefficients do not come from vague “complex textures”, but from microscopic phase obstructions in the heavy-mediated overlap products

$$K_{a\alpha}^{(I)} \overline{K_{b\alpha}^{(J)}}.$$

Therefore:

a. Exact criterion. If there exists a common basis in which all

$$K_{a\alpha}^{(I)}$$

are real and all $V_{ab}^{(I)}$ are real, then

$$a_{12} = a_{13} = a_{23} = 0$$

to the corresponding perturbative order.

So CP-odd reduced coefficients require a genuine microscopic complex obstruction.

LXIII. IDENTIFICATION OF

B_0, B_1, B_H

Suppose the perturbations are grouped as

$$\mathbb{V} = \epsilon_0 \mathbb{V}^{(0)} + \epsilon_Y \mathbb{V}^{(Y)} + \epsilon_H \mathbb{V}^{(H)} + \dots$$

Then the reduced effective operator admits the perturbative decomposition

$$\mathcal{Y}_{\text{eff}} = L^{(0)} + \epsilon_0 \mathcal{B}_0^{(1)} + \epsilon_Y \mathcal{B}_1^{(1)} + \epsilon_H \mathcal{B}_H^{(1)} + \mathcal{Q}^{(2)} + \dots,$$

where

$$\mathcal{B}_0^{(1)} = P_L \mathbb{V}^{(0)} P_L, \quad \mathcal{B}_1^{(1)} = P_L \mathbb{V}^{(Y)} P_L, \quad \mathcal{B}_H^{(1)} = P_L \mathbb{V}^{(H)} P_L,$$

and

$$\mathcal{Q}^{(2)} = - \sum_{I,J} \epsilon_I \epsilon_J P_L \mathbb{V}^{(I)} P_H (H^{(0)})^{-1} P_H \mathbb{V}^{(J)} P_L.$$

So, microscopically, the objects previously called B_0, B_1, B_H are not guessed matrices. They are the projected first-order spurion responses together with the systematic heavy-mediated corrections.

LXIV. WHAT IS RIGOROUSLY CLOSED

The following items are now exact:

1. The family space is the isolated three-dimensional spectral subspace

$$\mathcal{H}_L = \text{Ran } \mathbf{1}_{\sigma_L}(\mathbb{O}^{(0)}).$$

2. A localized family basis is only a $U(3)$ gauge choice inside $\mathcal{H}_L$.
3. The exact reduced operator is the Schur complement

$$\mathcal{Y}_{\text{eff}} = L^{(0)} + V_{LL} - V_{LH}(H^{(0)} + V_{HH})^{-1}V_{HL}.$$

4. To second order,

$$\mathcal{Y}_{\text{eff}} = L^{(0)} + V_{LL} - V_{LH}(H^{(0)})^{-1}V_{HL} + \mathcal{O}(\epsilon^3).$$

5. In the heavy eigenbasis,

$$(\mathcal{Y}_{\text{eff}}^{(2)})_{ab} = - \sum_{I,J} \epsilon_I \epsilon_J \sum_{\alpha} \frac{K_{a\alpha}^{(I)} \overline{K_{b\alpha}^{(J)}}}{E_{\alpha}}.$$

6. Therefore the reduced CP-odd coefficients satisfy

$$a_{ab}^{(2)} = - \sum_{I,J} \epsilon_I \epsilon_J \sum_{\alpha} \frac{\Im \left(K_{a\alpha}^{(I)} \overline{K_{b\alpha}^{(J)}} \right)}{E_{\alpha}}.$$

LXV. WHAT REMAINS OPEN

What is not yet closed is the explicit evaluation of

$$V_{ab}^{(I)}, \quad K_{a\alpha}^{(I)}, \quad E_{\alpha}$$

for the actual TECT defect PDE and its heavy-mode spectrum.

That is now the correct next frontier. The remaining work is no longer conceptual but computational: solve the defect spectral problem, construct the light projector, compute the overlap matrix elements, and project the resulting $\mathcal{Y}_{\text{eff}}$ onto the Hermitian family basis.

LXVI. FINAL SUMMARY

The reduced TECT flavor problem must be organized around the spectral projector of the microscopic reference defect operator.

The correct family space is

$$\mathcal{H}_L = \text{Ran } \mathbf{1}_{\sigma_L}(\mathbb{O}^{(0)}),$$

where σ_L is an isolated three-dimensional low-energy spectral sector. A localized family basis $\{\phi_a\}$ is then only a unitary choice inside this fixed space.

With respect to the decomposition

$$\mathcal{H}_{\text{micro}} = \mathcal{H}_L \oplus \mathcal{H}_H,$$

the full operator takes the block form

$$\mathbb{O} = \begin{pmatrix} L^{(0)} + V_{LL} & V_{LH} \\ V_{HL} & H^{(0)} + V_{HH} \end{pmatrix},$$

and the exact reduced family operator is

$$\mathcal{Y}_{\text{eff}} = L^{(0)} + V_{LL} - V_{LH}(H^{(0)} + V_{HH})^{-1}V_{HL}.$$

To second order in the perturbations,

$$\mathcal{Y}_{\text{eff}} = L^{(0)} + V_{LL} - V_{LH}(H^{(0)})^{-1}V_{HL} + \mathcal{O}(\epsilon^3).$$

If the heavy block is diagonalized as

$$\mathbb{O}^{(0)}\chi_\alpha = E_\alpha\chi_\alpha,$$

then

$$(\mathcal{Y}_{\text{eff}}^{(2)})_{ab} = - \sum_{I,J} \epsilon_I \epsilon_J \sum_{\alpha} \frac{K_{a\alpha}^{(I)} \overline{K_{b\alpha}^{(J)}}}{E_\alpha},$$

so the reduced CP-odd coefficients are

$$a_{ab}^{(2)} = - \sum_{I,J} \epsilon_I \epsilon_J \sum_{\alpha} \frac{\Im(K_{a\alpha}^{(I)} \overline{K_{b\alpha}^{(J)}})}{E_\alpha}.$$

Therefore the true microscopic origin of reduced CP structure is the imaginary heavy-mediated overlap of the light-heavy couplings.

The correct computational order is:

$$\Phi_{\text{def}} \rightarrow \mathbb{O}^{(0)} \rightarrow P_L, P_H \rightarrow \phi_a, \chi_\alpha \rightarrow V_{ab}^{(I)}, K_{a\alpha}^{(I)}, E_\alpha \rightarrow \mathcal{Y}_{\text{eff}} \rightarrow (D_i, S_{ab}, A_{ab}) \text{ coefficients.}$$

Hence the next real task is no longer exploratory fitting. It is the explicit evaluation of

$$V_{ab}^{(I)}, \quad K_{a\alpha}^{(I)}, \quad E_\alpha$$

from the actual TECT defect PDE and heavy-mode spectrum.

LXVII. CONSTRUCTION OF THE LIGHT FAMILY PROJECTOR FROM SINGLE-DEFECT MODES

We now give the principled construction of the reduced family space directly from the microscopic reference operator on a multi-defect background.

LXVIII. REFERENCE OPERATOR ON A SINGLE DEFECT

Let

$$\Phi_{1\text{def}}$$

be a single static TECT defect background centered at the origin, and let

$$\mathbb{O}_{1\text{def}}^{(0)}$$

be the corresponding self-adjoint reference operator on

$$\mathcal{H}_{\text{micro}} = L^2(\mathbb{R}^3) \otimes \mathcal{H}_{\text{int}}.$$

We assume that the single-defect problem has an isolated low-energy bound state (or quasi-zero mode)

$$\psi_\star \in \mathcal{H}_{\text{micro}}, \quad \mathbb{O}_{1\text{def}}^{(0)}\psi_\star = \lambda_\star\psi_\star,$$

with

$$\|\psi_\star\| = 1,$$

and that the rest of the spectrum is separated by a gap

$$\Delta_H > 0.$$

The simplest family scenario is the rank-one case:

$$\dim \ker_{\text{low}}(\mathbb{O}_{\text{1def}}^{(0)} - \lambda_\star) = 1.$$

If a single defect carried $q > 1$ protected low modes, then a three-defect cluster would produce $3q$ light states. Hence the observed family dimension 3 naturally corresponds to the rank-one single-defect case.

LXIX. TRANSLATED DEFECT MODES

Let $X_1, X_2, X_3 \in \mathbb{R}^3$ be three well-separated defect centers. Assume that the three-defect background

$$\Phi_{\text{3def}}$$

is a weakly interacting superposition of three single defects centered at the points X_a .

Define the translated approximate bound states

$$\psi_a(x) := \mathcal{T}_{X_a} \psi_\star(x) = \psi_\star(x - X_a), \quad a = 1, 2, 3.$$

If the internal bundle has a gauge/spin connection, then one should more properly use parallel-transported translates

$$\psi_a(x) = U_a(x) \psi_\star(x - X_a),$$

where U_a is the appropriate microscopic transport operator. For the abstract construction, this refinement does not change the algebra.

Because $\psi_\star$ is exponentially localized,

$$|\psi_\star(x)| \lesssim e^{-\mu|x|},$$

the overlaps satisfy

$$\langle \psi_a, \psi_b \rangle = \delta_{ab} + \mathcal{O}(e^{-\mu R_{ab}}), \quad R_{ab} := |X_a - X_b|.$$

Thus the Gram matrix

$$G_{ab} := \langle \psi_a, \psi_b \rangle$$

obeys

$$G = I + \mathcal{O}(e^{-\mu R_{\min}}), \quad R_{\min} := \min_{a \neq b} R_{ab},$$

and is invertible for sufficiently large separation.

LXX. APPROXIMATE LOW SUBSPACE OF THE THREE-DEFECT OPERATOR

Let

$$\mathbb{O}_{\text{3def}}^{(0)}$$

be the microscopic reference operator on the three-defect background. Define the span of the translated single-defect modes

$$\mathcal{S}_{\text{app}} := \text{span}\{\psi_1, \psi_2, \psi_3\}.$$

The first reduced matrix on this approximate subspace is

$$M_{ab}^{\text{app}} := \langle \psi_a, \mathbb{O}_{3\text{def}}^{(0)} \psi_b \rangle.$$

Because each ψ_a is an approximate localized eigenmode near the a -th defect, one expects

$$M_{ab}^{\text{app}} = \lambda_\star G_{ab} + T_{ab},$$

where

$$T_{ab} = \mathcal{O}(e^{-\mu R_{ab}})$$

encodes the tunneling / overlap correction.

If $\lambda_\star = 0$, then

$$M^{\text{app}} = T$$

is the effective tight-binding matrix of the low cluster.

LXXI. ORTHONORMALIZED LOCALIZED BASIS

Since the ψ_a are not exactly orthonormal, define

$$\phi_a := \sum_{b=1}^3 \psi_b (G^{-1/2})_{ba}.$$

Then

$$\langle \phi_a, \phi_b \rangle = \delta_{ab}.$$

The projected approximate light matrix becomes

$$L^{\text{app}} := G^{-1/2} M^{\text{app}} G^{-1/2},$$

that is,

$$L_{ab}^{\text{app}} = \langle \phi_a, \mathbb{O}_{3\text{def}}^{(0)} \phi_b \rangle.$$

This is the canonical orthonormalized localized family basis associated with the three-defect configuration.

LXXII. EXACT SPECTRAL PROJECTOR AND ITS RELATION TO THE APPROXIMATE SPAN

The true light family space is not merely $\mathcal{S}_{\text{app}}$, but the exact low spectral subspace of $\mathbb{O}_{3\text{def}}^{(0)}$.

Let Γ be a contour in the complex plane enclosing only the three low eigenvalues associated with the defect cluster and no other spectrum. Then the exact light projector is the Riesz projector

$$P_L = \frac{1}{2\pi i} \oint_{\Gamma} (z - \mathbb{O}_{3\text{def}}^{(0)})^{-1} dz.$$

Define

$$\mathcal{H}_L := \text{Ran}(P_L).$$

Then

$$\dim \mathcal{H}_L = 3$$

provided the three clustered eigenvalues remain isolated from the heavy spectrum.

Under standard exponentially localized multi-center perturbation theory, one has

$$\|P_L - P_{\text{app}}\| \lesssim e^{-\mu R_{\min}},$$

where P_{app} is the orthogonal projector onto $\mathcal{S}_{\text{app}}$. Hence the approximate span of translated single-defect modes converges exponentially to the exact light family space.

This is the precise meaning of the statement that the family sector is generated by three defect-localized modes.

LXXIII. BASIS-INDEPENDENCE AND LOCALIZED GAUGE CHOICE

The family space is the exact subspace

$$\mathcal{H}_L = \text{Ran}(P_L),$$

which is basis-independent. A localized family basis is any orthonormal basis of $\mathcal{H}_L$ chosen to maximize localization near the three defect cores.

In practice, one may use either:

1. the orthonormalized translated modes ϕ_a above;
2. the eigenbasis of $P_L \hat{x} P_L$ (or of another localization functional);
3. a Wannier-type minimization inside $\mathcal{H}_L$.

All such choices are related by a unitary matrix in $U(3)$, and therefore define the same reduced operator up to family-basis conjugation.

LXXIV. HEAVY SECTOR AND EXACT BLOCK SPLITTING

Let

$$P_H := 1 - P_L, \quad \mathcal{H}_H := \text{Ran}(P_H).$$

Then, with respect to

$$\mathcal{H}_{\text{micro}} = \mathcal{H}_L \oplus \mathcal{H}_H,$$

the reference operator is exactly block-diagonal:

$$\mathbb{O}_{3\text{def}}^{(0)} = \begin{pmatrix} L^{(0)} & 0 \\ 0 & H^{(0)} \end{pmatrix},$$

where

$$L^{(0)} := P_L \mathbb{O}_{3\text{def}}^{(0)} P_L, \quad H^{(0)} := P_H \mathbb{O}_{3\text{def}}^{(0)} P_H.$$

If the three low eigenvalues are nearly degenerate, then in a localized basis $\{\phi_a\}$,

$$L_{ab}^{(0)} = \langle \phi_a, \mathbb{O}_{3\text{def}}^{(0)} \phi_b \rangle$$

is exactly the reference family matrix induced by defect tunneling.

LXXV. PERTURBATIONS AND THE CORRECT COMPUTATION ORDER

Now introduce microscopic deformations

$$\mathbb{V}(\epsilon) = \sum_I \epsilon_I \mathbb{V}^{(I)} + \mathcal{O}(\epsilon^2), \quad \mathbb{O}(\epsilon) = \mathbb{O}_{3\text{def}}^{(0)} + \mathbb{V}(\epsilon).$$

The correct block matrix elements are

$$V_{LL}^{(I)} := P_L \mathbb{V}^{(I)} P_L, \quad V_{LH}^{(I)} := P_L \mathbb{V}^{(I)} P_H, \quad V_{HL}^{(I)} := P_H \mathbb{V}^{(I)} P_L, \quad V_{HH}^{(I)} := P_H \mathbb{V}^{(I)} P_H.$$

Choose an orthonormal localized family basis $\{\phi_a\} \subset \mathcal{H}_L$ and an orthonormal heavy basis $\{\chi_\alpha\} \subset \mathcal{H}_H$. Then

$$V_{ab}^{(I)} = \langle \phi_a, \mathbb{V}^{(I)} \phi_b \rangle, \quad K_{a\alpha}^{(I)} = \langle \phi_a, \mathbb{V}^{(I)} \chi_\alpha \rangle, \quad H_{\alpha\beta}^{(0)} = \langle \chi_\alpha, \mathbb{O}_{3\text{def}}^{(0)} \chi_\beta \rangle.$$

The exact reduced operator is

$$\mathcal{Y}_{\text{eff}} = L^{(0)} + V_{LL} - V_{LH}(H^{(0)} + V_{HH})^{-1}V_{HL}.$$

To second order,

$$\boxed{\mathcal{Y}_{\text{eff}} = L^{(0)} + V_{LL} - V_{LH}(H^{(0)})^{-1}V_{HL} + \mathcal{O}(\epsilon^3).}$$

So the correct computational pipeline is:

$$\boxed{(1) \text{ solve the reference three-defect spectral problem,}}$$

$$\boxed{(2) \text{ construct } P_L \text{ and a localized basis } \phi_a,}$$

$$\boxed{(3) \text{ compute } V_{ab}^{(I)} \text{ and } K_{a\alpha}^{(I)},}$$

$$\boxed{(4) \text{ evaluate } \mathcal{Y}_{\text{eff}}, \text{ then project to } D_i, S_{ab}, A_{ab}.}$$

LXXVI. HEAVY-EIGENBASIS FORMULA

If the heavy basis diagonalizes the reference heavy block,

$$\mathbb{Q}_{3\text{def}}^{(0)}\chi_\alpha = E_\alpha\chi_\alpha, \quad E_\alpha \geq \Delta_H > 0,$$

then

$$(H^{(0)})_{\alpha\beta}^{-1} = E_\alpha^{-1}\delta_{\alpha\beta},$$

and the second-order reduced matrix is

$$\boxed{(\mathcal{Y}_{\text{eff}}^{(2)})_{ab} = - \sum_{I,J} \epsilon_I \epsilon_J \sum_{\alpha} \frac{K_{a\alpha}^{(I)} \overline{K_{b\alpha}^{(J)}}}{E_\alpha}.$$

Hence the CP-odd reduced coefficients are

$$\boxed{a_{ab}^{(2)} = - \sum_{I,J} \epsilon_I \epsilon_J \sum_{\alpha} \frac{\Im(K_{a\alpha}^{(I)} \overline{K_{b\alpha}^{(J)}})}{E_\alpha}.$$

This is the exact microscopic source of CP structure in the reduced family operator.

LXXVII. WHAT IS RIGOROUSLY CLOSED AND WHAT IS NOT

The following points are now rigorously closed:

1. The family space is the exact low spectral subspace

$$\mathcal{H}_L = \text{Ran}(P_L)$$

of the reference three-defect operator.

2. A localized family basis is a unitary gauge choice inside $\mathcal{H}_L$, not an additional assumption.

3. For well-separated defects, the approximate span of translated single-defect modes converges exponentially to the exact family space.
4. The reduced operator is obtained by Schur complement after this spectral decomposition, not before it.
5. The CP-odd reduced coefficients arise from the imaginary heavy-mediated overlaps

$$\Im\left(K_{a\alpha}^{(I)}\overline{K_{b\alpha}^{(J)}}\right).$$

What is not yet closed is the concrete TECT-specific input:

1. the actual form of $\mathbb{O}_{1\text{def}}^{(0)}$ and $\mathbb{O}_{3\text{def}}^{(0)}$;
2. the proof that a single defect carries exactly one protected light mode;
3. the size of the heavy gap Δ_H ;
4. the explicit overlap integrals $V_{ab}^{(I)}$ and $K_{a\alpha}^{(I)}$.

These require the explicit TECT defect PDE.

LXXXVIII. FINAL SUMMARY

The correct family-space construction in the reduced TECT flavor problem is spectral, not ansatz-based. Start from the reference three-defect operator

$$\mathbb{O}_{3\text{def}}^{(0)}$$

and define the exact light family projector by the Riesz formula

$$P_L = \frac{1}{2\pi i} \oint_{\Gamma} (z - \mathbb{O}_{3\text{def}}^{(0)})^{-1} dz,$$

where Γ encloses the isolated three-state low-energy cluster.

Then

$$\mathcal{H}_L = \text{Ran}(P_L)$$

is the exact family space. A localized family basis $\{\phi_a\}$ is only a $U(3)$ choice inside this space.

For well-separated defects, the translated single-defect modes

$$\psi_a(x) = \psi_{\star}(x - X_a)$$

span an approximate low subspace whose projector converges exponentially to P_L .

After this spectral splitting, one computes the perturbation blocks

$$V_{LL}^{(I)}, \quad V_{LH}^{(I)}, \quad V_{HH}^{(I)},$$

and obtains the reduced operator by Schur complement:

$$\mathcal{Y}_{\text{eff}} = L^{(0)} + V_{LL} - V_{LH}(H^{(0)} + V_{HH})^{-1}V_{HL}.$$

To second order,

$$\mathcal{Y}_{\text{eff}} = L^{(0)} + V_{LL} - V_{LH}(H^{(0)})^{-1}V_{HL} + \mathcal{O}(\epsilon^3).$$

If the heavy basis diagonalizes $H^{(0)}$, then

$$(\mathcal{Y}_{\text{eff}}^{(2)})_{ab} = - \sum_{I,J} \epsilon_I \epsilon_J \sum_{\alpha} \frac{K_{a\alpha}^{(I)} \overline{K_{b\alpha}^{(J)}}}{E_{\alpha}},$$

so the CP-odd coefficients are

$$a_{ab}^{(2)} = - \sum_{I,J} \epsilon_I \epsilon_J \sum_{\alpha} \frac{\Im \left(K_{a\alpha}^{(I)} \overline{K_{b\alpha}^{(J)}} \right)}{E_{\alpha}}.$$

Thus the family problem is now fully organized. What remains is not conceptual but TECT-specific: write the actual defect operator, solve the single-defect mode problem, build the three-defect projector, and evaluate the overlap matrix elements.

LXXIX. MINIMAL SINGLE-DEFECT REFERENCE OPERATOR AND THE STATUS OF THE ONE-MODE ASSUMPTION

We now specify the minimal principled form of the single-defect reference operator that should underlie the reduced TECT flavor construction.

LXXX. WHY A FIRST-ORDER BUNDLE OPERATOR IS THE NATURAL REFERENCE CHOICE

At the reduced flavor stage, the reference operator should satisfy four requirements:

1. it must be self-adjoint, so that spectral projectors and Schur complements are well-defined;
2. it must admit localized defect-bound states;
3. it must naturally separate light and heavy sectors;
4. it must allow topological protection of low modes, if such protection is present.

The mathematically natural minimal class satisfying these requirements is a first-order Dirac/Callias-type operator on a bundle over $\mathbb{R}^3$:

$$\mathbb{O}_{1\text{def}}^{(0)} = i\Gamma^i \left(\partial_i + \Omega_i^{\text{def}}(x) \right) + \Phi_{\text{def}}(x).$$

Here:

- Γ^i are Hermitian Clifford generators acting on an internal fiber;
- $\Omega_i^{\text{def}}(x)$ is a defect-induced connection-type field;
- $\Phi_{\text{def}}(x)$ is a Hermitian endomorphism (mass/Higgs-type field) determined by the single-defect background.

This is the minimal reference form because:

1. it is compatible with self-adjoint elliptic theory;
2. it is the standard setting for localized zero modes bound to topological defects;
3. it fits naturally with the later light/heavy block elimination;
4. it isolates the topological input in the asymptotic structure of Φ_{def} .

LXXXI. ABSTRACT ASSUMPTIONS UNDER WHICH ONE LOCALIZED LIGHT MODE CAN BE JUSTIFIED

We now separate what can be established abstractly from what remains TECT-specific. Assume the following.

a. *Assumption A1 (self-adjoint Callias form).* The operator

$$\mathbb{O}_{\text{def}}^{(0)} = i\Gamma^i(\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}}$$

is essentially self-adjoint on a natural core and defines a Fredholm operator after suitable domain completion.

b. *Assumption A2 (asymptotic gap).* There exists $R_0 > 0$ and $m_\infty > 0$ such that for $|x| > R_0$,

$$\Phi_{\text{def}}(x)^2 \geq m_\infty^2 \mathbf{1}.$$

Equivalently, the defect background approaches a gapped bulk at infinity.

c. *Assumption A3 (topological nontriviality).* The asymptotic map

$$\widehat{\Phi}_{\text{def}}(x) := \frac{\Phi_{\text{def}}(x)}{|\Phi_{\text{def}}(x)|} \quad (|x| \rightarrow \infty)$$

defines a nontrivial topological class on the sphere at infinity.

d. *Assumption A4 (index one).* The associated Callias-type index is

$$\boxed{\text{Ind}(\mathbb{O}_{\text{def}}^{(0)}) = 1.}$$

Under these assumptions, the single-defect problem has exactly one protected zero mode (counted with sign/chirality in the appropriate grading).

LXXXII. ABSTRACT THEOREM-LEVEL CONSEQUENCE

Under A1–A4, one can state the following abstract theorem.

a. *Theorem (abstract single-defect one-mode criterion).* Let

$$\mathbb{O}_{\text{def}}^{(0)} = i\Gamma^i(\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}}$$

satisfy A1–A4. Then:

1. the essential spectrum is gapped away from zero by the asymptotic gap;
2. zero is an isolated eigenvalue of finite multiplicity;
3. the defect supports exactly one normalizable protected light mode (in the index-counting sector);
4. this mode is exponentially localized.

b. *Reason.* A2 implies that outside a compact region the operator is asymptotically invertible, so the essential spectrum is separated from zero. A1 and A3 place the problem in the standard Callias/Fredholm framework. A4 fixes the net number of protected zero modes to one. Standard Agmon-type estimates then yield exponential localization of isolated defect-bound states. $\square$

This theorem is abstract but rigorous once the assumptions are verified.

LXXXIII. WHAT IS GENUINELY PROVEN AND WHAT IS NOT

It is crucial to distinguish the following two levels.

A. What is abstractly closed

If the TECT defect operator can be brought into the above Callias-type class and its index is one, then the existence of one localized light mode per defect is mathematically justified. Consequently, three well-separated defects naturally generate a three-dimensional light family sector.

So the following implication is rigorously valid:

$$\boxed{\text{Callias-type structure} + \text{asymptotic gap} + \text{index one} \implies \text{one protected light mode per defect.}}$$

B. What is still open and TECT-specific

What has *not* yet been established is that the actual TECT microscopic defect operator satisfies A1–A4. More specifically, the following items remain open:

1. the exact microscopic linearized operator of TECT;
2. the precise identification of the connection term Ω_i^{def} ;
3. the precise identification of the endomorphism Φ_{def} ;
4. proof that the asymptotic bulk is invertible/gapped;
5. computation of the relevant topological class;
6. proof that the corresponding index is exactly 1.

Therefore the statement

“one defect carries exactly one family mode”

is not yet a theorem of TECT itself. At present it is an *abstract theorem conditional on verifying A1–A4 for the TECT defect PDE*.

LXXXIV. ALTERNATIVE SECOND-ORDER REFERENCE OPERATOR AND WHY IT IS LESS FUNDAMENTAL

One could also consider a second-order Schrödinger-type operator of the form

$$\mathbb{H}_{1\text{def}}^{(0)} = -\Delta + U_{\text{def}}(x).$$

Such an operator can certainly support localized bound states and may be convenient for numerics.

However, it is less fundamental for the present structural program because:

1. it does not encode chirality/topological index as transparently;
2. zero-mode protection is less natural than in the first-order Callias framework;
3. the later CP-carrying overlap structure is more naturally expressed in a first-order bundle language.

Thus a second-order effective operator may be useful as a derived auxiliary object, but the correct minimal reference operator for the TECT flavor program should be first-order.

LXXXV. THREE-DEFECT CONSEQUENCE IF THE SINGLE-DEFECT CRITERION HOLDS

Suppose the abstract single-defect theorem applies and each isolated defect supports exactly one exponentially localized light mode

$$\psi_{\star}(x - X_a), \quad a = 1, 2, 3.$$

Then for sufficiently well-separated defects, the three-defect reference operator

$$\mathbb{O}_{3\text{def}}^{(0)}$$

has an isolated low-energy three-state cluster, and the exact light projector

$$P_L = \frac{1}{2\pi i} \oint_{\Gamma} (z - \mathbb{O}_{3\text{def}}^{(0)})^{-1} dz$$

has rank three. Thus the family space

$$\mathcal{H}_L = \text{Ran}(P_L)$$

is not assumed; it is derived.

This closes the abstract family-space construction, conditional on the single-defect theorem.

LXXXVI. MINIMAL TECT-SPECIFIC TARGET FOR THE NEXT DERIVATION

The real next task is therefore sharply defined: derive the actual TECT linearized defect operator and show that it can be cast into the form

$$\mathbb{O}_{1\text{def}}^{(0)} = i\Gamma^i(\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}},$$

then verify:

1. self-adjointness/Fredholmness;
2. asymptotic invertibility of Φ_{def} ;
3. the relevant topological charge;
4. the index value

$$\text{Ind}(\mathbb{O}_{1\text{def}}^{(0)}) = 1.$$

Only after this is done will the statement

“one defect $\leftrightarrow$ one family mode”

become a first-principles theorem inside TECT rather than a conditional structural hypothesis.

LXXXVII. FINAL SUMMARY

The natural minimal reference operator for the single-defect TECT flavor problem is a first-order Dirac/Callias-type bundle operator

$$\boxed{\mathbb{O}_{1\text{def}}^{(0)} = i\Gamma^i(\partial_i + \Omega_i^{\text{def}}(x)) + \Phi_{\text{def}}(x).}$$

This choice is structurally preferred because it is self-adjoint, supports spectral projectors, and is the standard framework for topologically protected localized defect modes.

At the abstract level, the following implication is rigorous:

self-adjoint Callias-type operator + asymptotic gap + nontrivial topological class + $\text{Ind} = 1 \implies$ one exponentially localized prot

Hence, if the TECT single-defect operator satisfies these conditions, then three well-separated defects naturally generate a rank-three light family projector.

What is not yet proved is that the actual TECT microscopic defect operator does satisfy these hypotheses. Thus the statement

“one defect carries one family mode”

is currently a conditional theorem schema, not yet a derived theorem of TECT itself.

The correct next task is therefore:

derive the actual TECT linearized defect operator and verify the Callias-type hypotheses and index 1.

LXXXVIII. FROM A MICROSCOPIC TECT PDE TO A DIRAC/CALLIAS-TYPE REFERENCE OPERATOR

We now formulate the precise theorem schema that connects a general microscopic TECT field equation to the single-defect reference operator used in the reduced flavor program.

LXXXIX. GENERAL FIRST-ORDER MICROSCOPIC FLUCTUATION EQUATION

Let

$$\mathfrak{X}(t, x)$$

denote the microscopic field multiplet of TECT, and let

$$\mathfrak{B}_{\text{def}}(x)$$

be a static single-defect background solution of the full nonlinear field equations.

We consider small fluctuations

$$\mathfrak{X}(t, x) = \mathfrak{X}_{\text{def}}(x) + \psi(t, x),$$

and assume that the linearized fluctuation equation takes the general first-order form

$$\boxed{i G_{\text{def}}^0(x) \partial_t \psi + i G_{\text{def}}^i(x) \partial_i \psi + M_{\text{def}}(x) \psi = 0.}$$

Here:

- $G_{\text{def}}^0(x)$ is the time-kinetic Hermitian matrix;
- $G_{\text{def}}^i(x)$ are the spatial principal-symbol matrices;
- $M_{\text{def}}(x)$ is the defect-induced zeroth-order endomorphism.

This is the minimal general linearized form for a first-order microscopic PDE.

XC. STRUCTURAL ASSUMPTIONS

We now state the assumptions under which this linearized equation can be cast into the desired Dirac/Callias-type form.

a. Assumption B1 (positive kinetic metric). For every x ,

$$G_{\text{def}}^0(x) = G_{\text{def}}^0(x)^\dagger, \quad G_{\text{def}}^0(x) > 0.$$

b. Assumption B2 (Hermitian symmetric-hyperbolic structure). The spatial coefficient matrices satisfy

$$G_{\text{def}}^i(x) = G_{\text{def}}^i(x)^\dagger,$$

and the linearized evolution is symmetric-hyperbolic with respect to the inner product induced by G_{def}^0 .

c. Assumption B3 (regularity and asymptotic bulk limit). There exist bulk-limit matrices

$$G_\infty^0, \quad G_\infty^i, \quad M_\infty$$

such that

$$G_{\text{def}}^0(x) \rightarrow G_\infty^0, \quad G_{\text{def}}^i(x) \rightarrow G_\infty^i, \quad M_{\text{def}}(x) \rightarrow M_\infty \quad (|x| \rightarrow \infty),$$

with sufficient decay of the defect perturbation.

d. Assumption B4 (bulk gap). The asymptotic bulk operator is gapped at zero. Equivalently, after reduction to canonical form below, the asymptotic mass endomorphism is invertible.

These assumptions are abstract and do not yet use any special TECT details.

XCI. CANONICAL HERMITIAN REDUCTION

Define the positive Hermitian square root

$$S(x) := \left(G_{\text{def}}^0(x)\right)^{1/2}.$$

Introduce the canonically normalized fluctuation

$$\chi(t, x) := S(x) \psi(t, x).$$

Since

$$\psi = S^{-1}\chi,$$

the fluctuation equation becomes

$$i\partial_t\chi = \mathbb{H}_{\text{def}}\chi,$$

with

$$\boxed{\mathbb{H}_{\text{def}} = -i\Gamma_{\text{def}}^i(x)\partial_i + W_{\text{def}}(x),}$$

where

$$\boxed{\Gamma_{\text{def}}^i(x) = S(x)^{-1}G_{\text{def}}^i(x)S(x)^{-1},}$$

and

$$\boxed{W_{\text{def}}(x) = -iS^{-1}G_{\text{def}}^i(\partial_i S^{-1}) - S^{-1}M_{\text{def}}S^{-1}.}$$

This is an exact algebraic identity.

a. Proof. Substitute $\psi = S^{-1}\chi$ into

$$iG_{\text{def}}^0\partial_t\psi + iG_{\text{def}}^i\partial_i\psi + M_{\text{def}}\psi = 0.$$

Using

$$G_{\text{def}}^0 = S^2$$

and

$$\partial_i(S^{-1}\chi) = (\partial_i S^{-1})\chi + S^{-1}\partial_i\chi,$$

one obtains

$$iS\partial_t\chi + iG_{\text{def}}^i(\partial_i S^{-1})\chi + iG_{\text{def}}^i S^{-1}\partial_i\chi + M_{\text{def}}S^{-1}\chi = 0.$$

Multiplying by S^{-1} from the left yields

$$i\partial_t\chi = -iS^{-1}G_{\text{def}}^i S^{-1}\partial_i\chi - iS^{-1}G_{\text{def}}^i(\partial_i S^{-1})\chi - S^{-1}M_{\text{def}}S^{-1}\chi,$$

which is exactly the announced formula. $\square$

XCII. DIRAC/CALLIAS-TYPE DECOMPOSITION

The operator

$$\mathbb{H}_{\text{def}} = -i\Gamma_{\text{def}}^i(x)\partial_i + W_{\text{def}}(x)$$

is already in first-order Hermitian form. To cast it into the geometric form used previously, one writes

$$\mathbb{O}_{1\text{def}}^{(0)} = i \tilde{\Gamma}_{\text{def}}^i(x) (\partial_i + \Omega_i^{\text{def}}(x)) + \Phi_{\text{def}}(x),$$

where

$$\tilde{\Gamma}_{\text{def}}^i := -\Gamma_{\text{def}}^i,$$

and the zeroth-order term is split as

$$W_{\text{def}} = i \tilde{\Gamma}_{\text{def}}^i \Omega_i^{\text{def}} + \Phi_{\text{def}}.$$

This splitting is not unique until one chooses a bundle frame and a covariant derivative convention. However, once such a convention is fixed, the decomposition has a natural interpretation:

- $\tilde{\Gamma}_{\text{def}}^i$ is the defect-renormalized principal symbol;
- Ω_i^{def} is the induced connection term;
- Φ_{def} is the Hermitian endomorphism (mass/Higgs-type field).

Thus the Dirac/Callias-type reference form is not a guess: it is the canonical first-order Hermitian form of the linearized microscopic PDE.

XCIII. BULK LIMIT AND ASYMPTOTIC MASS GAP

Assume the defect background approaches a bulk vacuum at infinity. Then

$$\Gamma_{\text{def}}^i(x) \rightarrow \Gamma_{\infty}^i, \quad W_{\text{def}}(x) \rightarrow W_{\infty}, \quad \Phi_{\text{def}}(x) \rightarrow \Phi_{\infty} \quad (|x| \rightarrow \infty).$$

If

$$\det \Phi_{\infty} \neq 0,$$

then zero lies in a bulk gap. This is the precise spectral meaning of the asymptotic gap assumption.

Under standard elliptic/Fredholm hypotheses, this implies that zero can only occur as an isolated defect-bound state of finite multiplicity.

XCIV. WHEN A CALLIAS-TYPE INDEX EXISTS

To define a protected index, one needs an additional grading structure.

Assume there exists a Hermitian involution

$$\mathcal{C} = \mathcal{C}^\dagger, \quad \mathcal{C}^2 = 1,$$

such that

$$\{\mathcal{C}, \tilde{\Gamma}_{\text{def}}^i\} = 0, \quad [\mathcal{C}, \Omega_i^{\text{def}}] = 0, \quad \{\mathcal{C}, \Phi_{\text{def}}\} = 0.$$

Then the operator is odd with respect to the grading $\mathcal{C}$ and decomposes into chiral blocks

$$\mathbb{O}_{1\text{def}}^{(0)} = \begin{pmatrix} 0 & \mathbb{O}_- \\ \mathbb{O}_+ & 0 \end{pmatrix}.$$

If, in addition, Φ_{∞} is invertible, the operator falls into the standard Callias-type Fredholm class. Then one may define

$$\text{Ind}(\mathbb{O}_+) = \dim \ker \mathbb{O}_+ - \dim \ker \mathbb{O}_-.$$

Abstractly, if

$$\text{Ind}(\mathbb{O}_+) = 1,$$

then there is exactly one protected light mode in the relevant chiral sector.

So the correct theorem schema is:

$$\boxed{\text{first-order Hermitian linearization} + \text{asymptotic bulk gap} + \text{grading} + \text{Callias index } 1 \implies \text{one protected single-defect light mode}}$$

XCV. EXPONENTIAL LOCALIZATION

Suppose zero (or $\lambda_\star$ near zero) is an isolated eigenvalue inside the bulk gap. Then standard Agmon-type estimates imply exponential localization:

$$\|\psi_\star(x)\| \leq C e^{-\kappa|x|}$$

for every

$$0 < \kappa < m_\infty - |\lambda_\star|,$$

where m_∞ is the bulk spectral gap scale determined by Φ_∞ .

Thus the single-defect mode, if it exists, is not only normalizable but exponentially localized. This is exactly what is required for the later three-defect family projector construction.

XCVI. WHAT IS ABSTRACTLY CLOSED

Under B1–B4 and the grading/index hypotheses above, the following implications are rigorously valid:

1. the linearized microscopic PDE can be canonically reduced to the form

$$i\partial_t \chi = \mathbb{H}_{\text{def}} \chi, \quad \mathbb{H}_{\text{def}} = i\tilde{\Gamma}_{\text{def}}^i (\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}};$$

2. the asymptotic bulk gap is controlled by the invertibility of Φ_∞ ;
3. isolated defect-bound light modes are exponentially localized;
4. if the corresponding Callias index is 1, then a single defect supports exactly one protected light mode in the relevant sector.

So the reference operator program is mathematically coherent.

XCVII. WHAT IS STILL TECT-SPECIFIC AND OPEN

The following items are not closed abstractly and must be derived from the actual TECT microscopic PDE:

1. the precise microscopic field content $\mathfrak{X}$;
2. the actual linearized coefficients $G_{\text{def}}^0, G_{\text{def}}^i, M_{\text{def}}$;
3. the concrete identification of $\tilde{\Gamma}_{\text{def}}^i, \Omega_i^{\text{def}}, \Phi_{\text{def}}$;
4. proof that the asymptotic bulk is genuinely gapped;
5. proof that a grading $\mathcal{C}$ exists;
6. computation of the actual index, and specifically verification that it equals 1.

Therefore, at the present stage, the statement

“one TECT defect carries one family mode”

is still conditional. It becomes a theorem of TECT only after these six items are checked.

XCVIII. MINIMAL PRACTICAL TEMPLATE FOR THE ACTUAL TECT DERIVATION

Once the real TECT PDE is written, the correct derivation pipeline is:

1. find a static single-defect solution $\mathfrak{B}_{\text{def}}(x)$;

2. linearize the microscopic equations around it and read off

$$G_{\text{def}}^0, \quad G_{\text{def}}^i, \quad M_{\text{def}};$$

3. perform the canonical reduction with

$$S = (G_{\text{def}}^0)^{1/2};$$

4. identify the induced

$$\tilde{\Gamma}_{\text{def}}^i, \quad \Omega_i^{\text{def}}, \quad \Phi_{\text{def}};$$

5. compute the asymptotic bulk limit and check

$$\det \Phi_{\infty} \neq 0;$$

6. identify the grading $\mathcal{C}$, if present;

7. compute the corresponding index and test whether it equals 1.

Only after that does the family-projector construction become a first-principles theorem of TECT itself.

XCIX. FINAL SUMMARY

The correct minimal route from a microscopic TECT PDE to the single-defect reference operator is as follows. Start from the linearized fluctuation equation around a static defect background:

$$i G_{\text{def}}^0(x) \partial_t \psi + i G_{\text{def}}^i(x) \partial_i \psi + M_{\text{def}}(x) \psi = 0,$$

with $G_{\text{def}}^0 > 0$ Hermitian.

After the canonical field redefinition

$$\chi = (G_{\text{def}}^0)^{1/2} \psi,$$

this becomes

$$i \partial_t \chi = -i \Gamma_{\text{def}}^i(x) \partial_i \chi + W_{\text{def}}(x) \chi,$$

where

$$\Gamma_{\text{def}}^i = (G_{\text{def}}^0)^{-1/2} G_{\text{def}}^i (G_{\text{def}}^0)^{-1/2},$$

$$W_{\text{def}} = -i (G_{\text{def}}^0)^{-1/2} G_{\text{def}}^i \partial_i (G_{\text{def}}^0)^{-1/2} - (G_{\text{def}}^0)^{-1/2} M_{\text{def}} (G_{\text{def}}^0)^{-1/2}.$$

Equivalently, after choosing a bundle frame and covariant derivative convention, one obtains the Dirac/Callias-type reference form

$$\boxed{\mathbb{O}_{1\text{def}}^{(0)} = i \tilde{\Gamma}_{\text{def}}^i (\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}}.}$$

If the asymptotic bulk mass endomorphism is invertible,

$$\det \Phi_{\infty} \neq 0,$$

then zero lies in a bulk gap. If, furthermore, a grading exists and the corresponding Callias index is

$$\text{Ind}(\mathbb{O}_+) = 1,$$

then a single defect supports exactly one protected exponentially localized light mode.

Therefore the statement

$$\text{“one defect} \leftrightarrow \text{one family mode”}$$

is rigorously valid provided these operator-theoretic hypotheses hold.

What remains open is entirely TECT-specific: derive the actual linearized coefficients from the microscopic TECT PDE, identify $\tilde{\Gamma}_{\text{def}}^i, \Omega_i^{\text{def}}, \Phi_{\text{def}}$, check the bulk gap, and compute the relevant index.

C. MINIMAL MICROSCOPIC ACTION TEMPLATE FOR THE TECT FLAVOR PROGRAM

We now formulate the minimal microscopic action/PDE ansatz that is sufficient to generate the defect reference operator used in the reduced flavor program.

CI. WHY A FIRST-ORDER HERMITIAN FLUCTUATION SYSTEM IS THE CORRECT MINIMAL TEMPLATE

The reduced flavor program requires a microscopic fluctuation operator with the following properties:

1. a well-defined self-adjoint time evolution;
2. a natural light/heavy spectral decomposition;
3. compatibility with defect-localized bound states;
4. the possibility of a topological index for protected light modes.

The minimal general structure that satisfies all four requirements is a first-order Hermitian quadratic action for a fluctuation multiplet Ψ :

$$S[\Psi; \mathfrak{B}] = \int dt d^3x \left[\frac{i}{2} \Psi^\dagger K^0(\mathfrak{B}) \overleftrightarrow{\partial}_t \Psi + \frac{i}{2} \Psi^\dagger K^i(\mathfrak{B}) \overleftrightarrow{\partial}_i \Psi - \Psi^\dagger \mathcal{M}(\mathfrak{B}) \Psi \right].$$

Here:

- $\mathfrak{B}$ denotes the microscopic background fields of TECT;
- Ψ is the fluctuation multiplet on the relevant internal bundle;
- $K^0(\mathfrak{B})$ is the kinetic Hermitian matrix;
- $K^i(\mathfrak{B})$ are the spatial principal-symbol matrices;
- $\mathcal{M}(\mathfrak{B})$ is the Hermitian zeroth-order endomorphism.

This action is not yet claimed to be the unique TECT action. It is the minimal operator-theoretic template that can support the later spectral construction.

CII. EULER-LAGRANGE EQUATION OF THE QUADRATIC FIRST-ORDER ACTION

Varying the action with respect to $\Psi^\dagger$, one obtains

$$\delta S = \int dt d^3x \delta \Psi^\dagger \left[i K^0(\mathfrak{B}) \partial_t \Psi + i K^i(\mathfrak{B}) \partial_i \Psi + \mathcal{N}(\mathfrak{B}, \partial \mathfrak{B}) \Psi \right],$$

where $\mathcal{N}$ includes the zeroth-order mass term and also the derivative contributions coming from integration by parts when the coefficients vary in space.

More explicitly, the Euler–Lagrange equation can always be written in the canonical first-order form

$$i G^0(\mathfrak{B}, x) \partial_t \Psi + i G^i(\mathfrak{B}, x) \partial_i \Psi + M(\mathfrak{B}, \partial \mathfrak{B}; x) \Psi = 0,$$

with

$$G^0 = K^0, \quad G^i = K^i,$$

and where M contains both the original $\mathcal{M}$ and the lower-order pieces induced by derivatives of K^i .

Thus the desired linearized PDE is not an extra assumption: it is the universal Euler–Lagrange form of the first-order Hermitian quadratic action.

CIII. EXPANSION AROUND A STATIC DEFECT BACKGROUND

Let

$$\mathfrak{B}_{\text{def}}(x)$$

be a static single-defect solution of the full microscopic TECT equations.

Write the full microscopic field as

$$\mathfrak{B}(t, x) = \mathfrak{B}_{\text{def}}(x) + \delta\mathfrak{B}(t, x),$$

and expand the quadratic fluctuation action around $\mathfrak{B}_{\text{def}}$. Then the fluctuation multiplet ψ satisfies

$$\boxed{i G_{\text{def}}^0(x) \partial_t \psi + i G_{\text{def}}^i(x) \partial_i \psi + M_{\text{def}}(x) \psi = 0,}$$

where

$$G_{\text{def}}^0(x) := G^0(\mathfrak{B}_{\text{def}}(x), x), \quad G_{\text{def}}^i(x) := G^i(\mathfrak{B}_{\text{def}}(x), x),$$

$$M_{\text{def}}(x) := M(\mathfrak{B}_{\text{def}}(x), \partial\mathfrak{B}_{\text{def}}(x); x).$$

This is exactly the reference fluctuation equation needed for the single-defect spectral problem.

CIV. PRECISE FORMULAS FOR

$G_{\text{def}}^0, G_{\text{def}}^i, M_{\text{def}}$
From the action

$$S = \int dt d^3x \left[\frac{i}{2} \Psi^\dagger K^0 \overleftrightarrow{\partial}_t \Psi + \frac{i}{2} \Psi^\dagger K^i \overleftrightarrow{\partial}_i \Psi - \Psi^\dagger \mathcal{M} \Psi \right],$$

the field equation is

$$i K^0 \partial_t \Psi + i K^i \partial_i \Psi + \left(\mathcal{M} + \frac{i}{2} \partial_i K^i + \frac{i}{2} \partial_t K^0 \right) \Psi = 0.$$

For a static background,

$$\partial_t K^0(\mathfrak{B}_{\text{def}}) = 0,$$

so

$$\boxed{G_{\text{def}}^0 = K^0(\mathfrak{B}_{\text{def}}), \quad G_{\text{def}}^i = K^i(\mathfrak{B}_{\text{def}}),}$$

and

$$\boxed{M_{\text{def}} = \mathcal{M}(\mathfrak{B}_{\text{def}}) + \frac{i}{2} \partial_i K^i(\mathfrak{B}_{\text{def}}).}$$

Thus the three objects that enter the defect spectral problem are read directly from the quadratic action.

CV. CANONICAL NORMALIZATION AND REFERENCE HAMILTONIAN

Assume

$$G_{\text{def}}^0(x) = G_{\text{def}}^0(x)^\dagger > 0.$$

Define

$$S(x) := (G_{\text{def}}^0(x))^{1/2}, \quad \chi := S\psi.$$

Then the fluctuation equation becomes

$$i\partial_t \chi = \mathbb{H}_{\text{def}} \chi,$$

with

$$\mathbb{H}_{\text{def}} = -i \Gamma_{\text{def}}^i(x) \partial_i + W_{\text{def}}(x),$$

where

$$\Gamma_{\text{def}}^i = S^{-1} G_{\text{def}}^i S^{-1},$$

and

$$W_{\text{def}} = -i S^{-1} G_{\text{def}}^i \partial_i (S^{-1}) - S^{-1} M_{\text{def}} S^{-1}.$$

This is the exact canonical Hermitian Hamiltonian associated with the defect background.

CVI. DIRAC/CALLIAS DECOMPOSITION

After fixing a bundle frame and covariant-derivative convention, the Hamiltonian may be written as

$$\mathbb{O}_{1\text{def}}^{(0)} = i \tilde{\Gamma}_{\text{def}}^i (\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}},$$

with

$$\tilde{\Gamma}_{\text{def}}^i := -\Gamma_{\text{def}}^i,$$

and

$$W_{\text{def}} = i \tilde{\Gamma}_{\text{def}}^i \Omega_i^{\text{def}} + \Phi_{\text{def}}.$$

Thus the desired reference operator is not guessed: it is the canonically normalized first-order fluctuation operator of the microscopic defect background.

CVII. MINIMAL THEOREM SCHEMA FOR ONE PROTECTED DEFECT MODE

Suppose the following hold:

1. $G_{\text{def}}^0 > 0$, so the canonical Hamiltonian is well-defined;
2. the asymptotic bulk limit exists:

$$\tilde{\Gamma}_{\text{def}}^i(x) \rightarrow \tilde{\Gamma}_{\infty}^i, \quad \Phi_{\text{def}}(x) \rightarrow \Phi_{\infty} \quad (|x| \rightarrow \infty);$$

3. the asymptotic mass endomorphism is invertible:

$$\det \Phi_{\infty} \neq 0;$$

4. there exists a grading $\mathcal{C}$ such that the operator is of Callias type;
5. the corresponding index is

$$\text{Ind}(\mathbb{O}_+) = 1.$$

Then:

1. zero lies in a bulk gap;
2. the defect supports one protected normalizable light mode;
3. that mode is exponentially localized;
4. three well-separated defects generate a rank-three low spectral cluster.

Therefore the family-space construction follows once these operator-theoretic hypotheses are verified.

CVIII. WHAT IS ABSTRACTLY DERIVED AND WHAT IS STILL OPEN

The following are now abstractly derived:

1. a first-order Hermitian quadratic microscopic action yields the canonical fluctuation PDE

$$iG_{\text{def}}^0 \partial_t \psi + iG_{\text{def}}^i \partial_i \psi + M_{\text{def}} \psi = 0;$$

2. the coefficients are read directly as

$$G_{\text{def}}^0 = K^0(\mathfrak{B}_{\text{def}}), \quad G_{\text{def}}^i = K^i(\mathfrak{B}_{\text{def}}), \quad M_{\text{def}} = \mathcal{M}(\mathfrak{B}_{\text{def}}) + \frac{i}{2} \partial_i K^i(\mathfrak{B}_{\text{def}});$$

3. canonical normalization produces the reference operator

$$\mathbb{O}_{1\text{def}}^{(0)} = i\tilde{\Gamma}_{\text{def}}^i (\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}}.$$

The following remain TECT-specific and open:

1. the actual microscopic field multiplet $\mathfrak{B}$;
2. the actual coefficient matrices $K^0, K^i, \mathcal{M}$;
3. the explicit defect background $\mathfrak{B}_{\text{def}}$;
4. the asymptotic bulk gap;
5. the existence of a grading;
6. the actual index value, and in particular whether it equals 1.

So the operator-theoretic skeleton is now fixed. What remains is to insert the actual TECT microscopic model into that skeleton.

CIX. FINAL SUMMARY

A natural minimal microscopic ansatz for the TECT flavor program is a first-order Hermitian quadratic fluctuation action

$$S[\Psi; \mathfrak{B}] = \int dt d^3x \left[\frac{i}{2} \Psi^\dagger K^0(\mathfrak{B}) \overleftrightarrow{\partial}_t \Psi + \frac{i}{2} \Psi^\dagger K^i(\mathfrak{B}) \overleftrightarrow{\partial}_i \Psi - \Psi^\dagger \mathcal{M}(\mathfrak{B}) \Psi \right].$$

Around a static defect background $\mathfrak{B}_{\text{def}}$, this yields the canonical linearized PDE

$$iG_{\text{def}}^0 \partial_t \psi + iG_{\text{def}}^i \partial_i \psi + M_{\text{def}} \psi = 0,$$

with

$$G_{\text{def}}^0 = K^0(\mathfrak{B}_{\text{def}}), \quad G_{\text{def}}^i = K^i(\mathfrak{B}_{\text{def}}), \quad M_{\text{def}} = \mathcal{M}(\mathfrak{B}_{\text{def}}) + \frac{i}{2} \partial_i K^i(\mathfrak{B}_{\text{def}}).$$

After canonical normalization by

$$\chi = (G_{\text{def}}^0)^{1/2} \psi,$$

one obtains the Hermitian first-order defect operator

$$\mathbb{O}_{1\text{def}}^{(0)} = i\tilde{\Gamma}_{\text{def}}^i (\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}}.$$

Therefore the desired reference operator is not guessed: it is the canonical defect fluctuation operator of the microscopic theory.

If, furthermore, the asymptotic bulk is gapped, a grading exists, and the corresponding Callias index equals 1, then a single defect carries exactly one protected exponentially localized light mode.

What remains open is entirely TECT-specific: the actual microscopic field content, the actual coefficient matrices $K^0, K^i, \mathcal{M}$, the explicit defect solution, the bulk gap, the grading, and the index computation.

CX. DIAGNOSTIC REVIEW OF THE REFINED LOCKED + CLASS II MICROSCOPIC EXTRACTION BRANCH

The proposed branch is structurally correct and is presently the strongest explicit working microscopic route. However, as written, it is not yet fully sufficient to justify the phrase

“actual exact microscopic coefficient extraction.”

It should instead be presented as the strongest currently available *working extraction branch*, with several points made more precise.

CXI. STEP 1: MICROSCOPIC BRANCH — WHAT IS CORRECT AND WHAT MUST BE CLARIFIED

The expression

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi, P] + \beta_X X_i^a K_i^a[P, \Psi]$$

is a microscopic Lagrangian density (or action density), not yet a PDE. So the wording should be:

“We fix the strongest explicit microscopic action branch currently available.”

Moreover, the notation must be unified:

$$\Psi \quad \text{vs.} \quad \psi$$

should not be mixed.

Also, with no kinetic term for X_i^a , the current branch should be interpreted as a heavy auxiliary mediator truncation. Then integrating out X is indeed exact, but only at the quadratic fluctuation level:

“integrated out exactly at the quadratic fluctuation level by Schur complement.”

Finally, the currents J_i^a and K_i^a must be specified precisely enough that their index structure, Hermiticity, and lattice-locking transformation properties are unambiguous.

CXII. STEP 2: BLOCH EXPANSION AND SCHUR COMPLEMENT

The Bloch reduction is the right move, but the operator should be defined as the quadratic fluctuation operator around the locked BCC branch, not just informally as a Hessian.

Let

$$\mathfrak{B}_{\text{lock}}$$

denote the locked 12-mode BCC background, and let

$$\delta\Phi = (\delta\Psi, \delta X)$$

be the fluctuation multiplet. Then the quadratic Bloch fiber operator is

$$\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix}.$$

The exact quadratic effective operator after eliminating the heavy mediator is

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k),$$

provided $H_{XX}(k)$ is invertible in the momentum region of interest. Since the branch assumes

$$M_X^2 \gg 1,$$

this invertibility is plausible, but should be stated explicitly.

The shell-projected operator is then

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*},$$

where $k_* = G_*/2$ is the folded Bragg point and Π_{S_*} is the shell projector associated with the near-degenerate shell.

CXIII. STEP 3: FIRST-ORDER TRANSPORT MATRIX

The first-order transport matrix should be written with the correct evaluation point:

$$K_i = \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*}.$$

However, to make this mathematically clean, the low valley doublet must be spectrally isolated at $p = 0$. Then the valley projector should be defined as the Riesz projector

$$P_* = \frac{1}{2\pi i} \oint_{\Gamma} (z - H_{\text{shell}}(0))^{-1} dz,$$

where Γ encloses only the two low valley eigenvalues.

The linearized valley operator is then

$$H_{\text{val}}(p) = E_* P_* + \sum_i p_i M_i + \mathcal{O}(|p|^2), \quad M_i := P_* K_i P_*.$$

This is the correct first-order Dirac block extraction.

CXIV. STEP 4: DEFECT-FRAME DECOMPOSITION

Given an adapted defect frame

$$(\hat{n}, e_1, e_2),$$

define

$$M_{\parallel} := \hat{n}_i M_i, \quad M_1 := e_{1i} M_i, \quad M_2 := e_{2i} M_i.$$

At this stage, every M_{μ} is an arbitrary Hermitian 2×2 matrix, so one should first write the full Pauli expansion

$$M_{\mu} = t_{\mu} \mathbf{1} + \sum_{a=1}^3 v_{\mu}^a \sigma_a, \quad v_{\mu}^a = \frac{1}{2} (\sigma_a M_{\mu}).$$

Only after imposing the locked branch symmetry may one reduce this to the special form

$$M_{\parallel} = \lambda_{\parallel} \sigma_3, \quad M_1 = \alpha \sigma_1 + \beta \sigma_2, \quad M_2 = \alpha \sigma_2 - \beta \sigma_1.$$

This is the exact place where the symmetry theorem must be cited or proved.

CXV. STEP 5: TRACE EXTRACTION OF THE REDUCED COEFFICIENTS

Once the special locked form above is established, the correct coefficient extraction is

$$\lambda_{\parallel} = \frac{1}{2} (\sigma_3 M_{\parallel}),$$

$$\alpha = \frac{1}{4} (\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4} (\sigma_2 M_1 - \sigma_1 M_2).$$

These formulas are always correct under the locked branch selection rule. By contrast, the simpler equalities

$$\alpha = \frac{1}{2} (\sigma_1 M_1), \quad \beta = \frac{1}{2} (\sigma_2 M_1)$$

are not automatic unless the additional locked relations have already been proved:

$$(\sigma_2 M_2) = (\sigma_1 M_1), \quad (\sigma_1 M_2) = -(\sigma_2 M_1),$$

and all forbidden Pauli components vanish.

So the symmetric 1/4-trace formulas are the safer exact expressions.

CXVI. STEP 6: DICTIONARY TO

(g_c, g_E, g_O)

The proposed dictionary

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}$$

is plausible as a working Class II specialization, but it should be written more cautiously.

The exact statement should first be

$$\boxed{\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O,}$$

where the coefficients C_c, C_E, C_O are explicit microscopic overlap integrals computed from the locked branch wave-functions.

Only after an actual overlap evaluation may one conclude, if true, that

$$C_c = \frac{1}{2} I_{\theta^3}, \quad C_E = I_{\theta}^{(E)}, \quad C_O = I_{\theta}^{(O)},$$

and only under additional symmetry can one set

$$I_{\theta}^{(E)} = I_{\theta}^{(O)} =: I_{\theta}.$$

So the current text is slightly too compressed at this point.

CXVII. CORRECT HONEST STATUS

The correct current status is therefore:

the refined locked + Class II branch provides the strongest explicit microscopic working route to coefficient extraction,

but

it is not yet a fully closed exact extraction theorem until the projector, symmetry selection rules, and overlap dictionary are written

In particular, the following items still need to be made explicit:

1. the exact definitions of J_i^a and K_i^a ;
2. the precise quadratic fluctuation operator about the locked background;
3. the spectral definition of P_* ;
4. the locked symmetry theorem enforcing the special Pauli structure of $M_{\parallel}, M_1, M_2$;
5. the explicit overlap-integral derivation of C_c, C_E, C_O .

Once these are inserted, the branch becomes a clean and nearly submission-ready extraction framework.

CXVIII. FINAL SUMMARY

The proposed refined locked + Class II branch is structurally correct and is presently the strongest explicit microscopic working branch for coefficient extraction.

However, in its current form it should not yet be advertised as a fully exact extraction theorem.

The main points that still need to be made explicit are:

1. the branch is an action/Lagrangian branch, not yet a PDE;

2. the heavy mediator is integrated out exactly only at the quadratic fluctuation level;
3. the valley projector P_* should be defined spectrally as a Riesz projector;
4. the special Pauli structure of the projected matrices $M_{||}, M_1, M_2$ must be justified by the locked symmetry;
5. the coefficient dictionary to (g_c, g_E, g_O) should first be written with general overlap constants (C_c, C_E, C_O) , and only later reduced to specific integrals once explicitly computed.

Thus the correct honest statement is:

this branch already provides a very strong explicit working microscopic extraction framework,

but

it still requires a few precise definitions and symmetry statements before it becomes a fully closed exact extraction theorem.

CXIX. SPECIALIZATION OF THE ABSTRACT SCHUR-COMPLEMENT FORMALISM TO THE REFINED LOCKED + CLASS II WORKING BRANCH

We now specialize the general light/heavy block formalism to the explicit microscopic working branch

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi, P] + \beta_X X_i^a K_i^a[P, \Psi].$$

This is not yet the full microscopic TECT action, but it is the strongest currently available explicit working branch for actual coefficient extraction.

CXX. STATIONARY LOCKED BACKGROUND AND FLUCTUATION VARIABLES

Let

$$(\bar{\Psi}, \bar{P}, \bar{X})$$

be a static locked BCC branch solving the stationarity conditions of the above working action. Because X appears as a heavy mediator in the present truncation, its background value is determined algebraically by

$$M_X^2 \bar{X}_i^a + \alpha_X J_i^a[\bar{\Psi}, \bar{P}] + \beta_X K_i^a[\bar{P}, \bar{\Psi}] = 0.$$

Thus

$$\bar{X}_i^a = -\frac{\alpha_X J_i^a[\bar{\Psi}, \bar{P}] + \beta_X K_i^a[\bar{P}, \bar{\Psi}]}{M_X^2}.$$

Now write the fluctuations

$$\Psi = \bar{\Psi} + \delta\Psi, \quad X = \bar{X} + \delta X.$$

The quadratic fluctuation operator around the locked branch is the object that must be Bloch decomposed and shell projected.

CXXI. QUADRATIC BLOCH FIBER OPERATOR

Pass to the reciprocal-harmonic Bloch basis near the folded Bragg point

$$k_* = G_*/2, \quad k = k_* + p.$$

Let $\delta\Phi = (\delta\Psi, \delta X)$. Then the quadratic Bloch fiber operator takes the block form

$$\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix}.$$

The blocks are defined as the second variations of the microscopic working branch at the locked background:

$$H_{\Psi\Psi}(k) = \frac{\delta^2 \mathcal{L}_{\text{mic}}}{\delta\Psi^\dagger \delta\Psi} \Big|_{\bar{\Psi}, \bar{P}, \bar{X}}^{\text{Bloch}}(k),$$

$$H_{\Psi X}(k) = \frac{\delta^2 \mathcal{L}_{\text{mic}}}{\delta\Psi^\dagger \delta X} \Big|_{\bar{\Psi}, \bar{P}, \bar{X}}^{\text{Bloch}}(k), \quad H_{X\Psi}(k) = H_{\Psi X}(k)^\dagger,$$

$$H_{XX}(k) = \frac{\delta^2 \mathcal{L}_{\text{mic}}}{\delta X \delta X} \Big|_{\bar{\Psi}, \bar{P}, \bar{X}}^{\text{Bloch}}(k).$$

In the present Class II truncation, where X has no independent kinetic term at this stage,

$$H_{XX}(k) = M_X^2 \mathbf{1}_X$$

up to possible subleading background-induced algebraic corrections. So at the working level H_{XX} is heavy and invertible.

CXXII. EXPLICIT CURRENT VERTICES

Define the linearized current vertices

$$\mathcal{J}_i^a(k) := \frac{\delta J_i^a[\Psi, P]}{\delta\Psi} \Big|_{\bar{\Psi}, \bar{P}}^{\text{Bloch}}(k), \quad \mathcal{K}_i^a(k) := \frac{\delta K_i^a[P, \Psi]}{\delta\Psi} \Big|_{\bar{P}, \bar{\Psi}}^{\text{Bloch}}(k).$$

Then the mixed block is

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k),$$

where $\mathcal{J}(k)$ and $\mathcal{K}(k)$ denote the appropriately bundled collections of $\mathcal{J}_i^a(k)$ and $\mathcal{K}_i^a(k)$ over the mediator indices (i, a) .

Therefore the heavy-mediated correction is

$$\Delta H_{\text{eff}}(k) = -(\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)) H_{XX}(k)^{-1} (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger.$$

If $H_{XX}(k) = M_X^2 \mathbf{1}_X$, this simplifies to

$$\Delta H_{\text{eff}}(k) = -\frac{1}{M_X^2} (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)) (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger.$$

Hence the full effective shell operator is

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) + \Delta H_{\text{eff}}(k).$$

CXXIII. SHELL PROJECTION AND EXACT VALLEY PROJECTOR

Let Π_{S_*} denote the shell projector around the resonant folded shell S_* . Then

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}.$$

Let Γ be a contour surrounding only the two low valley eigenvalues of $H_{\text{shell}}(0)$. Then the exact valley projector is the Riesz projector

$$P_* = \frac{1}{2\pi i} \oint_{\Gamma} (z - H_{\text{shell}}(0))^{-1} dz.$$

This is the correct basis-independent definition of the valley doublet. Only after this spectral projection may one choose a Pauli basis $\{\sigma_1, \sigma_2, \sigma_3\}$ on $\text{Ran}(P_*)$.

CXXIV. FIRST-ORDER TRANSPORT OPERATOR

Define the first momentum derivative at the Bragg point

$$K_i = \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*}.$$

Since

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k),$$

one obtains the exact derivative formula

$$\partial_{k_i} H_{\text{eff}} = \partial_{k_i} H_{\Psi\Psi} - (\partial_{k_i} H_{\Psi X}) H_{XX}^{-1} H_{X\Psi} - H_{\Psi X} H_{XX}^{-1} (\partial_{k_i} H_{X\Psi}) + H_{\Psi X} H_{XX}^{-1} (\partial_{k_i} H_{XX}) H_{XX}^{-1} H_{X\Psi}.$$

In the pure auxiliary Class II truncation,

$$\partial_{k_i} H_{XX} = 0,$$

so this simplifies to

$$\partial_{k_i} H_{\text{eff}} = \partial_{k_i} H_{\Psi\Psi} - (\partial_{k_i} H_{\Psi X}) H_{XX}^{-1} H_{X\Psi} - H_{\Psi X} H_{XX}^{-1} (\partial_{k_i} H_{X\Psi}).$$

Projecting onto the valley subspace,

$$M_i = P_* K_i P_*.$$

This is the exact microscopic transport matrix entering the reduced Dirac block.

CXXV. LOCKED BRANCH PAULI SELECTION RULE

At this stage M_i is an arbitrary Hermitian 2×2 matrix. In an adapted defect frame

$$(\hat{n}, e_1, e_2),$$

define

$$M_{\parallel} = \hat{n}_i M_i, \quad M_1 = e_{1i} M_i, \quad M_2 = e_{2i} M_i.$$

The locked branch symmetry is precisely the statement that, after an appropriate Pauli basis choice on the valley doublet,

$$M_{\parallel} = \lambda_{\parallel} \sigma_3, \quad M_1 = \alpha \sigma_1 + \beta \sigma_2, \quad M_2 = \alpha \sigma_2 - \beta \sigma_1.$$

This is the exact reduced transport form that isolates the Dirac block.

CXXVI. TRACE EXTRACTION OF

$(\lambda_{\parallel}, \alpha, \beta)$

Once the locked Pauli form is established, the reduced coefficients are extracted exactly by

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}),$$

$$\alpha = \frac{1}{4}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4}(\sigma_2 M_1 - \sigma_1 M_2).$$

These formulas are exact under the locked symmetry selection rule. They are safer than the shorter unsymmetrized formulas, because they do not presuppose extra equalities beyond the locked form itself.

CXXVII. EXACT MICROSCOPIC RENORMALIZATION OF

(g_c, g_E, g_O)

Let

$$M_i = M_i^{\text{ref}} + \Delta M_i, \quad \Delta M_i = P_*(\partial_{k_i} \Delta H_{\text{eff}})P_*.$$

Then the Class II branch renormalizes the reduced invariants through

$$\lambda_{\parallel} = \lambda_{\parallel}^{\text{ref}} + \Delta \lambda_{\parallel}, \quad \alpha = \alpha^{\text{ref}} + \Delta \alpha, \quad \beta = \beta^{\text{ref}} + \Delta \beta,$$

with

$$\Delta \lambda_{\parallel} = \frac{1}{2}(\sigma_3 \Delta M_{\parallel}),$$

$$\Delta \alpha = \frac{1}{4}(\sigma_1 \Delta M_1 + \sigma_2 \Delta M_2), \quad \Delta \beta = \frac{1}{4}(\sigma_2 \Delta M_1 - \sigma_1 \Delta M_2).$$

Thus the exact Class II microscopic renormalization of the Dirac data is carried entirely by the Schur-complement correction ΔH_{eff} .

Now define the microscopic overlap constants C_c, C_E, C_O by

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.$$

Then the extracted Class II coefficients are

$$g_c = \frac{\lambda_{\parallel}}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}.$$

Only after an explicit overlap evaluation may one specialize

$$C_c = \frac{1}{2}I_{\theta'^3}, \quad C_E = I_{\theta}^{(E)}, \quad C_O = I_{\theta}^{(O)}.$$

If the locked branch symmetry further implies

$$I_{\theta}^{(E)} = I_{\theta}^{(O)} =: I_{\theta},$$

then one recovers the compressed working dictionary

$$\lambda_{\parallel} = \frac{g_c}{2}I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.$$

CXXVIII. WHAT IS NOW ACTUALLY CLOSED

Within the explicit refined locked + Class II working branch, the following points are now fixed:

1. the heavy mediator enters as a light/heavy block system whose quadratic elimination is exact at the fluctuation level;
2. the mediator correction to the shell operator is the Schur complement

$$\Delta H_{\text{eff}}(k) = -H_{\Psi X}(k)H_{XX}(k)^{-1}H_{X\Psi}(k);$$

3. the transport matrices are

$$M_i = P_* \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*} P_*;$$

4. the reduced locked coefficients are obtained by Pauli traces;
5. the microscopic Class II renormalization of (g_c, g_E, g_O) is read directly from those projected traces.

So this branch is now fully integrated into the general Schur-complement operator formalism.

CXXIX. WHAT REMAINS EXPLICIT BUT NOT YET FULLY EVALUATED

What still remains to be inserted from the concrete branch data is:

1. the fully explicit formulas for the current vertices $\mathcal{J}_i^a(k)$ and $\mathcal{K}_i^a(k)$;
2. the explicit locked-branch proof of the Pauli selection rule;
3. the actual overlap evaluation of C_c, C_E, C_O .

These are no longer conceptual gaps. They are the remaining branch-specific algebraic computations.

CXXX. FINAL SUMMARY

The refined locked + Class II working branch can now be cleanly inserted into the general reduced-operator formalism.

Starting from

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_{\Psi}^{\text{ref}} + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a + \beta_X X_i^a K_i^a,$$

the quadratic Bloch fiber operator takes the block form

$$\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix},$$

with

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k).$$

The heavy mediator is eliminated exactly at quadratic level by the Schur complement

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k)H_{XX}(k)^{-1}H_{X\Psi}(k).$$

After shell projection and spectral valley projection

$$P_* = \frac{1}{2\pi i} \oint_{\Gamma} (z - H_{\text{shell}}(0))^{-1} dz,$$

the transport matrices are

$$M_i = P_* \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*} P_*.$$

If the locked branch symmetry enforces

$$M_{\parallel} = \lambda_{\parallel} \sigma_3, \quad M_1 = \alpha \sigma_1 + \beta \sigma_2, \quad M_2 = \alpha \sigma_2 - \beta \sigma_1,$$

then

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}), \quad \alpha = \frac{1}{4}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4}(\sigma_2 M_1 - \sigma_1 M_2).$$

Thus the microscopic Class II branch renormalizes the reduced invariants entirely through the projected derivative of the Schur-complement correction.

Finally, introducing overlap constants

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O,$$

one obtains the exact extraction formulas

$$g_c = \frac{\lambda_{\parallel}}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}.$$

So the explicit working branch is now fully connected to the general microscopic coefficient-extraction formalism. What remains is only the branch-specific algebra: the explicit vertices $\mathcal{J}, \mathcal{K}$, the locked Pauli selection rule, and the overlap integrals C_c, C_E, C_O .

CXXXI. BLOCH-SPACE CURRENT VERTICES AND EXPLICIT CLASS II COEFFICIENT EXTRACTION

We now continue the microscopic derivation by writing the explicit Bloch-space structure of the mixed vertices

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)$$

for the refined locked + Class II working branch, and then deriving the corresponding renormalization formulas for

$$(g_c, g_E, g_O).$$

CXXXII. WORKING MICROSCOPIC BRANCH AND FLUCTUATION SPACE

We start from the explicit working branch

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi, P] + \beta_X X_i^a K_i^a[P, \Psi].$$

Let

$$(\bar{\Psi}, \bar{P}, \bar{X})$$

be a locked BCC stationary branch. We write

$$\Psi = \bar{\Psi} + \delta\Psi, \quad X = \bar{X} + \delta X.$$

If the quadratic fluctuation problem mixes $\delta\Psi$ and $\delta\Psi^\dagger$, then one should pass to the doubled Nambu field

$$\Upsilon = \begin{pmatrix} \delta\Psi \\ \delta\Psi^\dagger \end{pmatrix}.$$

All formulas below remain valid after replacing $\delta\Psi$ by Υ . For notational simplicity we denote the linear fluctuation field again by Ψ .

CXXXIII. LINEARIZED CURRENT OPERATORS

The mixed Hessian block is determined by the linearization of the currents around the locked background. Define the Fréchet derivatives

$$\mathbb{J}_i^a[\bar{\Psi}, \bar{P}] \delta \Psi := \left. \frac{d}{d\varepsilon} J_i^a[\bar{\Psi} + \varepsilon \delta \Psi, \bar{P}] \right|_{\varepsilon=0},$$

$$\mathbb{K}_i^a[\bar{P}, \bar{\Psi}] \delta \Psi := \left. \frac{d}{d\varepsilon} K_i^a[\bar{P}, \bar{\Psi} + \varepsilon \delta \Psi] \right|_{\varepsilon=0}.$$

Then the mixed quadratic vertex is

$$H_{\Psi X}(x) = \alpha_X \mathbb{J}(x) + \beta_X \mathbb{K}(x),$$

where $\mathbb{J}(x)$ and $\mathbb{K}(x)$ bundle the indices (i, a) .

For the working Class II branch, the natural general assumption is that these linearized current operators are local and at most first-order in spatial derivatives:

$$\mathbb{J}_i^a = \mathbb{A}_i^a(x) + \mathbb{B}_{ij}^a(x)(-i\partial_j), \quad \mathbb{K}_i^a = \mathbb{C}_i^a(x) + \mathbb{D}_{ij}^a(x)(-i\partial_j).$$

Here $\mathbb{A}, \mathbb{B}, \mathbb{C}, \mathbb{D}$ are periodic matrix-valued coefficient fields on the locked BCC background. This is the correct universal form for the Bloch calculation. No stronger assumption is needed at this stage.

CXXXIV. BLOCH-HARMONIC EXPANSION

Let Λ^* be the reciprocal lattice of the locked BCC branch. A Bloch fluctuation at quasi-momentum k is written as

$$\Psi_k(x) = e^{ik \cdot x} u_k(x), \quad u_k(x) = \sum_{G \in \Lambda^*} u_G(k) e^{iG \cdot x}.$$

Likewise, if needed, the mediator fluctuation is expanded as

$$X_i^a(x; k) = e^{ik \cdot x} \sum_{G \in \Lambda^*} x_i^a(G; k) e^{iG \cdot x}.$$

Expand the periodic coefficient fields in Fourier series:

$$\mathbb{A}_i^a(x) = \sum_{Q \in \Lambda^*} \mathbb{A}_i^a(Q) e^{iQ \cdot x}, \quad \mathbb{B}_{ij}^a(x) = \sum_{Q \in \Lambda^*} \mathbb{B}_{ij}^a(Q) e^{iQ \cdot x},$$

$$\mathbb{C}_i^a(x) = \sum_{Q \in \Lambda^*} \mathbb{C}_i^a(Q) e^{iQ \cdot x}, \quad \mathbb{D}_{ij}^a(x) = \sum_{Q \in \Lambda^*} \mathbb{D}_{ij}^a(Q) e^{iQ \cdot x}.$$

Then the Bloch-space matrix elements of the current vertices are

$$(\mathcal{J}_i^a(k))_{GG'} = \sum_{Q \in \Lambda^*} \delta_{G, G' + Q} \left[\mathbb{A}_i^a(Q) + \mathbb{B}_{ij}^a(Q)(k_j + G'_j) \right],$$

$$(\mathcal{K}_i^a(k))_{GG'} = \sum_{Q \in \Lambda^*} \delta_{G, G' + Q} \left[\mathbb{C}_i^a(Q) + \mathbb{D}_{ij}^a(Q)(k_j + G'_j) \right].$$

Therefore the exact mixed Bloch block is

$$(H_{\Psi X}(k))_{GG'; ia} = \alpha_X (\mathcal{J}_i^a(k))_{GG'} + \beta_X (\mathcal{K}_i^a(k))_{GG'}.$$

This is the explicit universal harmonic-space vertex formula.

CXXXV. BRAGG-POINT DERIVATIVE OF THE MIXED VERTEX

Because the dependence on k is affine at first-order differential level, the k -derivative is immediate:

$$(\partial_{k_\ell} \mathcal{J}_i^a(k))_{GG'} = \sum_{Q \in \Lambda^*} \delta_{G, G' + Q} \mathbb{B}_{i\ell}^a(Q),$$

$$(\partial_{k_\ell} \mathcal{K}_i^a(k))_{GG'} = \sum_{Q \in \Lambda^*} \delta_{G, G' + Q} \mathbb{D}_{i\ell}^a(Q).$$

Hence

$$\partial_{k_\ell} H_{\Psi X}(k) = \alpha_X \partial_{k_\ell} \mathcal{J}(k) + \beta_X \partial_{k_\ell} \mathcal{K}(k).$$

Evaluating at the folded Bragg point $k_* = G_*/2$ gives the exact first-order transport data required for the valley reduction.

CXXXVI. SCHUR-COMPLEMENT CORRECTION IN THE PURE AUXILIARY CLASS II TRUNCATION

At the working level, the heavy mediator block is

$$H_{XX}(k) = M_X^2 \mathbf{1}_X$$

up to subleading algebraic background corrections. Thus

$$H_{XX}^{-1}(k) = M_X^{-2} \mathbf{1}_X$$

and the heavy-mediated correction is

$$\Delta H_{\text{eff}}(k) = -\frac{1}{M_X^2} (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)) (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger.$$

Expanding by channels,

$$\Delta H_{\text{eff}}(k) = -\frac{\alpha_X^2}{M_X^2} \mathcal{J} \mathcal{J}^\dagger - \frac{\alpha_X \beta_X}{M_X^2} (\mathcal{J} \mathcal{K}^\dagger + \mathcal{K} \mathcal{J}^\dagger) - \frac{\beta_X^2}{M_X^2} \mathcal{K} \mathcal{K}^\dagger.$$

This is the exact Class II quadratic renormalization of the shell operator.

CXXXVII. DERIVATIVE OF THE EFFECTIVE OPERATOR

Differentiating the Schur correction gives

$$\partial_{k_\ell} \Delta H_{\text{eff}}(k) = -\frac{1}{M_X^2} [(\partial_{k_\ell} \mathcal{V}) \mathcal{V}^\dagger + \mathcal{V} (\partial_{k_\ell} \mathcal{V}^\dagger)],$$

where

$$\mathcal{V}(k) := \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k).$$

Expanding by channels,

$$\partial_{k_\ell} \Delta H_{\text{eff}} = -\frac{\alpha_X^2}{M_X^2} \mathfrak{D}_\ell^{JJ} - \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{D}_\ell^{JK} - \frac{\beta_X^2}{M_X^2} \mathfrak{D}_\ell^{KK},$$

with

$$\mathfrak{D}_\ell^{JJ} := (\partial_{k_\ell} \mathcal{J}) \mathcal{J}^\dagger + \mathcal{J} (\partial_{k_\ell} \mathcal{J}^\dagger),$$

$$\mathfrak{D}_\ell^{JK} := (\partial_{k_\ell} \mathcal{J}) \mathcal{K}^\dagger + \mathcal{J} (\partial_{k_\ell} \mathcal{K}^\dagger) + (\partial_{k_\ell} \mathcal{K}) \mathcal{J}^\dagger + \mathcal{K} (\partial_{k_\ell} \mathcal{J}^\dagger),$$

$$\mathfrak{D}_\ell^{KK} := (\partial_{k_\ell} \mathcal{K}) \mathcal{K}^\dagger + \mathcal{K} (\partial_{k_\ell} \mathcal{K}^\dagger).$$

The total transport operator is therefore

$$K_\ell = \partial_{k_\ell} H_{\Psi\Psi}(k)|_{k=k_*} - \frac{\alpha_X^2}{M_X^2} \mathfrak{D}_\ell^{JJ}(k_*) - \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{D}_\ell^{JK}(k_*) - \frac{\beta_X^2}{M_X^2} \mathfrak{D}_\ell^{KK}(k_*).$$

CXXXVIII. VALLEY PROJECTION AND PAULI DECOMPOSITION

Let

$$P_* = \frac{1}{2\pi i} \oint_\Gamma (z - H_{\text{shell}}(0))^{-1} dz$$

be the exact Riesz projector onto the low valley doublet. Define

$$M_\ell := P_* K_\ell P_*.$$

In an adapted defect frame $(\hat{n}, e_1, e_2)$, let

$$M_\parallel = \hat{n}_\ell M_\ell, \quad M_1 = e_{1\ell} M_\ell, \quad M_2 = e_{2\ell} M_\ell.$$

Under the locked branch symmetry selection rule,

$$M_\parallel = \lambda_\parallel \sigma_3, \quad M_1 = \alpha \sigma_1 + \beta \sigma_2, \quad M_2 = \alpha \sigma_2 - \beta \sigma_1.$$

Thus the exact reduced coefficients are

$$\lambda_\parallel = \frac{1}{2} (\sigma_3 M_\parallel),$$

$$\alpha = \frac{1}{4} (\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4} (\sigma_2 M_1 - \sigma_1 M_2).$$

CXXXIX. EXACT CLASS II RENORMALIZATION OF

$(\lambda_\parallel, \alpha, \beta)$
Split

$$M_\mu = M_\mu^{\text{ref}} + \Delta M_\mu.$$

Then

$$\lambda_\parallel = \lambda_\parallel^{\text{ref}} + \Delta \lambda_\parallel, \quad \alpha = \alpha^{\text{ref}} + \Delta \alpha, \quad \beta = \beta^{\text{ref}} + \Delta \beta,$$

with

$$\Delta \lambda_\parallel = -\frac{\alpha_X^2}{M_X^2} \Lambda_c^{JJ} - \frac{\alpha_X \beta_X}{M_X^2} \Lambda_c^{JK} - \frac{\beta_X^2}{M_X^2} \Lambda_c^{KK},$$

$$\Delta\alpha = -\frac{\alpha_X^2}{M_X^2}\Lambda_E^{JJ} - \frac{\alpha_X\beta_X}{M_X^2}\Lambda_E^{JK} - \frac{\beta_X^2}{M_X^2}\Lambda_E^{KK},$$

$$\Delta\beta = -\frac{\alpha_X^2}{M_X^2}\Lambda_O^{JJ} - \frac{\alpha_X\beta_X}{M_X^2}\Lambda_O^{JK} - \frac{\beta_X^2}{M_X^2}\Lambda_O^{KK},$$

where the channel overlaps are defined by the projected Pauli traces

$$\Lambda_c^{UV} := \frac{1}{2}(\sigma_3 P_*(\hat{n}_\ell \mathfrak{D}_\ell^{UV}(k_*))P_*),$$

$$\Lambda_E^{UV} := \frac{1}{4}(\sigma_1 P_*(e_{1\ell} \mathfrak{D}_\ell^{UV}(k_*))P_* + \sigma_2 P_*(e_{2\ell} \mathfrak{D}_\ell^{UV}(k_*))P_*),$$

$$\Lambda_O^{UV} := \frac{1}{4}(\sigma_2 P_*(e_{1\ell} \mathfrak{D}_\ell^{UV}(k_*))P_* - \sigma_1 P_*(e_{2\ell} \mathfrak{D}_\ell^{UV}(k_*))P_*),$$

with

$$(U, V) \in \{(J, J), (J, K), (K, K)\}.$$

This is the exact microscopic Class II renormalization formula for the reduced locked coefficients.

CXL. EXTRACTION OF

(g_c, g_E, g_O)

Introduce the exact overlap constants

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.$$

Then

$$g_c = \frac{\lambda_{\parallel}}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}.$$

Substituting the Class II renormalized coefficients,

$$g_c = \frac{\lambda_{\parallel}^{\text{ref}}}{C_c} - \frac{\alpha_X^2}{M_X^2} \frac{\Lambda_c^{JJ}}{C_c} - \frac{\alpha_X\beta_X}{M_X^2} \frac{\Lambda_c^{JK}}{C_c} - \frac{\beta_X^2}{M_X^2} \frac{\Lambda_c^{KK}}{C_c},$$

$$g_E = \frac{\alpha^{\text{ref}}}{C_E} - \frac{\alpha_X^2}{M_X^2} \frac{\Lambda_E^{JJ}}{C_E} - \frac{\alpha_X\beta_X}{M_X^2} \frac{\Lambda_E^{JK}}{C_E} - \frac{\beta_X^2}{M_X^2} \frac{\Lambda_E^{KK}}{C_E},$$

$$g_O = \frac{\beta^{\text{ref}}}{C_O} - \frac{\alpha_X^2}{M_X^2} \frac{\Lambda_O^{JJ}}{C_O} - \frac{\alpha_X\beta_X}{M_X^2} \frac{\Lambda_O^{JK}}{C_O} - \frac{\beta_X^2}{M_X^2} \frac{\Lambda_O^{KK}}{C_O}.$$

Only after explicit overlap evaluation may one specialize

$$C_c = \frac{1}{2}I_{\theta^3}, \quad C_E = I_{\theta}^{(E)}, \quad C_O = I_{\theta}^{(O)},$$

and, if the locked branch symmetry enforces it,

$$I_{\theta}^{(E)} = I_{\theta}^{(O)} =: I_{\theta}.$$

CXLI. WHAT IS NOW CLOSED

At the working branch level, the following are now explicit:

1. the universal Bloch-harmonic form of the mixed current vertices;
2. the exact Schur-complement correction generated by the heavy Class II mediator;
3. the exact k -derivative formula for the effective transport operator;
4. the projected Pauli-trace extraction of $(\lambda_{\parallel}, \alpha, \beta)$;
5. the exact channel decomposition of the Class II renormalization of (g_c, g_E, g_O) .

CXLII. WHAT REMAINS BRANCH-SPECIFIC

Only three branch-specific algebraic ingredients remain to be inserted:

1. the explicit Fourier coefficients

$$\mathbb{A}_i^a(Q), \mathbb{B}_{ij}^a(Q), \mathbb{C}_i^a(Q), \mathbb{D}_{ij}^a(Q);$$

2. the locked-branch proof that the projected transport matrices have the special Pauli form;
3. the actual overlap evaluation of

$$C_c, C_E, C_O.$$

These are no longer conceptual gaps. They are the remaining explicit calculations of the chosen working branch.

CXLIII. FINAL SUMMARY

For the refined locked + Class II working branch, the mixed Bloch vertex has the universal harmonic-space form

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k),$$

where, under the natural first-order local-current assumption,

$$(\mathcal{J}_i^a(k))_{GG'} = \sum_Q \delta_{G, G'+Q} \left[\mathbb{A}_i^a(Q) + \mathbb{B}_{ij}^a(Q)(k_j + G'_j) \right],$$

$$(\mathcal{K}_i^a(k))_{GG'} = \sum_Q \delta_{G, G'+Q} \left[\mathbb{C}_i^a(Q) + \mathbb{D}_{ij}^a(Q)(k_j + G'_j) \right].$$

In the pure auxiliary Class II truncation,

$$H_{XX}(k) = M_X^2 \mathbf{1}_X,$$

so the heavy-mediated correction is exactly

$$\Delta H_{\text{eff}}(k) = -\frac{1}{M_X^2} (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)) (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger.$$

Differentiating at the Bragg point and projecting to the exact valley doublet yields

$$M_i = P_* \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*} P_*.$$

Under the locked Pauli selection rule,

$$M_{\parallel} = \lambda_{\parallel} \sigma_3, \quad M_1 = \alpha \sigma_1 + \beta \sigma_2, \quad M_2 = \alpha \sigma_2 - \beta \sigma_1,$$

so

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}), \quad \alpha = \frac{1}{4}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4}(\sigma_2 M_1 - \sigma_1 M_2).$$

Thus the Class II mediator renormalizes the reduced invariants by

$$\Delta\lambda_{\parallel} = -\frac{\alpha_X^2}{M_X^2}\Lambda_c^{JJ} - \frac{\alpha_X\beta_X}{M_X^2}\Lambda_c^{JK} - \frac{\beta_X^2}{M_X^2}\Lambda_c^{KK},$$

$$\Delta\alpha = -\frac{\alpha_X^2}{M_X^2}\Lambda_E^{JJ} - \frac{\alpha_X\beta_X}{M_X^2}\Lambda_E^{JK} - \frac{\beta_X^2}{M_X^2}\Lambda_E^{KK},$$

$$\Delta\beta = -\frac{\alpha_X^2}{M_X^2}\Lambda_O^{JJ} - \frac{\alpha_X\beta_X}{M_X^2}\Lambda_O^{JK} - \frac{\beta_X^2}{M_X^2}\Lambda_O^{KK}.$$

Finally,

$$g_c = \frac{\lambda_{\parallel}}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O},$$

with C_c, C_E, C_O determined by explicit overlap integrals.

Therefore the only remaining work is branch-specific algebra: the explicit Fourier-space current coefficients, the locked Pauli selection-rule proof, and the overlap integrals.

CXLIV. BLOCK-NORM AND GERSHGORIN BUDGET THEOREM FOR THE FINAL REMOTE-GAP SLOT

CXLIV. SETUP

Let

$$H_R(p) = D_R(p) + R_R(p), \quad |p| \leq \rho,$$

act on the remote sector $\mathcal{H}_R$, and assume that for every $|p| \leq \rho$,

$$H_R(p) = H_R(p)^\dagger, \quad D_R(p) = D_R(p)^\dagger, \quad R_R(p) = R_R(p)^\dagger.$$

We assume that $D_R(p)$ is the benchmark remote operator and $R_R(p)$ is the residual remote mixing/remainder. Define the benchmark gap

$$\delta_{0,\rho} := \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_R(p))).$$

Define also the residual norm budget

$$\eta_{R,\rho} := \sup_{|p| \leq \rho} \|R_R(p)\|.$$

Then the basic perturbative spectral isolation criterion is:

a. Lemma (uniform perturbative remote-gap criterion). If

$$\delta_{0,\rho} > \eta_{R,\rho},$$

then

$$\text{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset, \quad \Delta_\rho := \delta_{0,\rho} - \eta_{R,\rho} > 0,$$

for all $|p| \leq \rho$.

b. Proof. By standard spectral stability for self-adjoint bounded perturbations,

$$\text{dist}(0, \text{spec}(D_R(p) + R_R(p))) \geq \text{dist}(0, \text{spec}(D_R(p))) - \|R_R(p)\|.$$

Taking the infimum in p gives the claim. $\square$

Thus the only remaining task is to produce usable upper bounds on $\eta_{R,\rho}$.

CXLVI. SECTORWISE BLOCK DECOMPOSITION

At the minimal truncation level, suppose

$$\mathcal{H}_R^{\text{tr}} = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non}} \oplus \mathcal{H}_{\text{wrong}}.$$

Write the residual block in sector form

$$R_R(p) = \begin{pmatrix} 0 & R_{\text{vol,high}} & R_{\text{vol,non}} & R_{\text{vol,wrong}} \\ R_{\text{high,vol}} & 0 & R_{\text{high,non}} & R_{\text{high,wrong}} \\ R_{\text{non,vol}} & R_{\text{non,high}} & 0 & R_{\text{non,wrong}} \\ R_{\text{wrong,vol}} & R_{\text{wrong,high}} & R_{\text{wrong,non}} & 0 \end{pmatrix} (p),$$

where

$$R_{ab}(p) : \mathcal{H}_b \rightarrow \mathcal{H}_a.$$

Define the pairwise block budgets

$$\eta_{ab,\rho} := \sup_{|p| \leq \rho} \|R_{ab}(p)\|.$$

CXLVII. BASIC BLOCK-NORM BUDGET

Since

$$R_R(p) = \sum_{a \neq b} E_{ab} R_{ab}(p),$$

where $E_{ab} R_{ab}(p)$ denotes the operator with only the (a, b) block nonzero, the triangle inequality gives

$$\|R_R(p)\| \leq \sum_{a \neq b} \|R_{ab}(p)\|.$$

Hence

$$\boxed{\eta_{R,\rho} \leq \sum_{a \neq b} \eta_{ab,\rho}.}$$

Therefore a sufficient explicit criterion for remote isolation is

$$\boxed{\delta_{0,\rho} > \sum_{a \neq b} \eta_{ab,\rho}.}$$

This bound is crude but rigorous.

CXLVIII. GERSHGORIN BUDGET FOR THE FINITE TRUNCATED REMOTE MATRIX

Assume now that $H_R^{\text{tr}}(p)$ is represented in a finite orthonormal basis of the truncated remote space. Let

$$(H_R^{\text{tr}}(p))_{mn}$$

be its matrix elements, and define the row sums

$$\mathcal{R}_m(p) := \sum_{n \neq m} |(H_R^{\text{tr}}(p))_{mn}|.$$

Define the Gershgorin lower bound

$$\Delta_{G,\rho} := \inf_{|p| \leq \rho} \min_m \left(|(H_R^{\text{tr}}(p))_{mm}| - \mathcal{R}_m(p) \right).$$

If

$$\boxed{\Delta_{G,\rho} > 0,}$$

then by Gershgorin's theorem and self-adjointness,

$$\boxed{\text{spec}(H_R^{\text{tr}}(p)) \cap (-\Delta_{G,\rho}, \Delta_{G,\rho}) = \emptyset \quad (|p| \leq \rho).}$$

This route is often sharper than the block-norm budget when the truncated remote matrix is sparse.

CXLIX. FULL REMOTE COMPLETION

For the full remote sector, let

$$H_R^{\text{full}}(p) = D_{\text{rem}}(p) + R_{\text{rem}}(p),$$

where $D_{\text{rem}}(p)$ is the full benchmark operator and $R_{\text{rem}}(p)$ contains:

1. all off-diagonal remote-remote couplings,
2. all couplings to tail sectors,
3. all bounded residual corrections to the diagonal benchmark.

Define

$$\eta_{\text{rem},\rho} := \sup_{|p| \leq \rho} \|R_{\text{rem}}(p)\|,$$

and let

$$\Delta_{\text{bench},\rho}^{\text{full}}$$

be any explicit lower bound such that

$$\boxed{\Delta_{\text{bench},\rho}^{\text{full}} \leq \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_{\text{rem}}(p)))}.}$$

Then the sufficient final Task IV criterion is

$$\boxed{\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.}$$

Under this condition,

$$\boxed{\text{spec}(H_R^{\text{full}}(p)) \cap (-\Delta_{\text{full},\rho}, \Delta_{\text{full},\rho}) = \emptyset, \quad \Delta_{\text{full},\rho} \geq \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho} > 0.}$$

CL. EXPLICIT POWER-COUNTING / HEAVY-SUPPRESSION BUDGET

By the previously established tail-coupling estimate theorem, one may assume

$$\eta_{\text{rem},\rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}, \quad r_a \geq 1.$$

Thus all remote mixing is at least $O(\rho)$ near the crossing, and heavy-mediated sectors carry additional inverse-mass suppression.

CLI. FINAL CLOSURE STATEMENT

Therefore the final remote-gap slot is closed once one supplies an explicit benchmark lower bound

$$\Delta_{\text{bench},\rho}^{\text{full}}$$

for the benchmark operator $D_{\text{rem}}(p)$ and verifies

$$\Delta_{\text{bench},\rho}^{\text{full}} > \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}.$$

This is the cleanest final Task IV criterion.

CLII. MODEL-SPECIFIC BENCHMARK LOWER BOUND

Assume the benchmark operator $D_{\text{rem}}(p)$ is block-diagonal across the remote sectors and that the following explicit lower bounds have already been established on $|p| \leq \rho$:

$$\Delta_{\text{vol},\rho}, \quad \Delta_{\text{high},\rho}, \quad \Delta_{\text{wrong-int},\rho}, \quad \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}.$$

Then one may take

$$\Delta_{\text{bench},\rho}^{\text{full}} = \min \left\{ \Delta_{\text{vol},\rho}, \Delta_{\text{high},\rho}, \Delta_{\text{wrong-int},\rho}, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}.$$

If, for example, the previously derived explicit lower bounds are

$$\Delta_{\text{vol},\rho} = 4K - |r_*|, \quad \Delta_{\text{high},\rho} = \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2},$$

$$\Delta_{\text{wrong-int},\rho} = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|,$$

then these may be inserted directly into the minimum above, provided each term is already known to be positive on the regime of interest.

CLIII. FINAL SUMMARY

The proposed block-norm / Gershgorin budget strategy for the final remote-gap slot is mathematically correct in structure.

The main logic is:

$$H_R(p) = D_R(p) + R_R(p), \quad \delta_{0,\rho} = \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_R(p))), \quad \eta_{R,\rho} = \sup_{|p| \leq \rho} \|R_R(p)\|.$$

If

$$\delta_{0,\rho} > \eta_{R,\rho},$$

then

$$\text{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset, \quad \Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.$$

The block budget

$$\eta_{R,\rho} \leq \sum_{a \neq b} \eta_{ab,\rho}$$

is a valid crude norm estimate.

The Gershgorin route is also valid, but should be stated explicitly for the finite truncated remote matrix.

For the full remote completion, the clean final condition is

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho},$$

where

$$\Delta_{\text{bench},\rho}^{\text{full}} \leq \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_{\text{rem}}(p))).$$

Thus the theorem is essentially correct, but to be fully closed and submission-safe it should explicitly include:

1. self-adjointness of H_R, D_R, R_R ;
2. the benchmark gap defined as a spectral distance, not merely as a diagonal heuristic;
3. a clear statement that the Gershgorin part applies to the finite truncated remote matrix;
4. a cleaned-up model-specific formula for $\Delta_{\text{bench},\rho}^{\text{full}}$.

CLIV. BLOCK-NORM AND GERSHGORIN BUDGET THEOREM FOR THE FINAL REMOTE-GAP SLOT

CLV. SETUP

Let

$$H_R(p) = D_R(p) + R_R(p), \quad |p| \leq \rho,$$

be the remote-sector operator on a Hilbert space $\mathcal{H}_R$. Assume that for each $|p| \leq \rho$,

$$H_R(p) = H_R(p)^\dagger, \quad D_R(p) = D_R(p)^\dagger, \quad R_R(p) = R_R(p)^\dagger.$$

Here $D_R(p)$ is the benchmark remote operator and $R_R(p)$ is the residual remote mixing/remainder. Define the benchmark spectral distance

$$\delta_{0,\rho} := \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_R(p))),$$

and the residual norm budget

$$\eta_{R,\rho} := \sup_{|p| \leq \rho} \|R_R(p)\|.$$

CLVI. UNIFORM PERTURBATIVE REMOTE-GAP CRITERION

a. *Theorem 1 (uniform remote-gap criterion).* If

$$\delta_{0,\rho} > \eta_{R,\rho},$$

then

$$\text{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset \quad (|p| \leq \rho),$$

where

$$\Delta_\rho := \delta_{0,\rho} - \eta_{R,\rho} > 0.$$

b. *Proof.* Fix $|p| \leq \rho$. Since $D_R(p)$ and $R_R(p)$ are self-adjoint, standard spectral stability under bounded self-adjoint perturbations gives

$$\text{dist}(0, \text{spec}(D_R(p) + R_R(p))) \geq \text{dist}(0, \text{spec}(D_R(p))) - \|R_R(p)\|.$$

Hence

$$\text{dist}(0, \text{spec}(H_R(p))) \geq \delta_{0,\rho} - \eta_{R,\rho} = \Delta_\rho > 0.$$

Therefore

$$\text{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset.$$

Since the bound is uniform in $|p| \leq \rho$, the claim follows. $\square$

CLVII. SECTORWISE BLOCK DECOMPOSITION

Assume now that the minimal truncated remote space splits as

$$\mathcal{H}_R^{\text{tr}} = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}.$$

Write the off-diagonal residual block as

$$R_R(p) = \begin{pmatrix} 0 & R_{\text{vol,high}} & R_{\text{vol,non}} & R_{\text{vol,wrong}} \\ R_{\text{high,vol}} & 0 & R_{\text{high,non}} & R_{\text{high,wrong}} \\ R_{\text{non,vol}} & R_{\text{non,high}} & 0 & R_{\text{non,wrong}} \\ R_{\text{wrong,vol}} & R_{\text{wrong,high}} & R_{\text{wrong,non}} & 0 \end{pmatrix} (p),$$

where

$$R_{ab}(p) : \mathcal{H}_b \rightarrow \mathcal{H}_a$$

are the pairwise off-diagonal couplings.

Define the pairwise budgets

$$\eta_{ab,\rho} := \sup_{|p| \leq \rho} \|R_{ab}(p)\|.$$

a. *Proposition 2 (basic block-norm budget).* One has the crude but rigorous estimate

$$\eta_{R,\rho} \leq \sum_{a \neq b} \eta_{ab,\rho}.$$

b. *Proof.* For each $|p| \leq \rho$,

$$R_R(p) = \sum_{a \neq b} E_{ab} R_{ab}(p),$$

where $E_{ab} R_{ab}(p)$ denotes the operator with only the (a, b) -block nonzero. By the triangle inequality,

$$\|R_R(p)\| \leq \sum_{a \neq b} \|R_{ab}(p)\|.$$

Taking the supremum over $|p| \leq \rho$ yields

$$\eta_{R,\rho} \leq \sum_{a \neq b} \eta_{ab,\rho}.$$

□

c. *Corollary 3 (block-budget remote isolation).* If

$$\delta_{0,\rho} > \sum_{a \neq b} \eta_{ab,\rho},$$

then

$$\text{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset \quad (|p| \leq \rho),$$

with

$$\Delta_\rho \geq \delta_{0,\rho} - \sum_{a \neq b} \eta_{ab,\rho} > 0.$$

CLVIII. GERSHGORIN BUDGET FOR THE FINITE TRUNCATED REMOTE MATRIX

We now assume that $H_R^{\text{tr}}(p)$ is represented in a finite orthonormal basis of the truncated remote space. Let

$$(H_R^{\text{tr}}(p))_{mn}$$

denote the corresponding matrix elements.

Define the row sums

$$\mathcal{R}_m(p) := \sum_{n \neq m} |(H_R^{\text{tr}}(p))_{mn}|.$$

Define the uniform Gershgorin budget

$$\Delta_{G,\rho} := \inf_{|p| \leq \rho} \min_m \left(|(H_R^{\text{tr}}(p))_{mm}| - \mathcal{R}_m(p) \right).$$

a. *Theorem 4 (Gershgorin remote-gap criterion for the finite truncation).* If

$$\Delta_{G,\rho} > 0,$$

then

$$\text{spec}(H_R^{\text{tr}}(p)) \cap (-\Delta_{G,\rho}, \Delta_{G,\rho}) = \emptyset \quad (|p| \leq \rho).$$

b. Proof. For each fixed $|p| \leq \rho$, Gershgorin's circle theorem implies that every eigenvalue $\lambda \in \text{spec}(H_R^{\text{tr}}(p))$ belongs to at least one disk

$$|\lambda - (H_R^{\text{tr}}(p))_{mm}| \leq \mathcal{R}_m(p).$$

Hence

$$|\lambda| \geq |(H_R^{\text{tr}}(p))_{mm}| - \mathcal{R}_m(p)$$

for some m , and therefore

$$|\lambda| \geq \min_m \left(|(H_R^{\text{tr}}(p))_{mm}| - \mathcal{R}_m(p) \right) \geq \Delta_{G,\rho} > 0.$$

Thus

$$\text{spec}(H_R^{\text{tr}}(p)) \cap (-\Delta_{G,\rho}, \Delta_{G,\rho}) = \emptyset.$$

Since the bound is uniform in $|p| \leq \rho$, the claim follows. $\square$

CLIX. FULL REMOTE COMPLETION WITH TAILS

For the full remote sector, write

$$H_R^{\text{full}}(p) = D_{\text{rem}}(p) + R_{\text{rem}}(p),$$

where:

- $D_{\text{rem}}(p)$ is the global benchmark remote operator;
- $R_{\text{rem}}(p)$ contains all off-diagonal remote-remote mixing, all couplings to tail sectors, and all bounded residual corrections to the benchmark part.

Assume

$$H_R^{\text{full}}(p) = H_R^{\text{full}}(p)^\dagger, \quad D_{\text{rem}}(p) = D_{\text{rem}}(p)^\dagger, \quad R_{\text{rem}}(p) = R_{\text{rem}}(p)^\dagger.$$

Define

$$\eta_{\text{rem},\rho} := \sup_{|p| \leq \rho} \|R_{\text{rem}}(p)\|,$$

and let

$$\Delta_{\text{bench},\rho}^{\text{full}}$$

be any explicit lower bound satisfying

$$\Delta_{\text{bench},\rho}^{\text{full}} \leq \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_{\text{rem}}(p))).$$

a. Theorem 5 (final Task IV remote isolation criterion). If

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho},$$

then

$$\text{spec}(H_R^{\text{full}}(p)) \cap (-\Delta_{\text{full},\rho}, \Delta_{\text{full},\rho}) = \emptyset \quad (|p| \leq \rho),$$

with

$$\Delta_{\text{full},\rho} \geq \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho} > 0.$$

b. Proof. Apply Theorem 1 with

$$D_R = D_{\text{rem}}, \quad R_R = R_{\text{rem}}, \quad \delta_{0,\rho} = \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_{\text{rem}}(p))).$$

Since

$$\Delta_{\text{bench},\rho}^{\text{full}} \leq \delta_{0,\rho},$$

the inequality

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}$$

implies

$$\delta_{0,\rho} > \eta_{\text{rem},\rho}.$$

The claim follows immediately from Theorem 1. $\square$

CLX. EXPLICIT POWER-COUNTING / HEAVY-SUPPRESSION BUDGET

Assume the previously established tail-coupling estimate theorem gives

$$\eta_{\text{rem},\rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}, \quad r_a \geq 1.$$

Then all remote mixing is at least $O(\rho)$ near the crossing, and heavy-mediated sectors carry additional inverse-mass suppression.

Hence a fully explicit sufficient final criterion is

$$\Delta_{\text{bench},\rho}^{\text{full}} > \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}.$$

CLXI. MODEL-SPECIFIC BENCHMARK LOWER BOUND

Assume the benchmark operator $D_{\text{rem}}(p)$ is block-diagonal across the remote sectors, and that the following explicit lower bounds have already been established for $|p| \leq \rho$:

$$\Delta_{\text{vol},\rho}, \quad \Delta_{\text{high},\rho}, \quad \Delta_{\text{wrong-int},\rho}, \quad \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}.$$

Then one may take

$$\Delta_{\text{bench},\rho}^{\text{full}} = \min \left\{ \Delta_{\text{vol},\rho}, \Delta_{\text{high},\rho}, \Delta_{\text{wrong-int},\rho}, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}.$$

If, for example, the explicit lower bounds are

$$\Delta_{\text{vol},\rho} = 4K - |r_*|,$$

$$\Delta_{\text{high},\rho} = \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2},$$

$$\Delta_{\text{wrong-int},\rho} = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|,$$

then these may be inserted directly into the minimum above, provided each quantity is already known to be positive in the parameter regime under consideration.

CLXII. FINAL CLOSURE STATEMENT

Therefore the final remote-gap slot is closed once one has:

1. a self-adjoint benchmark decomposition

$$H_R^{\text{full}}(p) = D_{\text{rem}}(p) + R_{\text{rem}}(p);$$

2. an explicit lower bound

$$\Delta_{\text{bench},\rho}^{\text{full}} \leq \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_{\text{rem}}(p)));$$

3. an explicit residual budget bound

$$\eta_{\text{rem},\rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}.$$

Under the single inequality

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho},$$

the complete remote block is uniformly spectrally isolated from zero.

CLXIII. FINAL SUMMARY

We rewrote the final remote-gap slot in a fully submission-safe form.

The key definitions are:

$$H_R(p) = D_R(p) + R_R(p), \quad \delta_{0,\rho} = \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_R(p))), \quad \eta_{R,\rho} = \sup_{|p| \leq \rho} \|R_R(p)\|.$$

If

$$\delta_{0,\rho} > \eta_{R,\rho},$$

then

$$\text{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset, \quad \Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.$$

For the finite truncated remote matrix, a sharper alternative is the Gershgorin bound

$$\Delta_{G,\rho} = \inf_{|p| \leq \rho} \min_m \left(|(H_R^{\text{tr}}(p))_{mm}| - \mathcal{R}_m(p) \right),$$

which implies spectral isolation whenever $\Delta_{G,\rho} > 0$.

For the full remote completion, let

$$H_R^{\text{full}}(p) = D_{\text{rem}}(p) + R_{\text{rem}}(p), \quad \eta_{\text{rem},\rho} = \sup_{|p| \leq \rho} \|R_{\text{rem}}(p)\|.$$

If

$$\Delta_{\text{bench},\rho}^{\text{full}} \leq \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_{\text{rem}}(p)))$$

and

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho},$$

then

$$\text{spec}(H_R^{\text{full}}(p)) \cap (-\Delta_{\text{full},\rho}, \Delta_{\text{full},\rho}) = \emptyset, \quad \Delta_{\text{full},\rho} \geq \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho} > 0.$$

Thus the final Task IV closure reduces to an explicit benchmark lower bound plus an explicit residual mixing budget.

CLXIV. STATUS REPORT: WHAT THE MAIN TASK ACTUALLY WAS, AND WHAT IS AND IS NOT FINISHED

The main program was never merely the remote-gap theorem or the exploratory CKM fitting by itself. The actual main line has been

TECT microscopic defect PDE $\longrightarrow$ single-defect light mode $\longrightarrow$ three-defect family projector $\longrightarrow$ Schur-complement reduction

Equivalently, the main target has been

$$\mathbb{O}_{1\text{def}}^{(0)} \rightarrow P_L \rightarrow \mathcal{Y}_{\text{eff}} \rightarrow (B_0, B_1, B_H) \rightarrow (H_Q, H_u, H_d, \dots) \rightarrow (g_c, g_E, g_O) \rightarrow \text{flavor/CKM structure.}$$

CLXV. WHAT IS ESSENTIALLY FINISHED

Several important auxiliary layers are now essentially closed.

a. (i) Exploratory flavor existence stage. At the reduced-model level, one already has strong numerical evidence that the flavor problem is nonempty: hierarchical CKM-like mixing exists, and CP-violating branches with

$$J \neq 0, \quad \delta \neq 0$$

exist as well. Thus the existential stage has done its job.

b. (ii) Reduced operator formalism. The abstract Schur-complement structure is already in place:

$$\mathcal{Y}_{\text{eff}} = L - KH^{-1}K^\dagger,$$

together with the Hermitian basis decomposition

$$\{D_0, D_1, D_2, S_{12}, S_{13}, S_{23}, A_{12}, A_{13}, A_{23}\}.$$

The reduced CP-odd coefficients have already been identified as heavy-mediated imaginary overlaps.

c. (iii) Remote-gap / Task IV closure theorem. The remote spectral isolation module is now essentially complete as an auxiliary theorem: the block-norm and Gershgorin budget strategy has been rewritten in a submission-safe form. So this part should be regarded as an almost-finished support theorem, not as the main unfinished bottleneck.

CLXVI. WHAT IS NOT FINISHED

The main program is not finished, because the genuinely TECT-specific microscopic input has not yet been derived.

a. (a) Actual TECT reference operator. The form

$$\mathbb{O}_{1\text{def}}^{(0)} = i\widetilde{\Gamma}_{\text{def}}^i(\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}}$$

has been established only as an abstract operator-theoretic template. It has not yet been derived from the actual microscopic TECT PDE.

b. (b) One defect $\leftrightarrow$ one light mode. The abstract theorem schema

$$\text{Callias-type structure} + \text{bulk gap} + \text{grading} + \text{Ind} = 1 \implies \text{one protected light mode}$$

has been identified, but the actual TECT index computation has not been done. So the statement

$$\text{“one defect carries one family mode”}$$

is still conditional, not yet a first-principles theorem of TECT.

c. (c) *Explicit Class II microscopic coefficient extraction.* The explicit working branch

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_{\Psi}^{\text{ref}} + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a + \beta_X X_i^a K_i^a$$

has already been connected to the abstract Schur-complement formalism, and the universal Bloch-vertex structure has been written down.

However, the actual branch-specific Fourier coefficients

$$\mathbb{A}, \mathbb{B}, \mathbb{C}, \mathbb{D}$$

have not yet been inserted and evaluated. So this stage is structurally organized but not numerically/computationally completed.

CLXVII. CORRECT CURRENT CONCLUSION

Therefore the correct status is:

many supporting layers are finished or nearly finished,

but

the main TECT-specific microscopic extraction problem is not yet finished.

In more operational terms:

existence, abstract structure, and auxiliary isolation theorems are largely in place;

actual microscopic coefficient extraction from the TECT defect background remains the main unfinished task.

CLXVIII. IMMEDIATE NEXT STEP

The correct next step is no longer another auxiliary theorem. It is to return to the main line and specify a minimal actual TECT microscopic field content, write the corresponding

$$K^0(\mathfrak{B}), \quad K^i(\mathfrak{B}), \quad \mathcal{M}(\mathfrak{B}),$$

linearize around a defect background, and then compute

$$G_{\text{def}}^0, \quad G_{\text{def}}^i, \quad M_{\text{def}},$$

followed by

$$P_L, \quad V_{ab}^{(I)}, \quad K_{a\alpha}^{(I)}, \quad E_{\alpha}.$$

Only that step will turn the current framework into a first-principles microscopic coefficient extraction.

CLXIX. FINAL SUMMARY

The main task was not merely the remote-gap theorem or the exploratory CKM fit. The actual main program has been

TECT defect PDE $\rightarrow$ single-defect light mode $\rightarrow$ three-defect family projector $\rightarrow$ Schur-complement reduced operator $\rightarrow$ micros

At present:

1. the exploratory flavor existence stage is essentially finished;
2. the abstract reduced-operator formalism is in place;
3. the remote-gap / Task IV theorem is essentially finished as an auxiliary support result;
4. but the genuinely TECT-specific microscopic coefficient extraction is not finished.

The unfinished core is:

derive the actual TECT defect fluctuation operator, verify the one-mode structure, and compute the explicit microscopic overla

So the honest status is:

the support theorems are largely in place, but the main microscopic extraction problem is still open.

CLXX. MINIMAL MICROSCOPIC MULTIPLY AND EXPLICIT COEFFICIENT EXTRACTION TEMPLATE

We now fix a minimal structurally correct microscopic working ansatz for the TECT coefficient-extraction program.

CLXXI. MINIMAL WORKING MICROSCOPIC MULTIPLY

The smallest multiplet that can support the previously derived operator-theoretic program is

$$\mathfrak{B} = (P^A, \Psi, X_i^a).$$

Here:

- $P^A(x, t)$ is the locked condensate / order-parameter background field;
- $\Psi(x, t)$ is the bundle-valued fluctuation field whose low spectral sector is ultimately projected onto the family space;
- $X_i^a(x, t)$ is the heavy Class II mediator field.

This is not yet claimed to be the unique microscopic TECT field content. It is the minimal working multiplet sufficient for actual coefficient extraction.

CLXXII. MINIMAL FIRST-ORDER HERMITIAN MICROSCOPIC ACTION

We take the microscopic working action to be

$$S[P, \Psi, X] = \int dt d^3x \left(\mathcal{L}_P[P] + \mathcal{L}_\Psi^{\text{ref}}[\Psi, P] + \mathcal{L}_X[X] + \mathcal{L}_{\Psi X}[\Psi, P, X] \right),$$

with the pieces defined as follows.

a. Reference Ψ -sector.

$$\mathcal{L}_\Psi^{\text{ref}} = \frac{i}{2} \Psi^\dagger Z^0(P) \overleftrightarrow{\partial}_t \Psi + \frac{i}{2} \Psi^\dagger Z^i(P) \overleftrightarrow{\partial}_i \Psi - \Psi^\dagger M_0(P) \Psi.$$

b. Heavy mediator sector. At the present working truncation level,

$$\mathcal{L}_X = \frac{M_X^2}{2} X_i^a X_i^a.$$

This is the auxiliary heavy Class II truncation used throughout the current extraction branch.

c. *Class II couplings.* We now choose the explicit local-current ansatz

$$J_i^a[\Psi, P] = \Psi^\dagger A_i^a(P) \Psi + \frac{i}{2} \Psi^\dagger B_{ij}^a(P) \overset{\leftrightarrow}{\partial}_j \Psi,$$

$$K_i^a[P, \Psi] = \Psi^\dagger C_i^a(P) \Psi + \frac{i}{2} \Psi^\dagger D_{ij}^a(P) \overset{\leftrightarrow}{\partial}_j \Psi.$$

Hence

$$\mathcal{L}_{\Psi X} = \alpha_X X_i^a J_i^a[\Psi, P] + \beta_X X_i^a K_i^a[P, \Psi].$$

CLXXIII. HERMITICITY CONDITIONS

To ensure a well-defined Hermitian fluctuation operator, we assume

$$Z^0(P) = Z^0(P)^\dagger, \quad Z^i(P) = Z^i(P)^\dagger, \quad M_0(P) = M_0(P)^\dagger,$$

and likewise

$$A_i^a(P) = A_i^a(P)^\dagger, \quad B_{ij}^a(P) = B_{ij}^a(P)^\dagger,$$

$$C_i^a(P) = C_i^a(P)^\dagger, \quad D_{ij}^a(P) = D_{ij}^a(P)^\dagger.$$

Under these conditions, the total quadratic Ψ -sector is Hermitian.

CLXXIV. UNIFIED FIRST-ORDER COEFFICIENT FIELDS

It is convenient to define the combined Class II coefficient fields

$$U_i^a(P) := \alpha_X A_i^a(P) + \beta_X C_i^a(P),$$

$$V_{ij}^a(P) := \alpha_X B_{ij}^a(P) + \beta_X D_{ij}^a(P).$$

Then

$$\mathcal{L}_{\Psi X} = X_i^a \Psi^\dagger U_i^a(P) \Psi + \frac{i}{2} X_i^a \Psi^\dagger V_{ij}^a(P) \overset{\leftrightarrow}{\partial}_j \Psi.$$

So the working microscopic action becomes

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_P[P] + \frac{i}{2} \Psi^\dagger Z^0(P) \overset{\leftrightarrow}{\partial}_t \Psi + \frac{i}{2} \Psi^\dagger \left(Z^i(P) + X_j^a V_{ji}^a(P) \right) \overset{\leftrightarrow}{\partial}_i \Psi - \Psi^\dagger \left(M_0(P) - X_i^a U_i^a(P) \right) \Psi + \frac{M_X^2}{2} X_i^a X_i^a.$$

This is the explicit minimal microscopic coefficient template.

CLXXV. EULER-LAGRANGE EQUATION FOR THE Ψ SECTOR

Varying with respect to $\Psi^\dagger$ gives the first-order field equation

$$i K^0(P, X) \partial_t \Psi + i K^i(P, X) \partial_i \Psi + M(P, X, \partial P, \partial X) \Psi = 0,$$

with

$$K^0(P, X) = Z^0(P),$$

$$K^i(P, X) = Z^i(P) + X_j^a V_{ji}^a(P),$$

and

$$M(P, X, \partial P, \partial X) = \frac{i}{2} \partial_i K^i(P, X) - \left(M_0(P) - X_i^a U_i^a(P) \right).$$

This is the exact coefficient identification for the working ansatz.

CLXXVI. DEFECT-BACKGROUND SPECIALIZATION

Let

$$(\bar{P}(x), \bar{X}_i^a(x))$$

be a static single-defect background of the working branch. Then the linearized reference fluctuation equation for the Ψ -sector is

$$i K_{\text{def}}^0(x) \partial_t \psi + i K_{\text{def}}^i(x) \partial_i \psi + M_{\text{def}}(x) \psi = 0,$$

where

$$K_{\text{def}}^0(x) = Z^0(\bar{P}(x)),$$

$$K_{\text{def}}^i(x) = Z^i(\bar{P}(x)) + \bar{X}_j^a(x) V_{ji}^a(\bar{P}(x)),$$

$$M_{\text{def}}(x) = \frac{i}{2} \partial_i K_{\text{def}}^i(x) - \left(M_0(\bar{P}(x)) - \bar{X}_i^a(x) U_i^a(\bar{P}(x)) \right).$$

These are the actual microscopic coefficient fields entering the defect spectral problem.

CLXXVII. TWO IMPORTANT SPECIAL CASES

a. Case 1: reference branch with $\bar{X} = 0$. If the locked reference background carries no mediator expectation value, then

$$K_{\text{def}}^0 = Z^0(\bar{P}), \quad K_{\text{def}}^i = Z^i(\bar{P}), \quad M_{\text{def}} = \frac{i}{2} \partial_i Z^i(\bar{P}) - M_0(\bar{P}).$$

In this case the Class II mediator enters only through the quadratic Schur-complement correction and not through the reference operator itself.

b. Case 2: algebraically induced $\bar{X} \neq 0$. If the background stationarity equation produces a nonzero mediator profile

$$M_X^2 \bar{X}_i^a + \alpha_X J_i^a[\bar{\Psi}, \bar{P}] + \beta_X K_i^a[\bar{P}, \bar{\Psi}] = 0,$$

then $\bar{X}$ shifts the reference transport and mass coefficients through the explicit formulas above. So the Class II branch contributes both:

1. to the reference first-order coefficients through $\bar{X}$;
2. to the quadratic effective operator through heavy-field elimination.

CLXXVIII. CANONICAL NORMALIZED DEFECT OPERATOR

Assume

$$K_{\text{def}}^0(x) = K_{\text{def}}^0(x)^\dagger > 0.$$

Define

$$S(x) := (K_{\text{def}}^0(x))^{1/2}, \quad \chi = S\psi.$$

Then the defect fluctuation equation becomes

$$i\partial_t \chi = \mathbb{H}_{\text{def}} \chi,$$

with

$$\boxed{\mathbb{H}_{\text{def}} = -i \Gamma_{\text{def}}^i(x) \partial_i + W_{\text{def}}(x),}$$

where

$$\boxed{\Gamma_{\text{def}}^i = S^{-1} K_{\text{def}}^i S^{-1},}$$

$$\boxed{W_{\text{def}} = -i S^{-1} K_{\text{def}}^i \partial_i (S^{-1}) - S^{-1} M_{\text{def}} S^{-1}.}$$

Equivalently, after choosing a bundle frame and covariant derivative convention,

$$\boxed{\mathbb{O}_{1\text{def}}^{(0)} = i \tilde{\Gamma}_{\text{def}}^i (\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}}.}$$

Thus the reference operator is now extracted from an explicit minimal microscopic ansatz rather than from a purely abstract placeholder.

CLXXIX. CONNECTION TO THE BLOCH/CLASS II EXTRACTION PROGRAM

The same local-current ansatz also determines the Bloch-space mixed vertices used previously. Indeed, after harmonic expansion on the locked BCC branch,

$$\mathcal{J}_i^a(k) \sim A_i^a(Q) + B_{ij}^a(Q)(k_j + G'_j), \quad \mathcal{K}_i^a(k) \sim C_i^a(Q) + D_{ij}^a(Q)(k_j + G'_j),$$

so

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)$$

is precisely the Bloch-space realization of the same microscopic coefficients A, B, C, D .

Therefore the single-defect reference operator program and the refined locked + Class II Bloch extraction program are no longer separate constructions: they are two specializations of the same minimal microscopic coefficient template.

CLXXX. WHAT IS NOW RIGOROUSLY FIXED

At the level of the working microscopic ansatz, the following are now explicit:

1. the minimal microscopic multiplet

$$(P^A, \Psi, X_i^a);$$

2. the local-current realization of J_i^a and K_i^a ;

3. the exact coefficient fields

$$K^0(P, X), \quad K^i(P, X), \quad M(P, X, \partial P, \partial X);$$

4. the defect-background coefficients

$$K_{\text{def}}^0, \quad K_{\text{def}}^i, \quad M_{\text{def}};$$

5. the canonically normalized reference operator

$$\mathbb{O}_{1\text{def}}^{(0)} = i \tilde{\Gamma}_{\text{def}}^i (\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}}.$$

CLXXXI. WHAT STILL REMAINS OPEN

What remains TECT-specific and still open is not the coefficient-extraction skeleton, but the actual concrete choice of

$$Z^0(P), \quad Z^i(P), \quad M_0(P), \quad A_i^a(P), \quad B_{ij}^a(P), \quad C_i^a(P), \quad D_{ij}^a(P),$$

together with:

1. the explicit locked BCC defect background $\bar{P}, \bar{X}$;
2. the proof of asymptotic bulk invertibility;
3. the grading and index analysis;
4. the explicit overlap integrals for the reduced family coefficients.

So the operator-theoretic and coefficient-extraction framework is now fixed. What remains is to insert the actual TECT constitutive data into it.

CLXXXII. FINAL SUMMARY

We fixed a minimal working microscopic multiplet for the TECT flavor program:

$$\mathfrak{B} = (P^A, \Psi, X_i^a).$$

The corresponding minimal first-order Hermitian microscopic action is

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_P[P] + \frac{i}{2} \Psi^\dagger Z^0(P) \overset{\leftrightarrow}{\partial}_t \Psi + \frac{i}{2} \Psi^\dagger \left(Z^i(P) + X_j^a V_{ji}^a(P) \right) \overset{\leftrightarrow}{\partial}_i \Psi - \Psi^\dagger \left(M_0(P) - X_i^a U_i^a(P) \right) \Psi + \frac{M_X^2}{2} X_i^a X_i^a,$$

where

$$U_i^a = \alpha_X A_i^a + \beta_X C_i^a, \quad V_{ij}^a = \alpha_X B_{ij}^a + \beta_X D_{ij}^a,$$

and

$$J_i^a = \Psi^\dagger A_i^a \Psi + \frac{i}{2} \Psi^\dagger B_{ij}^a \overset{\leftrightarrow}{\partial}_j \Psi, \quad K_i^a = \Psi^\dagger C_i^a \Psi + \frac{i}{2} \Psi^\dagger D_{ij}^a \overset{\leftrightarrow}{\partial}_j \Psi.$$

From this action, the exact first-order coefficient fields are

$$K^0(P, X) = Z^0(P), \quad K^i(P, X) = Z^i(P) + X_j^a V_{ji}^a(P),$$

$$M(P, X, \partial P, \partial X) = \frac{i}{2} \partial_i K^i(P, X) - \left(M_0(P) - X_i^a U_i^a(P) \right).$$

Therefore, on a static defect background $(\bar{P}, \bar{X})$, the defect fluctuation equation is

$$iK_{\text{def}}^0 \partial_t \psi + iK_{\text{def}}^i \partial_i \psi + M_{\text{def}} \psi = 0,$$

with

$$K_{\text{def}}^0 = Z^0(\bar{P}), \quad K_{\text{def}}^i = Z^i(\bar{P}) + \bar{X}_j^a V_{ji}^a(\bar{P}),$$

$$M_{\text{def}} = \frac{i}{2} \partial_i K_{\text{def}}^i - \left(M_0(\bar{P}) - \bar{X}_i^a U_i^a(\bar{P}) \right).$$

After canonical normalization this yields the reference operator

$$\mathbb{O}_{\text{ldef}}^{(0)} = i\widetilde{\Gamma}_{\text{def}}^i (\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}}.$$

So the microscopic coefficient-extraction skeleton is now explicit. The remaining open input is purely TECT-specific: the actual constitutive matrices $Z^0, Z^i, M_0, A, B, C, D$, the explicit defect background, and the corresponding gap/index analysis.

CLXXXIII. MINIMAL CONSTITUTIVE CHOICE COMPATIBLE WITH THE LOCKED BCC BACKGROUND

We now fix a minimal working constitutive ansatz compatible with the locked BCC branch and with the previously derived operator-extraction formalism.

CLXXXIV. LOCKED GEOMETRIC DATA EXTRACTED FROM THE CONDENSATE FIELD

Let the condensate field $P^A(x)$ determine the following local geometric data:

1. a scalar amplitude

$$\sigma(x) := \sigma(P(x));$$

2. a unit defect-axis field

$$\hat{n}_i(x) := \hat{n}_i(P(x)), \quad \hat{n}_i \hat{n}_i = 1;$$

3. a local locked frame

$$E_i^a(x) := E_i^a(P(x)) \in SO(3), \quad E_i^a E_i^b = \delta^{ab}, \quad E_i^a E_j^a = \delta_{ij}.$$

The field E_i^a expresses the locking between spatial and internal directions, while $\hat{n}_i$ selects the distinguished defect axis.

The corresponding transverse projector is

$$\Pi_{ij}^\perp := \delta_{ij} - \hat{n}_i \hat{n}_j.$$

CLXXXV. INTERNAL CLIFFORD DATA

Let the fluctuation field Ψ take values in an internal bundle carrying Hermitian matrices

$$\Gamma^i, \quad i = 1, 2, 3,$$

satisfying the Euclidean Clifford relations

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij} \mathbf{1}.$$

Assume also the existence of a Hermitian matrix

$$\Gamma^\sharp = (\Gamma^\sharp)^\dagger, \quad (\Gamma^\sharp)^2 = \mathbf{1}, \quad \{\Gamma^\sharp, \Gamma^i\} = 0,$$

which will play the role of the mass/Higgs endomorphism.

Define the defect-axis gamma

$$\Gamma_{\hat{n}} := \hat{n}_i \Gamma^i.$$

CLXXXVI. MINIMAL REFERENCE CONSTITUTIVE CHOICE

We take the reference Ψ -sector coefficients to depend only on the scalar amplitude σ :

$$\boxed{Z^0(P) = z_0(\sigma) \mathbf{1}, \quad z_0(\sigma) > 0,}$$

$$\boxed{Z^i(P) = v_\perp(\sigma) \Gamma^i + v_c(\sigma) \hat{n}^i \Gamma_{\hat{n}},}$$

$$\boxed{M_0(P) = m(\sigma) \Gamma^\sharp.}$$

This is the minimal isotropic-plus-axis-anisotropic Dirac form. The coefficient $v_\perp$ controls the transverse transport, while v_c is the longitudinal defect-axis correction.

CLXXXVII. MINIMAL CLASS II CURRENT CONSTITUTIVE CHOICE

We now choose the Class II local-current coefficients in the simplest locked form.

a. *Mass-type current coupling.*

$$\boxed{A_i^a(P) = u(\sigma) E_i^a \Gamma^\sharp, \quad C_i^a(P) = \tilde{u}(\sigma) E_i^a \Gamma^\sharp.}$$

b. *Transport-type current coupling.* We take

$$\boxed{B_{ij}^a(P) = E_i^a \left[b_E(\sigma) \delta_{j\ell} + b_O(\sigma) \epsilon_{j\ell m} \hat{n}_m + b_c(\sigma) \hat{n}_j \hat{n}_\ell \right] \Gamma^\ell,}$$

$$\boxed{D_{ij}^a(P) = E_i^a \left[d_E(\sigma) \delta_{j\ell} + d_O(\sigma) \epsilon_{j\ell m} \hat{n}_m + d_c(\sigma) \hat{n}_j \hat{n}_\ell \right] \Gamma^\ell.}$$

All coefficient functions

$$z_0, v_\perp, v_c, m, u, \tilde{u}, b_E, b_O, b_c, d_E, d_O, d_c$$

are real scalar functions of σ .

This is the minimal constitutive ansatz that is simultaneously:

1. local;
2. Hermitian;
3. compatible with spatial/internal locking;
4. rich enough to generate the three transport channels later identified as (g_E, g_O, g_c) .

CLXXXVIII. UNIFIED CLASS II TRANSPORT AND MASS COEFFICIENTS

Recall

$$U_i^a = \alpha_X A_i^a + \beta_X C_i^a, \quad V_{ij}^a = \alpha_X B_{ij}^a + \beta_X D_{ij}^a.$$

Define the combined scalar functions

$$\boxed{u_X(\sigma) := \alpha_X u(\sigma) + \beta_X \tilde{u}(\sigma),}$$

$$\boxed{v_E^{(X)}(\sigma) := \alpha_X b_E(\sigma) + \beta_X d_E(\sigma),}$$

$$\boxed{v_O^{(X)}(\sigma) := \alpha_X b_O(\sigma) + \beta_X d_O(\sigma),}$$

$$\boxed{v_c^{(X)}(\sigma) := \alpha_X b_c(\sigma) + \beta_X d_c(\sigma).}$$

Then

$$\boxed{U_i^a(P) = u_X(\sigma) E_i^a \Gamma^\sharp,}$$

and

$$\boxed{V_{ij}^a(P) = E_i^a \left[v_E^{(X)}(\sigma) \delta_{j\ell} + v_O^{(X)}(\sigma) \epsilon_{j\ell m} \hat{n}_m + v_c^{(X)}(\sigma) \hat{n}_j \hat{n}_\ell \right] \Gamma^\ell.}$$

CLXXXIX. LOCKED CLASS II BACKGROUND CONTRACTION

Define the locked mediator scalar projection

$$\chi_X(x) := \frac{1}{3} \bar{X}_i^a(x) E_i^a(x),$$

on the static defect background $(\bar{P}, \bar{X})$.

If the background is strictly locked,

$$\bar{X}_i^a(x) = X_*(x) E_i^a(x),$$

then

$$\chi_X(x) = X_*(x).$$

Under this locked contraction,

$$\bar{X}_j^a V_{ji}^a = 3\chi_X \left[v_E^{(X)} \delta_{i\ell} + v_O^{(X)} \epsilon_{i\ell m} \hat{n}_m + v_c^{(X)} \hat{n}_i \hat{n}_\ell \right] \Gamma^\ell,$$

and

$$\bar{X}_i^a U_i^a = 3\chi_X u_X \Gamma^\sharp.$$

CXC. EXPLICIT DEFECT-BACKGROUND COEFFICIENTS

Therefore the defect-background first-order coefficients are

$$K_{\text{def}}^0(x) = z_0(\bar{\sigma}(x)) \mathbf{1},$$

$$K_{\text{def}}^i(x) = v_\perp(\bar{\sigma}) \Gamma^i + v_c(\bar{\sigma}) \hat{n}^i \Gamma_{\hat{n}} + 3\chi_X \left[v_E^{(X)}(\bar{\sigma}) \delta_{i\ell} + v_O^{(X)}(\bar{\sigma}) \epsilon_{i\ell m} \hat{n}_m + v_c^{(X)}(\bar{\sigma}) \hat{n}_i \hat{n}_\ell \right] \Gamma^\ell,$$

and

$$M_{\text{def}}(x) = \frac{i}{2} \partial_i K_{\text{def}}^i(x) - m_{\text{eff}}(x) \Gamma^\sharp,$$

with

$$m_{\text{eff}}(x) := m(\bar{\sigma}(x)) - 3\chi_X(x) u_X(\bar{\sigma}(x)).$$

Thus the mediator modifies:

1. the transverse even transport channel through $v_E^{(X)}$;
2. the transverse odd channel through $v_O^{(X)}$;
3. the longitudinal/c-axis channel through $v_c^{(X)}$;
4. the local mass gap through u_X .

CXCI. EFFECTIVE TRANSPORT COEFFICIENTS AND THEIR MICROSCOPIC MEANING

It is convenient to group the effective transport coefficients as

$$g_E^{\text{eff}}(x) := v_\perp(\bar{\sigma}(x)) + 3\chi_X(x) v_E^{(X)}(\bar{\sigma}(x)),$$

$$g_O^{\text{eff}}(x) := 3\chi_X(x) v_O^{(X)}(\bar{\sigma}(x)),$$

$$g_c^{\text{eff}}(x) := v_c(\bar{\sigma}(x)) + 3\chi_X(x) v_c^{(X)}(\bar{\sigma}(x)).$$

Then

$$K_{\text{def}}^i(x) = g_E^{\text{eff}}(x)\Gamma^i + g_O^{\text{eff}}(x)\epsilon_{ilm}\hat{n}_m\Gamma^\ell + g_c^{\text{eff}}(x)\hat{n}_i\Gamma_{\hat{n}}.$$

This is the exact minimal constitutive origin of the three reduced transport channels.

CXCII. CANONICAL NORMALIZED DEFECT OPERATOR

Since

$$K_{\text{def}}^0(x) = z_0(\bar{\sigma}(x))\mathbf{1} > 0,$$

define

$$S(x) = \sqrt{z_0(\bar{\sigma}(x))}\mathbf{1}, \quad \chi = S\psi.$$

Then

$$i\partial_t\chi = \mathbb{H}_{\text{def}}\chi,$$

with

$$\mathbb{H}_{\text{def}} = -i\Gamma_{\text{can}}^i(x)\partial_i + W_{\text{def}}(x),$$

where

$$\Gamma_{\text{can}}^i(x) = \frac{1}{z_0(\bar{\sigma}(x))} \left[g_E^{\text{eff}}(x)\Gamma^i + g_O^{\text{eff}}(x)\epsilon_{ilm}\hat{n}_m\Gamma^\ell + g_c^{\text{eff}}(x)\hat{n}_i\Gamma_{\hat{n}} \right],$$

and

$$W_{\text{def}}(x) = -\frac{i}{z_0(\bar{\sigma})}K_{\text{def}}^i\partial_i(z_0(\bar{\sigma})^{-1/2}) - \frac{1}{z_0(\bar{\sigma})}\left(\frac{i}{2}\partial_i K_{\text{def}}^i - m_{\text{eff}}\Gamma^\sharp\right).$$

Equivalently,

$$\mathbb{O}_{\text{1def}}^{(0)} = i\tilde{\Gamma}_{\text{def}}^i(\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}},$$

with

$$\tilde{\Gamma}_{\text{def}}^i = -\Gamma_{\text{can}}^i,$$

and $\Omega_i^{\text{def}}, \Phi_{\text{def}}$ defined by the decomposition of W_{def} into connection-type and endomorphism-type pieces.

CXCIII. INTERPRETATION

This minimal constitutive choice has three important consequences.

a. (1) The three reduced transport channels are now explicit. The coefficients later read as (g_E, g_O, g_c) are no longer abstract placeholders. They arise directly from the constitutive functions

$$v_\perp, v_c, v_E^{(X)}, v_O^{(X)}, v_c^{(X)}$$

and the locked mediator scalar χ_X .

b. (2) The same coefficients control both the single-defect reference operator and the Bloch/Class II extraction branch. The Fourier-space vertex coefficients $\mathbb{A}, \mathbb{B}, \mathbb{C}, \mathbb{D}$ introduced previously are exactly the harmonic expansions of

$$A_i^a, B_{ij}^a, C_i^a, D_{ij}^a$$

from the present constitutive ansatz.

c. (3) The remaining ambiguity is now only constitutive, not structural. The operator skeleton is fixed. What remains is to determine the actual TECT constitutive functions

$$z_0, v_\perp, v_c, m, u, \tilde{u}, b_E, b_O, b_c, d_E, d_O, d_c.$$

CXCIV. WHAT IS NOW CLOSED AND WHAT REMAINS OPEN

The following are now fixed within the working microscopic program:

1. the minimal microscopic multiplet (P^A, Ψ, X_i^a) ;
2. a locked BCC-compatible local constitutive ansatz;
3. the explicit formulas for

$$K_{\text{def}}^0, K_{\text{def}}^i, M_{\text{def}};$$

4. the explicit constitutive origin of the reduced transport channels

$$(g_E^{\text{eff}}, g_O^{\text{eff}}, g_c^{\text{eff}});$$

5. the corresponding canonical single-defect reference operator.

What remains open is TECT-specific:

1. the actual constitutive functions of the theory;
2. the explicit defect solution $(\bar{P}, \bar{X})$;
3. the asymptotic bulk gap condition;
4. the grading/index computation for the single-defect light mode;
5. the later overlap and projector integrals for the three-defect reduction.

CXCV. FINAL SUMMARY

We fixed a minimal locked-BCC-compatible constitutive ansatz for the TECT working microscopic program. The condensate field P determines:

$$\sigma(P), \quad \hat{n}_i(P), \quad E_i^a(P) \in SO(3).$$

The reference Ψ -sector is taken as

$$Z^0(P) = z_0(\sigma)\mathbf{1}, \quad Z^i(P) = v_\perp(\sigma)\Gamma^i + v_c(\sigma)\hat{n}^i\Gamma_{\hat{n}}, \quad M_0(P) = m(\sigma)\Gamma^\sharp.$$

The Class II couplings are chosen as

$$\begin{aligned} A_i^a &= u(\sigma) E_i^a \Gamma^\sharp, & C_i^a &= \tilde{u}(\sigma) E_i^a \Gamma^\sharp, \\ B_{ij}^a &= E_i^a \left[b_E(\sigma) \delta_{j\ell} + b_O(\sigma) \epsilon_{j\ell m} \hat{n}_m + b_c(\sigma) \hat{n}_j \hat{n}_\ell \right] \Gamma^\ell, \\ D_{ij}^a &= E_i^a \left[d_E(\sigma) \delta_{j\ell} + d_O(\sigma) \epsilon_{j\ell m} \hat{n}_m + d_c(\sigma) \hat{n}_j \hat{n}_\ell \right] \Gamma^\ell. \end{aligned}$$

On a locked defect background, defining

$$\chi_X = \frac{1}{3} \bar{X}_i^a E_i^a,$$

the defect coefficients become

$$K_{\text{def}}^0 = z_0(\bar{\sigma}) \mathbf{1},$$

$$K_{\text{def}}^i = g_E^{\text{eff}} \Gamma^i + g_O^{\text{eff}} \epsilon_{ilm} \hat{n}_m \Gamma^\ell + g_c^{\text{eff}} \hat{n}_i \Gamma_{\hat{n}},$$

with

$$g_E^{\text{eff}} = v_\perp + 3\chi_X v_E^{(X)}, \quad g_O^{\text{eff}} = 3\chi_X v_O^{(X)}, \quad g_c^{\text{eff}} = v_c + 3\chi_X v_c^{(X)},$$

and

$$M_{\text{def}} = \frac{i}{2} \partial_i K_{\text{def}}^i - m_{\text{eff}} \Gamma^\sharp, \quad m_{\text{eff}} = m - 3\chi_X u_X.$$

After canonical normalization, this yields the explicit single-defect reference operator

$$\mathbb{O}_{\text{def}}^{(0)} = i \tilde{\Gamma}_{\text{def}}^i (\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}}.$$

Therefore the operator skeleton is no longer abstract: the reduced transport channels (g_E, g_O, g_c) now have an explicit microscopic constitutive origin.

The remaining open input is purely TECT-specific: the actual constitutive functions, the explicit defect profile, the bulk gap, and the index analysis.

CXCVI. RADIAL SINGLE-DEFECT ANSATZ AND EXPLICIT DEFECT-FRAME FORM OF THE REFERENCE OPERATOR

We now specialize the minimal constitutive ansatz to a radial single-defect background and derive the explicit defect-frame form of the single-defect reference operator.

CXCVII. RADIAL SINGLE-DEFECT BACKGROUND ANSATZ

We assume a static radial defect centered at the origin. Let

$$r := |x|, \quad \hat{n}_i(x) := \frac{x_i}{r}.$$

We take the condensate amplitude to be radial:

$$\bar{\sigma}(x) = \bar{\sigma}(r).$$

For the locked frame, we choose a local orthonormal spatial/internal frame adapted to the defect:

$$E_i^a(x) \in SO(3), \quad E_i^a E_i^b = \delta^{ab}, \quad E_i^a E_j^a = \delta_{ij},$$

with the distinguished radial direction aligned,

$$E_i^3(x) = \hat{n}_i(x).$$

Equivalently, one may think of $E_i^a(x)$ as a local rotation sending the internal axis $a = 3$ to the spatial radial unit vector $\hat{n}_i$. The remaining two columns span the local tangent frame on S^2 .

CXCVIII. LOCKED MEDIATOR RADIAL DECOMPOSITION

The most general rotationally covariant locked mediator profile has the form

$$\bar{X}_i^a(x) = X_\perp(r)(\delta_i^a - \hat{n}_i \hat{n}^a) + X_\parallel(r)\hat{n}_i \hat{n}^a.$$

Equivalently, in terms of the locked frame,

$$\bar{X}_i^a(x) = X_\perp(r) \sum_{\alpha=1}^2 E_i^\alpha E_r^\alpha + X_\parallel(r) E_i^3 E_r^3,$$

where $E_r^3 = 1$ in the adapted frame. The scalar locked contraction defined previously becomes

$$\chi_X(r) = \frac{1}{3} \bar{X}_i^a E_i^a = \frac{1}{3} (2X_\perp(r) + X_\parallel(r)).$$

CXCIX. RADIAL SPECIALIZATION OF THE CONSTITUTIVE COEFFICIENTS

From the previous minimal constitutive choice, define

$$g_E^{\text{eff}}(r) := v_\perp(\bar{\sigma}(r)) + 3\chi_X(r) v_E^{(X)}(\bar{\sigma}(r)),$$

$$g_O^{\text{eff}}(r) := 3\chi_X(r) v_O^{(X)}(\bar{\sigma}(r)),$$

$$g_c^{\text{eff}}(r) := v_c(\bar{\sigma}(r)) + 3\chi_X(r) v_c^{(X)}(\bar{\sigma}(r)),$$

and

$$m_{\text{eff}}(r) := m(\bar{\sigma}(r)) - 3\chi_X(r) u_X(\bar{\sigma}(r)).$$

Thus the first-order coefficients are radial functions only:

$$K_{\text{def}}^0(r) = z_0(\bar{\sigma}(r)) \mathbf{1},$$

$$K_{\text{def}}^i(r) = g_E(r) \Gamma^i + g_O(r) \epsilon_{i\ell m} \hat{n}_m \Gamma^\ell + g_c(r) \hat{n}_i \Gamma_{\hat{n}},$$

where for brevity we write

$$g_E(r) := g_E^{\text{eff}}(r), \quad g_O(r) := g_O^{\text{eff}}(r), \quad g_c(r) := g_c^{\text{eff}}(r), \quad \Gamma_{\hat{n}} := \hat{n}_i \Gamma^i.$$

CC. RADIAL/TRANSVERSE DECOMPOSITION OF THE TRANSPORT TENSOR

Introduce the transverse projector

$$\Pi_{ij}^\perp := \delta_{ij} - \hat{n}_i \hat{n}_j,$$

and the transverse Clifford matrices

$$\Gamma_i^\perp := \Pi_{i\ell}^\perp \Gamma^\ell.$$

Since

$$\Gamma^i = \Gamma_i^\perp + \hat{n}_i \Gamma_{\hat{n}},$$

we obtain

$$K_{\text{def}}^i(r) = g_E(r) \Gamma_i^\perp + (g_E(r) + g_c(r)) \hat{n}_i \Gamma_{\hat{n}} + g_O(r) \epsilon_{i\ell m} \hat{n}_m \Gamma^\ell.$$

It is convenient to define the odd transverse operator

$$J_i(\hat{n}) := \epsilon_{ilm} \hat{n}_m \Gamma^\ell.$$

This satisfies

$$\hat{n}_i J_i(\hat{n}) = 0,$$

so it is purely tangential. Therefore the transport tensor is split into exactly three channels:

$$K_{\text{def}}^i(r) = g_E(r) \Gamma_i^\perp + g_R(r) \hat{n}_i \Gamma_{\hat{n}} + g_O(r) J_i(\hat{n}), \quad g_R(r) := g_E(r) + g_c(r).$$

This is the precise radial meaning of the three reduced transport coefficients.

CCI. EXPLICIT ZERO-ORDER TERM $M_{\text{def}}(r)$

Recall

$$M_{\text{def}} = \frac{i}{2} \partial_i K_{\text{def}}^i - m_{\text{eff}}(r) \Gamma^\sharp.$$

We now compute $\partial_i K_{\text{def}}^i$.

First,

$$\partial_i \hat{n}_j = \frac{1}{r} \Pi_{ij}^\perp, \quad \partial_i \hat{n}_i = \frac{2}{r}.$$

Next,

$$\partial_i \Gamma_{\hat{n}} = (\partial_i \hat{n}_j) \Gamma^j = \frac{1}{r} \Pi_{ij}^\perp \Gamma^j.$$

For the radial part,

$$\partial_i (g_R(r) \hat{n}_i \Gamma_{\hat{n}}) = g'_R(r) \Gamma_{\hat{n}} + \frac{2g_R(r)}{r} \Gamma_{\hat{n}} + \frac{g_R(r)}{r} \Pi_{ij}^\perp \hat{n}_i \Gamma^j.$$

Since $\Pi_{ij}^\perp \hat{n}_i = 0$, the last term vanishes, hence

$$\partial_i (g_R(r) \hat{n}_i \Gamma_{\hat{n}}) = \left(g'_R(r) + \frac{2g_R(r)}{r} \right) \Gamma_{\hat{n}}.$$

For the even tangential part,

$$\partial_i (g_E(r) \Gamma_i^\perp) = g'_E(r) \hat{n}_i \Gamma_i^\perp + g_E(r) \partial_i \Gamma_i^\perp.$$

Because $\hat{n}_i \Gamma_i^\perp = 0$, only the second term survives. Now

$$\Gamma_i^\perp = \Gamma^i - \hat{n}_i \Gamma_{\hat{n}},$$

so

$$\partial_i \Gamma_i^\perp = -\partial_i (\hat{n}_i \Gamma_{\hat{n}}) = -\frac{2}{r} \Gamma_{\hat{n}} - \frac{1}{r} \Pi_{ij}^\perp \hat{n}_i \Gamma^j = -\frac{2}{r} \Gamma_{\hat{n}}.$$

Therefore

$$\partial_i (g_E(r) \Gamma_i^\perp) = -\frac{2g_E(r)}{r} \Gamma_{\hat{n}}.$$

For the odd tangential part,

$$\partial_i(g_O(r)J_i(\hat{n})) = g'_O(r)\hat{n}_iJ_i(\hat{n}) + g_O(r)\partial_iJ_i(\hat{n}).$$

The first term vanishes because $\hat{n}_iJ_i(\hat{n}) = 0$. Moreover,

$$J_i(\hat{n}) = \epsilon_{ilm}\hat{n}_m\Gamma^\ell,$$

so

$$\partial_iJ_i(\hat{n}) = \epsilon_{ilm}(\partial_i\hat{n}_m)\Gamma^\ell = \frac{1}{r}\epsilon_{ilm}\Pi_{im}^\perp\Gamma^\ell = 0,$$

since $\Pi_{im}^\perp$ is symmetric in i, m while ϵ_{ilm} is antisymmetric. Hence

$$\partial_i(g_O(r)J_i(\hat{n})) = 0.$$

Combining everything, we obtain

$$\partial_iK_{\text{def}}^i(r) = \left(g'_R(r) + \frac{2g_R(r)}{r} - \frac{2g_E(r)}{r}\right)\Gamma_{\hat{n}} = \left(g'_R(r) + \frac{2g_c(r)}{r}\right)\Gamma_{\hat{n}}.$$

Therefore the zeroth-order term is

$$M_{\text{def}}(r) = \frac{i}{2}\left(g'_R(r) + \frac{2g_c(r)}{r}\right)\Gamma_{\hat{n}} - m_{\text{eff}}(r)\Gamma^\sharp.$$

This is already a strong structural result: the odd transport channel $g_O(r)$ does not contribute directly to $\partial_iK_{\text{def}}^i$, but only to the first-order angular transport operator.

CCII. CANONICAL NORMALIZATION

Since

$$K_{\text{def}}^0(r) = z_0(\bar{\sigma}(r))\mathbf{1} > 0,$$

define

$$S(r) = \sqrt{z_0(\bar{\sigma}(r))}\mathbf{1}, \quad \chi = S(r)\psi.$$

Then

$$i\partial_t\chi = \mathbb{H}_{\text{def}}\chi,$$

with

$$\mathbb{H}_{\text{def}} = -i\Gamma_{\text{can}}^i(r, \hat{n})\partial_i + W_{\text{def}}(r, \hat{n}),$$

where

$$\Gamma_{\text{can}}^i(r, \hat{n}) = \frac{1}{z_0(r)}\left[g_E(r)\Gamma_i^\perp + g_R(r)\hat{n}_i\Gamma_{\hat{n}} + g_O(r)J_i(\hat{n})\right].$$

The zeroth-order piece is

$$W_{\text{def}} = -\frac{i}{z_0(r)}K_{\text{def}}^i\partial_i(z_0(r)^{-1/2}) - \frac{1}{z_0(r)}M_{\text{def}}(r).$$

Since z_0 is radial,

$$\partial_i(z_0(r)^{-1/2}) = (z_0^{-1/2})'(r)\hat{n}_i,$$

hence only the radial transport part contributes:

$$K_{\text{def}}^i \partial_i (z_0^{-1/2}) = g_R(r) \Gamma_{\hat{n}} (z_0^{-1/2})'(r).$$

Therefore

$$W_{\text{def}}(r, \hat{n}) = -\frac{i g_R(r)}{z_0(r)} (z_0^{-1/2})'(r) \Gamma_{\hat{n}} - \frac{i}{2z_0(r)} \left(g_R'(r) + \frac{2g_c(r)}{r} \right) \Gamma_{\hat{n}} + \frac{m_{\text{eff}}(r)}{z_0(r)} \Gamma^{\sharp}.$$

So the canonically normalized single-defect operator is explicitly

$$\mathbb{H}_{\text{def}} = -\frac{i}{z_0(r)} \left[g_E(r) \Gamma_i^{\perp} + g_R(r) \hat{n}_i \Gamma_{\hat{n}} + g_O(r) J_i(\hat{n}) \right] \partial_i + W_{\text{def}}(r, \hat{n}).$$

CCIII. RADIAL/ANGULAR DECOMPOSITION

Using

$$\partial_i = \hat{n}_i \partial_r + \frac{1}{r} \nabla_i^{S^2},$$

one gets

$$\Gamma_i^{\perp} \partial_i = \frac{1}{r} \Gamma_i^{\perp} \nabla_i^{S^2}, \quad \hat{n}_i \Gamma_{\hat{n}} \partial_i = \Gamma_{\hat{n}} \partial_r,$$

$$J_i(\hat{n}) \partial_i = \frac{1}{r} J_i(\hat{n}) \nabla_i^{S^2}.$$

Hence

$$\mathbb{H}_{\text{def}} = -\frac{i g_R(r)}{z_0(r)} \Gamma_{\hat{n}} \partial_r - \frac{i g_E(r)}{z_0(r) r} \Gamma_i^{\perp} \nabla_i^{S^2} - \frac{i g_O(r)}{z_0(r) r} J_i(\hat{n}) \nabla_i^{S^2} + W_{\text{def}}(r, \hat{n}).$$

This is the final radial single-defect reference form. It cleanly separates:

1. radial transport through $g_R(r)$;
2. even tangential Dirac transport through $g_E(r)$;
3. odd tangential/torsion-like transport through $g_O(r)$;
4. mass/gap structure through $m_{\text{eff}}(r)$.

CCIV. INTERPRETATION AND WHAT IS NOW CLOSED

The explicit radial form shows the precise microscopic meaning of the three transport channels:

$$g_E(r) \text{ controls even tangential transport,}$$

$$g_O(r) \text{ controls odd tangential transport,}$$

$$g_c(r) \text{ enters only through } g_R(r) = g_E(r) + g_c(r) \text{ and through the radial divergence term.}$$

Moreover, the mass profile is

$$m_{\text{eff}}(r) = m(\bar{\sigma}(r)) - 3\chi_X(r)u_X(\bar{\sigma}(r)).$$

So the asymptotic bulk gap is controlled by

$$m_{\text{eff}}(\infty),$$

while the localized defect distortion enters through the radial profiles

$$\bar{\sigma}(r), \quad \chi_X(r).$$

What is now rigorously fixed is the operator form. What remains open is the TECT-specific input: the actual radial profiles, the asymptotic behavior, and the grading/index computation.

CCV. FINAL SUMMARY

Using the radial single-defect ansatz

$$\hat{n}_i = \frac{x_i}{r}, \quad \bar{\sigma} = \bar{\sigma}(r), \quad \bar{X}_i^a = X_\perp(r)(\delta_i^a - \hat{n}_i \hat{n}^a) + X_\parallel(r)\hat{n}_i \hat{n}^a,$$

we obtained the explicit defect-background coefficients

$$K_{\text{def}}^0(r) = z_0(\bar{\sigma}(r))\mathbf{1},$$

$$K_{\text{def}}^i(r) = g_E(r)\Gamma_i^\perp + g_R(r)\hat{n}_i\Gamma_{\hat{n}} + g_O(r)J_i(\hat{n}), \quad g_R(r) = g_E(r) + g_c(r),$$

and

$$M_{\text{def}}(r) = \frac{i}{2} \left(g'_R(r) + \frac{2g_c(r)}{r} \right) \Gamma_{\hat{n}} - m_{\text{eff}}(r)\Gamma^\sharp,$$

with

$$m_{\text{eff}}(r) = m(\bar{\sigma}(r)) - 3\chi_X(r)u_X(\bar{\sigma}(r)).$$

After canonical normalization, the single-defect reference operator becomes

$$\mathbb{H}_{\text{def}} = -\frac{i g_R(r)}{z_0(r)} \Gamma_{\hat{n}} \partial_r - \frac{i g_E(r)}{z_0(r)r} \Gamma_i^\perp \nabla_i^{S^2} - \frac{i g_O(r)}{z_0(r)r} J_i(\hat{n}) \nabla_i^{S^2} + W_{\text{def}}(r, \hat{n}),$$

where

$$W_{\text{def}}(r, \hat{n}) = -\frac{i g_R(r)}{z_0(r)} (z_0^{-1/2})'(r) \Gamma_{\hat{n}} - \frac{i}{2z_0(r)} \left(g'_R(r) + \frac{2g_c(r)}{r} \right) \Gamma_{\hat{n}} + \frac{m_{\text{eff}}(r)}{z_0(r)} \Gamma^\sharp.$$

Thus the single-defect operator is now in an explicit radial form. The three transport channels have clear microscopic meanings: g_E is even tangential transport, g_O is odd tangential transport, and g_c modifies the radial channel and radial divergence term.

What remains open is no longer the operator form, but the TECT-specific constitutive profiles

$$\bar{\sigma}(r), \quad X_\parallel(r), \quad X_\perp(r),$$

their asymptotic gap behavior, and the eventual grading/index computation.

CCVI. PARTIAL-WAVE DECOMPOSITION OF THE RADIAL SINGLE-DEFECT OPERATOR

We now reduce the explicit radial single-defect reference operator to a one-dimensional first-order radial system.

CCVII. RADIAL FORM OF THE CANONICALLY NORMALIZED OPERATOR

From the previous step, the canonically normalized single-defect operator has the form

$$\mathbb{H}_{\text{def}} = -\frac{i g_R(r)}{z_0(r)} \Gamma_{\hat{n}} \partial_r - \frac{i g_E(r)}{z_0(r)r} \Gamma_i^\perp \nabla_i^{S^2} - \frac{i g_O(r)}{z_0(r)r} J_i(\hat{n}) \nabla_i^{S^2} + W_{\text{def}}(r, \hat{n}),$$

with

$$J_i(\hat{n}) := \epsilon_{i\ell m} \hat{n}_m \Gamma^\ell, \quad \Gamma_{\hat{n}} := \hat{n}_i \Gamma^i.$$

It is convenient to define the radial coefficients

$$a(r) := \frac{g_R(r)}{z_0(r)}, \quad b(r) := \frac{g_E(r)}{z_0(r)}, \quad c(r) := \frac{g_O(r)}{z_0(r)}, \quad \mu(r) := \frac{m_{\text{eff}}(r)}{z_0(r)}.$$

Also define the radial connection-like coefficient

$$\omega_r(r) := \frac{1}{2z_0(r)} \left(g'_R(r) + \frac{2g_c(r)}{r} - g_R(r) \frac{z'_0(r)}{z_0(r)} \right).$$

Then the zeroth-order term can be written as

$$W_{\text{def}}(r, \hat{n}) = -i \omega_r(r) \Gamma_{\hat{n}} + \mu(r) \Gamma^\sharp.$$

Hence the operator becomes

$$\mathbb{H}_{\text{def}} = -i a(r) \Gamma_{\hat{n}} \partial_r - \frac{i b(r)}{r} \Gamma_i^\perp \nabla_i^{S^2} - \frac{i c(r)}{r} J_i(\hat{n}) \nabla_i^{S^2} - i \omega_r(r) \Gamma_{\hat{n}} + \mu(r) \Gamma^\sharp.$$

CCVIII. ANGULAR DIRAC OPERATOR ON THE SPHERE

Define the spherical Dirac operator

$$\mathcal{D}_{S^2} := -i \Gamma_{\hat{n}} \Gamma_i^\perp \nabla_i^{S^2}.$$

Then one has the identities

$$\Gamma_i^\perp \nabla_i^{S^2} = i \Gamma_{\hat{n}} \mathcal{D}_{S^2},$$

and, using

$$J_i(\hat{n}) = i \Gamma_{\hat{n}} \Gamma_i^\perp,$$

also

$$J_i(\hat{n}) \nabla_i^{S^2} = -\mathcal{D}_{S^2}.$$

Therefore the radial operator takes the compact form

$$\mathbb{H}_{\text{def}} = -i a(r) \Gamma_{\hat{n}} \partial_r + \frac{b(r)}{r} \Gamma_{\hat{n}} \mathcal{D}_{S^2} + \frac{i c(r)}{r} \mathcal{D}_{S^2} - i \omega_r(r) \Gamma_{\hat{n}} + \mu(r) \Gamma^\sharp.$$

CCIX. ANGULAR EIGENBASIS

We now assume the standard spin-covariant spectral decomposition on S^2 . Let

$$Y_{\kappa m \varsigma}(\Omega)$$

be a simultaneous eigenbasis satisfying

$$\mathcal{D}_{S^2} Y_{\kappa m \varsigma} = \kappa Y_{\kappa m \varsigma},$$

$$\Gamma^\sharp Y_{\kappa m \varsigma} = \varsigma Y_{\kappa m \varsigma}, \quad \varsigma = \pm 1.$$

Because $\Gamma^\sharp$ anticommutes with each Γ^i , it commutes with $\mathcal{D}_{S^2}$, so the simultaneous diagonalization is consistent. Also, since

$$\{\Gamma_{\hat{n}}, \mathcal{D}_{S^2}\} = 0, \quad \{\Gamma_{\hat{n}}, \Gamma^\sharp\} = 0,$$

one has

$$\mathcal{D}_{S^2}(\Gamma_{\hat{n}} Y_{\kappa m \varsigma}) = -\kappa \Gamma_{\hat{n}} Y_{\kappa m \varsigma},$$

$$\Gamma^\sharp(\Gamma_{\hat{n}} Y_{\kappa m \varsigma}) = -\varsigma \Gamma_{\hat{n}} Y_{\kappa m \varsigma}.$$

Thus each angular sector closes on the two-dimensional span

$$\text{span}\{Y_{\kappa m \varsigma}, \Gamma_{\hat{n}} Y_{\kappa m \varsigma}\}.$$

CCX. PARTIAL-WAVE ANSATZ

We therefore expand a stationary mode of energy ω as

$$\chi(r, \Omega) = \sum_{\kappa, m, \varsigma} \left[f_{\kappa m \varsigma}(r) Y_{\kappa m \varsigma}(\Omega) + g_{\kappa m \varsigma}(r) \Gamma_{\hat{n}} Y_{\kappa m \varsigma}(\Omega) \right].$$

Because the background is radial, the different (κ, m, ς) channels decouple. So it is enough to study one fixed channel and suppress the indices:

$$\chi(r, \Omega) = f(r) Y(\Omega) + g(r) \Gamma_{\hat{n}} Y(\Omega),$$

with

$$\mathcal{D}_{S^2} Y = \kappa Y, \quad \Gamma^\sharp Y = \varsigma Y.$$

CCXI. RADIAL FIRST-ORDER SYSTEM

Applying $\mathbb{H}_{\text{def}} \chi = \omega \chi$ and projecting onto the basis

$$\{Y, \Gamma_{\hat{n}} Y\},$$

one obtains the exact radial system

$$\omega f = -i a(r) g'(r) - i \omega_r(r) g(r) - \frac{b(r)\kappa}{r} g(r) + i \frac{c(r)\kappa}{r} f(r) + \varsigma \mu(r) f(r),$$

$$\omega g = -i a(r) f'(r) - i \omega_r(r) f(r) + \frac{b(r)\kappa}{r} f(r) - i \frac{c(r)\kappa}{r} g(r) - \varsigma \mu(r) g(r).$$

This is the desired one-dimensional first-order system. All single-defect bound-state and zero-mode questions are now reduced to this radial ODE problem.

CCXII. EQUIVALENT REAL FORM FOR THE ZERO-MODE SECTOR

For $\omega = 0$, define

$$g(r) = i v(r), \quad f(r) = u(r).$$

Then the system becomes

$$a(r) v'(r) + \omega_r(r) v(r) - \frac{b(r)\kappa}{r} i v(r) + \frac{c(r)\kappa}{r} i u(r) + \varsigma \mu(r) u(r) = 0,$$

$$a(r) u'(r) + \omega_r(r) u(r) + \frac{b(r)\kappa}{r} u(r) + \frac{c(r)\kappa}{r} v(r) + \varsigma \mu(r) v(r) = 0,$$

or, keeping the original complex form, one may simply work with the previous pair. The key point is that the zero-mode problem is a closed radial system with no remaining angular variables.

CCXIII. GAUGE-REMOVAL OF THE RADIAL CONNECTION TERM

The term $\omega_r(r)$ has the same matrix structure as the radial derivative term. Indeed,

$$-i a(r) \Gamma_{\hat{n}} \partial_r - i \omega_r(r) \Gamma_{\hat{n}} = -i a(r) \Gamma_{\hat{n}} \left(\partial_r + \frac{\omega_r(r)}{a(r)} \right).$$

Therefore, introducing the scalar integrating factor

$$\mathcal{I}(r) := \exp \left(\int^r \frac{\omega_r(s)}{a(s)} ds \right),$$

and redefining

$$f(r) = \mathcal{I}(r)^{-1} \tilde{f}(r), \quad g(r) = \mathcal{I}(r)^{-1} \tilde{g}(r),$$

one removes the ω_r -term completely. The reduced radial system becomes

$$\omega \tilde{f} = -i a(r) \tilde{g}' - \frac{b(r)\kappa}{r} \tilde{g} + i \frac{c(r)\kappa}{r} \tilde{f} + \varsigma \mu(r) \tilde{f},$$

$$\omega \tilde{g} = -i a(r) \tilde{f}' + \frac{b(r)\kappa}{r} \tilde{f} - i \frac{c(r)\kappa}{r} \tilde{g} - \varsigma \mu(r) \tilde{g}.$$

So the genuinely dynamical coefficients are only

$$a(r), \quad b(r), \quad c(r), \quad \mu(r).$$

CCXIV. ASYMPTOTIC LARGE- r CRITERION

Assume the profiles approach bulk limits

$$a(r) \rightarrow a_\infty > 0, \quad b(r) \rightarrow b_\infty, \quad c(r) \rightarrow c_\infty, \quad \mu(r) \rightarrow \mu_\infty, \quad |\mu_\infty| > 0.$$

Then the $1/r$ angular terms vanish as $r \rightarrow \infty$, and the zero-mode equations asymptotically reduce to

$$-i a_\infty \tilde{g}'(r) + \varsigma \mu_\infty \tilde{f}(r) \approx 0,$$

$$-i a_\infty \tilde{f}'(r) - \varsigma \mu_\infty \tilde{g}(r) \approx 0.$$

Equivalently, after one more constant phase redefinition, this yields the second-order asymptotic equation

$$\tilde{f}''(r) \approx \frac{\mu_\infty^2}{a_\infty^2} \tilde{f}(r), \quad \tilde{g}''(r) \approx \frac{\mu_\infty^2}{a_\infty^2} \tilde{g}(r).$$

Hence the normalizable asymptotic branch behaves as

$$\boxed{\tilde{f}(r), \tilde{g}(r) \sim \exp\left(-\frac{|\mu_\infty|}{a_\infty} r\right)}.$$

This gives the explicit radial origin of exponential localization: the bulk gap is controlled by $|\mu_\infty|$, while the effective radial velocity is a_∞ .

CCXV. NEAR-CORE BEHAVIOR

Near $r = 0$, the $1/r$ angular terms dominate. The regularity condition is therefore controlled by the pair

$$\frac{\kappa b(r)}{r}, \quad \frac{\kappa c(r)}{r}.$$

If the profiles are smooth at the origin and

$$a(0), b(0), c(0), \mu(0)$$

are finite, then Frobenius analysis gives power-law behavior

$$f(r), g(r) \sim r^\nu$$

with exponents determined by the angular matrix

$$\kappa \begin{pmatrix} i c(0) & -b(0) \\ +b(0) & -i c(0) \end{pmatrix}.$$

Thus the actual existence of a regular zero mode is reduced to matching:

1. a regular near-core branch,
2. a decaying large- r branch.

This is exactly the standard defect-bound-state shooting problem.

CCXVI. WHAT IS NOW RIGOROUSLY CLOSED

The following are now fully fixed at the operator level:

1. the radial single-defect operator admits a clean partial-wave decomposition;
2. each angular channel reduces to a 2×2 first-order radial system;
3. the radial connection term can be gauged away by a scalar integrating factor;
4. the normalizable large- r behavior is governed by the ratio $|\mu_\infty|/a_\infty$;
5. the existence of a single-defect light mode is now a concrete ODE boundary-value problem.

CCXVII. WHAT REMAINS OPEN

What remains TECT-specific is:

1. the actual radial profiles

$$\bar{\sigma}(r), \quad X_{\parallel}(r), \quad X_{\perp}(r);$$

2. the actual constitutive functions entering

$$a(r), \quad b(r), \quad c(r), \quad \mu(r);$$

3. the grading/index analysis that identifies which (κ, ς) channel carries the protected mode;
4. the proof that exactly one regular decaying zero mode exists.

So the single-defect one-mode problem has now been reduced from a PDE to a concrete radial ODE classification problem.

CCXVIII. FINAL SUMMARY

The radial single-defect reference operator admits a full partial-wave decomposition.

Defining

$$a(r) = \frac{g_R(r)}{z_0(r)}, \quad b(r) = \frac{g_E(r)}{z_0(r)}, \quad c(r) = \frac{g_O(r)}{z_0(r)}, \quad \mu(r) = \frac{m_{\text{eff}}(r)}{z_0(r)},$$

the canonically normalized operator can be written as

$$\mathbb{H}_{\text{def}} = -i a(r) \Gamma_{\hat{n}} \partial_r + \frac{b(r)}{r} \Gamma_{\hat{n}} \mathcal{D}_{S^2} + \frac{i c(r)}{r} \mathcal{D}_{S^2} - i \omega_r(r) \Gamma_{\hat{n}} + \mu(r) \Gamma^{\sharp}.$$

For each angular channel

$$\mathcal{D}_{S^2} Y_{\kappa m \varsigma} = \kappa Y_{\kappa m \varsigma}, \quad \Gamma^{\sharp} Y_{\kappa m \varsigma} = \varsigma Y_{\kappa m \varsigma},$$

the mode ansatz

$$\chi(r, \Omega) = f(r) Y_{\kappa m \varsigma}(\Omega) + g(r) \Gamma_{\hat{n}} Y_{\kappa m \varsigma}(\Omega)$$

reduces the defect problem to a 2×2 radial first-order system:

$$\omega f = -i a(r) g' - i \omega_r(r) g - \frac{b(r) \kappa}{r} g + i \frac{c(r) \kappa}{r} f + \varsigma \mu(r) f,$$

$$\omega g = -i a(r) f' - i \omega_r(r) f + \frac{b(r) \kappa}{r} f - i \frac{c(r) \kappa}{r} g - \varsigma \mu(r) g.$$

After removing the radial connection term by an integrating factor, the dynamical coefficients are only

$$a(r), \quad b(r), \quad c(r), \quad \mu(r).$$

As $r \rightarrow \infty$, if

$$a(r) \rightarrow a_{\infty} > 0, \quad \mu(r) \rightarrow \mu_{\infty} \neq 0,$$

then the normalizable branch decays like

$$\exp\left(-\frac{|\mu_{\infty}|}{a_{\infty}} r\right).$$

Therefore the single-defect light-mode problem is now reduced to a concrete radial boundary-value problem: one must determine whether there exists a regular near-core solution that matches onto the decaying large- r branch in exactly one protected channel.

CCXIX. GRADING STRUCTURE AND CLASSIFICATION OF THE INDEX-CARRYING RADIAL CHANNEL

We now connect the radial single-defect ODE to the Callias/graded structure and classify which partial-wave channels can carry protected zero modes.

CCXX. CALLIAS CORE VERSUS THE ODD TANGENTIAL PERTURBATION

From the previous step, the canonically normalized radial operator is

$$\mathbb{H}_{\text{def}} = -i a(r) \Gamma_{\hat{n}} \partial_r + \frac{b(r)}{r} \Gamma_{\hat{n}} \mathcal{D}_{S^2} + \frac{i c(r)}{r} \mathcal{D}_{S^2} - i \omega_r(r) \Gamma_{\hat{n}} + \mu(r) \Gamma^{\sharp}.$$

We separate it into

$$\boxed{\mathbb{H}_{\text{def}} = \mathbb{H}_C + \mathbb{V}_O},$$

where the Callias core is

$$\boxed{\mathbb{H}_C := -i a(r) \Gamma_{\hat{n}} \partial_r + \frac{b(r)}{r} \Gamma_{\hat{n}} \mathcal{D}_{S^2} - i \omega_r(r) \Gamma_{\hat{n}} + \mu(r) \Gamma^{\sharp}},$$

and the odd tangential term is

$$\boxed{\mathbb{V}_O := \frac{i c(r)}{r} \mathcal{D}_{S^2}}.$$

CCXXI. GRADING OPERATOR

Assume there exists a Hermitian involution

$$\mathcal{C} = \mathcal{C}^{\dagger}, \quad \mathcal{C}^2 = \mathbf{1},$$

such that

$$\boxed{\{\mathcal{C}, \Gamma_{\hat{n}}\} = 0, \quad \{\mathcal{C}, \Gamma^{\sharp}\} = 0, \quad [\mathcal{C}, \mathcal{D}_{S^2}] = 0.}$$

Then

$$\{\mathcal{C}, \mathbb{H}_C\} = 0,$$

so $\mathbb{H}_C$ is the genuine graded Callias operator. By contrast,

$$[\mathcal{C}, \mathbb{V}_O] = 0,$$

so the $c(r)$ -term breaks the exact grading of the full operator.

This yields the correct conceptual split:

$$\boxed{\text{the index is defined by the Callias core } \mathbb{H}_C},$$

while

$$\boxed{\mathbb{V}_O \text{ is a lower-order grading-breaking perturbation.}}$$

CCXXII. ANGULAR BASIS ADAPTED TO THE GRADING

Because

$$[\mathcal{C}, \mathcal{D}_{S^2}] = 0,$$

we may choose angular modes

$$Y_{\kappa m s}(\Omega)$$

satisfying

$$\boxed{\mathcal{D}_{S^2} Y_{\kappa m s} = \kappa Y_{\kappa m s}, \quad \mathcal{C} Y_{\kappa m s} = s Y_{\kappa m s}, \quad s = \pm 1.}$$

Since

$$\{\Gamma_{\hat{n}}, \mathcal{C}\} = 0, \quad \{\Gamma_{\hat{n}}, \mathcal{D}_{S^2}\} = 0,$$

one has

$$\mathcal{D}_{S^2}(\Gamma_{\hat{n}} Y_{\kappa m s}) = -\kappa \Gamma_{\hat{n}} Y_{\kappa m s}, \quad \mathcal{C}(\Gamma_{\hat{n}} Y_{\kappa m s}) = -s \Gamma_{\hat{n}} Y_{\kappa m s}.$$

Thus a fixed radial channel is naturally spanned by

$$\boxed{\text{span}\{Y_{\kappa m s}, \Gamma_{\hat{n}} Y_{\kappa m s}\}.$$

CCXXIII. GAUGE REMOVAL AND CHIRAL BLOCK FORM

As already shown, the radial connection term can be removed by the scalar integrating factor

$$\mathcal{I}(r) = \exp\left(\int^r \frac{\omega_r(s)}{a(s)} ds\right).$$

After this gauge removal and the standard unitary radial measure transformation

$$(f, g) \mapsto r(f, g),$$

the Callias core in a fixed (κ, m) -channel takes the chiral block form

$$\boxed{\mathbb{H}_{C, \kappa} = \begin{pmatrix} 0 & Q_{\kappa, -}^{\text{core}} \\ Q_{\kappa, +}^{\text{core}} & 0 \end{pmatrix},}$$

with first-order scalar operators

$$\boxed{Q_{\kappa, \pm}^{\text{core}} = \pm a(r) \frac{d}{dr} + W_{\kappa}^{\text{core}}(r).}$$

In the minimal radial constitutive ansatz, the real superpotential is

$$\boxed{W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r},}$$

up to the sign convention that labels the channel κ .

CCXXIV. KERNEL DIMENSION AND EXPLICIT ZERO-MODE GENERATORS

Since $Q_{\kappa,\pm}^{\text{core}}$ are first-order scalar ODE operators, their kernels are at most one-dimensional:

$$\dim \ker Q_{\kappa,\pm}^{\text{core}} \leq 1.$$

Indeed, the formal zero-mode generators are

$$u_{\kappa,+}(r) = \exp\left(-\int^r \frac{W_{\kappa}^{\text{core}}(s)}{a(s)} ds\right),$$

$$u_{\kappa,-}(r) = \exp\left(+\int^r \frac{W_{\kappa}^{\text{core}}(s)}{a(s)} ds\right).$$

Thus the channel index is

$$\text{ind}_{\kappa} := \dim \ker Q_{\kappa,+}^{\text{core}} - \dim \ker Q_{\kappa,-}^{\text{core}} \in \{-1, 0, 1\}.$$

The contribution of a given channel is therefore completely determined by the L^2 -normalizability of these two explicit generators.

CCXXV. EFFECT OF THE g_O CHANNEL

Now include the grading-breaking term

$$\mathbb{V}_O = \frac{i c(r)}{r} \mathcal{D}_{S^2}.$$

On a fixed κ -channel this becomes

$$\mathbb{V}_{O,\kappa} = i \frac{\kappa c(r)}{r} \mathbf{1}.$$

Hence the full chiral radial operators are

$$Q_{\kappa,\pm}^{\text{full}} = \pm a(r) \frac{d}{dr} + W_{\kappa}^{\text{core}}(r) + i \frac{\kappa c(r)}{r}.$$

This is a decisive observation: the g_O -term enters as a purely imaginary scalar addition to the superpotential. Therefore the formal zero-mode generators become

$$u_{\kappa,\pm}^{\text{full}}(r) = \exp\left(\mp \int^r \frac{W_{\kappa}^{\text{core}}(s)}{a(s)} ds\right) \cdot \exp\left(\mp i \kappa \int^r \frac{c(s)}{a(s)s} ds\right).$$

The second factor is a pure phase. Hence

$$\text{the } g_O\text{-channel changes the phase of the radial zero-mode generator,}$$

but

$$\text{it does not change its } L^2\text{-normalizability.}$$

So the existence or nonexistence of a normalizable protected channel is controlled entirely by the real Callias core superpotential W_{κ}^{core} .

CCXXVI. CHANNEL-SELECTION CRITERION

We may now state the exact channel criterion.

a. Proposition (index-carrying channel criterion). Fix a channel (κ, m) . Then the corresponding protected zero-mode contribution is nonzero if and only if exactly one of the two generators

$$u_{\kappa,+}^{\text{core}}, \quad u_{\kappa,-}^{\text{core}}$$

belongs to the radial L^2 -space. Equivalently,

$$\text{ind}_\kappa \neq 0 \iff \text{exactly one of } Q_{\kappa,+}^{\text{core}}, Q_{\kappa,-}^{\text{core}} \text{ has a normalizable kernel.}$$

b. Proof. Because the kernels are at most one-dimensional and the full g_O -term only adds a phase, the only possibilities are:

$$(\dim \ker Q_{\kappa,+}, \dim \ker Q_{\kappa,-}) \in \{(0, 0), (1, 0), (0, 1)\}.$$

Hence

$$\text{ind}_\kappa \in \{0, \pm 1\},$$

and the index is nonzero exactly when one and only one chiral generator is L^2 -normalizable. $\square$

CCXXVII. LOWEST-CHANNEL DOMINANCE AND THE ONE-PROTECTED-CHANNEL CRITERION

In most radial Dirac/Callias problems, the angular spectrum obeys

$$|\kappa| \geq \kappa_{\min} > 0.$$

Typically the strongest candidate for a protected mode is the lowest channel

$$|\kappa| = \kappa_{\min},$$

because the centrifugal contribution

$$\frac{\kappa b(r)}{r}$$

is smallest there.

This leads to the following conditional theorem schema.

a. Theorem (conditional one-protected-channel criterion). Assume:

1. for every channel with $|\kappa| > \kappa_{\min}$, neither $u_{\kappa,+}^{\text{core}}$ nor $u_{\kappa,-}^{\text{core}}$ is L^2 -normalizable;
2. for exactly one lowest channel $\kappa = \kappa_*$ (after accounting for any residual symmetry/projector reduction), exactly one of

$$u_{\kappa_*,+}^{\text{core}}, \quad u_{\kappa_*, -}^{\text{core}}$$

is L^2 -normalizable.

Then

$$\boxed{\text{Ind}(\mathbb{H}_C) = \pm 1,}$$

and the single defect carries exactly one protected radial channel.

If, moreover, the relevant lowest-channel multiplicity is one, then

$$\boxed{\text{one defect} \leftrightarrow \text{one protected light mode.}}$$

b. Comment. The point here is complete honesty: the reduction to channel counting is now exact, but the actual verification of the hypotheses depends on the concrete TECT profiles $a(r), b(r), \mu(r)$ and on the angular multiplicity structure of the chosen bundle.

CCXXVIII. ASYMPTOTIC DIAGNOSTIC FOR THE CHANNEL INDEX

Since

$$u_{\kappa, \pm}^{\text{core}}(r) = \exp\left(\mp \int^r \frac{W_{\kappa}^{\text{core}}(s)}{a(s)} ds\right), \quad W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r},$$

the channel selection is controlled by the asymptotics of

$$\frac{W_{\kappa}^{\text{core}}(r)}{a(r)}.$$

At infinity, if

$$a(r) \rightarrow a_{\infty} > 0, \quad \mu(r) \rightarrow \mu_{\infty} \neq 0,$$

then

$$\frac{W_{\kappa}^{\text{core}}(r)}{a(r)} \rightarrow \frac{\mu_{\infty}}{a_{\infty}},$$

so one of the two chiral generators decays and the other grows exponentially.

Near the core, the sign and magnitude of

$$\frac{\kappa b(r)}{a(r)r}$$

control regularity. Thus the protected-channel problem is precisely a matching problem between:

1. the near-core regular branch selected by the centrifugal/superpotential term;
2. the large- r decaying branch selected by μ_{∞}/a_{∞} .

This is the exact radial meaning of the index-carrying channel criterion.

CCXXIX. WHAT IS NOW RIGOROUSLY CLOSED

The following points are now fixed:

1. the exact graded Callias core is obtained by dropping the g_O -term from the full radial operator;
2. each fixed κ -channel reduces to a one-dimensional chiral supercharge pair

$$Q_{\kappa, \pm}^{\text{core}} = \pm a(r) \partial_r + W_{\kappa}^{\text{core}}(r);$$

3. the channel index is nonzero iff exactly one chiral generator is L^2 -normalizable;
4. the g_O -term contributes only a phase and therefore does not change the L^2 -normalizability classification;
5. the one-defect/one-channel statement reduces to a lowest-channel uniqueness question.

CCXXX. WHAT REMAINS OPEN

What remains TECT-specific is:

1. the explicit grading operator $\mathcal{C}$;
2. the actual functions $a(r), b(r), c(r), \mu(r)$;
3. the actual angular spectrum and its multiplicities;
4. the proof that exactly one lowest channel has a normalizable chiral generator.

So the problem has now been reduced to its final concrete form: determine the real superpotential

$$W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}$$

for the actual TECT defect profiles and classify which channels produce L^2 -normalizable zero-mode generators.

CCXXXI. FINAL SUMMARY

We introduced a grading operator $\mathcal{C}$ satisfying

$$\{\mathcal{C}, \Gamma_{\hat{n}}\} = 0, \quad \{\mathcal{C}, \Gamma^{\sharp}\} = 0, \quad [\mathcal{C}, \mathcal{D}_{S^2}] = 0.$$

With this grading, the radial single-defect operator splits into:

1. the genuine graded Callias core

$$\mathbb{H}_C = -i a(r) \Gamma_{\hat{n}} \partial_r + \frac{b(r)}{r} \Gamma_{\hat{n}} \mathcal{D}_{S^2} - i \omega_r(r) \Gamma_{\hat{n}} + \mu(r) \Gamma^{\sharp},$$

2. the grading-breaking odd tangential perturbation

$$\mathbb{V}_O = \frac{i c(r)}{r} \mathcal{D}_{S^2}.$$

After gauge removal and radial reduction, each fixed κ -channel of the Callias core takes the chiral block form

$$\mathbb{H}_{C,\kappa} = \begin{pmatrix} 0 & Q_{\kappa,-}^{\text{core}} \\ Q_{\kappa,+}^{\text{core}} & 0 \end{pmatrix}, \quad Q_{\kappa,\pm}^{\text{core}} = \pm a(r) \frac{d}{dr} + W_{\kappa}^{\text{core}}(r),$$

with

$$W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.$$

Hence the channel index is

$$\text{ind}_{\kappa} = \dim \ker Q_{\kappa,+}^{\text{core}} - \dim \ker Q_{\kappa,-}^{\text{core}},$$

and it is nonzero iff exactly one of the two chiral generators is L^2 -normalizable.

The g_O -term only adds the purely imaginary shift

$$i \frac{\kappa c(r)}{r}$$

to the superpotential, so it changes only the phase of the zero-mode generator and not its normalizability.

Therefore the protected-channel classification is controlled entirely by the real superpotential

$$W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.$$

So the single-defect one-mode problem has now been reduced to its final concrete form: determine which κ -channel has exactly one L^2 -normalizable chiral generator. If exactly one lowest channel does so, then one defect carries one protected light mode.

CCXXXII. A SUFFICIENT CRITERION FOR A UNIQUE PROTECTED LOWEST- κ CHANNEL

We now turn the single-defect light-mode problem into an explicit sufficient criterion on the real Callias-core superpotential

$$W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.$$

CCXXXIII. CALLIAS-CORE CHANNEL GENERATORS

After gauge removal and radial measure normalization, the graded core operator in a fixed partial-wave channel κ is

$$\mathbb{H}_{C,\kappa} = \begin{pmatrix} 0 & Q_{\kappa,-}^{\text{core}} \\ Q_{\kappa,+}^{\text{core}} & 0 \end{pmatrix}, \quad Q_{\kappa,\pm}^{\text{core}} = \pm a(r) \frac{d}{dr} + W_{\kappa}^{\text{core}}(r),$$

with

$$a(r) > 0, \quad W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.$$

The formal zero-mode generators are

$$u_{\kappa,+}(r) = \exp\left(-\int^r \frac{W_{\kappa}^{\text{core}}(s)}{a(s)} ds\right), \quad u_{\kappa,-}(r) = \exp\left(+\int^r \frac{W_{\kappa}^{\text{core}}(s)}{a(s)} ds\right).$$

Since each $Q_{\kappa,\pm}^{\text{core}}$ is first-order scalar, its kernel is at most one-dimensional. Hence

$$\text{ind}_{\kappa} = \dim \ker Q_{\kappa,+}^{\text{core}} - \dim \ker Q_{\kappa,-}^{\text{core}} \in \{-1, 0, 1\},$$

and it is nonzero iff exactly one of $u_{\kappa,+}, u_{\kappa,-}$ is L^2 -normalizable.

CCXXXIV. ASYMPTOTIC LARGE- r DECAY CRITERION

Assume

$$a(r) \rightarrow a_{\infty} > 0, \quad \mu(r) \rightarrow \mu_{\infty} > 0, \quad b(r) = O(1) \quad (r \rightarrow \infty).$$

Then

$$W_{\kappa}^{\text{core}}(r) \rightarrow \mu_{\infty} > 0,$$

so

$$u_{\kappa,+}(r) \sim \exp\left(-\frac{\mu_{\infty}}{a_{\infty}} r\right), \quad u_{\kappa,-}(r) \sim \exp\left(+\frac{\mu_{\infty}}{a_{\infty}} r\right).$$

Therefore:

$u_{\kappa,+}$ is the unique asymptotically decaying branch for every admissible κ ,

while

$u_{\kappa,-}$ is never normalizable at infinity.

So the only question is whether $u_{\kappa,+}$ is regular enough at the core.

CCXXXV. NEAR-CORE CRITERION

Assume that near $r = 0$,

$$a(r) \rightarrow a_0 > 0, \quad b(r) \rightarrow b_0 > 0, \quad \mu(r) = O(1).$$

Then

$$W_\kappa^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r} = \frac{\kappa b_0}{r} + O(1).$$

Hence

$$u_{\kappa,+}(r) \sim r^{-\kappa b_0/a_0}, \quad u_{\kappa,-}(r) \sim r^{+\kappa b_0/a_0}.$$

After the standard unitary radial measure reduction, the radial L^2 -space is $L^2((0, \infty), dr)$. Thus a sufficient near-core criterion is:

$$u_{\kappa,+} \in L^2(0, r_0) \quad \Leftarrow \quad -\frac{\kappa b_0}{a_0} > -\frac{1}{2},$$

that is,

$$\frac{\kappa b_0}{a_0} < \frac{1}{2}.$$

Likewise, a sufficient exclusion criterion is

$$u_{\kappa,+} \notin L^2(0, r_0) \quad \Leftarrow \quad \frac{\kappa b_0}{a_0} \geq \frac{1}{2}.$$

This already shows how channel exclusion can occur: channels with too large positive effective centrifugal coefficient cannot support the decaying chiral generator.

CCXXXVI. MONOTONE SIGN-FLIP CRITERION FOR A PROTECTED CHANNEL

We now state a sharper sufficient criterion that uses the full sign structure of W_κ^{core} .

a. Assumptions. Let $\mathfrak{K}_{\text{adm}}$ be the discrete set of admissible angular channels after all bundle/grading/projector reductions. Assume:

1. $a(r)$ is continuous and strictly positive on $(0, \infty)$;
2. $\mu(r)$ is continuous, monotone increasing, and satisfies

$$\mu(0) \leq 0 < \mu_\infty := \lim_{r \rightarrow \infty} \mu(r);$$

3. $b(r)$ is continuous, nonnegative, and $b(r)/r$ is monotone decreasing on $(0, \infty)$;
4. there exists a unique channel $\kappa_* \in \mathfrak{K}_{\text{adm}}$ such that

$$W_{\kappa_*}^{\text{core}}(r) < 0 \quad \text{for } 0 < r < r_c, \quad W_{\kappa_*}^{\text{core}}(r) > 0 \quad \text{for } r > R_c,$$

for some $0 < r_c < R_c < \infty$;

5. for every $\kappa \neq \kappa_*$ in $\mathfrak{K}_{\text{adm}}$, one has

$$W_\kappa^{\text{core}}(r) \geq \frac{\gamma_\kappa}{r} \quad \text{for } 0 < r < r_c, \quad \gamma_\kappa > \frac{a_0}{2},$$

so that $u_{\kappa,+} \notin L^2(0, r_c)$.

b. Theorem (unique protected channel by monotone sign-flip). Under the above assumptions, exactly one channel contributes a protected chiral zero mode:

$$\boxed{\text{ind}_{\kappa_*} = +1, \quad \text{ind}_{\kappa} = 0 \quad (\kappa \neq \kappa_*).}$$

Consequently,

$$\boxed{\text{Ind}(\mathbb{H}_C) = +1.}$$

c. Proof. For every channel, $u_{\kappa,-}$ is excluded by the positive asymptotic mass gap $\mu_\infty > 0$, hence

$$u_{\kappa,-} \notin L^2(R, \infty).$$

So only $u_{\kappa,+}$ can possibly contribute.

For $\kappa = \kappa_*$, the near-core negativity of $W_{\kappa_*}^{\text{core}}$ implies

$$u_{\kappa_*,+}(r) = \exp\left(-\int^r \frac{W_{\kappa_*}^{\text{core}}(s)}{a(s)} ds\right)$$

is regular near the origin, while the asymptotic positivity implies exponential decay at infinity. Hence $u_{\kappa_*,+} \in L^2(0, \infty)$, so

$$\dim \ker Q_{\kappa_*,+}^{\text{core}} = 1, \quad \dim \ker Q_{\kappa_*, -}^{\text{core}} = 0.$$

Therefore

$$\text{ind}_{\kappa_*} = 1.$$

For any $\kappa \neq \kappa_*$, the near-core lower bound

$$W_{\kappa}^{\text{core}}(r) \geq \frac{\gamma_{\kappa}}{r}, \quad \gamma_{\kappa} > \frac{a_0}{2},$$

implies

$$u_{\kappa,+}(r) \lesssim r^{-\gamma_{\kappa}/a_0},$$

which is not L^2 near $r = 0$. So

$$\dim \ker Q_{\kappa,+}^{\text{core}} = 0$$

for all $\kappa \neq \kappa_*$. Since $u_{\kappa,-}$ is already excluded at infinity, one gets

$$\text{ind}_{\kappa} = 0 \quad (\kappa \neq \kappa_*).$$

Summing over channels yields

$$\text{Ind}(\mathbb{H}_C) = 1.$$

□

CCXXXVII. LOWEST-CHANNEL DOMINANCE AS A PRACTICAL COROLLARY

The previous theorem is abstractly clean but still depends on the admissible channel set. In practice, one usually has an ordered channel family

$$|\kappa_1| < |\kappa_2| < |\kappa_3| < \cdots,$$

and $b(r) \geq 0$ makes the centrifugal contribution larger for higher channels.

This yields the practical corollary:

a. Corollary (lowest-channel dominance). Suppose there exists a lowest admissible channel $\kappa_{\min}$ such that

1. $W_{\kappa_{\min}}^{\text{core}}$ has the required sign-flip structure:

$$W_{\kappa_{\min}}^{\text{core}}(r) < 0 \text{ near } r = 0, \quad W_{\kappa_{\min}}^{\text{core}}(r) > 0 \text{ for large } r;$$

2. for every higher channel $\kappa \neq \kappa_{\min}$, the centrifugal coefficient is sufficiently larger so that

$$\frac{\kappa b_0}{a_0} \geq \frac{1}{2}$$

near the origin.

Then only the lowest admissible channel can carry a protected zero mode.
This is the mathematically clean version of the slogan:

the protected defect mode lives in the lowest angular channel.

CCXXXVIII. ROLE OF THE ODD TANGENTIAL COEFFICIENT g_O

Now restore the full operator by including

$$\mathbb{V}_O = \frac{i c(r)}{r} \mathcal{D}_{S^2}.$$

On a fixed κ -channel, this adds

$$i \frac{\kappa c(r)}{r}$$

to the radial superpotential. Thus

$$u_{\kappa,+}^{\text{full}}(r) = u_{\kappa,+}^{\text{core}}(r) \cdot \exp\left(-i\kappa \int^r \frac{c(s)}{a(s)s} ds\right).$$

Hence:

$g_O(r)$ modifies the channel wavefunction by a phase only,

and therefore

$g_O(r)$ does not change which channel is L^2 -normalizable.

So the protected-channel classification is entirely governed by the real core data

$$a(r), \quad b(r), \quad \mu(r),$$

or equivalently by the real superpotential

$$W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.$$

CCXXXIX. WHAT IS NOW RIGOROUSLY CLOSED

The following points are now fixed:

1. the protected-channel problem reduces to the scalar superpotential

$$W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r};$$

2. the asymptotic bulk gap selects $u_{\kappa,+}$ as the only decaying branch;
3. channel exclusion is determined by the near-core behavior of $u_{\kappa,+}$;
4. under a unique sign-flip condition, exactly one admissible channel carries the index;
5. the odd tangential channel g_O affects only the phase and not the normalizability classification.

CCXL. WHAT REMAINS OPEN

What is still TECT-specific is:

1. the actual admissible channel set $\mathfrak{K}_{\text{adm}}$;
2. the actual profiles $a(r), b(r), \mu(r)$;
3. the proof that exactly one admissible lowest channel satisfies the sign-flip criterion.

So the one-defect/one-mode problem has now been reduced to a concrete sufficient criterion on the real radial superpotential.

CCXLI. FINAL SUMMARY

The single-defect protected-channel problem is controlled by the real Callias-core superpotential

$$W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.$$

For each channel,

$$Q_{\kappa, \pm}^{\text{core}} = \pm a(r) \frac{d}{dr} + W_{\kappa}^{\text{core}}(r),$$

so the formal generators are

$$u_{\kappa, +}(r) = \exp\left(-\int^r \frac{W_{\kappa}^{\text{core}}(s)}{a(s)} ds\right), \quad u_{\kappa, -}(r) = \exp\left(+\int^r \frac{W_{\kappa}^{\text{core}}(s)}{a(s)} ds\right).$$

If $\mu_{\infty} > 0$, then only $u_{\kappa, +}$ can decay at infinity. Hence channel selection reduces entirely to near-core regularity of $u_{\kappa, +}$.

Under a monotone sign-flip hypothesis — namely, that there exists a unique admissible channel κ_* for which

$$W_{\kappa_*}^{\text{core}}(r) < 0 \text{ near } r = 0, \quad W_{\kappa_*}^{\text{core}}(r) > 0 \text{ for large } r,$$

while all other admissible channels have $u_{\kappa, +} \notin L^2$ near the core — one obtains

$$\text{ind}_{\kappa_*} = 1, \quad \text{ind}_{\kappa} = 0 \quad (\kappa \neq \kappa_*),$$

and therefore

$$\text{Ind}(\mathbb{H}_C) = 1.$$

Moreover, the odd tangential coefficient $g_O(r)$ only adds the phase

$$\exp\left(-i\kappa \int^r \frac{c(s)}{a(s)s} ds\right)$$

to the radial generator, so it does not affect normalizability and does not change the protected-channel counting.

Thus the one-defect/one-protected-mode problem has been reduced to a sufficient criterion on the real radial profiles $a(r), b(r), \mu(r)$.

CCXLII. A PROFILE-LEVEL SUFFICIENT CRITERION FOR THE ONE-DEFECT/ONE-MODE STATEMENT

We now combine the radial Callias-core reduction with the previously established remote-gap isolation to obtain a concrete sufficient criterion for

$$\text{one defect} \leftrightarrow \text{one protected light mode}.$$

CCXLIII. STEP 1: SPECTRAL ISOLATION OF THE LOW ANGULAR SLOT

Let the full single-defect fluctuation operator be decomposed into

$$\mathbb{H}_{\text{full}} = \mathbb{H}_{\text{low}} \oplus \mathbb{H}_{\text{rem}} + \text{off-diagonal couplings},$$

where $\mathbb{H}_{\text{low}}$ denotes the candidate lowest angular/valley slot and $\mathbb{H}_{\text{rem}}$ contains all higher angular channels, wrong-internal sectors, and other remote pieces.

Assume the previously established remote-gap criterion holds uniformly:

$$\boxed{\text{spec}(\mathbb{H}_{\text{rem}}) \cap (-\Delta_{\text{rem}}, \Delta_{\text{rem}}) = \emptyset, \quad \Delta_{\text{rem}} > 0.}$$

Then any zero or near-zero protected mode must lie entirely in the low slot. Hence the protected-mode problem reduces to the projected lowest channel.

CCXLIV. STEP 2: PROJECTED LOWEST-CHANNEL CALLIAS CORE

Let the lowest projected channel be labeled by κ_* . After gauge removal and radial measure normalization, its graded Callias-core operator is

$$\mathbb{H}_{C, \kappa_*} = \begin{pmatrix} 0 & Q_{\kappa_*, -}^{\text{core}} \\ Q_{\kappa_*, +}^{\text{core}} & 0 \end{pmatrix}, \quad Q_{\kappa_*, \pm}^{\text{core}} = \pm a(r) \frac{d}{dr} + W_*(r),$$

where

$$\boxed{W_*(r) := W_{\kappa_*}^{\text{core}}(r) = \mu(r) + \frac{\kappa_* b(r)}{r}, \quad a(r) > 0.}$$

The formal zero-mode generators are

$$u_+(r) = \exp\left(-\int^r \frac{W_*(s)}{a(s)} ds\right), \quad u_-(r) = \exp\left(+\int^r \frac{W_*(s)}{a(s)} ds\right).$$

CCXLIV. PROFILE INEQUALITIES

We now impose explicit sufficient inequalities directly on $W_*(r)$.

a. Assumption P1 (uniform positive radial velocity). There exists $a_* > 0$ such that

$$a(r) \geq a_* > 0 \quad (r > 0).$$

b. Assumption P2 (core negativity with L^2 margin). There exist $r_0 > 0$ and $\varepsilon_0 > 0$ such that

$$\boxed{W_*(r) \leq -\left(\frac{1}{2} + \varepsilon_0\right) \frac{a(r)}{r} \quad (0 < r < r_0).}$$

c. Assumption P3 (bulk positivity). There exist $R_0 > r_0$ and $m_* > 0$ such that

$$\boxed{W_*(r) \geq m_* \quad (r > R_0).}$$

d. Assumption P4 (no competing low channels). For every other admissible channel $\kappa \neq \kappa_*$ that survives the projector reduction, there exist $r_\kappa > 0$ and $\varepsilon_\kappa > 0$ such that

$$\boxed{W_\kappa^{\text{core}}(r) \geq \left(\frac{1}{2} + \varepsilon_\kappa\right) \frac{a(r)}{r} \quad (0 < r < r_\kappa).}$$

This ensures that the decaying chiral generator in those channels is not L^2 -regular near the core.

CCXLVI. MAIN THEOREM

a. Theorem (profile-level one-defect/one-mode sufficient criterion). Assume:

1. the remote-gap isolation holds, so that only the projected lowest slot can host a zero mode;
2. the projected lowest channel κ_* satisfies P1–P3;
3. all competing projected channels satisfy P4.

Then

$$\dim \ker Q_{\kappa_*,+}^{\text{core}} = 1, \quad \dim \ker Q_{\kappa_*, -}^{\text{core}} = 0,$$

and for every $\kappa \neq \kappa_*$,

$$\dim \ker Q_{\kappa,+}^{\text{core}} = \dim \ker Q_{\kappa,-}^{\text{core}} = 0.$$

Consequently,

$$\text{Ind}(\mathbb{H}_C) = +1,$$

and the defect carries exactly one protected light mode in the low slot.

b. Proof. From P3,

$$u_+(r) \sim \exp\left(-\int^r \frac{W_*(s)}{a(s)} ds\right)$$

decays exponentially for $r > R_0$, while

$$u_-(r) \sim \exp\left(+\int^r \frac{W_*(s)}{a(s)} ds\right)$$

grows exponentially, so $u_- \notin L^2(R_0, \infty)$.

From P2,

$$-\frac{W_*(r)}{a(r)} \geq \left(\frac{1}{2} + \varepsilon_0\right) \frac{1}{r} \quad (0 < r < r_0),$$

hence

$$u_+(r) = \exp\left(-\int^r \frac{W_*(s)}{a(s)} ds\right) \lesssim r^{1/2+\varepsilon_0},$$

so $u_+ \in L^2(0, r_0)$. Therefore $u_+ \in L^2(0, \infty)$, and since the kernel of a first-order scalar operator is at most one-dimensional,

$$\dim \ker Q_{\kappa_*,+}^{\text{core}} = 1.$$

Also $\dim \ker Q_{\kappa_*, -}^{\text{core}} = 0$ by the large- r exclusion of u_- .

Now let $\kappa \neq \kappa_*$. By P4,

$$\frac{W_\kappa^{\text{core}}(r)}{a(r)} \geq \left(\frac{1}{2} + \varepsilon_\kappa\right) \frac{1}{r} \quad (0 < r < r_\kappa),$$

so

$$u_{\kappa,+}(r) = \exp\left(-\int^r \frac{W_\kappa^{\text{core}}(s)}{a(s)} ds\right) \gtrsim r^{-(1/2+\varepsilon_\kappa)},$$

which is not L^2 near $r = 0$. Thus

$$\dim \ker Q_{\kappa,+}^{\text{core}} = 0.$$

As before, $u_{\kappa,-}$ is excluded by the positive bulk gap. Hence no competing channel contributes.

Therefore the total graded index is

$$\text{Ind}(\mathbb{H}_C) = 1.$$

Because the remote sector is uniformly gapped away from zero, this index is realized by exactly one protected low mode. $\square$

CCXLVII. MONOTONE-PROFILE COROLLARY

The previous theorem is stated directly in terms of $W_*(r)$. A more physical corollary is obtained from monotone profile assumptions.

a. Corollary (monotone mass profile version). Assume:

1. $a(r) \geq a_* > 0$;
2. $b(r) \geq 0$ and $b(r)$ is bounded;
3. $\mu(r)$ is continuous and monotone increasing;
4. there exists a unique projected channel κ_* such that

$$\mu(r) + \frac{\kappa_* b(r)}{r}$$

is negative near the core and positive for large r ;

5. for every other projected channel $\kappa \neq \kappa_*$,

$$\mu(r) + \frac{\kappa b(r)}{r}$$

is positive near the core with the L^2 -exclusion margin of P4.

Then the conclusion of the theorem holds.

b. Comment. This is the operational form most likely to be checked in a concrete TECT profile calculation. It says that the unique protected channel is the unique one whose real superpotential crosses from sufficiently negative near the core to strictly positive in the bulk.

CCXLVIII. ROLE OF THE ODD TANGENTIAL COEFFICIENT

Restore the grading-breaking odd tangential term

$$\mathbb{V}_O = \frac{i c(r)}{r} \mathcal{D}_{S^2}.$$

On the fixed projected channel κ_* , this contributes

$$i \frac{\kappa_* c(r)}{r}$$

to the radial superpotential, so the full generator becomes

$$u_+^{\text{full}}(r) = u_+(r) \exp\left(-i\kappa_* \int^r \frac{c(s)}{a(s)s} ds\right).$$

Hence

$$\boxed{|u_+^{\text{full}}(r)| = |u_+(r)|.}$$

Therefore the g_O -channel does not affect the L^2 -normalizability criterion and does not alter the protected-mode counting. It only changes the phase profile of the mode.

CCXLIX. WHAT IS NOW ACTUALLY CLOSED

The one-defect/one-mode problem has now been reduced to a genuine profile inequality problem. The following are now fixed:

1. the remote-gap theorem removes all higher sectors from the zero-energy window;
2. the protected channel is determined entirely by the projected real superpotential

$$W_*(r) = \mu(r) + \frac{\kappa_* b(r)}{r};$$

3. explicit core and bulk inequalities P2–P3 are sufficient to produce one L^2 -normalizable chiral generator;
4. explicit core positivity inequalities P4 exclude all competing projected channels;
5. the odd transport coefficient g_O changes only the phase, not the channel counting.

CCL. WHAT REMAINS TECT-SPECIFIC

What remains to be verified from the actual TECT defect solution is:

1. the concrete projected channel κ_* ;
2. the actual radial profiles

$$a(r), \quad b(r), \quad \mu(r);$$

3. the validity of the profile inequalities P2–P4;
4. the actual remote-gap isolation of the higher channels in the same parameter regime.

So the full conceptual chain is now closed: the remaining work is a concrete profile verification problem.

CCLI. FINAL SUMMARY

We combined the radial Callias-core reduction with the remote-gap theorem to obtain a concrete sufficient criterion for

$$\text{one defect} \leftrightarrow \text{one protected light mode.}$$

Let the projected lowest channel be κ_* , and define the real superpotential

$$W_*(r) = \mu(r) + \frac{\kappa_* b(r)}{r}.$$

If:

1. the remote sector is uniformly gapped away from zero;
2. $a(r) \geq a_* > 0$;
3. near the core,

$$W_*(r) \leq -\left(\frac{1}{2} + \varepsilon_0\right) \frac{a(r)}{r};$$

4. in the bulk,

$$W_*(r) \geq m_* > 0;$$

5. every competing projected channel $\kappa \neq \kappa_*$ satisfies

$$W_\kappa^{\text{core}}(r) \geq \left(\frac{1}{2} + \varepsilon_\kappa\right) \frac{a(r)}{r}$$

near the core,

then exactly one chiral generator is L^2 -normalizable, all competing channels are excluded, and

$$\text{Ind}(\mathbb{H}_C) = 1.$$

Thus the one-defect/one-mode problem has been reduced to verifying explicit inequalities for the real radial profiles $a(r), b(r), \mu(r)$.

Moreover, the odd tangential coefficient g_O only contributes a phase and therefore does not affect the protected-channel counting.

CCLII. WORKING RADIAL PROFILE ANSATZ AND A CONCRETE SUFFICIENT CRITERION FOR ONE DEFECT / ONE PROTECTED MODE

We now impose a minimal working radial profile ansatz for the defect background and derive explicit sufficient inequalities on the resulting coefficients

$$a(r), \quad b(r), \quad \mu(r)$$

that guarantee exactly one protected channel.

CCLIII. WORKING RADIAL PROFILE ANSATZ

We assume a smooth radial single-defect background of the form

$$\bar{\sigma}(r), \quad X_\perp(r), \quad X_\parallel(r),$$

with the regularity conditions

$$\bar{\sigma}(r) = \sigma_0 + \sigma_2 r^2 + O(r^4),$$

$$X_\perp(r) = X_0 + x_{\perp,2} r^2 + O(r^4), \quad X_\parallel(r) = X_0 + x_{\parallel,2} r^2 + O(r^4),$$

as $r \rightarrow 0$. The equality

$$X_\perp(0) = X_\parallel(0) = X_0$$

is the smoothness condition required for the tensor

$$\bar{X}_i{}^a = X_\perp(r)(\delta_i{}^a - \hat{n}_i \hat{n}^a) + X_\parallel(r) \hat{n}_i \hat{n}^a$$

to be regular at the origin.

At large radius, assume exponential approach to the bulk vacuum:

$$\bar{\sigma}(r) = \sigma_\infty - \Sigma_\infty e^{-m_\sigma r} + o(e^{-m_\sigma r}),$$

$$X_\perp(r) = X_{\perp,\infty} + A_\perp e^{-M_X r} + o(e^{-M_X r}), \quad X_\parallel(r) = X_{\parallel,\infty} + A_\parallel e^{-M_X r} + o(e^{-M_X r}).$$

In the simplest vacuum branch one has

$$X_{\perp,\infty} = X_{\parallel,\infty} = 0,$$

but we keep the formulas general.

Define

$$\chi_X(r) := \frac{1}{3}(2X_\perp(r) + X_\parallel(r)).$$

Then

$$\chi_X(r) = X_0 + O(r^2) \quad (r \rightarrow 0),$$

and

$$\chi_X(r) \rightarrow \chi_\infty := \frac{1}{3}(2X_{\perp,\infty} + X_{\parallel,\infty}) \quad (r \rightarrow \infty).$$

CCLIV. RADIAL CONSTITUTIVE COEFFICIENTS

From the previously fixed minimal constitutive choice,

$$g_E(r) = v_\perp(\bar{\sigma}(r)) + 3\chi_X(r) v_E^{(X)}(\bar{\sigma}(r)),$$

$$g_c(r) = v_c(\bar{\sigma}(r)) + 3\chi_X(r) v_c^{(X)}(\bar{\sigma}(r)),$$

$$\mu_{\text{num}}(r) = m(\bar{\sigma}(r)) - 3\chi_X(r) u_X(\bar{\sigma}(r)),$$

and

$$a(r) = \frac{g_R(r)}{z_0(\bar{\sigma}(r))}, \quad b(r) = \frac{g_E(r)}{z_0(\bar{\sigma}(r))}, \quad \mu(r) = \frac{\mu_{\text{num}}(r)}{z_0(\bar{\sigma}(r))},$$

where

$$g_R(r) = g_E(r) + g_c(r).$$

Thus

$$a(r) = \frac{v_\perp(\bar{\sigma}(r)) + v_c(\bar{\sigma}(r)) + 3\chi_X(r)(v_E^{(X)}(\bar{\sigma}(r)) + v_c^{(X)}(\bar{\sigma}(r)))}{z_0(\bar{\sigma}(r))},$$

$$b(r) = \frac{v_\perp(\bar{\sigma}(r)) + 3\chi_X(r) v_E^{(X)}(\bar{\sigma}(r))}{z_0(\bar{\sigma}(r))},$$

$$\mu(r) = \frac{m(\bar{\sigma}(r)) - 3\chi_X(r) u_X(\bar{\sigma}(r))}{z_0(\bar{\sigma}(r))}.$$

CCLV. CORE AND BULK COEFFICIENTS

By smoothness of the profiles and constitutive functions, the core limits exist:

$$a_0 := \lim_{r \rightarrow 0} a(r), \quad b_0 := \lim_{r \rightarrow 0} b(r), \quad \mu_0 := \lim_{r \rightarrow 0} \mu(r),$$

with

$$a_0 = \frac{v_\perp(\sigma_0) + v_c(\sigma_0) + 3X_0(v_E^{(X)}(\sigma_0) + v_c^{(X)}(\sigma_0))}{z_0(\sigma_0)},$$

$$b_0 = \frac{v_\perp(\sigma_0) + 3X_0 v_E^{(X)}(\sigma_0)}{z_0(\sigma_0)},$$

$$\mu_0 = \frac{m(\sigma_0) - 3X_0 u_X(\sigma_0)}{z_0(\sigma_0)}.$$

Likewise, the bulk limits exist:

$$a_\infty := \lim_{r \rightarrow \infty} a(r), \quad b_\infty := \lim_{r \rightarrow \infty} b(r), \quad \mu_\infty := \lim_{r \rightarrow \infty} \mu(r),$$

with

$$a_\infty = \frac{v_\perp(\sigma_\infty) + v_c(\sigma_\infty) + 3\chi_\infty(v_E^{(X)}(\sigma_\infty) + v_c^{(X)}(\sigma_\infty))}{z_0(\sigma_\infty)},$$

$$b_\infty = \frac{v_\perp(\sigma_\infty) + 3\chi_\infty v_E^{(X)}(\sigma_\infty)}{z_0(\sigma_\infty)},$$

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

In the simplest vacuum branch $\chi_\infty = 0$, these reduce to

$$a_\infty = \frac{v_\perp(\sigma_\infty) + v_c(\sigma_\infty)}{z_0(\sigma_\infty)}, \quad b_\infty = \frac{v_\perp(\sigma_\infty)}{z_0(\sigma_\infty)}, \quad \mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

CCLVI. NEAR-CORE AND BULK ASYMPTOTICS OF THE SUPERPOTENTIAL

For a projected angular channel κ , the Callias-core superpotential is

$$W_\kappa^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.$$

Near the core,

$$a(r) = a_0 + O(r^2), \quad b(r) = b_0 + O(r^2), \quad \mu(r) = \mu_0 + O(r^2),$$

so

$$W_\kappa^{\text{core}}(r) = \mu_0 + \frac{\kappa b_0}{r} + O(r).$$

Thus the $1/r$ term dominates, and its sign is governed entirely by κb_0 .

At large radius,

$$a(r) = a_\infty + o(1), \quad b(r) = b_\infty + o(1), \quad \mu(r) = \mu_\infty + o(1),$$

hence

$$W_\kappa^{\text{core}}(r) = \mu_\infty + \frac{\kappa b_\infty}{r} + o(1).$$

Therefore

$$W_\kappa^{\text{core}}(r) \rightarrow \mu_\infty \quad (r \rightarrow \infty).$$

CCLVII. PROFILE-LEVEL SUFFICIENT INEQUALITIES

We now formulate a directly checkable sufficient criterion.

a. Assumption S1 (positive bulk gap).

$$\mu_\infty > 0.$$

b. Assumption S2 (positive radial velocity).

$$a_0 > 0, \quad a_\infty > 0.$$

c. *Assumption S3 (positive core angular coefficient).*

$$b_0 > 0.$$

d. *Assumption S4 (unique projected signed lowest channel).* After the remote-gap and projector reduction, the admissible low-slot channel set contains a unique signed lowest candidate

$$\kappa_* < 0,$$

with multiplicity one. All competing projected channels are either excluded by the remote-gap theorem or satisfy

$$\kappa \geq \kappa_{\text{comp}} > 0.$$

e. *Assumption S5 (core regularity margin for the protected channel).* There exists $\varepsilon_* > 0$ such that

$$\frac{|\kappa_*| b_0}{a_0} \geq \frac{1}{2} + \varepsilon_*.$$

f. *Assumption S6 (core exclusion margin for competing channels).* For every competing projected channel $\kappa \neq \kappa_*$,

$$\frac{\kappa b_0}{a_0} \geq \frac{1}{2} + \varepsilon_\kappa$$

for some $\varepsilon_\kappa > 0$.

CCLVIII. MAIN PROFILE THEOREM

a. *Theorem (working one-defect/one-mode sufficient criterion).* Assume S1–S6. Then the projected lowest channel κ_* carries exactly one L^2 -normalizable chiral zero-mode generator, while all competing projected channels carry none. Consequently,

$$\text{Ind}(\mathbb{H}_C) = 1,$$

and the defect carries exactly one protected light mode.

b. *Proof.* Because $\mu_\infty > 0$, the large- r asymptotic generator

$$u_{\kappa,+}(r) = \exp\left(-\int^r \frac{W_\kappa^{\text{core}}(s)}{a(s)} ds\right)$$

is the unique decaying branch for every channel, while

$$u_{\kappa,-}(r) = \exp\left(+\int^r \frac{W_\kappa^{\text{core}}(s)}{a(s)} ds\right)$$

is never L^2 -normalizable at infinity.

For the protected channel κ_* , near the core

$$W_{\kappa_*}^{\text{core}}(r) = \mu_0 - \frac{|\kappa_*| b_0}{r} + O(r).$$

Since the $1/r$ term dominates, S5 implies that for sufficiently small r ,

$$W_{\kappa_*}^{\text{core}}(r) \leq -\left(\frac{1}{2} + \frac{\varepsilon_*}{2}\right) \frac{a(r)}{r}.$$

Thus the decaying branch behaves like

$$u_{\kappa_*,+}(r) \lesssim r^{1/2+\varepsilon_*/2},$$

and is L^2 -regular near the origin. Combined with large- r decay, this gives

$$\dim \ker Q_{\kappa_*,+}^{\text{core}} = 1, \quad \dim \ker Q_{\kappa_*, -}^{\text{core}} = 0.$$

For a competing channel $\kappa \neq \kappa_*$, S6 implies that near the core

$$W_{\kappa}^{\text{core}}(r) \geq \left(\frac{1}{2} + \frac{\varepsilon_{\kappa}}{2} \right) \frac{a(r)}{r},$$

so

$$u_{\kappa,+}(r) \gtrsim r^{-(1/2+\varepsilon_{\kappa}/2)},$$

which is not L^2 -regular at the origin. Since $u_{\kappa,-}$ is already excluded at infinity, no competing channel contributes.

Hence exactly one projected channel contributes one protected chiral generator, and

$$\text{Ind}(\mathbb{H}_{\text{C}}) = 1.$$

□

CCLIX. PRACTICAL MONOTONE-PROFILE COROLLARY

The theorem above can be rewritten in a more physical form.

a. Corollary (monotone working-profile version). Assume:

1. $\bar{\sigma}(r)$ is monotone increasing from σ_0 to σ_{∞} ;
2. $X_{\perp}(r)$ and $X_{\parallel}(r)$ are monotone decreasing from X_0 to their bulk limits;
3. $z_0(\sigma) > 0$, $v_{\perp}(\sigma) > 0$, and $v_{\perp}(\sigma) + v_c(\sigma) > 0$;
4. the constitutive combination $m(\sigma) - 3\chi_X u_X(\sigma)$ tends to a strictly positive bulk value;
5. the projected low-slot contains a unique signed lowest channel $\kappa_* < 0$.

If, in addition, the core ratios satisfy

$$\frac{|\kappa_*| b_0}{a_0} > \frac{1}{2}, \quad \frac{\kappa b_0}{a_0} > \frac{1}{2} \quad \text{for all projected competitors } \kappa > 0,$$

then the defect carries exactly one protected light mode.

b. Comment. This is the form most useful for an actual TECT profile check. Everything is reduced to the core constants

$$a_0, \quad b_0,$$

the bulk mass parameter

$$\mu_{\infty},$$

and the projected channel structure.

CCLX. ROLE OF THE ODD TANGENTIAL PROFILE $g_O(r)$

The odd tangential profile enters the full radial generator as a phase factor only:

$$u_{\kappa,+}^{\text{full}}(r) = u_{\kappa,+}^{\text{core}}(r) \exp\left(-i\kappa \int^r \frac{c(s)}{a(s)s} ds\right).$$

Therefore

$$\boxed{|u_{\kappa,+}^{\text{full}}(r)| = |u_{\kappa,+}^{\text{core}}(r)|.}$$

So $g_O(r)$ does not alter the protected-channel counting. It only changes the phase profile of the wavefunction.

CCLXI. WHAT IS NOW ACTUALLY REDUCED TO EXPLICIT VERIFICATION

The one-defect/one-mode statement is now reduced to verifying the following concrete TECT data:

1. the core constants a_0, b_0 ;
2. the bulk gap $\mu_\infty > 0$;
3. the projected admissible low-slot channel structure;
4. the remote-gap isolation of all higher sectors.

These are now profile-check problems rather than structural unknowns.

CCLXII. FINAL SUMMARY

Using the working radial defect ansatz

$$\bar{\sigma}(r), \quad X_\perp(r), \quad X_\parallel(r),$$

the Callias-core coefficients are

$$a(r) = \frac{g_R(r)}{z_0(\bar{\sigma}(r))}, \quad b(r) = \frac{g_E(r)}{z_0(\bar{\sigma}(r))}, \quad \mu(r) = \frac{m(\bar{\sigma}(r)) - 3\chi_X(r)u_X(\bar{\sigma}(r))}{z_0(\bar{\sigma}(r))},$$

with

$$\chi_X(r) = \frac{1}{3}(2X_\perp(r) + X_\parallel(r)).$$

Their core and bulk limits are

$$a_0, \quad b_0, \quad \mu_0, \quad a_\infty, \quad b_\infty, \quad \mu_\infty.$$

For the projected lowest channel κ_* , the real superpotential is

$$W_*(r) = \mu(r) + \frac{\kappa_* b(r)}{r}.$$

A sufficient one-defect/one-mode criterion is:

1. $\mu_\infty > 0$ (bulk gap);
2. $a_0 > 0$ and $b_0 > 0$;
3. after projector reduction there is a unique signed lowest channel $\kappa_* < 0$ of multiplicity one;
- 4.

$$\frac{|\kappa_*| b_0}{a_0} > \frac{1}{2};$$

5. every competing projected channel satisfies

$$\frac{\kappa b_0}{a_0} > \frac{1}{2}.$$

Under these conditions, exactly one chiral generator is L^2 -normalizable, all competing channels are excluded, and

$$\text{Ind}(\mathbb{H}_C) = 1.$$

Thus the one-defect/one-mode problem has been reduced to explicit verification of the core coefficients, the bulk gap, the projected channel structure, and the remote-gap isolation.

CCLXIII. FINAL UNIFIED CHAIN: REMOTE-GAP ISOLATION + BLOCH EXTRACTION + RADIAL ONE-MODE CRITERION

We now combine the three previously separate modules into a single working sufficient criterion for

$$\text{one defect} \leftrightarrow \text{one protected light mode.}$$

The three modules are:

1. the remote-gap isolation theorem;
2. the Bloch/valley/Class II coefficient extraction of (g_c, g_E, g_O) ;
3. the radial Callias-core one-channel criterion.

CCLXIV. MICROSCOPIC TRANSPORT TRIPLET AND ITS TWO PROJECTIONS

From the locked BCC/Class II coefficient extraction, the reduced valley transport coefficients are defined by

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O,$$

with

$$g_c = \frac{\lambda_{\parallel}}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}.$$

These are the valley-projected transport coefficients.

From the radial single-defect constitutive ansatz, the corresponding core transport coefficients are

$$g_E^{\text{core}} := g_E^{\text{eff}}(0), \quad g_c^{\text{core}} := g_c^{\text{eff}}(0), \quad g_O^{\text{core}} := g_O^{\text{eff}}(0),$$

and the bulk mass coefficient is

$$\mu_{\infty} = \lim_{r \rightarrow \infty} \frac{m(\bar{\sigma}(r)) - 3\chi_X(r)u_X(\bar{\sigma}(r))}{z_0(\bar{\sigma}(r))}.$$

To connect the two descriptions, we introduce the minimal local-to-valley matching hypothesis:

a. Matching hypothesis. There exist positive constants

$$\xi_E > 0, \quad \xi_c > 0, \quad \xi_O > 0$$

such that

$$\boxed{g_E^{\text{core}} = \xi_E g_E, \quad g_c^{\text{core}} = \xi_c g_c, \quad g_O^{\text{core}} = \xi_O g_O.}$$

In the strict local-core matching approximation one has simply

$$\xi_E = \xi_c = \xi_O = 1,$$

but the theorem below does not require this stronger specialization.

CCLXV. CORE/BULK COEFFICIENTS ENTERING THE RADIAL CRITERION

Recall that the radial Callias-core coefficients are

$$a(r) = \frac{g_R(r)}{z_0(r)}, \quad b(r) = \frac{g_E(r)}{z_0(r)}, \quad \mu(r) = \frac{m_{\text{eff}}(r)}{z_0(r)}, \quad g_R(r) = g_E(r) + g_c(r).$$

At the core,

$$a_0 = \frac{g_E^{\text{core}} + g_c^{\text{core}}}{z_{0,0}}, \quad b_0 = \frac{g_E^{\text{core}}}{z_{0,0}}, \quad c_0 = \frac{g_O^{\text{core}}}{z_{0,0}},$$

where

$$z_{0,0} := z_0(\sigma_0) > 0.$$

Under the matching hypothesis, this becomes

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \quad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \quad c_0 = \frac{\xi_O g_O}{z_{0,0}}.$$

At infinity,

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

In the simplest vacuum branch $\chi_\infty = 0$,

$$\mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

CCLXVI. REMOTE-GAP ISOLATION AND REDUCTION TO THE LOWEST PROJECTED CHANNEL

Let $\mathbb{H}_{\text{full}}$ be the full single-defect fluctuation operator decomposed into

$$\mathbb{H}_{\text{full}} = \mathbb{H}_{\text{low}} \oplus \mathbb{H}_{\text{rem}} + \text{off-diagonal couplings}.$$

Assume the previously established full remote-gap condition:

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

Then the remote-gap theorem implies

$$\text{spec}(\mathbb{H}_{\text{rem}}) \cap (-\Delta_{\text{rem}}, \Delta_{\text{rem}}) = \emptyset$$

for some $\Delta_{\text{rem}} > 0$. Therefore any protected zero or near-zero mode must lie in the projected low slot.

We assume that after this low-slot projection there remains a unique signed lowest admissible channel

$$\kappa_* < 0$$

of multiplicity one.

CCLXVII. RADIAL ONE-MODE INEQUALITIES IN TERMS OF

(g_c, g_E, g_O)

For the lowest projected channel, the real Callias-core superpotential is

$$W_*(r) = \mu(r) + \frac{\kappa_* b(r)}{r}.$$

The protected-channel criterion obtained earlier requires:

1. bulk positivity:

$$\mu_\infty > 0;$$

2. positive radial velocity:

$$a_0 > 0;$$

3. positive core angular coefficient:

$$b_0 > 0;$$

4. lowest-channel core regularity:

$$\frac{|\kappa_*| b_0}{a_0} > \frac{1}{2};$$

5. competitor exclusion: for every surviving projected competitor $\kappa \neq \kappa_*$,

$$\frac{\kappa b_0}{a_0} > \frac{1}{2}.$$

Substituting the matching formulas for a_0, b_0 , these become

$$\boxed{\xi_E g_E + \xi_c g_c > 0},$$

$$\boxed{\xi_E g_E > 0},$$

$$\boxed{\mu_\infty > 0},$$

and

$$\boxed{2|\kappa_*| \xi_E g_E > \xi_E g_E + \xi_c g_c}.$$

Similarly, every projected competitor $\kappa \neq \kappa_*$ is excluded if

$$\boxed{2\kappa \xi_E g_E > \xi_E g_E + \xi_c g_c}.$$

Thus the radial one-mode criterion has been translated entirely into inequalities on the extracted transport coefficients together with the bulk mass parameter.

CCLXVIII. MAIN UNIFIED THEOREM

a. Theorem (final working one-defect/one-mode sufficient criterion). Assume:

1. the full remote-gap condition holds:

$$\Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho};$$

2. after remote-gap reduction and low-slot projection, there is a unique signed lowest admissible channel $\kappa_* < 0$ of multiplicity one;

3. the extracted Bloch/valley coefficients (g_c, g_E, g_O) satisfy the matching hypothesis with positive constants (ξ_c, ξ_E, ξ_O) ;

4. the bulk mass parameter satisfies

$$\mu_\infty > 0;$$

5. the core transport coefficients satisfy

$$\xi_E g_E + \xi_c g_c > 0, \quad \xi_E g_E > 0,$$

and

$$2|\kappa_*| \xi_E g_E > \xi_E g_E + \xi_c g_c;$$

6. every competing projected channel $\kappa \neq \kappa_*$ satisfies

$$2\kappa \xi_{EgE} > \xi_{EgE} + \xi_c g_c.$$

Then the full defect operator carries exactly one protected light mode:

$$\boxed{\text{Ind}(\mathbb{H}_C) = 1,}$$

and therefore

$$\boxed{\text{one defect} \leftrightarrow \text{one protected light mode.}}$$

b. Proof. By the remote-gap theorem, all remote sectors are uniformly excluded from the zero-energy window, so the protected-mode problem reduces to the low projected slot.

In that slot, the radial Callias-core criterion applies with

$$a_0 = \frac{\xi_{EgE} + \xi_c g_c}{z_{0,0}}, \quad b_0 = \frac{\xi_{EgE}}{z_{0,0}},$$

and $z_{0,0} > 0$. Hence the assumptions

$$\xi_{EgE} + \xi_c g_c > 0, \quad \xi_{EgE} > 0$$

imply

$$a_0 > 0, \quad b_0 > 0.$$

The lowest-channel inequality

$$2|\kappa_*| \xi_{EgE} > \xi_{EgE} + \xi_c g_c$$

is equivalent to

$$\frac{|\kappa_*| b_0}{a_0} > \frac{1}{2},$$

so the decaying chiral generator in the κ_* -channel is L^2 -regular at the core.

The bulk positivity $\mu_\infty > 0$ guarantees that the same generator decays at infinity, while the opposite chiral generator is excluded.

For every competitor $\kappa \neq \kappa_*$, the inequality

$$2\kappa \xi_{EgE} > \xi_{EgE} + \xi_c g_c$$

is equivalent to

$$\frac{\kappa b_0}{a_0} > \frac{1}{2},$$

which excludes the decaying generator at the core.

Thus exactly one projected channel contributes one normalizable chiral generator. Since the remote sectors are spectrally isolated, the total defect index is

$$\text{Ind}(\mathbb{H}_C) = 1.$$

Therefore the defect carries exactly one protected light mode. $\square$

CCLXIX. ROLE OF THE ODD TRANSPORT COEFFICIENT

The coefficient g_O remains physically relevant in the full operator, but it does not enter the protected-mode counting. Indeed, under the matching hypothesis,

$$c_0 = \frac{\xi_O g_O}{z_{0,0}},$$

and the odd tangential term contributes only the channel-dependent phase

$$\exp\left(-i\kappa \int^r \frac{c(s)}{a(s)s} ds\right)$$

to the radial generator. Thus

g_O affects the phase profile of the mode, but not the L^2 -normalizability criterion.

CCLXX. OPERATIONAL INTERPRETATION

The theorem shows that the entire one-defect/one-mode problem is reduced to checking the following explicit data:

1. the remote-gap inequality

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho};$$

2. the low-slot channel structure and uniqueness of the signed lowest channel κ_* ;
3. the signs and sizes of the extracted transport coefficients g_E, g_c ;
4. the bulk mass gap sign $\mu_\infty > 0$.

In particular, the decisive low-core inequality is

$$2|\kappa_*| \xi_E g_E > \xi_E g_E + \xi_c g_c.$$

This is the concrete bridge between the Bloch/Class II transport extraction and the radial one-mode criterion.

CCLXXI. WHAT IS NOW ACTUALLY CLOSED

The conceptual chain is now complete:

remote-gap theorem $\implies$ low-slot isolation $\implies$ radial projected channel problem $\implies$ one-defect/one-mode criterion in terms of

What remains is no longer structural. It is the actual verification of the inequalities and matching constants for the concrete TECT working branch.

CCLXXII. FINAL SUMMARY

We unified the three previously separate modules:

remote-gap isolation, Bloch/valley extraction of (g_c, g_E, g_O) , radial Callias-core one-mode criterion.

The defect-core coefficients are

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \quad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \quad c_0 = \frac{\xi_O g_O}{z_{0,0}},$$

while the bulk mass parameter is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

If:

1. the remote-gap condition

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}$$

holds;

2. the low-slot projection leaves a unique signed lowest channel $\kappa_* < 0$ of multiplicity one;
3. $\mu_\infty > 0$;
- 4.

$$\xi_{Eg_E} + \xi_{cg_c} > 0, \quad \xi_{Eg_E} > 0,$$

and

$$2|\kappa_*|\xi_{Eg_E} > \xi_{Eg_E} + \xi_{cg_c};$$

5. every competing projected channel satisfies

$$2\kappa \xi_{Eg_E} > \xi_{Eg_E} + \xi_{cg_c};$$

then

$$\text{Ind}(\mathbb{H}_C) = 1,$$

and therefore one defect carries exactly one protected light mode.

Moreover, g_O affects only the phase of the radial generator and does not alter the protected-mode counting.

Thus the remaining work is purely explicit verification: evaluate the extracted coefficients and check the inequalities in the concrete TECT working branch.

CCLXXIII. BENCHMARK DICTIONARY LINKING BLOCH/CLASS II EXTRACTION TO THE RADIAL ONE-MODE CRITERION

We now construct the practical benchmark dictionary that connects:

1. the Bloch/valley/Class II extracted coefficients

$$(\lambda_{\parallel}, \alpha, \beta),$$

2. the radial Callias-core coefficients

$$(a_0, b_0, c_0, \mu_\infty),$$

3. the final one-defect/one-protected-mode sufficient criterion.

CCLXXIV. EXACT VALLEY EXTRACTION LAYER

On the low valley doublet, the transport coefficients are extracted by

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}), \quad \alpha = \frac{1}{4}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4}(\sigma_2 M_1 - \sigma_1 M_2).$$

By definition of the overlap constants,

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O,$$

hence

$$g_c = \frac{\lambda_{\parallel}}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}.$$

This part is exact once C_c, C_E, C_O are fixed by the chosen Bloch/overlap convention.

CCLXXV. EXACT RADIAL CORE LAYER

From the radial constitutive reduction, the core coefficients are

$$a_0 = \frac{g_E^{\text{core}} + g_c^{\text{core}}}{z_{0,0}}, \quad b_0 = \frac{g_E^{\text{core}}}{z_{0,0}}, \quad c_0 = \frac{g_O^{\text{core}}}{z_{0,0}},$$

where

$$z_{0,0} := z_0(\sigma_0) > 0.$$

The bulk mass parameter is

$$\mu_{\infty} = \frac{m(\sigma_{\infty}) - 3\chi_{\infty} u_X(\sigma_{\infty})}{z_0(\sigma_{\infty})}.$$

In the simplest vacuum branch $\chi_{\infty} = 0$,

$$\mu_{\infty} = \frac{m(\sigma_{\infty})}{z_0(\sigma_{\infty})}.$$

CCLXXVI. MATCHING HYPOTHESIS BETWEEN VALLEY AND CORE TRANSPORT DATA

To connect the valley-extracted coefficients with the defect-core coefficients, introduce the matching constants

$$g_E^{\text{core}} = \xi_E g_E, \quad g_c^{\text{core}} = \xi_c g_c, \quad g_O^{\text{core}} = \xi_O g_O,$$

with

$$\xi_E > 0, \quad \xi_c > 0, \quad \xi_O > 0.$$

These constants encode the difference between the core-local projection and the valley/Bloch projection. They are not yet a theorem of the branch; they are the remaining matching data to be computed explicitly.

Under this hypothesis,

$$a_0 = \frac{1}{z_{0,0}} \left(\xi_E \frac{\alpha}{C_E} + \xi_c \frac{\lambda_{\parallel}}{C_c} \right),$$

$$b_0 = \frac{\xi_E}{z_{0,0}} \frac{\alpha}{C_E}, \quad c_0 = \frac{\xi_O}{z_{0,0}} \frac{\beta}{C_O}.$$

CCLXXVII. DIMENSIONLESS BENCHMARK RATIOS

Define the two dimensionless ratios

$$R_c := \frac{\xi_c C_E}{\xi_E C_c} \frac{\lambda_{\parallel}}{\alpha}, \quad R_O := \frac{\xi_O C_E}{\xi_E C_O} \frac{\beta}{\alpha},$$

whenever $\alpha \neq 0$.

Then the core coefficients simplify to

$$a_0 = b_0(1 + R_c), \quad c_0 = b_0 R_O.$$

Equivalently,

$$\frac{a_0}{b_0} = 1 + R_c, \quad \frac{c_0}{b_0} = R_O.$$

This is the cleanest practical dictionary. The parameter R_c controls the relative strength of the radial/longitudinal channel compared with the even tangential channel, while R_O controls the odd/torsion-like phase channel.

CCLXXVIII. RADIAL ONE-MODE CRITERION IN DICTIONARY FORM

For the projected lowest channel κ_* , the real Callias-core superpotential is

$$W_*(r) = \mu(r) + \frac{\kappa_* b(r)}{r}.$$

The previously derived sufficient criterion requires:

1. remote-gap isolation:

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho};$$

2. a unique signed lowest projected channel $\kappa_* < 0$ of multiplicity one;
3. bulk positivity:

$$\mu_{\infty} > 0;$$

4. positive core transport coefficients:

$$a_0 > 0, \quad b_0 > 0;$$

5. lowest-channel regularity:

$$\frac{|\kappa_*| b_0}{a_0} > \frac{1}{2};$$

6. competitor exclusion: for all projected competitors $\kappa \neq \kappa_*$,

$$\frac{\kappa b_0}{a_0} > \frac{1}{2}.$$

Using

$$a_0 = b_0(1 + R_c),$$

this becomes

$$\mu_\infty > 0, \quad b_0 > 0, \quad 1 + R_c > 0,$$

$$2|\kappa_*| > 1 + R_c,$$

and for every competitor,

$$2\kappa > 1 + R_c.$$

Since

$$b_0 = \frac{\xi_E}{z_{0,0}} \frac{\alpha}{C_E},$$

the sign condition $b_0 > 0$ is equivalent to

$$\frac{\xi_E}{C_E} \alpha > 0.$$

Under the standard convention $C_E > 0$ and $\xi_E > 0$, this reduces simply to

$$\alpha > 0.$$

Thus the one-mode criterion is controlled entirely by

$$\mu_\infty, \quad \alpha, \quad R_c, \quad \kappa_*.$$

CCLXXIX. MOST PRACTICAL LOWEST-CHANNEL SPECIALIZATION

The practically most relevant case is:

1. the projected lowest signed channel is

$$\kappa_* = -1;$$

2. all other surviving projected competitors satisfy

$$\kappa \geq 1.$$

Then the inequalities reduce to

$$-1 < R_c < 1,$$

together with

$$\alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

Indeed:

$$1 + R_c > 0 \iff R_c > -1,$$

and

$$2|\kappa_*| > 1 + R_c \quad \text{with} \quad |\kappa_*| = 1 \iff 2 > 1 + R_c \iff R_c < 1.$$

Hence the simplest operational criterion is precisely

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

CCLXXX. ROLE OF β AND R_O

The odd transport coefficient enters through

$$R_O = \frac{\xi_O C_E}{\xi_E C_O} \frac{\beta}{\alpha},$$

but, as shown previously, it contributes only a phase to the radial zero-mode generator. Therefore:

$$\boxed{R_O \text{ does not enter the protected-channel counting inequalities.}}$$

It is still physically meaningful, because it controls the phase/torsion-like structure of the wavefunction and will matter for finer transport and chirality diagnostics, but not for the one-mode index count itself.

CCLXXXI. EXACT VERSUS CONDITIONAL PARTS OF THE DICTIONARY

It is important to distinguish the exact parts from the still-conditional parts.

a. Exact. The following are exact within the working formalism:

1. the extraction of $(\lambda_{\parallel}, \alpha, \beta)$ from the projected Pauli traces;
2. the definitions

$$g_c = \lambda_{\parallel}/C_c, \quad g_E = \alpha/C_E, \quad g_O = \beta/C_O;$$

3. the core formulas

$$a_0 = \frac{g_E^{\text{core}} + g_c^{\text{core}}}{z_{0,0}}, \quad b_0 = \frac{g_E^{\text{core}}}{z_{0,0}}, \quad c_0 = \frac{g_O^{\text{core}}}{z_{0,0}};$$

4. the radial one-mode inequalities once $a_0, b_0, \mu_{\infty}, \kappa_*$ are known.

b. Conditional / branch-specific. The following remain to be verified in the actual branch:

1. the matching constants ξ_E, ξ_c, ξ_O ;
2. the unique signed lowest projected channel $\kappa_* < 0$;
3. the remote-gap inequality in the same parameter regime;
4. the sign of μ_{∞} .

Thus the dictionary is structurally complete, but still requires explicit branch evaluation of a small number of matching and spectral data.

CCLXXXII. FINAL WORKING THEOREM

a. Theorem (practical working one-defect/one-mode criterion). Assume:

1. the remote-gap inequality holds:

$$\Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho};$$

2. the projected low slot contains a unique signed lowest channel $\kappa_* < 0$ of multiplicity one;
3. the matching hypothesis with positive ξ_E, ξ_c, ξ_O holds;
4. $\mu_{\infty} > 0$;

5.

$$\frac{\xi_E}{C_E} \alpha > 0, \quad 1 + R_c > 0, \quad 2|\kappa_*| > 1 + R_c;$$

6. every competing projected channel satisfies

$$2\kappa > 1 + R_c.$$

Then

$$\boxed{\text{Ind}(\mathbb{H}_C) = 1,}$$

and therefore

$$\boxed{\text{one defect} \leftrightarrow \text{one protected light mode.}}$$

In the practically important case

$$\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1,$$

this reduces to the compact criterion

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.}$$

CCLXXXIII. FINAL SUMMARY

We constructed the practical benchmark dictionary connecting the Bloch/Class II coefficients

$$(\lambda_{\parallel}, \alpha, \beta)$$

to the radial one-mode coefficients

$$(a_0, b_0, c_0, \mu_\infty).$$

The exact valley extraction is

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.$$

The exact core formulas are

$$a_0 = \frac{g_E^{\text{core}} + g_c^{\text{core}}}{z_{0,0}}, \quad b_0 = \frac{g_E^{\text{core}}}{z_{0,0}}, \quad c_0 = \frac{g_O^{\text{core}}}{z_{0,0}}.$$

Under the matching hypothesis

$$g_E^{\text{core}} = \xi_E g_E, \quad g_c^{\text{core}} = \xi_c g_c, \quad g_O^{\text{core}} = \xi_O g_O,$$

this becomes

$$a_0 = \frac{1}{z_{0,0}} \left(\xi_E \frac{\alpha}{C_E} + \xi_c \frac{\lambda_{\parallel}}{C_c} \right), \quad b_0 = \frac{\xi_E}{z_{0,0}} \frac{\alpha}{C_E}, \quad c_0 = \frac{\xi_O}{z_{0,0}} \frac{\beta}{C_O}.$$

Defining

$$R_c := \frac{\xi_c C_E}{\xi_E C_c} \frac{\lambda_{\parallel}}{\alpha}, \quad R_O := \frac{\xi_O C_E}{\xi_E C_O} \frac{\beta}{\alpha},$$

one gets

$$a_0 = b_0(1 + R_c), \quad c_0 = b_0 R_O.$$

Hence the one-mode criterion is controlled by

$$\mu_\infty, \quad \alpha, \quad R_c, \quad \kappa_*,$$

while R_O affects only the phase profile and not the protected-mode counting.

In the practically most relevant case

$$\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1,$$

the working sufficient criterion reduces to

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.}$$

This is the cleanest operational form of the full chain: remote-gap isolation, Bloch/Class II transport extraction, and radial one-defect/one-mode protection are now expressed in a single set of inequalities.

CCLXXXIV. PRACTICAL CALCULATION CHECKLIST FOR THE REFINED LOCKED + CLASS II WORKING BRANCH

We now convert the final unified theorem into an explicit calculation protocol. The goal is to determine the data

$$C_c, C_E, C_O, \quad \xi_c, \xi_E, \xi_O, \quad \kappa_*, \quad \mu_\infty,$$

and then verify the working sufficient criterion for

$$\text{one defect} \leftrightarrow \text{one protected light mode}.$$

CCLXXXV. STEP 0: FIX CONVENTIONS ONCE AND FOR ALL

Before any extraction, one must fix:

1. the normalization of the low-valley Pauli basis $\{\sigma_1, \sigma_2, \sigma_3\}$;
2. the normalization of the radial core basis and the core coefficient convention $z_{0,0} > 0$;
3. the sign convention for the lowest signed projected channel κ_* ;
4. the overlap normalization convention used to define C_c, C_E, C_O .

Without these conventions, sign comparisons between Bloch-extracted and radial coefficients are ambiguous. Once the convention is fixed, all quantities below are unambiguously defined.

CCLXXXVI. STEP 1: REMOTE-GAP VALIDATION

Compute the explicit benchmark lower bound

$$\Delta_{\text{bench},\rho}^{\text{full}}$$

and the full residual mixing budget

$$\eta_{\text{rem},\rho}.$$

Then verify

$$\boxed{\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

If this inequality fails, the low slot is not spectrally isolated, and the one-mode criterion cannot yet be applied. If it holds, then all higher angular/remote sectors are uniformly excluded from the zero-energy window.

This is the mandatory first gate.

CCLXXXVII. STEP 2: LOW-VALLEY EXTRACTION OF

$(\lambda_{\parallel}, \alpha, \beta)$

On the locked BCC/Class II branch, compute:

1. the Bloch fiber operator $H_{\text{eff}}(k)$;
2. the shell projection $H_{\text{shell}}(p)$;
3. the exact low-valley Riesz projector

$$P_* = \frac{1}{2\pi i} \oint_{\Gamma} (z - H_{\text{shell}}(0))^{-1} dz;$$

4. the transport matrices

$$M_i = P_* \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*} P_*.$$

Then, in the adapted defect frame $(\hat{n}, e_1, e_2)$, define

$$M_{\parallel} = \hat{n}_i M_i, \quad M_1 = e_{1i} M_i, \quad M_2 = e_{2i} M_i.$$

Assuming the locked Pauli selection rule has been established, extract

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}),$$

$$\alpha = \frac{1}{4}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4}(\sigma_2 M_1 - \sigma_1 M_2).$$

This produces the Bloch/valley transport triplet.

CCLXXXVIII. STEP 3: DETERMINATION OF

C_c, C_E, C_O

The overlap constants

$$C_c, \quad C_E, \quad C_O$$

must be fixed in the same convention as the valley extraction.

The cleanest practical protocol is the calibration protocol:

a. Calibration run C. Impose a constitutive deformation such that, in the low-energy transport sector, only the *c*-type channel is switched on with unit normalization, while the *E*- and *O*-type channels are off. Then compute the extracted $\lambda_{\parallel}$. Set

$$C_c := \lambda_{\parallel}^{(c\text{-unit})}.$$

b. Calibration run E. Impose a constitutive deformation such that only the even tangential channel is switched on with unit normalization. Then compute the extracted α . Set

$$C_E := \alpha^{(E\text{-unit})}.$$

c. Calibration run O. Impose a constitutive deformation such that only the odd tangential channel is switched on with unit normalization. Then compute the extracted β . Set

$$C_O := \beta^{(O\text{-unit})}.$$

This protocol is exact within the chosen normalization convention. After it is completed, define

$$g_c = \frac{\lambda_{\parallel}}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}.$$

CCLXXXIX. STEP 4: DETERMINATION OF

$a_0, b_0, c_0, \mu_\infty$ from the radial constitutive profile

From the radial constitutive reduction,

$$a_0 = \frac{g_E^{\text{core}} + g_c^{\text{core}}}{z_{0,0}}, \quad b_0 = \frac{g_E^{\text{core}}}{z_{0,0}}, \quad c_0 = \frac{g_O^{\text{core}}}{z_{0,0}}.$$

Also,

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

So one must compute:

1. the core value $\sigma_0 = \bar{\sigma}(0)$;
2. the core mediator contraction $\chi_X(0)$;
3. the bulk values $\sigma_\infty, \chi_\infty$;
4. the constitutive functions z_0, m, u_X evaluated at these core/bulk values.

This produces the radial core/bulk data

$$a_0, \quad b_0, \quad c_0, \quad \mu_\infty.$$

CCXC. STEP 5: DETERMINATION OF

ξ_E, ξ_c, ξ_O

The matching constants are defined by

$$g_E^{\text{core}} = \xi_E g_E, \quad g_c^{\text{core}} = \xi_c g_c, \quad g_O^{\text{core}} = \xi_O g_O.$$

The clean practical protocol is again calibration-based.

a. Core/valley matching run E. Use the same E -unit calibration as in Step 3, compute both:

1. the valley-extracted g_E ;
2. the radial core coefficient g_E^{core} .

Then set

$$\xi_E := \frac{g_E^{\text{core}}}{g_E}.$$

Similarly,

$$\xi_c := \frac{g_c^{\text{core}}}{g_c}, \quad \xi_O := \frac{g_O^{\text{core}}}{g_O}.$$

This is the cleanest way to determine the matching constants without making any uncontrolled identification between local-core data and Bloch-projected data.

CCXCI. STEP 6: DETERMINATION OF THE PROJECTED LOWEST SIGNED CHANNEL

κ_*

Now determine the projected admissible angular channel set after:

1. remote-gap elimination of higher sectors;
2. internal/projector restriction to the low slot;
3. grading-compatible angular decomposition.

Operationally, one computes the spectrum and multiplicities of the projected angular operator

$$\mathcal{D}_{S^2}^{\text{proj}}$$

in the low slot and records the surviving signed channels

$$\mathfrak{K}_{\text{adm}}^{\text{low}}.$$

Then identify whether there exists a unique signed lowest channel

$$\boxed{\kappa_* < 0}$$

of multiplicity one.

If such a unique channel does not exist, the compact one-mode criterion cannot yet be applied. This is the second mandatory gate.

CCXCII. STEP 7: CONSTRUCT THE BENCHMARK RATIOS

Once the previous steps are completed, define

$$\boxed{R_c := \frac{\xi_c C_E}{\xi_E C_c} \frac{\lambda_{\parallel}}{\alpha}, \quad R_O := \frac{\xi_O C_E}{\xi_E C_O} \frac{\beta}{\alpha},}$$

provided $\alpha \neq 0$.

Then

$$\boxed{a_0 = b_0(1 + R_c), \quad c_0 = b_0 R_O.}$$

Thus the decisive radial one-mode inequalities can be written entirely in terms of

$$R_c, \quad \alpha, \quad \mu_{\infty}, \quad \kappa_*.$$

CCXCIII. STEP 8: FINAL INEQUALITY CHECK

The final working sufficient criterion is:

$$\boxed{\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho},}$$

$$\boxed{\mu_{\infty} > 0,}$$

$$\boxed{\frac{\xi_E}{C_E} \alpha > 0,}$$

$$1 + R_c > 0,$$

$$2|\kappa_*| > 1 + R_c,$$

and for every projected competitor $\kappa \neq \kappa_*$,

$$2\kappa > 1 + R_c.$$

In the practically most relevant case

$$\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1,$$

this reduces to

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho},$$

assuming the standard sign convention $C_E > 0$, $\xi_E > 0$.

CCXCIV. OPTIONAL STEP 9: DIRECT RADIAL SHOOTING VALIDATION

Even after the sufficient inequalities are verified, the most convincing final diagnostic is a direct radial ODE check in the projected lowest channel.

Using

$$Q_{\kappa_*, \pm}^{\text{core}} = \pm a(r) \frac{d}{dr} + W_{\kappa_*}^{\text{core}}(r), \quad W_{\kappa_*}^{\text{core}}(r) = \mu(r) + \frac{\kappa_* b(r)}{r},$$

one may numerically integrate the decaying generator and verify directly that:

1. it is L^2 -regular at the core;
2. the opposite chiral generator is excluded;
3. no competing channel admits a normalizable branch.

This does not replace the theorem, but it gives a strong numerical confirmation of the same conclusion.

CCXCV. EXACT VERSUS CONDITIONAL OUTPUTS OF THE PROTOCOL

After the full checklist is completed, the outputs separate as follows.

a. Exact within the working branch.

1. $\lambda_{\parallel}, \alpha, \beta$;
2. C_c, C_E, C_O once the calibration convention is fixed;
3. g_c, g_E, g_O ;
4. $a_0, b_0, c_0, \mu_\infty$;
5. R_c, R_O .

b. Still branch-verified rather than abstractly guaranteed.

1. ξ_c, ξ_E, ξ_O ;
2. the uniqueness and sign of κ_* ;
3. the positivity of μ_∞ ;
4. the remote-gap inequality in the same parameter regime.

Thus the remaining problem is no longer conceptual. It is a finite explicit verification problem on the chosen working branch.

CCXCVI. FINAL SUMMARY

The actual working calculation protocol is now fixed.
One must compute, in this order:

1. the remote-gap inequality

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho};$$

2. the low-valley transport coefficients

$$\lambda_{\parallel}, \alpha, \beta;$$

3. the overlap constants C_c, C_E, C_O by unit-channel calibration runs;

4. the radial core/bulk coefficients

$$a_0, b_0, c_0, \mu_{\infty};$$

5. the matching constants

$$\xi_c, \xi_E, \xi_O$$

by core/valley calibration runs;

6. the projected signed lowest channel κ_* ;

7. the benchmark ratios

$$R_c = \frac{\xi_c C_E}{\xi_E C_c} \frac{\lambda_{\parallel}}{\alpha}, \quad R_O = \frac{\xi_O C_E}{\xi_E C_O} \frac{\beta}{\alpha}.$$

Then the final practical sufficient criterion is:

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}, \quad \mu_{\infty} > 0, \quad \frac{\xi_E}{C_E} \alpha > 0, \quad 1 + R_c > 0, \quad 2|\kappa_*| > 1 + R_c,$$

together with competitor exclusion

$$2\kappa > 1 + R_c \quad (\kappa \neq \kappa_*).$$

In the most practical case

$$\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1,$$

this reduces to

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_{\infty} > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

Therefore the one-defect/one-mode problem has been reduced to a concrete finite checklist of computations on the refined locked + Class II working branch.

CCXCVII. AUDIT TABLE FOR THE REFINED LOCKED + CLASS II WORKING BRANCH

CCXCVIII. PURPOSE OF THE AUDIT

The current goal is not to introduce new abstract structure, but to classify every ingredient of the working one-defect/one-mode chain into exactly one of the following three statuses:

1. **Already available:** structurally derived or already established in the current working branch;
2. **Needs explicit computation:** not conceptually open, but still requires concrete evaluation from the branch data or code;
3. **Still an assumption / matching hypothesis:** not yet derived inside the branch, and must be treated honestly as an assumption until verified.

The chain being audited is

$\text{remote-gap isolation} \implies \text{low-slot projection} \implies (\lambda_{\parallel}, \alpha, \beta) \implies (g_c, g_E, g_O) \implies (a_0, b_0, c_0, \mu_{\infty}) \implies \text{radial one-mode criterion}.$
--

CCXCIX. I. ALREADY AVAILABLE

The following ingredients are already available in the current formalism.

A. A. Reduced-operator and Schur-complement skeleton

1. The light/heavy reduction

$$\mathcal{Y}_{\text{eff}} = L - KH^{-1}K^{\dagger}$$

is already established.

2. The explicit specialization to the refined locked + Class II working branch is already established:

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_{\Psi}^{\text{ref}} + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a + \beta_X X_i^a K_i^a.$$

3. The quadratic heavy elimination is already structurally fixed:

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k)H_{XX}(k)^{-1}H_{X\Psi}(k).$$

4. In the pure auxiliary working truncation,

$$H_{XX}(k) = M_X^2 \mathbf{1}_X$$

is already part of the explicit branch structure.

B. B. Valley/Bloch extraction layer

1. The exact projected transport matrices

$$M_i = P_* \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*} P_*$$

are already formally fixed.

2. The low-valley coefficient extraction formulas are already fixed:

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}),$$

$$\alpha = \frac{1}{4}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4}(\sigma_2 M_1 - \sigma_1 M_2).$$

3. The exact dictionary

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O$$

is already fixed as the correct normalization layer, even though the constants C_c, C_E, C_O themselves are not yet numerically evaluated.

C. C. Radial constitutive and one-mode layer

1. The minimal constitutive ansatz for the working microscopic multiplet

$$(P^A, \Psi, X_i^a)$$

is already fixed.

2. The defect-core coefficient definitions are already fixed:

$$a_0 = \frac{g_E^{\text{core}} + g_c^{\text{core}}}{z_{0,0}}, \quad b_0 = \frac{g_E^{\text{core}}}{z_{0,0}}, \quad c_0 = \frac{g_O^{\text{core}}}{z_{0,0}}.$$

3. The bulk mass parameter is already fixed by definition:

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

4. The radial Callias-core superpotential is already fixed:

$$W_\kappa^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.$$

5. The sufficient one-mode criterion is already structurally fixed:

$$\mu_\infty > 0, \quad a_0 > 0, \quad b_0 > 0, \quad \frac{|\kappa_*|b_0}{a_0} > \frac{1}{2},$$

with competitor exclusion for all other projected channels.

D. D. Remote-gap theorem layer

1. The full remote-gap sufficient criterion is already fixed:

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

2. The clean self-adjoint operator form and the corresponding perturbative spectral-isolation proof are already in place.
3. The block-budget and finite-truncation Gershgorin variants are already structurally closed as support theorems.

CCC. II. NEEDS EXPLICIT COMPUTATION

The following items are not conceptually open; they simply still require explicit branch evaluation.

A. A. Explicit evaluation of the Bloch-side constants

1. The overlap constants

$$C_c, \quad C_E, \quad C_O$$

still need to be computed by the calibration protocol.

2. Concretely, this requires the three unit-channel calibration runs:

$$(c\text{-unit}), \quad (E\text{-unit}), \quad (O\text{-unit}).$$

3. After those runs,

$$g_c = \frac{\lambda_\parallel}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}$$

will become explicit numerical/analytic outputs.

B. B. Explicit evaluation of the matching constants

1. The matching constants

$$\xi_c, \quad \xi_E, \quad \xi_O$$

still need to be computed.

2. This requires paired core/valley calibration: for each unit-channel run, compute both the valley-extracted coefficient and the radial core coefficient.
3. Then set

$$\xi_E = \frac{g_E^{\text{core}}}{g_E}, \quad \xi_c = \frac{g_c^{\text{core}}}{g_c}, \quad \xi_O = \frac{g_O^{\text{core}}}{g_O}.$$

C. C. Explicit low-slot spectral projection

1. The exact valley projector

$$P_* = \frac{1}{2\pi i} \oint_{\Gamma} (z - H_{\text{shell}}(0))^{-1} dz$$

is structurally fixed, but still needs explicit evaluation in the current branch.

2. The projected angular/channel set

$$\mathcal{R}_{\text{adm}}^{\text{low}}$$

still needs to be explicitly computed after remote-gap reduction and low-slot projection.

3. In particular, the signed lowest channel κ_* must still be read off from the actual projected operator.

D. D. Explicit core/bulk profile evaluation

1. The background profiles

$$\bar{\sigma}(r), \quad X_{\perp}(r), \quad X_{\parallel}(r)$$

still need to be supplied explicitly, numerically or analytically, in the working branch.

2. From them one must compute

$$\sigma_0, \quad X_0, \quad \sigma_{\infty}, \quad \chi_{\infty}.$$

3. Then one can explicitly evaluate

$$a_0, \quad b_0, \quad c_0, \quad \mu_{\infty}.$$

E. E. Final benchmark ratios

Once the previous layers are computed, the benchmark ratios

$$R_c = \frac{\xi_c C_E}{\xi_E C_c} \frac{\lambda_{\parallel}}{\alpha}, \quad R_O = \frac{\xi_O C_E}{\xi_E C_O} \frac{\beta}{\alpha}$$

still need to be explicitly evaluated.

Then the practically relevant criterion

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_{\infty} > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}$$

can be directly checked.

CCCI. III. STILL ASSUMPTIONS / BRANCH-SPECIFIC HYPOTHESES

The following items are not yet derived and must still be treated honestly as assumptions or matching hypotheses.

A. A. Matching hypothesis itself

At present, the relations

$$g_E^{\text{core}} = \xi_E g_E, \quad g_c^{\text{core}} = \xi_c g_c, \quad g_O^{\text{core}} = \xi_O g_O$$

are still a matching hypothesis until the calibration runs are actually completed.

So the existence of positive finite matching constants

$$\xi_E, \xi_c, \xi_O$$

is still a branch-specific assumption, not yet a theorem.

B. B. Unique signed lowest projected channel

The statement

$\text{there exists a unique signed lowest projected channel } \kappa_* < 0 \text{ of multiplicity one}$

is still an assumption until the projected channel spectrum is explicitly computed.

This is one of the most important remaining branch-specific checks.

C. C. Actual positivity of the bulk mass gap

The inequality

$$\mu_\infty > 0$$

is still a condition to be checked from the actual constitutive functions and asymptotic branch. At present it is structurally well-defined, but not yet verified in the concrete working branch.

D. D. Simultaneous parameter regime overlap

Even if each separate condition is individually plausible, it is still an assumption that all of the following can hold in the same parameter regime:

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}, \quad \mu_\infty > 0, \quad -1 < R_c < 1, \quad \kappa_* < 0 \text{ unique.}$$

The existence of a common regime is not yet proven; it must be checked explicitly.

CCCII. IV. MINIMAL DECISION TREE

The whole working program now reduces to the following decision tree.

a. Gate 1. Check

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

If false, the low slot is not isolated and the one-mode criterion cannot be applied.

b. Gate 2. Compute the projected admissible channel set and determine whether there exists a unique signed lowest $\kappa_* < 0$. If false, the compact one-mode criterion is not yet available.

c. *Gate 3.* Compute

$$\lambda_{\parallel}, \alpha, \beta, \quad C_c, C_E, C_O, \quad \xi_c, \xi_E, \xi_O.$$

Then construct

$$R_c, \quad R_O.$$

d. *Gate 4.* Compute

$$\mu_{\infty}.$$

If $\mu_{\infty} \leq 0$, the bulk gap does not support the protected-mode criterion.

e. *Gate 5.* Check

$$\frac{\xi_E}{C_E} \alpha > 0, \quad 1 + R_c > 0, \quad 2|\kappa_*| > 1 + R_c,$$

and all competitor inequalities.

If all gates pass, then

$$\text{Ind}(\mathbb{H}_C) = 1$$

follows by the working theorem.

CCCI. V. CURRENT HONEST STATUS

The current honest status is therefore:

1. The structural chain is already built.
2. The support theorems are largely complete.
3. The remaining work is finite and explicit.
4. The remaining bottleneck is no longer conceptual, but computational/verification-based.

In particular, the main unfinished tasks are exactly:

$$\boxed{C_c, C_E, C_O, \quad \xi_c, \xi_E, \xi_O, \quad \kappa_*, \quad \mu_{\infty}, \quad \Delta_{\text{bench}, \rho}^{\text{full}} - \eta_{\text{rem}, \rho}.}$$

Once these are evaluated, the working one-defect/one-mode criterion becomes a direct numerical/analytic verification problem.

CCCIV. FINAL SUMMARY

The audit divides the remaining ingredients into three classes.

a. *Already available.* The following are already structurally established:

$$\mathcal{Y}_{\text{eff}} = L - KH^{-1}K^{\dagger},$$

the refined locked + Class II branch, the remote-gap theorem, the valley extraction formulas for $(\lambda_{\parallel}, \alpha, \beta)$, the radial constitutive formulas for $(a_0, b_0, c_0, \mu_{\infty})$, and the abstract one-mode sufficient criterion.

b. *Needs explicit computation.* The following still need concrete evaluation:

$$C_c, C_E, C_O, \quad \xi_c, \xi_E, \xi_O, \quad \kappa_*, \quad \mu_{\infty}, \quad \Delta_{\text{bench}, \rho}^{\text{full}}, \eta_{\text{rem}, \rho}.$$

c. Still assumptions. Until explicitly verified in the working branch, the following remain assumptions: the existence of positive matching constants, the uniqueness of the signed lowest projected channel, the positivity of μ_∞ , and the existence of a common parameter regime where all inequalities hold simultaneously.

Therefore the main program is no longer conceptually open. It has become a finite explicit verification problem on the refined locked + Class II working branch.

CCCV. PRIORITY COMPUTATION I: EXPLICIT EVALUATION PROTOCOL FOR THE REMOTE-GAP GATE

The first mandatory gate in the working one-defect/one-mode program is the remote-gap isolation inequality

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

If this fails, the low slot is not spectrally isolated, and none of the later channel-counting arguments can be safely applied. So this gate must be evaluated first.

CCCVI. STEP 1: FIX THE LOW-SLOT WINDOW AND THE MOMENTUM RADIUS ρ

Choose the reference low slot (the candidate Dirac/valley slot) and fix a momentum radius

$$0 < \rho \leq \rho_{\text{max}}$$

small enough that the low-slot expansion remains valid. All estimates below are then carried out uniformly for

$$|p| \leq \rho.$$

The output of this step is a concrete candidate interval

$$0 < \rho \leq \rho_*,$$

inside which both the low-slot expansion and the remote benchmark formulas are trusted.

CCCVII. STEP 2: COMPUTE THE BENCHMARK LOWER BOUNDS FOR EACH REMOTE SECTOR

Let the full remote benchmark operator be block-diagonal across the remote sectors:

$$D_{\text{rem}}(p) = D_{\text{vol}}(p) \oplus D_{\text{high}}(p) \oplus D_{\text{wrong-int}}(p) \oplus D_{\text{non-enr}}^{(N)}(p) \oplus \cdots.$$

For each sector, compute an explicit lower bound on the spectral distance from zero:

$$\Delta_{\text{vol},\rho} \leq \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_{\text{vol}}(p))),$$

$$\Delta_{\text{high},\rho} \leq \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_{\text{high}}(p))),$$

$$\Delta_{\text{wrong-int},\rho} \leq \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_{\text{wrong-int}}(p))),$$

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \leq \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_{\text{non-enr}}^{(N)}(p))).$$

Then define

$$\Delta_{\text{bench},\rho}^{\text{full}} := \min \left\{ \Delta_{\text{vol},\rho}, \Delta_{\text{high},\rho}, \Delta_{\text{wrong-int},\rho}, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}, \dots \right\}.$$

This is the benchmark side of the gate.

a. Working-branch placeholders. In the current working branch, the intended sectorwise lower bounds are of the schematic form

$$\begin{aligned}\Delta_{\text{vol},\rho} &\sim 4K - |r_*|, \\ \Delta_{\text{high},\rho} &\sim \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, \\ \Delta_{\text{wrong-int},\rho} &\sim q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|,\end{aligned}$$

together with the previously established non-enriched lower bound. These expressions must be inserted only after their positivity regime is explicitly checked.

CCCVIII. STEP 3: COMPUTE THE RESIDUAL REMOTE MIXING BUDGET

Write the full remote remainder as

$$R_{\text{rem}}(p) = R_{\text{rem}}^{\text{offdiag}}(p) + R_{\text{rem}}^{\text{tail}}(p) + R_{\text{rem}}^{\text{diag corr}}(p).$$

Define

$$\eta_{\text{rem},\rho} := \sup_{|p| \leq \rho} \|R_{\text{rem}}(p)\|.$$

To bound this explicitly, split it into:

A. (a) finite sectorwise off-diagonal budget

For the truncated remote sector

$$\mathcal{H}_R^{\text{tr}} = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}},$$

define pairwise off-diagonal budgets

$$\eta_{ab,\rho} = \sup_{|p| \leq \rho} \|R_{ab}(p)\|.$$

Then

$$\boxed{\eta_{R,\rho}^{\text{tr}} \leq \sum_{a \neq b} \eta_{ab,\rho}.$$

B. (b) tail-coupling / power-counting budget

Using the previously established tail theorem, bound the tail part by

$$\boxed{\eta_{\text{tail},\rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}, \quad r_a \geq 1.}$$

C. (c) bounded diagonal correction budget

If diagonal benchmark corrections are present, define

$$\eta_{\text{diag},\rho} := \sup_{|p| \leq \rho} \|R_{\text{rem}}^{\text{diag corr}}(p)\|.$$

Then a clean total bound is

$$\boxed{\eta_{\text{rem},\rho} \leq \eta_{R,\rho}^{\text{tr}} + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho}.$$

This is the residual side of the gate.

CCCIX. STEP 4: DEFINE THE GATE MARGIN

Define the remote-gap margin

$$\mathfrak{M}_{\text{rem}}(\rho) := \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.$$

Then the pass/fail criterion is simply:

$$\mathfrak{M}_{\text{rem}}(\rho) > 0 \iff \text{remote-gap gate passes at radius } \rho.$$

The optimal practical strategy is to search for a window

$$0 < \rho \leq \rho_*$$

such that

$$\mathfrak{M}_{\text{rem}}(\rho) > 0$$

uniformly on that window.

CCCX. STEP 5: OPTIONAL SHARPER TRUNCATED CHECK BY GERSHGORIN

For the finite truncated remote matrix $H_R^{\text{tr}}(p)$, define

$$\mathcal{R}_m(p) := \sum_{n \neq m} |(H_R^{\text{tr}}(p))_{mn}|,$$

and

$$\Delta_{G,\rho} := \inf_{|p| \leq \rho} \min_m \left(|(H_R^{\text{tr}}(p))_{mm}| - \mathcal{R}_m(p) \right).$$

If

$$\Delta_{G,\rho} > 0,$$

then the truncated remote matrix is already spectrally isolated from zero. This does not replace the full operator-norm gate, but it is often a useful sharper diagnostic when the remote matrix is sparse.

CCCXI. STEP 6: OUTPUT CLASSIFICATION OF THE REMOTE-GAP GATE

The remote-gap audit should output exactly one of the following statuses:

a. Status A: passed. There exists a working window $0 < \rho \leq \rho_*$ such that

$$\mathfrak{M}_{\text{rem}}(\rho) > 0.$$

Then the low slot is spectrally isolated and the one-mode program may proceed.

b. Status B: inconclusive. The available lower/upper bounds are too crude to decide the sign of $\mathfrak{M}_{\text{rem}}(\rho)$. Then sharper benchmark estimates or sharper residual budgets are needed.

c. Status C: failed. The best available estimate gives

$$\mathfrak{M}_{\text{rem}}(\rho) \leq 0$$

throughout the candidate window. Then the current branch does not yet isolate the low slot and the later protected-mode counting cannot be trusted.

CCCXII. PRIORITY LIST OF ACTUAL QUANTITIES TO COMPUTE NEXT INSIDE GATE 1

To make Gate 1 executable in the current working branch, the next concrete computations should be ordered as follows:

1. compute or confirm explicit positive formulas for

$$\Delta_{\text{vol},\rho}, \Delta_{\text{high},\rho}, \Delta_{\text{wrong-int},\rho}, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)};$$

2. compute the pairwise budgets

$$\eta_{ab,\rho}$$

for the truncated remote off-diagonal blocks;

3. compute the tail constants

$$K_a, \tilde{K}_a, r_a, \sigma_a;$$

4. compute or bound $\eta_{\text{diag},\rho}$;

5. assemble

$$\Delta_{\text{bench},\rho}^{\text{full}} \quad \text{and} \quad \eta_{\text{rem},\rho};$$

6. inspect the sign of

$$\mathfrak{M}_{\text{rem}}(\rho).$$

Only after this gate is passed is it efficient to move on to

$$(\lambda_{\parallel}, \alpha, \beta) \rightarrow (C_c, C_E, C_O) \rightarrow \kappa_* \rightarrow \mu_{\infty} \rightarrow (\xi_c, \xi_E, \xi_O).$$

CCCXIII. CONCLUSION OF THE FIRST-PRIORITY PROTOCOL

The first-priority computational target is therefore not a new theorem but the explicit evaluation of

$$\boxed{\mathfrak{M}_{\text{rem}}(\rho) = \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.$$

Once this quantity is shown to be positive on a nontrivial ρ -window, the rest of the one-defect/one-mode chain becomes operationally meaningful.

CCCXIV. FINAL SUMMARY

The first mandatory gate of the working program is the remote-gap margin

$$\mathfrak{M}_{\text{rem}}(\rho) = \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.$$

To evaluate it, one must:

1. compute the sectorwise benchmark lower bounds and define

$$\Delta_{\text{bench},\rho}^{\text{full}} = \min\{\Delta_{\text{vol},\rho}, \Delta_{\text{high},\rho}, \Delta_{\text{wrong-int},\rho}, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}, \dots\};$$

2. compute the residual mixing budget

$$\eta_{\text{rem},\rho} \leq \eta_{R,\rho}^{\text{tr}} + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho},$$

with

$$\eta_{R,\rho}^{\text{tr}} \leq \sum_{a \neq b} \eta_{ab,\rho}, \quad \eta_{\text{tail},\rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a};$$

3. inspect the sign of

$$\mathfrak{M}_{\text{rem}}(\rho).$$

The gate passes iff

$$\boxed{\mathfrak{M}_{\text{rem}}(\rho) > 0}$$

on a nontrivial working window $0 < \rho \leq \rho_*$.

This is the highest-priority explicit computation, because if it fails, the later one-mode channel analysis is not yet trustworthy.

CCCXV. HONEST CURRENT STATUS OF THE MICROSCOPIC PDE

At the present stage, the full microscopic TECT condensate PDE is not yet uniquely derived from first principles. The current explicit model

$$\boxed{L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi, P] + \beta_X X_i^a K_i^a[P, \Psi]} \quad (4)$$

should therefore be described as the strongest explicit *working microscopic ansatz*, not yet as the final complete TECT action.

What is already structurally fixed is the reduced infrared Dirac-type form. The final finite arbiter of whether the current working branch is dynamically sufficient is the Bloch-frame calibration:

$$\boxed{C_{\parallel} \neq 0, \quad (C_E, C_O) \neq (0, 0).} \quad (5)$$

If this calibration succeeds, the current branch is dynamically sufficient inside this explicit setting. If it fails, then the current L_{mic} is structurally incomplete and the true complete microscopic TECT action must contain additional UV structure.

CCCXVI. CORRECTED TRANSVERSE MICROSCOPIC AUDIT

The transverse microscopic audit must be stated carefully.

A. Naive local JK does not suffice

In the minimal shell-adapted form, the J -sector is axial,

$$J \sim j \Sigma_3,$$

while the K -sector is transverse,

$$K \sim k_E \Sigma_1 + k_O \Sigma_2.$$

Then the simplest local real Hermitian JK bilinear reduces to

$$\Sigma_3(k_E \Sigma_1 + k_O \Sigma_2) + (k_E \Sigma_1 + k_O \Sigma_2) \Sigma_3 = k_E \{\Sigma_3, \Sigma_1\} + k_O \{\Sigma_3, \Sigma_2\} = 0.$$

Therefore:

$$\boxed{\text{the naive local real Hermitian } JK \text{ bilinear vanishes.}} \quad (6)$$

So the statement

$$JK \text{ automatically produces } c_E, c_O$$

is too strong and should not be used.

B. Actual transverse frontier

The genuinely load-bearing transverse audit is the shell-derivative audit

$$\boxed{\partial_{p\nu} \widehat{\Sigma}_{JK}(G_*; p)|_{p=0}, \quad \partial_{p\nu} \widehat{\Sigma}_{KK}(G_*; p)|_{p=0}.} \quad (7)$$

These derivative-generated shell tensors can avoid the simple Pauli anticommutation obstruction and generate nonzero transverse witnesses.

In the explicit folded-Bragg Class II branch, this derivative bookkeeping compresses to the Fourier-overlap problem

$$\boxed{\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \widehat{J}_i^A(Q) \widehat{K}_{i,\alpha}^A(Q').} \quad (8)$$

In the minimal first-shell-dominant reduction, the final transverse yes/no test is reduced to two explicit overlaps:

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.} \quad (9)$$

If these hold, then

$$(\alpha, \beta) \neq (0, 0)$$

is realized in the explicit Class II branch.

CCCXVII. HOW TO KEEP THE TD/DT RESULT WITHOUT OVERCLAIMING

The previously written TD/DT section can still be used, but only with the correct status label.

A. Branch-level transverse summary

Within the current explicit branch, the mixed shell-defect blocks TD and DT may be summarized as

$$V_\perp^\mu(\xi) = \theta'(\xi) \left[(A_E + B_E) E_a^\mu \Sigma_a + (A_O + B_O) J_a^\mu \Sigma_a \right]. \quad (10)$$

Define

$$C_E := A_E + B_E, \quad C_O := A_O + B_O.$$

Under the previously fixed projector normalization conventions, the reduced transverse coefficients are summarized by

$$\boxed{e_0^{(1)} = \frac{1}{2} C_E = \frac{1}{2} (A_E + B_E), \quad o_0^{(1)} = \frac{1}{2} C_O = \frac{1}{2} (A_O + B_O).} \quad (11)$$

Using

$$\alpha = I_\theta e_0^{(1)}, \quad \beta = I_\theta o_0^{(1)}, \quad I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

one obtains

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_E + B_E), \quad \beta = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_O + B_O).} \quad (12)$$

a. Correct status label. Equation (12) should be described as:

$$\boxed{\text{the branch-level reduced transverse summary within the current explicit LO/Class II realization.}} \quad (13)$$

It should *not* be described as the final unconditional microscopic closure of the full TECT PDE.

CCCXVIII. UNIFIED HONEST ENDPOINT

Putting everything together, the strongest honest current statement is:

$$\boxed{\text{The reduced Dirac-type PDE form is structurally fixed.}} \quad (14)$$

$$\boxed{\text{The current } L_{\text{mic}} \text{ is the strongest explicit working microscopic branch.}} \quad (15)$$

$$\boxed{\text{The transverse sector is not decided by a naive local } JK \text{ term, but by the derivative-resolved shell audit of } JK \text{ and/or } KK.} \quad (16)$$

$$\boxed{\text{Inside the current explicit branch, the TD/DT transverse summary may be used as a reduced branch-level coefficient summary}} \quad (17)$$

$$\boxed{\text{The final microscopic frontier remains the explicit Bloch calibration of } (\lambda_{\parallel}, \alpha, \beta), \text{ equivalently } (C_{\parallel}, C_E, C_O).} \quad (18)$$

CCCXIX. PRACTICAL CONSEQUENCE FOR THE MANUSCRIPT

Therefore the manuscript should distinguish three levels clearly:

1. **Algebraically closed:** the reduced anisotropic Dirac-type PDE and Clifford embedding;
2. **Explicitly realized in the current branch:** the branch-level transverse coefficient summary from TD/DT and the Class II shell-projection framework;
3. **Still open microscopically:** the final unconditional evaluation of the Bloch-frame overlaps and the full-action coefficient extraction.

This wording is both strong and honest.

CCCXX. FINAL SUMMARY

The attached material sharpens the honest logical status of the current program.

First, the current microscopic model

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi, P] + \beta_X X_i^a K_i^a[P, \Psi]$$

should be described as the strongest explicit working microscopic ansatz, not yet as the final complete TECT action.

Second, the reduced anisotropic Dirac-type PDE form is already structurally fixed, and the final finite arbiter of the current branch is the Bloch calibration

$$C_{\parallel} \neq 0, \quad (C_E, C_O) \neq (0, 0).$$

Third, the transverse microscopic frontier is not the naive local JK bilinear, which vanishes by Pauli anticommutation, but the derivative-resolved shell audit

$$\partial_{p_{\nu}} \widehat{\Sigma}_{JK}(G_*; p) \big|_{p=0}, \quad \partial_{p_{\nu}} \widehat{\Sigma}_{KK}(G_*; p) \big|_{p=0}.$$

Fourth, within the current explicit LO/Class II branch, the TD/DT block formulas may still be used as a branch-level reduced summary:

$$e_0^{(1)} = \frac{1}{2}(A_E + B_E), \quad o_0^{(1)} = \frac{1}{2}(A_O + B_O),$$

$$\alpha = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_E + B_E), \quad \beta = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_O + B_O).$$

Thus the correct final wording is:

$$\boxed{\text{the reduced PDE is structurally fixed, the current branch is an explicit working realization, and the final microscopic frontier}} \quad (19)$$

CCCXXI. CURRENT WORKING BENCHMARK PROPOSITION FOR GATE 1

We now fix the benchmark side of the remote-gap gate in the notation of the current refined locked + Class II working branch.

CCCXXII. GOAL

The first mandatory gate is the remote-gap inequality

$$\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}.$$

If this fails, the low slot is not spectrally isolated, and the later protected-channel counting is not yet trustworthy. So the first nonredundant task is to fix a clean working formula for

$$\Delta_{\text{bench},\rho}^{\text{fin}}.$$

CCCXXIII. SECTORWISE BENCHMARK INPUTS

In the current working truncation, the remote benchmark is built from four sectorwise lower bounds.

A. 1. Higher-shell benchmark

Let

$$\delta_{\text{shell}} = \inf_{Q \in \mathcal{Q}_{\text{high}}} ||Q|^2 - q_*^2| > 0,$$

and assume the shell energy is of the form

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2, \quad K > 0.$$

Then

$$\Delta_{\text{high},\rho}^{(0)} \geq K\delta_{\text{shell}}^2 - |r_*|.$$

For the current BCC reciprocal arithmetic,

$$\delta_{\text{shell}} = 2,$$

hence

$$\Delta_{\text{high},\rho}^{(0)} \geq 4K - |r_*|.$$

B. 2. Volumetric benchmark

For the heavy volumetric sector, the current working branch uses a positive lower bound of the form

$$\Delta_{\text{vol},\rho}^{(0)} = \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}.$$

This is the current refined heavy-locking benchmark. If one prefers a coarser but simpler benchmark, one may replace it by a positive lower bound such as

$$\Delta_{\text{vol},\rho}^{(0)} \geq c_{\text{vol}} M_\sigma, \quad c_{\text{vol}} > 0,$$

provided that this alternative lower bound has already been justified in the chosen truncation.

C. 3. Wrong-internal benchmark

At the shell point G_* , the wrong-internal splitting is

$$E_L - E_* = q_*^2 \left[(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right].$$

Therefore the corresponding benchmark is

$$\Delta_{\text{wrong-int},\rho}^{(0)} = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|.$$

Under a bounded perturbation

$$H = H_0 + \delta H, \quad \|Q\delta H Q\| \leq \varepsilon_{\text{int}},$$

the gap is stable:

$$m_{\text{int}} \geq \Delta_{\text{wrong-int},\rho}^{(0)} - \varepsilon_{\text{int}}.$$

D. 4. Same-shell non-enriched benchmark

Let

$$\Delta_{\text{non-enr},\rho}^{(N)}$$

denote the same-shell non-enriched lower bound produced by the finite truncated eigenvalue computation, and let

$$\varepsilon_{\text{non-enr},\rho}^{(N)}$$

be its truncation/error budget. Then the usable benchmark entry is

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}.$$

CCCXXIV. CURRENT WORKING MASTER BENCHMARK

Combining the four sectorwise lower bounds, define

$$\Delta_{\text{bench},\rho}^{\text{fin}} := \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}.$$

Equivalently, if the coarser volumetric benchmark is used, replace the second entry by $c_{\text{vol}} M_\sigma$.

CCCXXV. GATE-1 PROPOSITION

a. Proposition (current working benchmark gate). Assume the remote remainder satisfies

$$\eta_{R,\rho} = \sup_{|p| \leq \rho} \|R_R(p)\|.$$

If

$$\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho},$$

then the remote sector has a positive uniform spectral gap:

$$\Delta_\rho \geq \Delta_{\text{bench},\rho}^{\text{fin}} - \eta_{R,\rho} > 0.$$

b. Proof. By construction,

$$\Delta_{\text{bench},\rho}^{\text{fin}} \leq \delta_{0,\rho} := \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_R(p))).$$

Therefore

$$\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho} \implies \delta_{0,\rho} > \eta_{R,\rho}.$$

The uniform remote-gap criterion then gives

$$\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.$$

Hence the low slot is spectrally isolated from the remote sector. $\square$

CCCXXVI. HONEST CURRENT STATUS OF GATE 1

The current benchmark side is now compressed to a very small remaining frontier.

a. Already explicit or nearly explicit. The following are already available in usable working form:

1. the higher-shell benchmark

$$4K - |r_*|;$$

2. the wrong-internal benchmark

$$q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|;$$

3. the heavy volumetric benchmark in the current refined heavy-locking working form

$$\sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2}.$$

b. Still genuinely unresolved. The only genuinely model-specific unresolved benchmark entry is

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)},$$

together with the residual remote-mixing norm

$$\eta_{R,\rho}.$$

Thus Gate 1 is now reduced to exactly:

one finite same-shell non-enriched eigenvalue computation + one remote-mixing norm bound.

CCCXXVII. OPERATIONAL CONCLUSION

Therefore the next explicit computation inside Gate 1 should not be a new theorem. It should be:

1. compute

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)};$$

2. compute or bound

$$\eta_{R,\rho};$$

3. assemble

$$\mathfrak{M}_{\text{rem}}(\rho) = \Delta_{\text{bench},\rho}^{\text{fin}} - \eta_{R,\rho};$$

4. determine whether $\mathfrak{M}_{\text{rem}}(\rho) > 0$ on a nontrivial ρ -window.

Only after this gate is passed is it maximally efficient to continue to

$$(\lambda_{||}, \alpha, \beta) \rightarrow (C_c, C_E, C_O) \rightarrow \kappa_* \rightarrow \mu_\infty.$$

CCCXXVIII. FINAL SUMMARY

The correct priority is to push Gate 1 rather than further polish the TD/DT section.
The current working remote benchmark is

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2}, q_* \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}.$$

If

$$\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho},$$

then

$$\Delta_\rho \geq \Delta_{\text{bench},\rho}^{\text{fin}} - \eta_{R,\rho} > 0,$$

so the low slot is spectrally isolated.

At the current stage, the higher-shell and wrong-internal entries are already explicit, and the volumetric entry is available in a usable working form. The only genuinely unresolved benchmark-side item is the same-shell non-enriched lower bound, together with the residual remote-mixing norm.

Therefore the real remaining Gate-1 frontier is:

$$\boxed{\text{one finite same-shell non-enriched eigenvalue computation} \quad + \quad \text{one remote-mixing norm bound.}}$$

CCCXXIX. FINITE TRUNCATED EIGENVALUE FORMULATION OF THE SAME-SHELL NON-ENRICHED BENCHMARK

We now isolate the last unresolved benchmark entry in Gate 1, namely the same-shell non-enriched lower bound

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}.$$

CCCXXX. 1. SAME-SHELL DECOMPOSITION

Let

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}, \quad |p| \leq \rho,$$

be the shell-projected effective operator near the folded Bragg point.

Assume the shell space admits the orthogonal decomposition

$$\boxed{\mathcal{H}_{S_*} = \mathcal{H}_{\text{enr}} \oplus \mathcal{H}_{\text{non-enr}},}$$

where:

- $\mathcal{H}_{\text{enr}}$ is the enriched low slot (the candidate Dirac/valley slot),
- $\mathcal{H}_{\text{non-enr}}$ is the orthogonal complement inside the same shell.

Let

$$P_{\text{enr}}$$

be the orthogonal projector onto $\mathcal{H}_{\text{enr}}$, and define

$$\boxed{Q_{\text{non}} := \Pi_{S_*} - P_{\text{enr}}.}$$

Then the exact same-shell non-enriched operator is

$$\boxed{H_{\text{non}}(p) := Q_{\text{non}} H_{\text{shell}}(p) Q_{\text{non}}.}$$

CCCXXXI. 2. EXACT BENCHMARK QUANTITY

The exact same-shell non-enriched benchmark gap is

$$\delta_{\text{non-enr},\rho} := \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(H_{\text{non}}(p))).$$

This is the exact quantity that must be inserted into the full remote benchmark minimum. However, it is generally not directly computable in closed form. So we replace it by a finite truncation plus an explicit error budget.

CCCXXXII. 3. FINITE TRUNCATION

Choose a finite-dimensional trial subspace

$$\mathcal{H}_{\text{non}}^{(N)} \subset \mathcal{H}_{\text{non-enr}}, \quad \dim \mathcal{H}_{\text{non}}^{(N)} = N,$$

with orthogonal projector

$$Q_{\text{non}}^{(N)}.$$

Define the finite truncated operator

$$H_{\text{non}}^{(N)}(p) := Q_{\text{non}}^{(N)} H_{\text{non}}(p) Q_{\text{non}}^{(N)}.$$

Equivalently, after choosing an orthonormal basis of $\mathcal{H}_{\text{non}}^{(N)}$, one obtains an $N \times N$ Hermitian matrix

$$\mathbf{H}_{\text{non}}^{(N)}(p).$$

Define the finite benchmark

$$\Delta_{\text{non-enr},\rho}^{(N)} := \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(H_{\text{non}}^{(N)}(p))).$$

This is the quantity obtained by the actual finite eigenvalue computation.

CCCXXXIII. 4. TRUNCATION ERROR BUDGET

Define the truncation remainder by

$$\varepsilon_{\text{non-enr},\rho}^{(N)} := \sup_{|p| \leq \rho} \left\| H_{\text{non}}(p) - \iota_N H_{\text{non}}^{(N)}(p) \iota_N^* \right\|,$$

where

$$\iota_N : \mathcal{H}_{\text{non}}^{(N)} \hookrightarrow \mathcal{H}_{\text{non-enr}}$$

is the canonical inclusion.

In practice, one may bound $\varepsilon_{\text{non-enr},\rho}^{(N)}$ by:

1. a tail norm estimate from omitted harmonic modes;
2. an operator-norm estimate from neglected off-shell couplings;
3. a numerical residual estimate if the truncated basis is obtained variationally.

The important point is only that $\varepsilon_{\text{non-enr},\rho}^{(N)}$ be an explicit uniform bound for $|p| \leq \rho$.

CCCXXXIV. 5. BENCHMARK PROPOSITION

a. Proposition (finite same-shell non-enriched benchmark). Assume $H_{\text{non}}(p)$ is self-adjoint for every $|p| \leq \rho$, and let

$$\Delta_{\text{non-enr},\rho}^{(N)}, \quad \varepsilon_{\text{non-enr},\rho}^{(N)}$$

be defined as above. Then

$$\delta_{\text{non-enr},\rho} \geq \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}.$$

b. Proof. For each fixed $|p| \leq \rho$, the self-adjoint perturbation inequality gives

$$\text{dist}(0, \text{spec}(H_{\text{non}}(p))) \geq \text{dist}(0, \text{spec}(H_{\text{non}}^{(N)}(p))) - \left\| H_{\text{non}}(p) - \iota_N H_{\text{non}}^{(N)}(p) \iota_N^* \right\|.$$

Taking the infimum over $|p| \leq \rho$ yields

$$\delta_{\text{non-enr},\rho} \geq \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}.$$

□

CCCXXXV. 6. USABLE BENCHMARK ENTRY

Therefore the usable same-shell non-enriched benchmark entry is

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}$$

This is exactly the object that enters the current working master benchmark:

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}.$$

CCCXXXVI. 7. COMPUTATIONAL REALIZATION

The actual finite computation now reduces to the following explicit task:

1. choose the truncated same-shell non-enriched basis $\mathcal{H}_{\text{non}}^{(N)}$;
2. compute the Hermitian matrix $\mathbf{H}_{\text{non}}^{(N)}(p)$;
3. compute its smallest absolute eigenvalue uniformly for $|p| \leq \rho$:

$$\Delta_{\text{non-enr},\rho}^{(N)} = \inf_{|p| \leq \rho} \min_j |\lambda_j^{(N)}(p)|;$$

4. compute or bound the truncation error

$$\varepsilon_{\text{non-enr},\rho}^{(N)};$$

5. form

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}.$$

This is the single remaining benchmark-side finite eigenvalue problem in Gate 1.

CCCXXXVII. 8. HONEST CURRENT STATUS

Within the current working branch, the higher-shell, volumetric, and wrong-internal benchmark entries already exist in usable explicit form. Therefore the benchmark-side frontier has now been reduced to exactly one unresolved finite computation:

$$\boxed{\text{evaluate } \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}}.$$

Once this is done, the benchmark side of Gate 1 is complete, and the only remaining Gate-1 task is the residual norm budget $\eta_{R,\rho}$.

CCCXXXVIII. FINAL SUMMARY

The unresolved benchmark entry in Gate 1 is the same-shell non-enriched gap.
Let

$$H_{\text{non}}(p) = Q_{\text{non}} H_{\text{shell}}(p) Q_{\text{non}}$$

be the exact same-shell non-enriched operator, and let

$$H_{\text{non}}^{(N)}(p)$$

be its finite truncation on an N -dimensional trial space.

Define

$$\Delta_{\text{non-enr},\rho}^{(N)} = \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(H_{\text{non}}^{(N)}(p))),$$

and

$$\varepsilon_{\text{non-enr},\rho}^{(N)} = \sup_{|p| \leq \rho} \left\| H_{\text{non}}(p) - \iota_N H_{\text{non}}^{(N)}(p) \iota_N^* \right\|.$$

Then the exact benchmark is bounded below by

$$\delta_{\text{non-enr},\rho} \geq \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}.$$

Therefore the usable same-shell non-enriched benchmark entry is

$$\boxed{\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}}.$$

This is the only unresolved benchmark-side finite eigenvalue computation still needed to complete the benchmark part of Gate 1.

CCCXXXIX. CLEAN DECOMPOSITION OF THE RESIDUAL REMOTE BUDGET IN GATE 1

We now fix the residual side of Gate 1 in the notation of the current refined locked + Class II working branch.

CCCXL. 1. GOAL

The first mandatory gate is

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}},$$

where

$$\eta_{R,\rho} := \sup_{|p| \leq \rho} \|R_R(p)\|.$$

The benchmark side has already been reduced to

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}.$$

So it remains to fix a clean and explicit decomposition of $\eta_{R,\rho}$.

CCCXLI. 2. FULL RESIDUAL DECOMPOSITION

Write the full remote operator as

$$H_R(p) = D_R(p) + R_R(p), \quad |p| \leq \rho,$$

where $D_R(p)$ is the benchmark remote operator and $R_R(p)$ is the full residual remote remainder.

We decompose the remainder into three pieces:

$$R_R(p) = R_R^{\text{tr}}(p) + R_R^{\text{tail}}(p) + R_R^{\text{diag}}(p).$$

Here:

1. $R_R^{\text{tr}}(p)$ is the off-diagonal remainder inside the finite truncated remote sector;
2. $R_R^{\text{tail}}(p)$ contains all couplings to omitted tail sectors and all induced tail-mediated corrections;
3. $R_R^{\text{diag}}(p)$ contains the bounded difference between the exact diagonal/reference part and the simplified benchmark diagonal used in $\Delta_{\text{bench},\rho}^{\text{fin}}$.

Define the corresponding budgets

$$\eta_{R,\rho}^{\text{tr}} := \sup_{|p| \leq \rho} \|R_R^{\text{tr}}(p)\|,$$

$$\eta_{\text{tail},\rho} := \sup_{|p| \leq \rho} \|R_R^{\text{tail}}(p)\|,$$

$$\eta_{\text{diag},\rho} := \sup_{|p| \leq \rho} \|R_R^{\text{diag}}(p)\|.$$

Then by the triangle inequality,

$$\eta_{R,\rho} \leq \eta_{R,\rho}^{\text{tr}} + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho}.$$

CCCXLII. 3. FINITE TRUNCATED REMOTE BUDGET

In the finite truncated remote sector, let

$$\mathcal{H}_R^{\text{tr}} = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}.$$

Write the truncated off-diagonal remainder as

$$R_R^{\text{tr}}(p) = \begin{pmatrix} 0 & R_{\text{vol},\text{high}} & R_{\text{vol},\text{non}} & R_{\text{vol},\text{wrong}} \\ R_{\text{high},\text{vol}} & 0 & R_{\text{high},\text{non}} & R_{\text{high},\text{wrong}} \\ R_{\text{non},\text{vol}} & R_{\text{non},\text{high}} & 0 & R_{\text{non},\text{wrong}} \\ R_{\text{wrong},\text{vol}} & R_{\text{wrong},\text{high}} & R_{\text{wrong},\text{non}} & 0 \end{pmatrix} (p).$$

Define the pairwise budgets

$$\eta_{ab,\rho} := \sup_{|p| \leq \rho} \|R_{ab}(p)\|.$$

Then

$$\eta_{R,\rho}^{\text{tr}} \leq \sum_{a \neq b} \eta_{ab,\rho}.$$

Explicitly,

$$\eta_{R,\rho}^{\text{tr}} \leq \eta_{\text{vol,high},\rho} + \eta_{\text{vol,non},\rho} + \eta_{\text{vol,wrong},\rho} + \eta_{\text{high,vol},\rho} + \eta_{\text{high,non},\rho} + \eta_{\text{high,wrong},\rho} + \eta_{\text{non,vol},\rho} + \eta_{\text{non,high},\rho} + \eta_{\text{non,wrong},\rho} + \eta_{\text{wrong,vol},\rho} + \eta_{\text{wrong,non},\rho} + \eta_{\text{wrong,wrong},\rho}.$$

If self-adjointness gives

$$R_{ab}(p) = R_{ba}(p)^\dagger,$$

then the symmetric budgets satisfy

$$\eta_{ab,\rho} = \eta_{ba,\rho},$$

and the previous bound may be rewritten more economically as

$$\eta_{R,\rho}^{\text{tr}} \leq 2 \left(\eta_{\text{vol,high},\rho} + \eta_{\text{vol,non},\rho} + \eta_{\text{vol,wrong},\rho} + \eta_{\text{high,non},\rho} + \eta_{\text{high,wrong},\rho} + \eta_{\text{non,wrong},\rho} \right).$$

This is still crude, but it is rigorous and directly computable.

CCCXLIII. 4. TAIL BUDGET

Let $R_R^{\text{tail}}(p)$ denote all couplings involving omitted harmonics, omitted heavy auxiliaries, and tail-mediated effective corrections. By the previously established power-counting/tail theorem, one may assume a bound of the form

$$\eta_{\text{tail},\rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}, \quad r_a \geq 1.$$

This expresses two structural facts:

1. every tail correction vanishes at least as $O(\rho)$ near the low crossing;
2. heavy-mediated tails carry additional inverse-mass suppression.

In the current branch, this budget includes in particular:

1. omitted same-shell harmonics outside the chosen N -dimensional non-enriched truncation;
2. omitted higher-shell harmonics coupled through residual off-shell transport;
3. omitted heavy-locking or mediator-induced tail corrections beyond the retained quadratic working truncation.

CCCXLIV. 5. DIAGONAL CORRECTION BUDGET

The benchmark operator $D_R(p)$ is not necessarily the exact diagonal block of the full operator; it is a simplified lower-bound model. Hence define the diagonal correction

$$R_R^{\text{diag}}(p) := D_R^{\text{exact}}(p) - D_R^{\text{bench}}(p),$$

where $D_R^{\text{bench}}(p)$ is the diagonal/reference operator whose spectrum is bounded below by $\Delta_{\text{bench},\rho}^{\text{fn}}$.

Its uniform budget is

$$\eta_{\text{diag},\rho} := \sup_{|p| \leq \rho} \|R_R^{\text{diag}}(p)\|.$$

Typical contributions to $\eta_{\text{diag},\rho}$ in the current working branch are:

1. bounded deviations of the actual volumetric branch from the coarse benchmark

$$\sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2};$$

2. bounded deviations of the actual wrong-internal splitting from the simple shell-point estimate

$$q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|;$$

3. residual bounded corrections to the same-shell non-enriched benchmark not already absorbed into

$$\varepsilon_{\text{non-enr}, \rho}^{(N)}.$$

If the benchmark entries are taken directly from exact diagonal sectors, then one may set

$$\eta_{\text{diag}, \rho} = 0.$$

Otherwise, $\eta_{\text{diag}, \rho}$ must be kept explicitly.

CCCXLV. 6. MASTER RESIDUAL PROPOSITION

- Proposition (current working residual budget).* For the current refined locked + Class II working branch,

$$\eta_{R, \rho} \leq \eta_{R, \rho}^{\text{tr}} + \eta_{\text{tail}, \rho} + \eta_{\text{diag}, \rho},$$

with

$$\eta_{R, \rho}^{\text{tr}} \leq \sum_{a \neq b} \eta_{ab, \rho},$$

and

$$\eta_{\text{tail}, \rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}.$$

Therefore the Gate-1 margin obeys

$$\mathfrak{M}_{\text{rem}}(\rho) := \Delta_{\text{bench}, \rho}^{\text{fin}} - \eta_{R, \rho} \geq \Delta_{\text{bench}, \rho}^{\text{fin}} - \eta_{R, \rho}^{\text{tr}} - \eta_{\text{tail}, \rho} - \eta_{\text{diag}, \rho}.$$

So a sufficient practical gate condition is

$$\Delta_{\text{bench}, \rho}^{\text{fin}} > \eta_{R, \rho}^{\text{tr}} + \eta_{\text{tail}, \rho} + \eta_{\text{diag}, \rho}.$$

- Proof.* This is immediate from the decomposition

$$R_R = R_R^{\text{tr}} + R_R^{\text{tail}} + R_R^{\text{diag}}$$

and the triangle inequality. $\square$

CCCXLVI. 7. GATE-1 CLOSURE STATEMENT

Combining the benchmark proposition from the previous step with the present residual proposition, the full Gate-1 pass criterion becomes

$$\min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr}, \rho}^{(N)} - \varepsilon_{\text{non-enr}, \rho}^{(N)} \right\} > \eta_{R, \rho}^{\text{tr}} + \eta_{\text{tail}, \rho} + \eta_{\text{diag}, \rho}.$$

If this holds on a nontrivial interval $0 < \rho \leq \rho_*$, then

$$\Delta_\rho > 0$$

and the low slot is uniformly isolated from the remote sector.

CCCXLVII. 8. HONEST CURRENT STATUS

At this point, Gate 1 is reduced to exactly two explicit finite tasks:

1. compute

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)};$$

2. compute or bound

$$\eta_{R,\rho}^{\text{tr}}, \quad \eta_{\text{tail},\rho}, \quad \eta_{\text{diag},\rho}.$$

Everything else in Gate 1 is already structurally organized.

Therefore the unresolved Gate-1 frontier is no longer conceptual. It is the explicit evaluation of a finite non-enriched benchmark together with an explicit residual norm budget.

CCCXLVIII. FINAL SUMMARY

The residual side of Gate 1 is cleanly decomposed as

$$R_R(p) = R_R^{\text{tr}}(p) + R_R^{\text{tail}}(p) + R_R^{\text{diag}}(p),$$

so that

$$\eta_{R,\rho} \leq \eta_{R,\rho}^{\text{tr}} + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho}.$$

The finite truncated budget is

$$\eta_{R,\rho}^{\text{tr}} \leq \sum_{a \neq b} \eta_{ab,\rho}, \quad \eta_{ab,\rho} = \sup_{|p| \leq \rho} \|R_{ab}(p)\|.$$

The tail budget is bounded by power counting:

$$\eta_{\text{tail},\rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}, \quad r_a \geq 1.$$

The diagonal correction budget is

$$\eta_{\text{diag},\rho} = \sup_{|p| \leq \rho} \|R_R^{\text{diag}}(p)\|.$$

Hence a sufficient Gate-1 condition is

$$\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}^{\text{tr}} + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho}.$$

Combining this with the previous benchmark result, the remaining unresolved Gate-1 tasks are exactly:

1. the finite same-shell non-enriched computation

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)};$$

2. the explicit evaluation of

$$\eta_{R,\rho}^{\text{tr}}, \quad \eta_{\text{tail},\rho}, \quad \eta_{\text{diag},\rho}.$$

Thus Gate 1 is now fully reduced to a finite benchmark computation plus a finite residual-norm computation.

CCCXLIX. PAIRWISE AUDIT TABLE FOR THE TRUNCATED REMOTE BUDGET

We now resolve the truncated remote norm budget

$$\eta_{R,\rho}^{\text{tr}}$$

into explicit pairwise source terms in the notation of the current refined locked + Class II working branch.

CCCL. 1. PROJECTORS AND UNIQUE PAIR SET

Let

$$Q_{\text{vol}}, \quad Q_{\text{high}}, \quad Q_{\text{non}}, \quad Q_{\text{wrong}}$$

denote the orthogonal projectors onto

$$\mathcal{H}_{\text{vol}}, \quad \mathcal{H}_{\text{high}}, \quad \mathcal{H}_{\text{non-enr}}, \quad \mathcal{H}_{\text{wrong-int}},$$

respectively, inside the finite truncated remote sector.

For $a \neq b$, define the pairwise off-diagonal blocks

$$R_{ab}(p) := Q_a H_R^{\text{tr}}(p) Q_b, \quad \eta_{ab,\rho} := \sup_{|p| \leq \rho} \|R_{ab}(p)\|.$$

By self-adjointness,

$$R_{ab}(p) = R_{ba}(p)^\dagger, \quad \eta_{ab,\rho} = \eta_{ba,\rho}.$$

Hence it suffices to track the six unordered pairs

$$(\text{vol}, \text{high}), (\text{vol}, \text{non}), (\text{vol}, \text{wrong}), (\text{high}, \text{non}), (\text{high}, \text{wrong}), (\text{non}, \text{wrong}).$$

The truncated remote budget obeys

$$\boxed{\eta_{R,\rho}^{\text{tr}} \leq 2 \left(\eta_{\text{vol,high},\rho} + \eta_{\text{vol,non},\rho} + \eta_{\text{vol,wrong},\rho} + \eta_{\text{high,non},\rho} + \eta_{\text{high,wrong},\rho} + \eta_{\text{non,wrong},\rho} \right)}.$$

CCCLI. 2. UNIVERSAL DIRECT/HEAVY DECOMPOSITION FOR EACH PAIR

Write the effective operator as

$$H_{\text{eff}}(k) = H_{\text{dir}}(k) + \Delta_X(k),$$

where

$$\Delta_X(k) = -H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k).$$

In the auxiliary Class II working truncation,

$$H_{XX}(k) = M_X^2 \mathbf{1}_X, \quad H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k).$$

Therefore

$$\Delta_X(k) = -\frac{1}{M_X^2} (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)) (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger.$$

For each pair $a \neq b$, define the direct budget

$$\boxed{\mathcal{D}_{ab,\rho} := \sup_{|p| \leq \rho} \|Q_a H_{\text{dir}}(k_* + p) Q_b\|}.$$

Define the heavy-mediated channel budgets

$$\mathcal{M}_{ab,\rho}^{JJ} := \sup_{|p| \leq \rho} \|Q_a \mathcal{J} \mathcal{J}^\dagger Q_b\|,$$

$$\mathcal{M}_{ab,\rho}^{JK} := \sup_{|p| \leq \rho} \|Q_a (\mathcal{J} \mathcal{K}^\dagger + \mathcal{K} \mathcal{J}^\dagger) Q_b\|,$$

$$\mathcal{M}_{ab,\rho}^{KK} := \sup_{|p| \leq \rho} \|Q_a \mathcal{K} \mathcal{K}^\dagger Q_b\|.$$

Then every pairwise budget satisfies the exact current-branch bound

$$\eta_{ab,\rho} \leq \mathcal{D}_{ab,\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{ab,\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{KK}.$$

This is the master pairwise audit formula.

CCCLII. 3. PAIRWISE SOURCE TABLE

We now identify the source content of each pair.

A. (i) Volumetric–higher-shell pair

$$\eta_{\text{vol,high},\rho} \leq \mathcal{D}_{\text{vol,high},\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{\text{vol,high},\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{\text{vol,high},\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{\text{vol,high},\rho}^{KK}.$$

a. Direct source. $\mathcal{D}_{\text{vol,high},\rho}$ comes from direct volumetric–inter-shell transport mixing inside H_{dir} , i.e. shell-number-changing couplings between the heavy volumetric block and higher-shell harmonics.

b. Heavy-mediated source. The three $\mathcal{M}_{\text{vol,high},\rho}^{UV}$ terms come from Class II mediation with one leg in the volumetric sector and the other in the higher-shell sector.

B. (ii) Volumetric–same-shell non-enriched pair

$$\eta_{\text{vol,non},\rho} \leq \mathcal{D}_{\text{vol,non},\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{\text{vol,non},\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{\text{vol,non},\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{\text{vol,non},\rho}^{KK}.$$

a. Direct source. $\mathcal{D}_{\text{vol,non},\rho}$ comes from the direct coupling between the volumetric branch and the same-shell non-enriched harmonics.

b. Heavy-mediated source. The $\mathcal{M}_{\text{vol,non},\rho}^{UV}$ terms come from mediator exchange with one leg on the volumetric sector and the other on the same-shell non-enriched sector.

C. (iii) Volumetric–wrong-internal pair

$$\eta_{\text{vol,wrong},\rho} \leq \mathcal{D}_{\text{vol,wrong},\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{\text{vol,wrong},\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{\text{vol,wrong},\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{\text{vol,wrong},\rho}^{KK}.$$

- a. Direct source.* $\mathcal{D}_{\text{vol,wrong},\rho}$ comes from direct volumetric coupling to the wrong-internal branch.
b. Heavy-mediated source. The $\mathcal{M}_{\text{vol,wrong},\rho}^{UV}$ terms come from mediated exchange between the volumetric and wrong-internal sectors.

D. (iv) Higher-shell–same-shell non-enriched pair

$$\eta_{\text{high,non},\rho} \leq \mathcal{D}_{\text{high,non},\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{\text{high,non},\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{\text{high,non},\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{\text{high,non},\rho}^{KK}.$$

- a. Direct source.* $\mathcal{D}_{\text{high,non},\rho}$ is the direct inter-shell transport coupling between higher-shell modes and same-shell non-enriched modes.
b. Heavy-mediated source. The $\mathcal{M}_{\text{high,non},\rho}^{UV}$ terms come from mediator exchange between these two shell sectors.

E. (v) Higher-shell–wrong-internal pair

$$\eta_{\text{high,wrong},\rho} \leq \mathcal{D}_{\text{high,wrong},\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{\text{high,wrong},\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{\text{high,wrong},\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{\text{high,wrong},\rho}^{KK}.$$

- a. Direct source.* $\mathcal{D}_{\text{high,wrong},\rho}$ comes from direct shell-changing internal misalignment couplings.
b. Heavy-mediated source. The $\mathcal{M}_{\text{high,wrong},\rho}^{UV}$ terms come from Class II-mediated transport with one leg in the higher-shell branch and one leg in the wrong-internal branch.

F. (vi) Same-shell non-enriched–wrong-internal pair

$$\eta_{\text{non,wrong},\rho} \leq \mathcal{D}_{\text{non,wrong},\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{\text{non,wrong},\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{\text{non,wrong},\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{\text{non,wrong},\rho}^{KK}.$$

- a. Direct source.* $\mathcal{D}_{\text{non,wrong},\rho}$ is the direct same-shell internal mixing between the non-enriched branch and the wrong-internal branch.
b. Heavy-mediated source. The $\mathcal{M}_{\text{non,wrong},\rho}^{UV}$ terms are the corresponding Class II-mediated internal/off-diagonal transport couplings.

CCCLIII. 4. COMPACT SIX-PAIR MASTER BOUND

Combining all six unique pairs, we obtain

$$\eta_{R,\rho}^{\text{tr}} \leq 2 \sum_{(ab) \in \mathfrak{P}_6} \left[\mathcal{D}_{ab,\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{ab,\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{KK} \right],$$

where

$$\mathfrak{P}_6 = \{(\text{vol}, \text{high}), (\text{vol}, \text{non}), (\text{vol}, \text{wrong}), (\text{high}, \text{non}), (\text{high}, \text{wrong}), (\text{non}, \text{wrong})\}.$$

This is the explicit pairwise audit formula for the truncated remote budget.

CCCLIV. 5. GATE-1 MASTER MARGIN IN FULLY EXPANDED PAIRWISE FORM

Substituting the pairwise audit into the residual decomposition, the Gate-1 sufficient condition becomes

$$\Delta_{\text{bench},\rho}^{\text{fin}} > 2 \sum_{(ab) \in \mathfrak{P}_6} \left[\mathcal{D}_{ab,\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{ab,\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{KK} \right] + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho}.$$

Equivalently, the remote-gap margin is

$$\mathfrak{M}_{\text{rem}}(\rho) = \Delta_{\text{bench},\rho}^{\text{fin}} - 2 \sum_{(ab) \in \mathfrak{P}_6} \left[\mathcal{D}_{ab,\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{ab,\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{KK} \right] - \eta_{\text{tail},\rho} - \eta_{\text{diag},\rho}.$$

The gate passes whenever

$$\mathfrak{M}_{\text{rem}}(\rho) > 0.$$

CCCLV. 6. HONEST CURRENT STATUS

The pairwise audit is now structurally complete.

What remains explicitly unevaluated is not the form of the decomposition, but only the concrete values of:

$$\mathcal{D}_{ab,\rho}, \quad \mathcal{M}_{ab,\rho}^{JJ}, \quad \mathcal{M}_{ab,\rho}^{JK}, \quad \mathcal{M}_{ab,\rho}^{KK}, \quad \eta_{\text{tail},\rho}, \quad \eta_{\text{diag},\rho}.$$

Thus the truncated remote budget is no longer conceptually open. It has been reduced to a finite six-pair source audit plus the tail/diagonal remainders.

CCCLVI. FINAL SUMMARY

The truncated remote budget is fully resolved into six unordered sector pairs:

$$(\text{vol}, \text{high}), (\text{vol}, \text{non}), (\text{vol}, \text{wrong}), (\text{high}, \text{non}), (\text{high}, \text{wrong}), (\text{non}, \text{wrong}).$$

For each pair,

$$\eta_{ab,\rho} \leq \mathcal{D}_{ab,\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{ab,\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{KK},$$

where:

- $\mathcal{D}_{ab,\rho}$ is the direct off-diagonal budget from H_{dir} ;
- $\mathcal{M}_{ab,\rho}^{JJ}$, $\mathcal{M}_{ab,\rho}^{JK}$, $\mathcal{M}_{ab,\rho}^{KK}$ are the three heavy-mediated Class II budgets.

Therefore

$$\eta_{R,\rho}^{\text{tr}} \leq 2 \sum_{(ab) \in \mathfrak{P}_6} \left[\mathcal{D}_{ab,\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{ab,\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{KK} \right].$$

Combining this with the tail and diagonal budgets, the full Gate-1 sufficient condition becomes

$$\Delta_{\text{bench},\rho}^{\text{fin}} > 2 \sum_{(ab) \in \mathfrak{P}_6} \left[\mathcal{D}_{ab,\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{ab,\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{KK} \right] + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho}.$$

Thus Gate 1 has now been reduced to a finite six-pair source audit plus tail/diagonal remainder bounds.

CCCLVII. DOMINANT-PAIR REDUCED GATE-1 CONDITION

Under the current structural assumptions, the dominant truncated residual budget is carried by the pairs

$$(\text{high}, \text{non}), \quad (\text{non}, \text{wrong}).$$

All other pairs are suppressed by either:

- volumetric mass gap M_σ ,
- shell mismatch gap,
- or double suppression.

Therefore,

$$\eta_{R,\rho}^{\text{tr}} \leq 2 \left(\eta_{\text{high},\text{non},\rho} + \eta_{\text{non},\text{wrong},\rho} \right) + \text{subleading}.$$

Hence the Gate-1 sufficient condition reduces to

$$\Delta_{\text{bench},\rho}^{\text{fin}} > 2 \left(\eta_{\text{high},\text{non},\rho} + \eta_{\text{non},\text{wrong},\rho} \right) + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho}.$$

CCCLVIII. FINAL PRACTICAL GATE-1 INEQUALITY

At leading order, the dominant contribution to the truncated residual budget is

$$\eta_{\text{non},\text{wrong},\rho}.$$

Therefore a practical sufficient condition for Gate 1 is

$$\Delta_{\text{bench},\rho}^{\text{fin}} > 2 \eta_{\text{non},\text{wrong},\rho} + O(\rho) + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho}.$$

In particular, in the small- ρ regime,

$$\Delta_{\text{bench},\rho}^{\text{fin}} > 2 \eta_{\text{non},\text{wrong},\rho}$$

is the leading-order control condition.

CCCLIX. DIRECT NON-WRONG MIXING

The direct mixing operator is

$$H_{\text{dir}}^{\text{non},\text{wrong}} = Q_{\text{non}} V_{\perp}^{\mu} \partial_{\mu} Q_{\text{wrong}}.$$

Using the TD/DT result,

$$V_{\perp}^{\mu} = \theta'(\xi) [C_E E_a^{\mu} \Sigma_a + C_O J_a^{\mu} \Sigma_a],$$

we obtain

$$H_{\text{dir}}^{\text{non},\text{wrong}} = \theta'(\xi) [C_E Q_{\text{non}} (E_a^{\mu} \Sigma_a) Q_{\text{wrong}} + C_O Q_{\text{non}} (J_a^{\mu} \Sigma_a) Q_{\text{wrong}}] \partial_{\mu}.$$

CCCLX. HEAVY-MEDIATED NON-WRONG MIXING

The mediated contribution is

$$Q_{\text{non}} \Delta_X Q_{\text{wrong}} = -\frac{1}{M_X^2} Q_{\text{non}} (\alpha_X \mathcal{J} + \beta_X \mathcal{K}) (\alpha_X \mathcal{J} + \beta_X \mathcal{K})^\dagger Q_{\text{wrong}}.$$

Expanding,

$$= -\frac{1}{M_X^2} \left[\alpha_X^2 Q_{\text{non}} \mathcal{J} \mathcal{J}^\dagger Q_{\text{wrong}} + \alpha_X \beta_X Q_{\text{non}} (\mathcal{J} \mathcal{K}^\dagger + \mathcal{K} \mathcal{J}^\dagger) Q_{\text{wrong}} + \beta_X^2 Q_{\text{non}} \mathcal{K} \mathcal{K}^\dagger Q_{\text{wrong}} \right].$$

CCCLXI. FINAL BOUND FOR NON-WRONG MIXING

Combining the direct and mediated contributions, we obtain

$$\eta_{\text{non,wrong},\rho} \leq \kappa \rho (|C_E| + |C_O|) + \frac{1}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2).$$

CCCLXII. FINAL GATE-1 CONDITION (FULLY EXPLICIT)

A sufficient condition for Gate 1 is

$$\Delta_{\text{bench},\rho}^{\text{fin}} > 2 \left[\kappa \rho (|C_E| + |C_O|) + \frac{1}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2) \right] + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho}.$$

CCCLXIII. ACTUAL NEXT TARGET: EXTRACT THE ENRICHED SHELL DIRAC CARRIER

The current most useful direction is not further polishing of the TD/DT summary or further expansion of the remote-gap bookkeeping. According to the current notes, the actual completion criterion is finite and concrete:

$$H_{\text{eff}}(k) \rightarrow H_{\text{shell}}(p) \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

Therefore the next nonredundant task is to construct the enriched shell carrier explicitly.

CCCLXIV. 1. SHELL-PROJECTED OPERATOR

Let

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k)$$

be the current refined locked + Class II effective Bloch operator.

Choose the resonant shell S_* around the folded Bragg point $k_* = G_*/2$, and define

$$H_{\text{shell}}^{(N)}(p) := P_{S_*}^{(N)} H_{\text{eff}}(k_* + p) P_{S_*}^{(N)}, \quad |p| \leq \rho.$$

Here $P_{S_*}^{(N)}$ is a finite shell truncation retaining the enriched harmonics needed for the low Dirac carrier.

CCCLXV. 2. ENRICHED SHELL DECOMPOSITION

The shell space is decomposed as

$$\mathcal{H}_{\text{shell}}^{(N)} = \mathcal{H}_D^{(N)} \oplus \mathcal{H}_{\text{non-enr}}^{(N)},$$

where

- $\mathcal{H}_D^{(N)}$ is the enriched 4-dimensional Dirac carrier candidate,
- $\mathcal{H}_{\text{non-enr}}^{(N)}$ is the same-shell complement.

This is exactly the practical route stated in the notes: construct $H_{\text{shell}}^{(N)}(p)$, identify the enriched 4-dimensional Dirac carrier, then treat the complement as the same-shell non-enriched slot.

Let

$$P_D^{(N)}$$

be the orthogonal projector onto $\mathcal{H}_D^{(N)}$, and

$$Q_{\text{non}}^{(N)} := P_{S_*}^{(N)} - P_D^{(N)}.$$

CCCLXVI. 3. THE SHELL-CORE HAMILTONIAN

Define the enriched shell-core block

$$H_{\text{core}}^{(N)}(p) := P_D^{(N)} H_{\text{shell}}^{(N)}(p) P_D^{(N)}.$$

By the current shell-Dirac notes, the correct canonical target form is

$$H_{\text{core}}^{\text{can}}(p) = E_*(p) \mathbf{1}_4 + m \tau^1 \otimes \mathbf{1}_2 + v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + \tau^1 \otimes (p_1 W^1 + p_2 W^2) + O(|p|^2).$$

The shell block becomes a massless Dirac principal symbol if and only if:

$$m = 0,$$

$$v_{\parallel} \neq 0,$$

$$W^1, W^2 \text{ are traceless Hermitian, nonzero, and linearly independent.}$$

After internal basis reduction this becomes

$$W^1 = \nu_1 s^1, \quad W^2 = \nu_2 s^2, \quad \nu_1 \nu_2 \neq 0.$$

These are exactly the shell-level Dirac carrier conditions isolated in the notes.

CCCLXVII. 4. WHY THIS IS THE REAL FRONTIER

The notes are explicit that the remaining open step is not another abstract witness theorem, but the finite enriched Bloch calculation:

$$\text{construct } H_{\text{eff}}(k), H_{\text{shell}}(p), P_* K_i P_*, \text{ and extract } (\lambda_{\parallel}, \alpha, \beta).$$

Moreover, the same-shell non-enriched slot is already reduced to one concrete finite matrix plus one perturbative radius estimate.

Therefore the real next theorem target is:

$$\text{construct } H_{\text{core}}^{(N)}(p) \text{ explicitly and show that it satisfies } m = 0, v_{\parallel} \neq 0, W^1, W^2 \text{ full-rank.}$$

CCCLXVIII. 5. FINITE COMPUTATION PROTOCOL

The actual finite computation is now:

1. construct $H_{\text{eff}}(k)$ from the refined locked + Class II branch;
2. choose the near-resonant harmonic set S_* ;
3. build $H_{\text{shell}}^{(N)}(p)$;
4. identify the enriched 4-dimensional candidate carrier $\mathcal{H}_D^{(N)}$;
5. compute

$$H_{\text{core}}^{(N)}(0)$$

and check whether the constant mass term m vanishes;

6. compute

$$K_i^{(N)} := \partial_{p_i} H_{\text{core}}^{(N)}(0);$$

7. decompose the linear symbol into

$$v_{\parallel}, \quad W^1, \quad W^2;$$

8. verify

$$v_{\parallel} \neq 0, \quad W^1, W^2 \text{ Pauli-full-rank.}$$

After that, one no longer reproves the general Dirac theorem. One simply reads off

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

from the three Pauli projections.

CCCLXIX. 6. REPOSITIONING GATE 1 CORRECTLY

Gate 1 should now be viewed as a support condition:

$$\Delta_{\text{bench}, \rho}^{\text{fin}} > \eta_{R, \rho} \implies \text{the enriched shell carrier is spectrally isolated from the remote sector.}$$

It is necessary, but it is not the actual last frontier. The actual frontier is the explicit calculation of the enriched shell-core symbol itself.

CCCLXX. 7. HONEST CURRENT STATUS

The correct current status is therefore:

1. the reduced anisotropic Dirac architecture is already known abstractly;
2. the refined locked + Class II branch is the correct current microscopic branch;
3. the same-shell non-enriched problem is already reduced to a finite Hermitian matrix benchmark;
4. the genuinely remaining central task is to compute the enriched shell-core block and extract

$$(\lambda_{\parallel}, \alpha, \beta).$$

So the program should now push the enriched shell carrier directly.

CCCLXXI. FINAL SUMMARY

The most useful next step is not further TD/DT polishing and not further expansion of Gate 1 bookkeeping. The actual completion criterion, according to the current notes, is the finite enriched Bloch calculation

$$H_{\text{eff}}(k) \rightarrow H_{\text{shell}}^{(N)}(p) \rightarrow \mathcal{H}_D^{(N)} \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

The enriched shell core must be put into the canonical form

$$H_{\text{core}}^{\text{can}}(p) = E_*(p) \mathbf{1}_4 + m \tau^1 \otimes \mathbf{1}_2 + v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + \tau^1 \otimes (p_1 W^1 + p_2 W^2) + O(|p|^2),$$

and the actual Dirac carrier conditions are

$$m = 0, \quad v_{\parallel} \neq 0, \quad W^1, W^2 \text{ traceless Hermitian and linearly independent.}$$

Gate 1 remains necessary, but only as a support theorem guaranteeing that the enriched shell carrier is spectrally isolated. The real frontier is the explicit evaluation of the enriched shell-core block and the extraction of the three coefficients

$$(\lambda_{\parallel}, \alpha, \beta).$$

CCCLXXII. PROJECT: TECT — MASTER CONTENTS FOR CONTINUATION (ENRICHED SHELL DIRAC CARRIER STAGE)

CCCLXXIII. CURRENT MAIN GOAL

The current main goal is no longer to polish support theorems in isolation. The actual frontier is now:

construct the enriched shell Dirac carrier explicitly and extract $(\lambda_{\parallel}, \alpha, \beta)$ from the refined locked + Class II branch.

(19)

Equivalently, the main completion chain is

$H_{\text{eff}}(k) \longrightarrow H_{\text{shell}}^{(N)}(p) \longrightarrow \mathcal{H}_D^{(N)} \longrightarrow P_* K_i P_* \longrightarrow (\lambda_{\parallel}, \alpha, \beta).$

(20)

This is the current nonredundant target.

CCCLXXIV. WHAT IS STRUCTURALLY ALREADY FIXED

CCCLXXV. WORKING MICROSCOPIC BRANCH

The current strongest explicit microscopic working branch is

$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi, P] + \beta_X X_i^a K_i^a[P, \Psi].$

(21)

This is *not* yet claimed to be the final complete TECT microscopic action. It is the strongest explicit working microscopic ansatz currently available.

CCCLXXVI. LIGHT/HEAVY REDUCTION

The exact reduced operator skeleton is already fixed:

$$\boxed{\mathcal{Y}_{\text{eff}} = L - KH^{-1}K^\dagger.} \quad (22)$$

In the Bloch/Class II realization this becomes

$$\boxed{H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k)H_{XX}(k)^{-1}H_{X\Psi}(k),} \quad (23)$$

with

$$\boxed{H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k).} \quad (24)$$

In the auxiliary heavy Class II truncation,

$$\boxed{H_{XX}(k) = M_X^2 \mathbf{1}_X.} \quad (25)$$

Therefore

$$\boxed{\Delta_X(k) = -\frac{1}{M_X^2} (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)) (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger.} \quad (26)$$

CCCLXXVII. SHELL PROJECTION AND VALLEY EXTRACTION

The shell-projected operator is

$$\boxed{H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}.} \quad (27)$$

The low-valley projector is defined spectrally as

$$\boxed{P_* = \frac{1}{2\pi i} \oint_{\Gamma} (z - H_{\text{shell}}(0))^{-1} dz.} \quad (28)$$

The projected transport matrices are

$$\boxed{M_i = P_* \partial_{k_i} H_{\text{eff}}(k)|_{k=k_*} P_*}. \quad (29)$$

In the adapted defect frame $(\hat{n}, e_1, e_2)$,

$$M_{\parallel} = \hat{n}_i M_i, \quad M_1 = e_{1i} M_i, \quad M_2 = e_{2i} M_i. \quad (30)$$

Under the locked Pauli selection rule,

$$\boxed{M_{\parallel} = \lambda_{\parallel} \sigma_3, \quad M_1 = \alpha \sigma_1 + \beta \sigma_2, \quad M_2 = \alpha \sigma_2 - \beta \sigma_1.} \quad (31)$$

Hence the extracted reduced coefficients are

$$\boxed{\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}),} \quad (32)$$

$$\boxed{\alpha = \frac{1}{4}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4}(\sigma_2 M_1 - \sigma_1 M_2).} \quad (33)$$

CCCLXXVIII. EXACT OVERLAP NORMALIZATION LAYER

The Bloch/transport overlap constants are defined by

$$\boxed{\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.} \quad (34)$$

So

$$\boxed{g_c = \frac{\lambda_{\parallel}}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}.} \quad (35)$$

This normalization layer is exact once C_c, C_E, C_O are calibrated.

CCCLXXIX. CURRENT HONEST STATUS OF THE TRANSVERSE SECTOR

The reduced anisotropic Dirac-type PDE form is already structurally fixed, but the microscopic transverse frontier must be stated honestly.

A naive local real Hermitian JK bilinear vanishes because

$$\{\Sigma_3, \Sigma_1\} = \{\Sigma_3, \Sigma_2\} = 0. \quad (36)$$

Therefore the genuine microscopic transverse frontier is the derivative-resolved shell audit:

$$\boxed{\partial_{p\nu} \widehat{\Sigma}_{JK}(G_*; p)|_{p=0}, \quad \partial_{p\nu} \widehat{\Sigma}_{KK}(G_*; p)|_{p=0}.} \quad (37)$$

However, within the current explicit LO/Class II realization, the TD/DT sector may still be used as a branch-level reduced summary:

$$\boxed{e_0^{(1)} = \frac{1}{2}(A_E + B_E), \quad o_0^{(1)} = \frac{1}{2}(A_O + B_O),} \quad (38)$$

and

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_E + B_E), \quad \beta = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_O + B_O).} \quad (39)$$

This TD/DT formula should be described only as a *branch-level reduced transverse summary*, not yet as the unconditional final microscopic closure of the full TECT PDE.

CCCLXXX. SINGLE-DEFECT CONSTITUTIVE/RADIAL REDUCTION

A minimal working constitutive ansatz compatible with the locked BCC background has already been fixed. The minimal microscopic multiplet is

$$\boxed{(P^A, \Psi, X_i^a).} \quad (40)$$

The defect-background first-order coefficients are

$$\boxed{K_{\text{def}}^0(x) = z_0(\bar{\sigma}(x))\mathbf{1},} \quad (41)$$

$$K_{\text{def}}^i(x) = g_E^{\text{eff}}(x)\Gamma^i + g_O^{\text{eff}}(x)\epsilon_{i\ell m}\hat{n}_m\Gamma^\ell + g_c^{\text{eff}}(x)\hat{n}_i\Gamma_{\hat{n}}, \quad (42)$$

with

$$g_E^{\text{eff}} = v_\perp + 3\chi_X v_E^{(X)}, \quad g_O^{\text{eff}} = 3\chi_X v_O^{(X)}, \quad g_c^{\text{eff}} = v_c + 3\chi_X v_c^{(X)}. \quad (43)$$

The effective mass profile is

$$m_{\text{eff}}(x) = m(\bar{\sigma}(x)) - 3\chi_X(x)u_X(\bar{\sigma}(x)). \quad (44)$$

After radial defect specialization, the canonically normalized single-defect operator becomes

$$\mathbb{H}_{\text{def}} = -\frac{i g_R(r)}{z_0(r)}\Gamma_{\hat{n}}\partial_r - \frac{i g_E(r)}{z_0(r)r}\Gamma_i^\perp\nabla_i^{S^2} - \frac{i g_O(r)}{z_0(r)r}J_i(\hat{n})\nabla_i^{S^2} + W_{\text{def}}(r, \hat{n}), \quad (45)$$

with

$$g_R(r) = g_E(r) + g_c(r). \quad (46)$$

CCCLXXXI. CALLIAS CORE AND ONE-DEFECT/ONE-MODE CRITERION

The graded Callias core is

$$\mathbb{H}_C = -i a(r)\Gamma_{\hat{n}}\partial_r + \frac{b(r)}{r}\Gamma_{\hat{n}}\mathcal{D}_{S^2} - i\omega_r(r)\Gamma_{\hat{n}} + \mu(r)\Gamma^\sharp, \quad (47)$$

where

$$a(r) = \frac{g_R(r)}{z_0(r)}, \quad b(r) = \frac{g_E(r)}{z_0(r)}, \quad \mu(r) = \frac{m_{\text{eff}}(r)}{z_0(r)}. \quad (48)$$

For each angular channel κ , the real superpotential is

$$W_\kappa^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}. \quad (49)$$

The protected-channel counting is controlled entirely by this real superpotential. The odd tangential coefficient g_O changes only the phase of the radial generator and does not affect L^2 -normalizability.

A working sufficient criterion for one defect / one protected mode is:

$$\mu_\infty > 0, \quad a_0 > 0, \quad b_0 > 0, \quad \frac{|\kappa_*|b_0}{a_0} > \frac{1}{2}, \quad (50)$$

together with competitor exclusion for all other projected channels.

CCCLXXXII. REMOTE-GAP THEOREM AND ITS CORRECT ROLE

The full remote-gap sufficient criterion is

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}.} \quad (51)$$

The current working benchmark side is

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}.} \quad (52)$$

The same-shell non-enriched entry has been reduced to a finite Hermitian eigenvalue problem:

$$\boxed{\delta_{\text{non-enr},\rho} \geq \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}} \quad (53)$$

The residual side has been cleanly decomposed as

$$\boxed{\eta_{R,\rho} \leq \eta_{R,\rho}^{\text{tr}} + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho},} \quad (54)$$

with

$$\boxed{\eta_{R,\rho}^{\text{tr}} \leq 2 \sum_{(ab) \in \mathfrak{P}_6} \left[\mathcal{D}_{ab,\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{ab,\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{KK} \right].} \quad (55)$$

Gate 1 is necessary, but it is now correctly understood as a *support theorem*: it guarantees that the enriched low slot is spectrally isolated from the remote sector. It is no longer the main frontier.

CCCLXXXIII. WHAT IS ACTUALLY FINISHED

The following are already structurally closed:

1. the light/heavy Schur-complement reduction;
2. the refined locked + Class II working microscopic branch;
3. the shell projection and valley projector formalism;
4. the Pauli extraction formulas for $(\lambda_{\parallel}, \alpha, \beta)$;
5. the minimal constitutive/radial reduction;
6. the Callias-core one-channel criterion;
7. the remote-gap theorem and its benchmark/residual decomposition;
8. the reduction of the same-shell non-enriched benchmark to a finite Hermitian matrix problem.

CCCLXXXIV. WHAT IS STILL NOT FINISHED

The actual unfinished frontier is now very specific:

$$\boxed{\text{explicitly construct the enriched shell core } H_{\text{core}}^{(N)}(p) \text{ and verify that it is a massless Dirac carrier.}} \quad (56)$$

This means:

1. build $H_{\text{shell}}^{(N)}(p)$;
2. identify the enriched 4-dimensional Dirac carrier candidate $\mathcal{H}_D^{(N)}$;
3. compute

$$H_{\text{core}}^{(N)}(p) := P_D^{(N)} H_{\text{shell}}^{(N)}(p) P_D^{(N)};$$

4. extract the constant mass term m ;
5. compute the linear symbol coefficients $v_{\parallel}, W^1, W^2$.

The canonical target form is

$$H_{\text{core}}^{\text{can}}(p) = E_*(p) \mathbf{1}_4 + m \tau^1 \otimes \mathbf{1}_2 + v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + \tau^1 \otimes (p_1 W^1 + p_2 W^2) + O(|p|^2). \quad (57)$$

The explicit Dirac-carrier conditions are

$$m = 0, \quad v_{\parallel} \neq 0, \quad W^1, W^2 \text{ traceless Hermitian, nonzero, and linearly independent.} \quad (58)$$

After that, the actual extracted coefficients $(\lambda_{\parallel}, \alpha, \beta)$ are simply read off from the Pauli projections.

CCCLXXXV. WHAT REMAINS ASSUMPTION-LIKE / BRANCH-SPECIFIC

The following are still branch-specific and not yet fully verified:

1. the explicit finite basis choice defining $\mathcal{H}_D^{(N)}$;
2. the actual explicit computation of $H_{\text{core}}^{(N)}(p)$;
3. the exact overlap constants C_c, C_E, C_O via calibration;
4. the matching constants ξ_c, ξ_E, ξ_O ;
5. the uniqueness of the signed lowest projected channel $\kappa_* < 0$;
6. the sign of the bulk mass gap μ_{∞} in the same working regime;
7. the simultaneous existence of a common parameter regime where

$$\Delta_{\text{bench}, \rho}^{\text{fin}} > \eta_{R, \rho}, \quad \mu_{\infty} > 0, \quad \text{and the Dirac-carrier conditions}$$

all hold together.

CCCLXXXVI. CURRENT HONEST SUMMARY

The correct current status is:

$$\boxed{\text{The reduced anisotropic Dirac architecture is structurally fixed.}} \quad (59)$$

$$\boxed{\text{The refined locked + Class II branch is the strongest current explicit microscopic realization.}} \quad (60)$$

$$\boxed{\text{The same-shell non-enriched and remote-gap layers are support problems, largely reduced to finite explicit computations.}} \quad (61)$$

$$\boxed{\text{The genuinely central unfinished task is the explicit enriched shell-core Dirac extraction.}} \quad (62)$$

CCCLXXXVII. IMMEDIATE NEXT TASK FOR THE NEXT CONVERSATION

The best nonredundant next step is:

$$\boxed{\text{fix the explicit block basis for } \mathcal{H}_D^{(N)} \text{ inside } H_{\text{shell}}^{(N)}(p), \text{ and write } H_{\text{core}}^{(N)}(p) \text{ explicitly.}} \quad (63)$$

Concretely, the next conversation should begin with:

1. choosing the enriched 4-dimensional shell basis;
2. writing the projected block matrix $H_{\text{core}}^{(N)}(p)$;
3. separating its constant and linear pieces;
4. checking

$$m = 0, \quad v_{\parallel} \neq 0, \quad W^1, W^2 \text{ full-rank.}$$

CCCLXXXVIII. BOTTOM-LINE CONTINUATION TAG

$$\boxed{\text{Next chat should start from: “construct the explicit enriched shell basis and compute } H_{\text{core}}^{(N)}(p).”} \quad (64)$$

CCCLXXXIX. FINAL SUMMARY

The current main frontier is no longer support-theorem polishing. It is the explicit enriched shell Dirac-carrier extraction:

$$H_{\text{eff}}(k) \rightarrow H_{\text{shell}}^{(N)}(p) \rightarrow \mathcal{H}_D^{(N)} \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

What is already fixed:

1. the refined locked + Class II working microscopic branch;
2. the Schur-complement reduction;
3. the shell/valley extraction formulas for $(\lambda_{\parallel}, \alpha, \beta)$;
4. the radial constitutive reduction and the Callias-core one-channel criterion;
5. the remote-gap theorem and the reduction of the same-shell non-enriched benchmark to a finite Hermitian problem.

What remains central and unfinished:

1. choose the explicit enriched 4-dimensional shell basis $\mathcal{H}_D^{(N)}$;
2. compute the projected core block

$$H_{\text{core}}^{(N)}(p) = P_D^{(N)} H_{\text{shell}}^{(N)}(p) P_D^{(N)};$$

3. verify the Dirac-carrier conditions

$$m = 0, \quad v_{\parallel} \neq 0, \quad W^1, W^2 \text{ traceless Hermitian and linearly independent;}$$

4. then read off the extracted transport coefficients

$$(\lambda_{\parallel}, \alpha, \beta).$$

Thus the honest current status is:

the architecture is built, the support layers are mostly reduced to finite checks, and the main unfinished frontier is the explicit enriched shell-core computation.